

GENERAL RELATIVITY

Y. Ding
March 27, 2026

CONTENTS

1	Introduction	8
1.1	Gravity vs. Spacetime Curvature	8
1.1.1	Gravity	8
1.1.2	Geometry and Curvature	8
1.2	Newtonian Mechanics and Gravity	10
1.2.1	Inertial Frames and Galilean Transformation	10
1.2.2	Newtonian Gravity	11
1.3	Relativistic Gravity	12
2	Special Relativity	13
2.1	Spacetime Geometry	13
2.1.1	Minkowski Spacetime	13
2.1.2	Time Dilation	14
2.2	Relativistic Kinematics	14
2.2.1	Lorentz Transformation	14
2.3	Relativistic Dynamics	15
2.3.1	Four-Vectors	15
2.3.2	4-Velocity and 4-Momentum	16
2.3.3	Action and Lagrangian for a Free Particle	17
2.3.4	Observers and Observations	19
2.4	Vectors and Tensors	21
2.4.1	Vectors	21
2.4.2	Dual Vectors	21
2.4.3	Tensors	22
2.4.4	Symmetric and Antisymmetric Tensors	23
2.4.5	The Electromagnetic Field Tensor	23
2.5	Classical Field Theory	24
2.5.1	Real Scalar Fields	25
2.5.2	The 4-Potential	25
2.6	The Stress-Energy Tensor	26
2.6.1	Perfect Fluids	27
2.6.2	Electromagnetic Fields	28
3	Differential Geometry	30

3.1	Manifolds	30
3.1.1	Sets and Maps	30
3.1.2	Formal Definition of a Manifold	31
3.2	Vectors and Tensors	31
3.2.1	Directional Derivatives	31
3.2.2	Commutators	33
3.2.3	Tensors	33
3.3	The Metric Tensor	33
3.3.1	Properties of the Metric Tensor	33
3.3.2	Local Inertial Frames	34
3.3.3	Tensor Densities	36
3.4	Differential Forms	37
3.4.1	Exterior Derivatives	37
3.4.2	Integration	39
4	Curvature	41
4.1	On the Relation of Gravitation to Curvature	41
4.1.1	The Principle of Equivalence	41
4.1.2	The Geodesic Equation	42
4.2	The Covariant Derivative*	43
4.2.1	Connection Coefficients	44
4.2.2	The Covariant Derivative	45
4.2.3	Coordinate Transformations	47
4.2.4	Commutator of Vector Fields	48
4.2.5	Stokes's Theorem on Curved Manifolds	49
4.3	Parallel-Transport and Geodesics	49
4.3.1	The Equation of Parallel Transport	49
4.3.2	The Geodesic Equation	50
4.4	Curvature*	50
4.4.1	The Riemann Curvature Tensor	50
4.4.2	Properties of the Curvature Tensor	52
4.4.3	Coordinate Component Representation	55
4.4.4	Geodesic Deviation	55
4.4.5	The Ricci Tensor, Ricci Scalar, Weyl Tensor, and Einstein Tensor	56
4.5	Symmetries and Conserved Quantities	57
4.5.1	The Lie Derivative	57
4.5.2	Killing Vectors	58

4.5.3	Killing Tensors	60
4.5.4	Relation to Curvature	60
5	Gravitation	62
5.1	Transition from Special Relativity to General Relativity	62
5.1.1	From Newton's Second Law to Geodesic Equation	62
5.1.2	The Stress-Energy Tensor	63
5.1.3	Geometrized Units	63
5.2	Einstein's Equation	63
5.2.1	From Poisson's Equation to Einstein's Equation	63
5.2.2	Properties of Einstein's Equation	65
5.3	The Lagrangian Formulation	66
5.3.1	The Einstein-Hilbert Action	66
5.3.2	Einstein's Equation in Vacuum	67
5.3.3	Einstein's Equation with Matter	70
5.4	The Cosmological Constant	71
6	Weak Fields and Gravitational Radiation	73
6.1	Linearized Gravity	73
6.1.1	Weak-Field Einstein's Equation	73
6.1.2	Gauge Transformations	74
6.2	The Newtonian Limit	76
6.2.1	The Newtonian Metric	76
6.2.2	Gravitational Lensing I: Newtonian Solution	77
6.3	The Gravitational Wave Solution	77
6.3.1	General Solutions	77
6.3.2	The Transverse-Traceless Gauge	77
6.3.3	Effects on Free Particles	79
6.3.4	Polarization	80
6.4	Generation of Gravitational Waves	81
6.4.1	The Quadrupole Moment Tensor	81
6.4.2	Energy Loss	82
6.4.3	Exact Solutions	82
7	The Schwarzschild Solution	83
7.1	The Schwarzschild Metric	83
7.1.1	Conditions on the Metric	83
7.1.2	The Schwarzschild Metric	84

7.1.3	Gravitational Redshift	85
7.2	Birkhoff's Theorem	86
7.3	Orbits in Schwarzschild Spacetime	86
7.3.1	The Effective Potential	87
7.3.2	Radial Plunge Orbits	88
7.3.3	Circular Orbits	90
7.3.4	Elliptical Orbits and Precession	91
7.3.5	Photon Orbits	92
7.3.6	Gravitational Lensing II: Schwarzschild Solution	93
7.3.7	The Shapiro Delay	95
7.4	Static, Spherically Symmetric Stars	95
7.4.1	Interior Solutions	96
7.4.2	Buchdahl's Theorem	98
7.4.3	Realistic Stars	98
8	Black Holes	99
8.1	Schwarzschild Black Holes	99
8.1.1	Eddington-Finkelstein Coordinates	99
8.1.2	Kruskal-Szekeres Coordinates	100
8.1.3	Other Coordinates	103
8.1.4	Schwarzschild Solution with $M < 0$	103
8.2	General Black Holes	103
8.2.1	Black Hole Theorems	103
8.2.2	Penrose Diagrams	104
8.2.3	Mass, Charge, and Angular Momentum	106
8.3	Charged (Reissner-Nordström) Black Holes	106
8.3.1	The Reissner-Nordström Metric	106
8.3.2	$Q^2 > M^2$ Solution	107
8.3.3	$Q^2 < M^2$ Solution	108
8.3.4	$Q^2 = M^2$ Solution	109
8.4	Rotating (Kerr) Black Holes	109
8.4.1	The Kerr Metric	109
8.4.2	The Event Horizon and Singularity	111
8.4.3	Ergosphere	111
8.4.4	Equatorial Photon Trajectories	112
8.4.5	The Penrose Process	113
8.5	Black Hole Thermodynamics	116

8.5.1	Killing Horizons	116
8.5.2	Surface Gravity	116
8.5.3	Laws of Black Hole Mechanics	119
9	Homogeneous Cosmology	121
9.1	Homogeneous, Isotropic Universe	121
9.1.1	Scales, Units, and the Cosmological Principle	121
9.1.2	Mathematical Definition of Homogeneity and Isotropy	121
9.2	The Robertson-Walker Metric	122
9.2.1	Spaces of Constant Curvature	122
9.2.2	The Robertson-Walker Metric	123
9.3	The Friedmann Equation	123
9.3.1	The Curvature Tensors	123
9.3.2	Evolution of Energy Density	124
9.3.3	The Friedmann Equation	125
9.3.4	Cosmological Parameters	126
9.4	Model Universes	127
9.4.1	Empty (Milne) Universe	127
9.4.2	Single-Component Universes	128
9.4.3	Multiple-Component Universes	129
9.4.4	The Λ CDM Model	131
9.5	The History of the Universe	132
10	Global Structures and Singularities	135
10.1	Futures and Pasts	135
10.1.1	Causality	135
10.1.2	Domain of Influence	136
10.1.3	Domain of Dependence	136
10.1.4	Cauchy Surface	137
10.2	Singularity Theorems	137
10.2.1	Local Argument	137
10.2.2	Global Argument	138
10.2.3	Singularity Theorems	138
11	Quantum Effects in Curved Spacetime	140
A	Proofs and Derivations	142
A.1	Special Relativity	142

A.1.1	Lorentz Transformation	142
A.2	Differential Geometry and Curvature	144
A.2.1	Determinant	144
A.2.2	Christoffel Symbols Transformation Law	145
A.3	The Geodesic Equation	146
A.3.1	Affinely Parametrized Geodesic Equation	146
A.4	Gravitational Waves	148
A.4.1	Circularly Polarized Gravitational Waves	148
A.5	The Schwarzschild Solution	148
A.5.1	The Periapsis Precession	148
A.5.2	The Shapiro Delay Integral	149
A.5.3	Static, Spherically Symmetric Perfect Fluid; Conservation of Stress-Energy Tensor	151
A.6	Black Holes	152
A.6.1	Kruskal-Szekeres Metric	152
A.6.2	Interesting Things about Black Holes	152
A.6.3	The Reissner-Nordström Metric	153

B	Curvature Tensors Reference	157
----------	------------------------------------	------------

To use this notes well, one should be familiar with various math tools and areas of physics:

1. Vector Calculus: partial derivatives, multiple integrals, the divergence theorem.
2. Analytical mechanics: calculus of variations, Lagrangian, principle of extremal action.
3. Other physics: Newtonian gravity, acquaintance with special relativity and electromagnetism.

Tensor analysis and some differential geometry will be covered in this notes.

1 INTRODUCTION

1.1 Gravity vs. Spacetime Curvature

1.1.1 Gravity

There are four fundamental forces or interactions in nature: gravity, electromagnetism (EM), strong, and weak. In terms of interaction range, gravity is a long-range force like EM. In terms of strength, gravity is the weakest among the four: about $\sim 10^{30}$ weaker than the other three. For example, the ratio of the gravitational force to electric repulsion between two protons is

$$\frac{F_g}{F_e} = \frac{Gm_p^2/r^2}{e^2/4\pi\epsilon_0 r^2} = \frac{4\pi\epsilon_0 Gm_p^2}{e^2} \sim 10^{-36}.$$

according to Newtonian gravity. Thus, gravity does not play a role in quantum physics because the masses of fundamental particles are very small. However, unlike charges of EM, which can be positive or negative, the charges of gravity, namely mass and energy, are always positive. On large length scales, objects are neutral, so the effect of EM weakens. Strong and weak interactions are short-range forces, so they don't play a role on these scales at all. But masses build up on these scales—as gravity is always attractive, it dominates on these scales and is important in astrophysics and cosmology.

Unlike the other three interactions, gravity does not have a well-posed, self-consistent, experimentally testable quantum theory. The understanding of gravity is a classical, but a large subject, known as Einstein's general theory of relativity, or **general relativity** (GR). In general relativity, gravity is no longer a force, but represents a curvature of spacetime. The mathematical tools necessary to study general relativity are tensor calculus and differential geometry. They are to general relativity as vector calculus is to Maxwell's theory of electromagnetism.

1.1.2 Geometry and Curvature

A curve is one-dimensional, meaning that it can be parametrized by only one parameter or variable. For example, in a three-dimensional Euclidean space with coordinates (x, y, z) , a parametric curve can be expressed as

$$\mathbf{r}(t) = x(t)\hat{\mathbf{x}} + y(t)\hat{\mathbf{y}} + z(t)\hat{\mathbf{z}}.$$

The **curvature** κ at any point with $\dot{\mathbf{r}} \neq 0$ is defined as

$$\kappa = \frac{|\ddot{\mathbf{r}} \times \dot{\mathbf{r}}|}{|\dot{\mathbf{r}}|^3}.$$

There is not much geometry going on in one dimension, so let's turn to two-dimensional spaces: a surface. A two-dimensional Euclidean surface is a flat space with zero curvature. The Euclidean axioms and geometry works on this surface. For example, the sum of the interior angles of a triangle is π ; the ratio of the circumference of a circle to its radius is 2π .

Then what are some surface with non-Euclidean geometry, or nonzero curvature?

Example 1.1. A 2-sphere

A **2-sphere** is just a fancy name of the surface of a sphere embedded in 3D Euclidean space. The “two” stands for two-dimensional—the surface of a sphere is parametrized by two variables. When we are working with curved space, we should be careful about what is meant by some geometric object. Common questions can be: what defines a straight line, or what defines a triangle?

A **straight line** is defined as the path of shortest distance between two points. A straight line on a 2-sphere is a segment of

a **great circle**. A great circle is the largest circle you can draw on a sphere. It is the intersection of the sphere with a plane containing the center of the sphere. Longitude lines are great circles; latitude lines except the equator are not great circles. A **triangle** is defined as a three-sided figure with three non-collinear points connected by straight lines. Thus, a triangle on a 2-sphere is called a spherical triangle. Its sides are paths along great circles.

All triangles on a 2-sphere has sum of the interior angles greater than π . For instance, you start from the north pole and walk south along the 0° longitude line. Once you reach the equator (call it point A), turn west (this is a 90° or $\pi/2$ turn) and walk towards the 90°W longitude along the equator. Once you reach the 90°W longitude (call it point B), you turn north (this is another 90° or $\pi/2$ turn). Eventually, you will return to the north pole along the 90°W longitude line, but you start with walking along the 0° longitude line. This means these two lines are also at angles 90° or $\pi/2$ (call this point C). The sum of the interior angles at A , B , and C is $3\pi/2 > \pi$. Note that this is a triangle because all its sides are segments of great circles (two longitudes and one equator).

In fact, the sum of the interior angles of a spherical triangle is related to its area S by

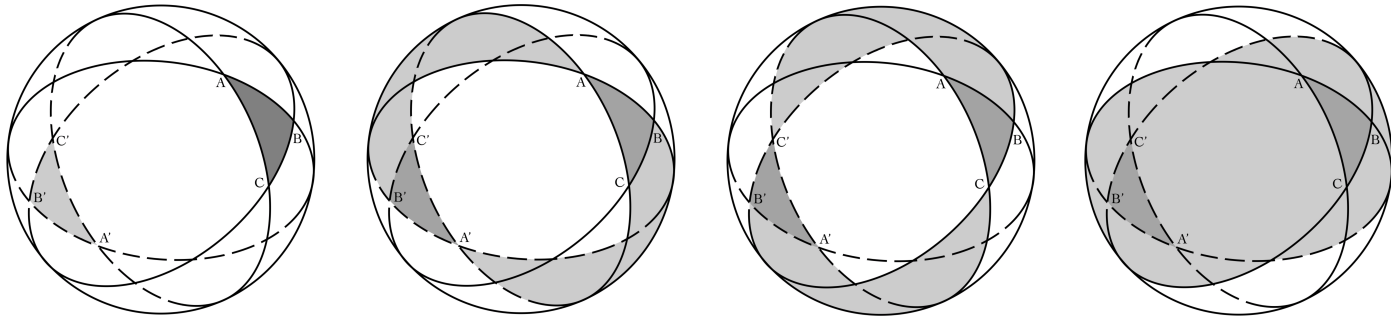
$$\sum_{\text{vertices}} \left(\begin{smallmatrix} \text{interior} \\ \text{angles} \end{smallmatrix} \right) = \pi + \frac{S}{R^2}, \quad (1.1)$$

where R is the radius of the 2-sphere, the distance from every point on the sphere to its center. All circles that are not great circles have the ratio of its circumference to radius

$$\frac{C}{r} = 2\pi \frac{\sin(r/R)}{r/R}.$$

Note that the radius r is defined as the distance of a straight line (defined on the 2-sphere) connecting one of the point on the circle to its center. The C/r ratio is easy to prove by making use of arc length formula $R\theta = r$ in spherical coordinates (θ, ϕ) and some trigonometry. Here we will prove (1.1).

Proof. First, we know that a spherical triangle is made from three great circles. The three great circles will intersect at six points, A , B , C , A' , B' , and C' . From the figure below, we can see that ABC forms a triangle, and $A'B'C'$ forms another congruent triangle opposite to $\triangle ABC$ by reflection symmetry around the center of the sphere.



Next, choose one vertex of $\triangle ABC$ and its corresponding reflection. For example, the second figure above chooses A and A' . The two great circles intersecting at A and A' enclose two congruent shaded area, each containing one of $\triangle ABC$ or $\triangle A'B'C'$. These shaded area are proportional to $\angle A$: if we take A to be the north pole ($\theta = 0$) and A' to be the south pole ($\theta = \pi$), then the shaded area is

$$S_A = S_{\text{sphere}} \cdot \frac{2\angle A}{2\pi} = 4\pi R^2 \frac{\angle A}{\pi} = 4R^2 \angle A.$$

We can do the same for B and B' (the fourth figure), and C and C' (the third figure),

$$S_B = 4R^2 \angle B, \quad S_C = 4R^2 \angle C.$$

If we sum up these shaded area, they will cover the entire sphere. But we are overcounting $\triangle ABC$ and $\triangle A'B'C'$ for additional two times, resulting in an additional area of $4S_{ABC}$. This means

$$S_A + S_B + S_C - 4S_{ABC} = S_{\text{sphere}} \implies 4R^2(\angle A + \angle B + \angle C) = 4\pi R^2 + 4S_{ABC}.$$

Divide both sides by $4R^2$, we have

$$\angle A + \angle B + \angle C = \pi + \frac{S_{ABC}}{R^2}.$$

□

Curvature and geometry in higher-dimensional spaces are hard to visualize. To study general relativity, one needs to get used to four-dimensional spacetime. Unfortunately, we are unable to visualize a four-dimensional object embedded in three-dimensional spaces, which is the space we live in. To work with four-dimensional spacetime, we need some intrinsic properties of curvature that does not require or depend on the existence of embeddings. We do this by studying the **Riemann curvature tensor**, $R^\mu{}_{\nu\alpha\beta}$, a rank-4 tensor whose indices μ , ν , α , and β can take on 1 to N for an N -dimensional space. This means there are N^4 components in this tensor, hence 256 components for a 4D spacetime. Luckily, most components are dependent on each other and are related by symmetry. In general, the number of independent components is $N^2(N^2 - 1)/12$. There is only 1 in 2D (which is called the *Gaussian curvature*), 6 in 3D, and 20 in 4D. We will see the Riemann curvature tensor in Chapter 4.

Mathematically, the Riemann curvature tensor is related to the second derivatives of the **metric tensor**, $g_{\mu\nu}$. The key thing about the metric tensor provides a way to define a line element. A **line element** ds^2 is a measure of the infinitesimal distance in terms of coordinates. In 3D Euclidean flat space with Cartesian coordinates (x, y, z) , the line element is given by

$$ds^2 = dx^2 + dy^2 + dz^2,$$

which is the famous Pythagorean theorem. Integrating $ds = \sqrt{dx^2 + dy^2 + dz^2}$ along a curve gives the length of the curve. In principle, with the line element, we are able to know everything about the geometry of a space. Of course we are not restricted to this coordinate system. Another useful coordinate to describe 3D Euclidean space is the spherical coordinates (r, θ, ϕ) . They are related to Cartesian coordinates by

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta.$$

Working out the differentials, one can find the line element to be

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

One thing to remember here is that even though ds^2 looks different in different coordinates, it is an invariant, scalar quantity that does not depend the coordinate you choose. One can derive the circumference of a circle of radius r to be $2\pi r$, no matter which coordinates they choose. Coordinates are *representations* of quantities, and we prefer one coordinate over the other for a certain problem because good coordinates make a problem simpler.

1.2 Newtonian Mechanics and Gravity

1.2.1 Inertial Frames and Galilean Transformation

Newtonian mechanics assumes that the geometry is flat and independent of time. Time is a universal quantity—all clocks tick at the same rate. In Cartesian coordinates (x, y, z) , the flat geometry is described by

$$ds^2 = dx^2 + dy^2 + dz^2.$$

The laws of Newtonian mechanics work the simplest in **inertial reference frames**. In an inertial frame, all free particles move in a straight line with constant velocity.

Suppose we find one inertial frame S and define an origin $\mathcal{O}$ in that frame. To measure time, we will also put a clock at the origin. We can find other inertial frames based on S . One (different) inertial frame S' can be a frame in which all axes (x', y', z') are parallel to axes (x, y, z) respectively in S . Moreover, the x' -axis is aligned with the x -axis. The clock in both frames are synchronized when their origins are at the same point. The frame S' is also moving with a velocity v with respect to S . All the above descriptions can be summarized to a coordinate transformation:

$$t' = t, \quad x' = x - vt, \quad y' = y, \quad z' = z.$$

This is a **Galilean transformation**.

In an inertial frame, if you do not have any communication with or observation from other frames, you cannot tell whether your frame is moving or not. Moreover, all laws of physics are the same in all inertial frames. This is known as the **principle of relativity**. By “same laws of physics” we mean the Lagrangian/Action is invariant, or the equations of motion are *covariant*. (Basically, covariant means that certain quantities change in the equations of motion, but the form of equations stays the same. For example, the electric or magnetic fields may look different in two inertial frames, but Maxwell’s equations are still true in both frames.)

1.2.2 Newtonian Gravity

In Newtonian gravity, we work with the law of universal gravitation:

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}}. \quad (1.2)$$

It describes the gravitational force $\mathbf{F}$ between two masses M and m separated by $\mathbf{r}$, and $G \approx 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the gravitational constant. The negative sign in $-\hat{\mathbf{r}} = -\mathbf{r}/r$ indicates that gravity is always attractive.

Newtonian gravity works nearly identical with electrostatics governed by Coulomb’s law, basically because they are both inverse-square laws.

1. Field: In electrostatics, we define the electric field $\mathbf{E}$ to be the force per unit charge on a particle. In Newtonian gravity, the gravitational field (or acceleration) $\mathbf{g}$ is defined as the force per unit mass:

$$\mathbf{g} = -\frac{GM}{r^2}\hat{\mathbf{r}}. \quad (1.3)$$

2. Potential: Both gravity and electrostatic force are conservative forces. In electrostatics, the electric potential ϕ is defined as the potential energy per unit charge, related to $\mathbf{E}$ by $-\nabla\phi = \mathbf{E}$. In Newtonian gravity, the gravitational potential Φ is defined as

$$\Phi = -\frac{GM}{r}, \quad (1.4)$$

with a similar relation to gravitational field:

$$\mathbf{g} = -\nabla\Phi \iff \Phi(\mathbf{r}_b) - \Phi(\mathbf{r}_a) = -\int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{g} \cdot d\boldsymbol{\ell}.$$

Since gravity is conservative, the path integral of Φ (which is the work done on the particle) from $\mathbf{r}_a$ to $\mathbf{r}_b$ is path independent.

3. Linearity: Newtonian gravity is linear, so the gravitational field obeys the principle of superposition. The total potential due to a mass distribution is equal to the sum of the potentials from individual small masses:

$$\Phi(\mathbf{r}) = -\sum_i \frac{GM}{|\mathbf{r} - \mathbf{r}_i|} \quad \text{and} \quad \Phi(\mathbf{r}) = -\int \frac{G\rho(\mathbf{r}')d^3\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|},$$

where ρ stands for the mass density of the distribution.

4. Gauss’s law and Poisson’s equation: these are the gravitational counterpart of Gauss’s law and Poisson’s equation for electrostatics. Gauss’s law:

$$\nabla \cdot \mathbf{g} = -4\pi G\rho \iff \oint_S \mathbf{g} \cdot d\mathbf{S} = -4\pi GM. \quad (1.5)$$

Poisson’s equation:

$$\nabla^2\Phi = 4\pi G\rho. \quad (1.6)$$

Later, we will see that principle of superposition does not hold for general relativity. Nonetheless, to be a good theory, general relativity agrees with Newtonian gravity for weak gravitational field limit.

1.3 Relativistic Gravity

General relativity incorporates special relativity and gravity. There are basically two fundamental constants involved in general relativity: the speed of light c (which characterizes relativity) and the gravitational constant G (which characterizes gravity),

$$c \approx 2.998 \times 10^8 \text{ m/s}, \quad G \approx 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

By dimensional analysis, we can guess the relativistic length scale of a mass M to be

$$R_g = \frac{GM}{c^2}, \tag{1.7}$$

which is known as the **gravitational radius** of M . When the typical size R of mass M is comparable to R_g (which is equivalent to the gravitational potential energy of some mass m being comparable to its rest mass energy), general relativity comes into play,

$$\frac{R_g}{R} = \frac{GM}{Rc^2} = \frac{GMm/R}{mc^2} \sim 1 \implies \text{general relativity matters.}$$

For example,

1. The Sun: $M_\odot = 1.989 \times 10^{30} \text{ m}$, $R_\odot = 6.96 \times 10^8 \text{ m}$,

$$\frac{GM_\odot}{R_\odot c^2} \simeq 2 \times 10^{-6}.$$

2. The Earth: $M_\oplus = 5.974 \times 10^{24} \text{ kg}$, $R_\oplus = 6.38 \times 10^6 \text{ m}$,

$$\frac{GM_\oplus}{R_\oplus c^2} \simeq 7 \times 10^{-10}.$$

This means Newtonian gravity is a good approximation of the Sun and the solar system.

3. Neutron stars: $M \simeq 1.4 M_\odot$, $R \simeq 10 \text{ km}$,

$$\frac{GM}{Rc^2} \sim 0.2.$$

General relativity is very important to the structure of neutron stars. In fact, the precise equation of state of neutron stars is not yet determined.

4. Black holes: a non-spinning, chargeless black hole has the size of the **Schwarzschild radius**, $R = 2GM/c^2$, so

$$\frac{GM}{Rc^2} = 0.5.$$

General relativity is very important.

5. The universe: the mass-energy density of the universe is about $\rho \simeq 9 \times 10^{-27} \text{ kg/m}^3$, while the observable universe has a size of $R \sim 10^{26} \text{ m}$. This gives

$$\frac{GM}{Rc^2} \sim \frac{4\pi GR^3 \rho}{3Rc^2} \sim 0.7.$$

The understanding of overall dynamics of the universe requires general relativity.

In general relativity, the Sun and the Earth are in the weak-field limit; neutron stars, black holes, and the universe are in the strong-field limit.

2 SPECIAL RELATIVITY

In 1887, the Michelson and Morley experiment shows the constancy of the speed of light. There are two postulates in special relativity:

1. The laws of physics take the same form in all inertial reference frames. All inertial frames are fundamentally indistinguishable based on experiments in that frame only.
2. The speed of light (and other massless particles) is the maximum speed in the universe and has the same value in all inertial reference frames,

$$c = 2.99792458 \times 10^8 \text{ m/s.}$$

Note that this number is exact because modern physics define the “meter” using the speed of light in vacuum. With these assumptions, many odd things would occur: Galilean transformation is no longer true, clocks do not run universally, etc. In this chapter, we will review special relativity using two approaches: one is using an introductory spacetime geometry, and the other is more mathematical, using Lorentz transformation formulas.

2.1 Spacetime Geometry

2.1.1 Minkowski Spacetime

In special relativity, time and space are somewhat mixed up in coordinate transformations. Thus, they are unified into a four-dimensional entity called **spacetime**. To be precise, let's list properties of an inertial frame in special relativity:

1. Clocks that have been synchronized at some time will remain synchronized.
2. The properties of points in *space* is described by Euclidean geometry.
3. The spatial separation between points in space with specified coordinates does not change with time.

When one transform one inertial frame to another, simultaneous events in one inertial reference frame need not be simultaneous in other frames. Events that occur at the same place need not be at the same place in other frames.

An event in an inertial frame can be labeled with four coordinates known as **Minkowski coordinates**: (ct, x, y, z) . (In spacetime coordinates, time is conventionally the zeroth coordinate and is multiplied by a c to ensure the dimension is right. Later, we will set $c = 1$ for convenience so that the time coordinate is just t , but for now let's stick with ct .) The geometry of spacetime in an inertial frame is given by the line element

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2. \quad (2.1)$$

(We will prove it later using Lorentz transformations.) It is the square of the **spacetime interval** ds between two infinitesimally separated events. This spacetime is called **Minkowski spacetime**. The line element is a scalar quantity, invariant under any coordinate transformations. In another frame with Minkowski coordinates (ct', x', y', z') , the line element has exactly the form

$$ds^2 = -c^2 dt'^2 + dx'^2 + dy'^2 + dz'^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2.$$

Note the minus sign in front of $c^2 dt^2$. This indicates that the spacetime is flat, but non-Euclidean. The line element ds^2 can be positive, negative, or zero. We can put any two infinitesimally separated events into these three categories:

1. $ds^2 > 0$: the two events are **spacelike** separated. The distance separation dominates over the time separation. For example, we can have $dt = 0$ in some frame. Then the two events occur at the same time, or simultaneously.

2. $ds^2 < 0$: the two events are **timelike** separated. The time separation dominates over the distance separation. For example, setting $dx = dy = dz = 0$ makes two events occur at the same point in space.
3. $ds^2 = 0$: in this case, $c^2 dt^2 = dx^2 + dy^2 + dz^2$, or $d\mathbf{x}^2/dt^2 = c^2$. This describes a photon, so such events are called **lightlike** separated or **null** separated.

Since ds^2 is invariant, these qualitative results (spacelike, null, or timelike) are also independent of reference frame or coordinate system. In other words, if two events are spacelike separated (not necessarily occur at the same time), one can always find a reference frame such that these two events occur simultaneously. Similarly, if two events are timelike separated, one can always find a reference frame such that they occur at the same place.

A particle's trajectory through spacetime is called its **world line**. Any material particle ($m \neq 0$) follows a timelike world line— $ds^2 < 0$ along the world line. This is equivalent to say that their speed cannot exceed the speed of light. Since $ds^2 < 0$, we will define the quantity $d\tau$, called the **proper time** of that particle, to be

$$d\tau^2 = -\frac{ds^2}{c^2}. \quad (2.2)$$

Note that in the particle's rest frame, $dx = dy = dz = 0$, so the proper time is $d\tau^2 = -(c^2 dt^2)/c^2$, or $d\tau = dt$. Thus, the proper time of a particle measures the infinitesimal time interval of a clock that is instantaneously comoving with the particle.

2.1.2 Time Dilation

With the proper time and the spacetime interval, we are able to produce a famous result of special relativity: time dilation. Suppose two events A and B occur along the world line of a particle as seen from our rest frame with coordinates (ct, x, y, z) . We would like to know in the particle's frame, what is the time separation between A and B . The time separation observed in the particle's frame is just its proper time separation integrated along the world line

$$\begin{aligned} \tau_{AB} &= \int_A^B d\tau = \int_A^B \frac{1}{c} \sqrt{-ds^2} = \frac{1}{c} \int_A^B \sqrt{c^2 dt^2 - dx^2 - dy^2 - dz^2} \\ &= \frac{1}{c} \int_{t_A}^{t_B} \sqrt{c^2 - \left(\frac{dx}{dt}\right)^2 - \left(\frac{dy}{dt}\right)^2 - \left(\frac{dz}{dt}\right)^2} dt \\ &= \int_{t_A}^{t_B} \left[1 - \frac{\mathbf{v}(t)^2}{c^2}\right]^{1/2} dt = \int_{t_A}^{t_B} \frac{dt}{\gamma(t)}, \end{aligned}$$

where

$$\gamma = \left[1 - \frac{\mathbf{v}(t)^2}{c^2}\right]^{-1/2}$$

is the instantaneous **Lorentz factor** of the particle measured in our rest frame. Because $|\mathbf{v}| < c$, the Lorentz factor is always greater than 1. Therefore, $\tau_{AB} = \tau_B - \tau_A \leq t_B - t_A$: moving clocks run slow. We derive this entirely based on spacetime geometry.

2.2 Relativistic Kinematics

2.2.1 Lorentz Transformation

In Newtonian mechanics, suppose an observer in frame S assigns an event A with coordinates (t, x, y, z) and another observer in a frame S' that has a relative velocity v in x -direction to S . Assume both observers are in inertial frames, and their x -axes are aligned. Also, assume that $t = 0$ represents the point when the origins of S and S' (denoted by $\mathcal{O}$ and $\mathcal{O}'$ respectively) overlap. Then

according to Galilean transformation, the coordinates (t', x', y', z') of the event A in S' frame will be

$$\begin{aligned} t' &= t, \\ x' &= x - vt, \\ y' &= y, \\ z' &= z. \end{aligned}$$

This is the usual coordinate transformation between inertial frames in classical mechanics. In special relativity, S' will assign the coordinates of A according to the **Lorentz transformation** from S to S' ,

$$\boxed{\begin{aligned} ct' &= \gamma(ct - \beta x), \\ x' &= \gamma(x - \beta ct), \\ y' &= y, \\ z' &= z, \end{aligned}} \quad (2.3)$$

where c is the speed of light, $\beta = v/c$, and $\gamma = 1/\sqrt{1 - \beta^2}$ is the Lorentz factor. A detailed derivation of the Lorentz transformation is given in Appendix A.1.1. Taking the non-relativistic limit ($v/c \rightarrow 0$) reduces the Lorentz transformation to the Galilean transformation. One can also show that the line element, $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$, is invariant under Lorentz transformation. The inverse Lorentz transformation (from S' to S) can evidently be obtained by replacing β with $-\beta$ (reversing the velocity).

Example 2.1. Time Dilation

The proper time τ of a clock fixed at $\mathcal{O}'$ is just the time t' measured by that clock in S' frame,

$$t' = \gamma \left(t - \frac{vx}{c^2} \right) \equiv \tau.$$

In S frame, the clock is moving at a velocity of v , so its position is $x = vt$. Hence the proper time of the clock is

$$\tau = \gamma \left(t - \frac{v^2 t}{c^2} \right) = \gamma \left(1 - \frac{v^2}{c^2} \right) t = \frac{t}{\gamma}.$$

This agrees with the integral $\int dt/\gamma(t)$ when γ is constant.

2.3 Relativistic Dynamics

From now on, we will use the natural units $c = 1$. The Lorentz transformation is given by

$$t' = \gamma(t - vx), \quad x' = \gamma(x - vt), \quad y' = y, \quad z' = z.$$

It looks more symmetric for t' and x' . Also, the line element and proper time are essentially the same,

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 = -d\tau^2.$$

To return to SI units, we shall use dimensional analysis to find out where should the c go. For now, let's develop some math tools that are useful for relativity.

2.3.1 Four-Vectors

In special relativity, we often use Minkowski coordinates with basis vectors $\{e_\mu\} = \{e_t, e_x, e_y, e_z\}$. The basis vectors always have *subscript* indices. Conventionally, Greek indices represent numbers $(0, 1, 2, 3) \rightarrow (t, x, y, z)$, while Latin indices represent $(1, 2, 3) \rightarrow$

(x, y, z) . The vector pointing from the origin to an event in S frame can be written as:

$$\mathbf{x} = x^0 \mathbf{e}_0 + x^1 \mathbf{e}_1 + x^2 \mathbf{e}_2 + x^3 \mathbf{e}_3 = \sum_{\mu=0}^3 x^\mu \mathbf{e}_\mu = (t, x, y, z).$$

Note that all components $\{x^\mu\}$ are labeled with *superscript* indices. To avoid writing the summation symbol everywhere, we will adopt the **Einstein summation convention**: repeated subscript and superscript indices are implicitly summed over. They are called dummy indices. Indices that does not repeat are known as free indices. Also in this notes, I will write four-vectors as bolded-italic (e.g. $\mathbf{x}, \mathbf{p}, \mathbf{u}$) or simply italic (x, p, u) to distinguish with just bolded three-vectors (e.g. $\mathbf{x}, \mathbf{p}, \mathbf{u}$). Therefore, a four-vector can be written compactly as

$$\mathbf{x} = x^\mu \mathbf{e}_\mu.$$

By definition, the component of any four-vector $\mathbf{a}$ (not just position) transform from one inertial frame to another according to the Lorentz transformation:

$$\begin{aligned} a'^0 &= \gamma(a^0 - va^1), \\ a'^1 &= \gamma(a^1 - va^0), \\ a'^2 &= a^2, \\ a'^3 &= a^3. \end{aligned}$$

A four-vector have a “length”, or **magnitude**, defined by

$$\mathbf{a} \cdot \mathbf{a} = -a^0 a^0 + a^1 a^1 + a^2 a^2 + a^3 a^3 \begin{cases} < 0, & \implies \text{timelike vector,} \\ = 0, & \implies \text{null/lightlike vector,} \\ > 0, & \implies \text{spacelike vector.} \end{cases}$$

The magnitude is an invariant quantity, just like the line element. (However, the position or displacement four-vectors can only be defined in flat spacetime. In curved spacetime, four-vectors need to be defined in a local, flat, tangent space to the curved spacetime.) The definition of a magnitude can be generalized to a **scalar product** of four-vectors:

$$\mathbf{a} \cdot \mathbf{b} = -a^0 b^0 + a^1 b^1 + a^2 b^2 + a^3 b^3 = \mathbf{b} \cdot \mathbf{a} \quad \text{or} \quad \mathbf{a} \cdot \mathbf{b} = \eta_{\mu\nu} a^\mu b^\nu.$$

Here $\eta_{\mu\nu}$ is called the **Minkowski metric**. The $\{\eta_{\mu\nu}\}$ are components of the **metric tensor** in the (t, x, y, z) Minkowski coordinate basis of an inertial frame in special relativity. It can be represented by a diagonal matrix:

$$\eta_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.4)$$

How is the metric tensor $\eta_{\alpha\beta}$ related to basis vectors? We can write $\mathbf{a} \cdot \mathbf{b} = (a^\mu \mathbf{e}_\mu) \cdot (b^\nu \mathbf{e}_\nu) = a^\mu b^\nu (\mathbf{e}_\mu \cdot \mathbf{e}_\nu)$. Hence

$$\eta_{\mu\nu} = \mathbf{e}_\mu \cdot \mathbf{e}_\nu.$$

This implies that the Minkowski coordinate basis vectors satisfy $\mathbf{e}_\mu \cdot \mathbf{e}_\nu = 0$ if $\mu \neq \nu$ —it is an orthogonal basis. Furthermore, they are also unit vectors because all nonzero components of $\eta_{\mu\nu}$ are only -1 or 1 . In conclusion, the Minkowski coordinate basis vectors form an *orthonormal* set of basis vectors. Now with the metric, we can write down the most usual notation of the line element:

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu.$$

This is why $\eta_{\mu\nu}$ is called the metric tensor. It enables one to compute the infinitesimal spacetime interval ds between events with infinitesimal differences in coordinates. As we will see later, all geometric properties are stored in the metric tensor.

2.3.2 4-Velocity and 4-Momentum

An important four-vector is the **4-velocity** of a particle along its world line. The conventional spatial three-velocity is tangent to the particle’s path that is parametrized by time t . Similarly, we define the 4-velocity u to be a vector tangent to the world line

parametrized by its proper time τ . Its component is

$$u^\mu \equiv \frac{dx^\mu}{d\tau}. \quad (2.5)$$

Writing them out explicitly:

$$u^0 = \frac{dt}{d\tau} = \gamma, \quad u^1 = \frac{dx}{d\tau} = \frac{dx}{dt} \frac{dt}{d\tau} = \gamma v^x, \quad u^2 = \gamma v^y, \quad u^3 = \gamma v^z.$$

Theorem 2.1.

The 4-velocity u is a unit timelike vector:

$$u \cdot u = \eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = \frac{\eta_{\mu\nu} dx^\mu dx^\nu}{d\tau^2} = \frac{ds^2}{d\tau^2} = -1.$$

As the Minkowski coordinate basis vectors for time is also a timelike unit vector, $e_0 \cdot e_0 = -1$, it suggests that the 4-velocity might be useful as a basis vector in some reference frame.

For a uniformly moving particle, we can define an inertial frame S' in which it is at rest. Then the 4-velocity is in the direction of e'_0 , the basis vector of time. For an accelerated particle, there is no inertial frame in which it is always at rest, but there is a frame called **momentarily comoving reference frame** (MCRF) which is a momentary inertial frame for the particle. The world line of the particle is curved, but the 4-velocity is still tangent to the world line, or tangent to e'_0 .

The three momentum in classical mechanics is defined as $m\mathbf{v}$. In special relativity, the **4-momentum** p is defined as the rest mass m times the 4-velocity,

$$p \equiv mu \quad \Longleftrightarrow \quad p^\mu = mu^\mu. \quad (2.6)$$

Since $u = (\gamma, \gamma\mathbf{v})$, the components of the 4-momentum are $(\gamma m, \gamma m\mathbf{v})$. The time component is the relativistic energy of an object $E = \gamma mc^2$ (with $c = 1$), while the spatial components are the relativistic three-momentum $\mathbf{p} = \gamma m\mathbf{v}$. The 4-momentum is a four-vector, so its magnitude should be invariant. In fact, its magnitude is just the rest mass,

$$p \cdot p = -E^2 + \mathbf{p}^2 = \eta_{\mu\nu} p^\mu p^\nu = m^2 \eta_{\mu\nu} u^\mu u^\nu = -m^2.$$

Putting the c 's back in, we recover the equation of relativistic energy:

$$E^2 = m^2 c^4 + \mathbf{p}^2 c^2. \quad (2.7)$$

2.3.3 Action and Lagrangian for a Free Particle

Some review: in classical mechanics, we define the Lagrangian to be $L = T - V$, where T and P are the kinetic and potential energy, respectively, of the system. The action along a trajectory from point a to point b is defined as

$$S[q_i(t)] = \int_{t_a}^{t_b} dt L(q_i, \dot{q}_i),$$

where q_i is some generalized coordinates that specify every point on the trajectory. By Hamilton's principle (or principle of stationary action), the action of a actual physical trajectory is extremized among all possible trajectories going from a to b . We then extremize the action using calculus of variations. It helps obtain the equation of motion for each $q_i(t)$, which is the Euler-Lagrange equation

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = 0.$$

For a non-relativistic free particle of mass m , if q_i represents (x, y, z) , then

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2), \quad V = 0.$$

The Euler-Lagrange equation says that

$$0 = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = \frac{d}{dt} (m \dot{q}_i) = m \ddot{q}_i.$$

This is just Newton's second law for a free particle $m\mathbf{a} = 0$. It says that a free particle should travel at constant velocity in space.

The Lagrangian/action is the most fundamental one among all physical quantities. It tells which physical path a particle will take. Even though observers in each frame can use different coordinates to represent the trajectory, every one should agree with the actual physical trajectory it takes. Hence the Lagrangian/action should be the “most” invariant quantity, so it needs to be Lorentz invariant in special relativity.

The goal now is to find L for a relativistic particle. The classical Lagrangian cannot do the job because: 1. it has no c and hence no speed limit; 2. it is not Lorentz invariant. Note that the action has dimensions of angular momentum, $[S] = [E \cdot T]$. We want to have some Lorentz-invariant quantities that make the units right. There are three fundamental constants we can use: the speed of light c , Planck constant $\hbar$, and gravitational constant G . (In this section we will write c 's out explicitly.) Planck constant $\hbar$ has the unit of angular momentum, but we cannot use it because there is no quantum mechanics going on yet. Of course G is not useful since there is no gravity. The only available constants now are c and m . We have no choice but to set $L \propto mc^2$. To get dimensions of T , the only Lorentz-invariant candidate is the proper time τ . In conclusion, the action for a relativistic free particle must be of the form

$$S \propto mc^2 \int_{\text{WL}} d\tau,$$

where WL stands for world line. For future convenience, we will choose the constant of proportionality to be -1 . To make it an integral over dt in general reference frame, use the relation $d\tau = dt/\gamma$. Then the action along the world line is

$$S = -mc^2 \int_{t_a}^{t_b} \frac{dt}{\gamma} = -mc^2 \sqrt{1 - \beta^2} \int_{t_a}^{t_b} dt,$$

where $\beta = v/c$ indicates the speed of this free particle in some inertial frame. We can identify the Lagrangian to be

$$L = -mc^2 \sqrt{1 - \beta^2}. \quad (2.8)$$

Checking the non-relativistic limit:

$$L = -mc^2 \left(1 - \frac{1}{2}\beta^2 + \dots \right) = -mc^2 + \frac{1}{2}mv^2 + \mathcal{O}(v^4).$$

Comparing to $L = T - V$ in classical mechanics, the rest mass energy acts like a potential energy of the particle.

In the last section we stated the definition of 4-momentum without a reason, and we did not prove its conservation. Now with the Lagrangian, it is time to compute the conserved quantities. We know that if a coordinate q_i is cyclic (which means it does not appear in the Lagrangian), then its conjugate generalized momentum $\partial L / \partial \dot{q}_i$ is conserved. In the free-particle case, there is no explicit x, y, z . The conserved momentum in x is

$$p_x = \frac{\partial L}{\partial \dot{x}} = -mc^2 \frac{\partial}{\partial \dot{x}} \left(1 - \frac{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}{c^2} \right)^{-1/2} = -mc^2 \left(\frac{\dot{x}/c^2}{\sqrt{1 - (v/c)^2}} \right) = mc \frac{\dot{x}/c}{\sqrt{1 - (v/c)^2}} = \gamma mc \beta_x.$$

The same holds for p_y and p_z . Note that the Lagrangian has no explicit time-dependence, which means the Hamiltonian/total energy is also conserved. The Hamiltonian is defined as

$$\begin{aligned} H &= \sum_i p_i \dot{q}_i - L \\ &= (mc\beta\gamma)(c\beta) - (-mc^2\sqrt{1 - \beta^2}) \\ &= mc^2 (\gamma\beta^2 + \sqrt{1 - \beta^2}) \\ &= mc^2 \left[\frac{\beta^2}{\sqrt{1 - \beta^2}} + \frac{1 - \beta^2}{\sqrt{1 - \beta^2}} \right] \\ &= \gamma mc^2. \end{aligned}$$

This is the conserved relativistic energy. We have proved almost everything. The only problem now is the massless particle, for which $L = 0$. By the invariant mass

$$p_\mu p^\mu = -E^2 + p^2 = -m^2 = 0.$$

This forces $E = p$ for a massless particle. Then how do we know E , for example, for a photon? It is given by Planck's formula

$$E = \hbar\omega = h\nu.$$

2.3.4 Observers and Observations

Consider an arbitrary observer moving on a generic timelike world line through spacetime. A clock and coordinates are well defined in their local frame S' . Assume the basis vectors in this frame are orthonormal $\mathbf{e}'_\mu \cdot \mathbf{e}'_\nu = \eta_{\mu\nu}$. Note that the observer's frame S' need not be an inertial frame. There will always be an inertial frame at any moment which is instantaneously at rest (the MCRF) with S' .

The observer is always at rest in S' , so their 4-velocity is $u = \mathbf{e}'_0$. Suppose the observer measures a particle with 4-momentum p , we want to know the energy and momentum of this particle in their frame. Recall that the energy is the zeroth *component* of the momentum. To obtain components, we just project a vector onto the basis vector, just like we do for ordinary 3-vectors. Thus, the energy of that particle in frame S' is

$$E' = -p \cdot \mathbf{e}'_0 = -p \cdot u.$$

Note that there is a minus sign before the dot product. It comes from the metric tensor $\eta_{\mu\nu}$ (see the following theorem).

Theorem 2.2. Projection of 4-momentum

An observer with an instantaneous orthonormal basis with $\mathbf{e}_\mu \cdot \mathbf{e}_\nu = \eta_{\mu\nu}$ measures the components p^ν of a particle's 4-momentum. This component p^ν satisfy

$$\eta_{\mu\nu} p^\nu = p \cdot \mathbf{e}_\mu \quad (2.9)$$

Explicitly,

$$-p^0 = p \cdot \mathbf{e}_0, \quad p^1 = p \cdot \mathbf{e}_1, \quad p^2 = p \cdot \mathbf{e}_2, \quad p^3 = p \cdot \mathbf{e}_3.$$

Proof. Let the 4-momentum be expressed as components and basis vectors:

$$p = p^0 \mathbf{e}_0 + p^1 \mathbf{e}_1 + p^2 \mathbf{e}_2 + p^3 \mathbf{e}_3 = p^\nu \mathbf{e}_\nu.$$

Take the inner product with $\mathbf{e}_\mu$,

$$p \cdot \mathbf{e}_\mu = p^\nu \mathbf{e}_\nu \cdot \mathbf{e}_\mu = p^\nu \eta_{\nu\mu}.$$

Since the metric tensor is symmetric, $\eta_{\mu\nu} = \eta_{\nu\mu}$, this proves the theorem,

$$\eta_{\mu\nu} p^\nu = p \cdot \mathbf{e}_\mu.$$

In fact, the projection method can be generalized to any four-vector. It is not limited to 4-momentum. □

Example 2.2. Projection vs. Lorentz transformation

Consider an inertial frame S with orthonormal basis $\mathbf{e}_\mu \cdot \mathbf{e}_\nu = \eta_{\mu\nu}$. The observer is traveling at an instantaneous 4-velocity

$$u = \gamma_{\text{ob}} \mathbf{e}_0 + \gamma_{\text{ob}} v_{\text{ob}} \mathbf{e}_1,$$

where γ_{ob} is the (instantaneous) Lorentz factor of the observer, and v_{ob} is their spatial velocity. (We also assume that at the instant the observer is traveling in the x -direction.) The particle's 4-momentum as seen in frame S is given by

$$p = m\gamma_p \mathbf{e}_0 + m\gamma_p v_p \mathbf{e}_1,$$

where γ_p and v_p are the particle's Lorentz factor and spatial velocity, respectively. Then according to Theorem 2.2, the observer will measure the energy of the particle to be

$$E' = -p \cdot u = -(-m\gamma_p \gamma_{\text{ob}} + m\gamma_p \gamma_{\text{ob}} v_p v_{\text{ob}}) = m\gamma_p \gamma_{\text{ob}} (1 - v_p v_{\text{ob}}) = \frac{m(1 - v_p v_{\text{ob}})}{\sqrt{1 - v_p^2} \sqrt{1 - v_{\text{ob}}^2}}.$$

As a check, we will use the Lorentz transformation to derive E' :

$$E' = \gamma_{\text{ob}}(p^0 - v_{\text{ob}}p^1) = \gamma_{\text{ob}}(m\gamma_p - v_{\text{ob}}m\gamma_pv_p) = m\gamma_p\gamma_{\text{ob}}(1 - v_pv_{\text{ob}}) = \frac{m(1 - v_pv_{\text{ob}})}{\sqrt{1 - v_p^2}\sqrt{1 - v_{\text{ob}}^2}}.$$

It seems like the Lorentz transformation is more direct and efficient. It turns out that the projection method works better in curved spacetime in general relativity.

Example 2.3. Photons

A photon with energy E has a 4-momentum $p = (E, \mathbf{p})$ and $|\mathbf{p}| = E$. It is possible to construct another 4-vector by dividing p by $\hbar$. According to quantum mechanics, the energy of a photon is related to its (angular) frequency ω by $E = \hbar\omega = h\nu$. The spatial momentum is related to its wave vector $\mathbf{k}$ by $\mathbf{p} = \hbar\mathbf{k}$. Hence

$$k = \frac{p}{\hbar} = (\omega, \mathbf{k})$$

is also a 4-vector, called the **wave 4-vector**. Since $k^2 = p^2/\hbar^2 = 0$, we must have $|\mathbf{k}| = \omega$. Suppose in frame S , an observer is moving at constant speed v along the x -axis. There is a source shooting photons in a straight line with an angle α with the x -axis and it meets the observer. We want to find the relativistic Doppler shift measured by the observer. Since the Doppler shift is directly related to ω , so the problem turns to finding the timelike component of the wave 4-vector.

We know the timelike basis vector of the observer is $u = (\gamma, \gamma v, 0, 0)$, and the wave 4-vector of this photon is

$$k = (\omega, \omega \cos \alpha, \omega \sin \alpha, 0),$$

both in frame S . Let the observer's basis vector be orthonormal, $e'_\mu \cdot e'_\nu = \eta_{\mu\nu}$. To find the timelike component of k in the observer's frame S' , just take the inner product with the timelike basis vector u of the observer and multiply by a minus sign:

$$\omega' = -k \cdot u = +\gamma\omega - \gamma v \omega \cos \alpha = \gamma\omega(1 - v \cos \alpha).$$

The minus sign comes from

$$k \cdot u = (k'^\mu e'_\mu) \cdot e'_0 = k'^\mu \eta_{\mu 0} = k'^0 \eta_{00} = -\omega'.$$

The same as in Theorem 2.2. The ratio of ω' and ω defines the Doppler shift:

$$\frac{\omega'}{\omega} = \gamma(1 - v \cos \alpha) = \frac{1 - v \cos \alpha}{\sqrt{1 - v^2}}.$$

To check whether this makes sense, let's consider some special cases. First, let $\alpha = 0$ and $v > 0$, i.e. the photon and the observer are moving in the same $+x$ -direction. The Doppler shift is $\sqrt{(1 - v)/(1 + v)} < 1$; the photon appears redshifted. If $\alpha = 0$ and $v < 0$, then $\sqrt{(1 - v)/(1 + v)} > 1$ and the photon is blueshifted. Next, let $\alpha = \pi/2$, which means the photon is in the transverse direction ($+y$ -direction). The Doppler shift is $1/\sqrt{1 - v^2} = \gamma$. This is because the observer is time-dilated relative to the photon source, $dt' = dt/\gamma$. The photon then appears blueshifted according to the observer as it receives more wave crests per unit time.

Example 2.4. Relativistic Beaming

Photon emitted at angle θ' with respect to e'_1 and energy E_0 .

$$p^\mu = E(u'^\mu + \cos \theta' e'_1{}^\mu + \sin \theta' e'_2{}^\mu) = E(u^\mu + \cos \theta e_1{}^\mu + \sin \theta e_2{}^\mu).$$

$$u'^\mu = u^\mu \cosh \alpha + e_1{}^\mu \sinh \alpha, \quad e'_1{}^\mu = e_1{}^\mu \cosh \alpha + u^\mu \sinh \alpha.$$

$$\cos \theta = \frac{\eta_{\mu\nu} p^\mu e_1{}^\nu}{-\eta_{\alpha\beta} p^\alpha u^\beta} = \frac{E_0(\sinh \alpha + \cos \theta' \cosh \alpha)}{E_0 \cosh \alpha + \cos \theta' \sinh \alpha} = \frac{\beta + \cos \theta'}{1 + \beta \cos \theta'}.$$

2.4 Vectors and Tensors

This section deals with more formal operations on vectors and tensors.

2.4.1 Vectors

A **vector space** is a set of objects called vectors that are linear under addition and scalar multiplication. Consider two vectors $\mathbf{u}$ and $\mathbf{v}$ in a vector space V and two scalars a and b . Then

$$(a + b)(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + b\mathbf{u} + a\mathbf{v} + b\mathbf{v}.$$

Vectors can be decomposed into components with respect to some set of basis vectors. A basis $\{\mathbf{e}_\mu\}$ is any set of linearly independent vectors that span the vector space. This means any vector can be written as a linear combination of basis vectors, and no vector in the basis is a linear combination of other basis vectors. We write the decomposition of a vector into basis vectors as

$$\mathbf{u} = u^\mu \mathbf{e}_\mu.$$

For a loose notation, we often (and will) call u^μ as the vector (but we should always remember that they are components). The number of vectors in a basis is the **dimension** of the vector space. We shall use Minkowski spacetime as an example.

Example 2.5. Minkowski Spacetime

Consider a vector $\mathbf{v}$ at a particular point p in spacetime. We write its coordinates as (v^0, v^1, v^2, v^3) . Here v^μ are the components. The abstract vector $\mathbf{v}$ can be decomposed into

$$\mathbf{v} = v^0 \mathbf{e}_0 + v^1 \mathbf{e}_1 + v^2 \mathbf{e}_2 + v^3 \mathbf{e}_3.$$

Hence the vector space (called the **tangent space** T_p) around a point in Minkowski spacetime is 4-dimensional. If we write $\mathbf{v}$ as column vectors, then the Lorentz transformation would be

$$v^{\mu'} = \Lambda^{\mu'}_{\nu} v^{\nu} \iff \begin{bmatrix} v^{0'} \\ v^{1'} \\ v^{2'} \\ v^{3'} \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v^0 \\ v^1 \\ v^2 \\ v^3 \end{bmatrix}.$$

Note that the abstract vector itself does not transform. It is the components v^μ that transform. Thus, we can work out the transformation rule of basis vectors,

$$\mathbf{v} = v^\mu \mathbf{e}_\mu = v^{\nu'} \mathbf{e}_{\nu'} = \Lambda^{\nu'}_{\mu} v^\mu \mathbf{e}_{\nu'}.$$

The basis vectors transform under the inverse of the Lorentz transformation:

$$\mathbf{e}_\mu = \Lambda^{\nu'}_{\mu} \mathbf{e}_{\nu'} \implies \mathbf{e}_{\nu'} = \Lambda^{\mu}_{\nu'} \mathbf{e}_\mu.$$

2.4.2 Dual Vectors

For a vector space V , there is another associated vector space called the **dual vector space** V^* . It is the space of all linear maps from the original vector space to real numbers. Let $\omega \in V^*$, $\mathbf{u}, \mathbf{v} \in V$, and $a, b \in \mathbb{R}$, then

$$\omega(a\mathbf{u} + b\mathbf{v}) = a\omega(\mathbf{u}) + b\omega(\mathbf{v}) \in \mathbb{R}.$$

The dual vector space is also a vector space, so

$$(a\omega + b\eta)(\mathbf{u}) = a\omega(\mathbf{u}) + b\eta(\mathbf{u}).$$

We can always find a set of basis dual vectors $\{e^{*\mu}\}$ such that

$$e^{*\nu}(e_\mu) = \delta^\nu_\mu.$$

Each dual vector in V^* can be written as a linear combination of basis vectors:

$$\omega = \omega_\mu e^{*\mu}.$$

Again, for a loose notation, we often call ω_μ as the dual vector (but they are in fact components). The action of a dual vector on a vector, in component notation, is

$$\omega(v) = \omega_\mu e^{*\mu}(v^\nu e_\nu) = \omega_\mu v^\nu e^{*\mu}(e_\nu) = \omega_\mu v^\nu \delta^\mu_\nu = \omega_\mu v^\mu \in \mathbb{R}.$$

Then the components can be equivalently written as $\omega_\mu = \omega(e_\mu)$ because

$$\omega(e_\mu) = \omega_\nu e^{*\nu}(e_\mu) = \omega_\nu \delta^\nu_\mu = \omega_\mu.$$

Because ω_μ and v^μ are just numbers, we can equally write $\omega(v) = v^\mu \omega_\mu$. This suggests that we can think of vectors as linear maps from the dual vector space to real numbers:

$$v(\omega) = v^\mu \omega_\mu = \omega_\mu v^\mu = \omega(v).$$

In other words, the dual space $(V^*)^*$ to the dual vector space V^* is just V . The transformation of dual vectors is the inverse of the transformation of original vectors. This is required such that the scalar $\omega_\mu v^\mu$ to remain unchanged in transformation. The basis dual vectors, transform the same way as the transformation of original vectors. We will conclude them here:

$$v^{\mu'} = \Lambda^{\mu'}_\nu v^\nu, \quad e_{\mu'} = \Lambda^\nu_{\mu'} e_\nu, \quad \omega_{\mu'} = \Lambda^\nu_{\mu'} \omega_\nu, \quad e^{*\mu'} = \Lambda^{\mu'}_\nu e^{*\nu}.$$

These relations are completely consistent with index notation.

2.4.3 Tensors

Tensors are generalization of vectors and dual vectors. A **tensor** T of rank (k, l) is a multilinear map from a collection of k dual vectors and l vectors to $\mathbb{R}$:

$$T(\omega_1, \omega_2, \dots, \omega_k, v_1, v_2, \dots, v_l) \in \mathbb{R}.$$

Note that $\omega_1, \dots, \omega_k$ and $v_1, \dots, v_l$ are distinct dual vectors and vectors, not components. Multilinearity means that a tensor acts linearly in each of its arguments. For example, a rank $(1, 1)$ tensor has multilinearity

$$T(a\omega + b\eta, c\mathbf{u} + d\mathbf{v}) = acT(\omega, \mathbf{u}) + adT(\omega, \mathbf{v}) + bcT(\eta, \mathbf{u}) + bdT(\eta, \mathbf{v}).$$

To be less abstract, let's view some examples:

- A $(0, 0)$ tensor is a scalar in $\mathbb{R}$.
- A $(0, 1)$ tensor is a dual vector: it maps 1 vector to real numbers.
- A $(1, 0)$ tensor is a vector: it maps 1 dual vector to real numbers.
- A $(0, 2)$ tensor can be the metric tensor, which takes 2 vectors and map them to real numbers by via inner product $\mathbf{u} \cdot \mathbf{v}$. The inner product, by definition, is linear in both its argument.

We can manipulate tensors by usual addition and scalar multiplication. Note that only tensors of the same rank can be added together, e.g. you cannot add a scalar to a vector. If T is a (k, l) tensor and S is an (m, n) tensor, then their **tensor product** is a $(k + m, l + n)$ tensor $T \otimes S$, defined by

$$T \otimes S(\omega_1, \dots, \omega_k, \dots, \omega_{k+m}, v_1, \dots, v_l, \dots, v_{l+n}) = T(\omega_1, \dots, \omega_k, v_1, \dots, v_l) \times S(\omega_{k+1}, \dots, \omega_{k+m}, v_{l+1}, \dots, v_{l+n}),$$

where $\times$ means ordinary multiplication of scalars. The tensor product helps us construct basis from vectors and dual vectors. In particular, the basis for a (k, l) tensor can be constructed by taking the tensor product of the basis vectors of k dual vectors and l

vectors:

$$\mathbf{e}_{\mu_1} \otimes \cdots \otimes \mathbf{e}_{\mu_k} \otimes \mathbf{e}^{*\nu_1} \otimes \cdots \otimes \mathbf{e}^{*\nu_l}.$$

The components of tensors then follows

$$T = T^{\mu_1 \cdots \mu_k}_{\nu_1 \cdots \nu_l} \mathbf{e}_{\mu_1} \otimes \cdots \otimes \mathbf{e}_{\mu_k} \otimes \mathbf{e}^{*\nu_1} \otimes \cdots \otimes \mathbf{e}^{*\nu_l}.$$

The components can be equivalently written as

$$T^{\mu_1 \cdots \mu_k}_{\nu_1 \cdots \nu_l} = T(\mathbf{e}^{*\mu_1}, \dots, \mathbf{e}^{*\mu_k}, \mathbf{e}_{\nu_1}, \dots, \mathbf{e}_{\nu_l}).$$

The transformation rule of tensors follows

$$T^{\mu'_1 \cdots \mu'_k}_{\nu'_1 \cdots \nu'_l} = \Lambda^{\mu'_1}_{\mu_1} \cdots \Lambda^{\mu'_k}_{\mu_k} \Lambda^{\nu_1}_{\nu'_1} \cdots \Lambda^{\nu_l}_{\nu'_l} T^{\mu_1 \cdots \mu_k}_{\nu_1 \cdots \nu_l}.$$

That is, each upper index transform like a vector, and each lower index transform like a dual vector. If we define tensors via the transformation rule, a $(1,1)$ tensor can also be viewed as a map from vectors to vectors:

$$v^\mu = T^\mu_{\nu} v^\nu,$$

and so on for other tensors because they all satisfy the transformation rule. A special operation of tensors is **contraction**. We can sum over one upper index and one lower index by

$$S^{\mu\alpha}_{\nu\alpha} = T^\mu_{\nu}.$$

2.4.4 Symmetric and Antisymmetric Tensors

A tensor can be symmetric or antisymmetric in its indices. For example, if the tensor is the same under exchange of two of its indices, $S_{\mu\nu\alpha} = S_{\nu\mu\alpha}$, it is considered **symmetric** in the first two indices. If the tensor is the same under exchange of any pair of its indices,

$$S_{\mu\nu\alpha} = S_{\mu\alpha\nu} = S_{\nu\alpha\mu} = S_{\nu\mu\alpha} = S_{\alpha\mu\nu} = S_{\alpha\nu\mu},$$

then we say it is **totally symmetric**. Similarly, if the tensor acquires a minus sign under exchange of two of its indices, $A_{\mu\nu\alpha} = -A_{\nu\mu\alpha}$, it is **antisymmetric** in the first two indices. A tensor that is antisymmetric in any pair of its indices is called **totally antisymmetric**. An example of totally antisymmetric object would be the Levi-Civita symbol $\epsilon_{\mu\nu\alpha\beta}$ (though it is in general not a tensor in curved spacetime).

We can **symmetrize** indices of a tensor. To do this, we sum over all permutations of the indices of interest and divide the number of terms:

$$T_{(\mu_1 \mu_2 \cdots \mu_n) \mu_{n+1} \cdots} = \frac{1}{n!} (T_{\mu_1 \mu_2 \cdots \mu_n \mu_{n+1} \cdots} + \text{sum over other permutations of indices } \mu_1 \cdots \mu_n).$$

The symmetrization is denoted as round brackets. The normalization factor $1/n!$ is just a convention: for a totally symmetric tensor, we expect

$$S_{(\mu_1 \cdots \mu_n)} = S_{\mu_1 \cdots \mu_n}.$$

We can also **antisymmetrize** indices of a tensor:

$$T_{[\mu_1 \mu_2 \cdots \mu_n] \mu_{n+1} \cdots} = \frac{1}{n!} (T_{\mu_1 \mu_2 \cdots \mu_n \mu_{n+1} \cdots} + \text{alternating sum over permutations of indices } \mu_1 \cdots \mu_n).$$

The alternating sum means that even permutation of indices acquires a plus sign, while odd permutation of indices acquires a minus sign. For example,

$$T_{[\mu\nu\alpha]\beta} = \frac{1}{6} (T_{\mu\nu\alpha\beta} - T_{\mu\alpha\nu\beta} + T_{\alpha\mu\nu\beta} - T_{\nu\mu\alpha\beta} + T_{\nu\alpha\mu\beta} - T_{\alpha\nu\mu\beta}).$$

2.4.5 The Electromagnetic Field Tensor

As an example of antisymmetric tensor and antisymmetrization, we should mention the electromagnetic field tensor $F_{\mu\nu}$. The field tensor is defined by

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (2.10)$$

where $A^\mu = (\phi, \mathbf{A})$ is the four-potential, and $A_\mu = (-\phi, \mathbf{A})$. The matrix elements of $F_{\mu\nu}$ are components of electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$. In classical electromagnetism, the electric and magnetic fields can be written in terms of potentials:

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A},$$

or in index notation

$$E^i = \partial^i A_t - \partial_t A^i, \quad B^i = \epsilon^{ijk} \partial_j A_k.$$

We can almost see that $F_{\mu\nu}$ are just electric and magnetic fields. One can find that

$$F_{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{bmatrix}.$$

Either by (2.10) or by the matrix above, it is clear that the field tensor is an antisymmetric tensor. Maxwell's equations, in the usual coordinate space and SI units, is

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} - \frac{\partial \mathbf{B}}{\partial t} &= 0, \\ \nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} &= \mu_0 \mathbf{J}, \end{aligned}$$

where $c^2 = 1/\mu_0\epsilon_0$. In natural units, Gauss's law and Maxwell-Ampère's law can be unified into one equation involving the field tensor:

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu,$$

where $J^\mu = (c\rho, \mathbf{J}) \rightarrow (\rho, \mathbf{J})$ is the four-current and

$$F^{\mu\nu} = F_{\alpha\beta} \eta^{\mu\alpha} \eta^{\nu\beta} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{bmatrix}.$$

$\nabla \cdot \mathbf{B} = 0$ and Faraday's law becomes

$$\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} = 0.$$

Since $F_{\mu\nu}$ is antisymmetric, we can also write

$$-\partial_\alpha F_{\nu\mu} - \partial_\mu F_{\alpha\nu} - \partial_\nu F_{\mu\alpha} = 0,$$

or, combining the two equations, we have

$$\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} - \partial_\alpha F_{\nu\mu} - \partial_\mu F_{\alpha\nu} - \partial_\nu F_{\mu\alpha} = \partial_{[\mu} F_{\nu\alpha]} = 0.$$

2.5 Classical Field Theory

In classical mechanics, the action S is given by the integral of the Lagrangian,

$$S = \int dt L(q, \dot{q}).$$

The equations of motion of a particle is given by extremizing the action, which results in the Euler-Lagrange equations:

$$\delta S = 0 \implies \frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = 0.$$

In classical field theory, the dynamical variable are the fields ϕ^i , where i indicates independent fields. We care more about the **Lagrangian density** $\mathcal{L}$, defined by

$$L = \int d^3x \mathcal{L}(\phi^i, \partial_\mu \phi^i).$$

In field theory, we often call the Lagrangian density just the Lagrangian. The action can be written as

$$S = \int dt L = \int d^4x \mathcal{L}(\phi^i, \partial_\mu \phi^i).$$

Extremizing the action gives the Euler-Lagrange equation for a field theory:

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi^i} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^i)} \right) = 0.} \quad (2.11)$$

2.5.1 Real Scalar Fields

A real scalar field $\phi(x)$ is the simplest example of a field. The Lagrangian for a real scalar field with a potential $V(\phi)$ is

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi).$$

To get the equation of motion for ϕ , we calculate

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \phi} &= -\frac{dV}{d\phi}. \\ \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} &= -\frac{1}{2} \frac{\partial}{\partial (\partial_\mu \phi)} (\eta^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi) = -\frac{1}{2} \eta^{\alpha\beta} (\delta^\mu_\alpha \partial_\beta \phi + \partial_\alpha \phi \delta^\mu_\beta) = -\frac{1}{2} (\eta^{\mu\beta} \partial_\beta \phi + \eta^{\alpha\mu} \partial_\alpha \phi) = -\partial^\mu \phi. \end{aligned}$$

Substituting these quantities into the Euler-Lagrange equation (2.11) gives

$$\partial_\mu \partial^\mu \phi - \frac{dV}{d\phi} = 0. \quad (2.12)$$

This is the equation of motion. A common choice of the potential is $V(\phi) = \frac{1}{2} m^2 \phi^2$, where m is the “mass” of the field. Equation (2.12) with this potential leads to the famous **Klein-Gordon equation**,

$$\partial_\mu \partial^\mu \phi - m^2 \phi = 0. \quad (2.13)$$

(In quantum field theory, it is more likely to see the Klein-Gordon equation written as $\partial_\mu \partial^\mu \phi + m^2 \phi = 0$. This is because gravity physicists use metric signature $(-, +, +, +)$, while high energy physicists use $(+, -, -, -)$ so that $\partial_\mu \partial^\mu = \partial_t^2 - \nabla^2$. We will stick to $(-, +, +, +)$ in general relativity.)

2.5.2 The 4-Potential

The theory of electromagnetism is also a field theory. However, the dynamical fields are not electric and magnetic fields, but the 4-potential $A_\mu = (\phi, \mathbf{A})$, where ϕ is the scalar potential and $\mathbf{A}$ is the vector potential. We already know the observable quantities are electric and magnetic fields, which are components of the field tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

It is well-known that there are additional gauge freedoms in A_μ . Under a **gauge transformation**,

$$A_\mu \rightarrow A_\mu + \partial_\mu \lambda(x)$$

for arbitrary function $\lambda(x)$, the field tensor is unchanged,

$$F_{\mu\nu} \rightarrow F_{\mu\nu} + \partial_\mu \partial_\nu \lambda - \partial_\nu \partial_\mu \lambda = F_{\mu\nu}.$$

Observable quantities must be gauge invariant, so components of $F_{\mu\nu}$ are observables. The Lagrangian of electromagnetism is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu J^\mu, \quad (2.14)$$

where $J^\mu = (\rho, \mathbf{J})$ is the 4-current. If we plug (2.14) into the Euler-Lagrange equation, we will get Maxwell’s equations. (We will get two of Maxwell’s equations, namely $\partial_\mu F^{\mu\nu} = -J^\nu$. These are Gauss’s law and Maxwell-Ampère’s law. The other two, $\partial_{[\mu} F_{\nu\alpha]} = 0$

are satisfied automatically.)

Proof. The dynamical fields A_μ has four components, so we expect to get four equations of motion,

$$\frac{\partial \mathcal{L}}{\partial A_\nu} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) = 0,$$

labeled by $\nu = 0, 1, 2, 3$. The first term is easy,

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = \delta^\nu_\mu J^\mu = J^\nu.$$

To compute the second term, we write

$$F_{\mu\nu} F^{\mu\nu} = F_{\alpha\beta} F^{\alpha\beta} = \eta^{\alpha\rho} \eta^{\beta\sigma} F_{\alpha\beta} F_{\rho\sigma}.$$

Using $F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha$, its derivative with respect to $\partial_\mu A_\nu$ is

$$\frac{\partial F_{\alpha\beta}}{\partial (\partial_\mu A_\nu)} = \frac{\partial}{\partial (\partial_\mu A_\nu)} (\partial_\alpha A_\beta - \partial_\beta A_\alpha) = \delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha.$$

Then the derivative of $F_{\alpha\beta} F^{\alpha\beta}$ with respect to $\partial_\mu A_\nu$ is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} &= \frac{\partial (F_{\alpha\beta} F^{\alpha\beta})}{\partial (\partial_\mu A_\nu)} = \frac{\partial}{\partial (\partial_\mu A_\nu)} (\eta^{\alpha\rho} \eta^{\beta\sigma} F_{\alpha\beta} F_{\rho\sigma}) \\ &= \eta^{\alpha\rho} \eta^{\beta\sigma} \left[\frac{\partial F_{\alpha\beta}}{\partial (\partial_\mu A_\nu)} F_{\rho\sigma} + F_{\alpha\beta} \frac{\partial F_{\rho\sigma}}{\partial (\partial_\mu A_\nu)} \right] \\ &= \eta^{\alpha\rho} \eta^{\beta\sigma} [(\delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha) F_{\rho\sigma} + (\delta^\mu_\rho \delta^\nu_\sigma - \delta^\mu_\sigma \delta^\nu_\rho) F_{\alpha\beta}] \\ &= (\eta^{\mu\rho} \eta^{\nu\sigma} - \eta^{\nu\rho} \eta^{\mu\sigma}) F_{\sigma\rho} + (\eta^{\alpha\mu} \eta^{\beta\nu} - \eta^{\alpha\nu} \eta^{\beta\mu}) F_{\alpha\beta} \\ &= (F^{\mu\nu} - F^{\nu\mu}) + (F^{\mu\nu} - F^{\nu\mu}) \\ &= 4F^{\mu\nu}. \end{aligned}$$

Substituting $\partial \mathcal{L} / \partial A_\nu$ and $\partial \mathcal{L} / \partial (\partial_\mu A_\nu)$ into the Euler-Lagrange equation, we arrive at

$$\partial_\mu F^{\mu\nu} = -J^\nu, \quad \text{where} \quad F^{\mu\nu} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{bmatrix}.$$

The $\nu = 0$ component is Gauss's law,

$$-\partial_i E^i = -\rho \implies \nabla \cdot \mathbf{E} = \rho.$$

The $\nu = 1, 2, 3$ components are components of Maxwell-Ampère's law,

$$\partial_t E^i - \epsilon^{ij}_k \partial_j B^k = -J^i \implies \nabla \times \mathbf{B} = \mathbf{J} + \frac{\partial \mathbf{E}}{\partial t}.$$

□

2.6 The Stress-Energy Tensor

Before we go into general relativity, which relates matter to curvature of spacetime, we should get familiar with different forms of matter content. The idea of statistical mechanics carries to relativity: instead of describing individual particle's 4-momentum in a system, we describe the system as a whole. A **fluid** is a continuum characterized by macroscopic quantities like density, pressure, entropy, viscosity, etc. In order to describe a fluid, we define the **stress-energy tensor** $T_{\mu\nu}$, which contains all information about energy density, momentum density, pressure, and stress. The stress energy tensor is a symmetric tensor,

$$T_{\mu\nu} = T_{\nu\mu}.$$

Its components is defined as “the flux of four-momentum p^μ across a surface of constant x^ν .” The following is a more detailed definition: as measured by an observer moving along the fluid with an orthonormal basis $\{\mathbf{e}_\mu\} = (u, \mathbf{e}_i)$,

$$\begin{aligned} T_{\mu\nu}u^\mu u^\nu &= \text{energy density,} \\ T_{\mu\nu}u^\mu (\mathbf{e}_i)^\nu &= \text{momentum density in the } i\text{th direction,} \\ T_{\mu\nu}(\mathbf{e}_i)^\mu (\mathbf{e}_i)^\nu &= \text{pressure in the } i\text{th direction,} \\ T_{\mu\nu}(\mathbf{e}_i)^\mu (\mathbf{e}_j)^\nu &= \text{shear in the } i, j\text{th direction.} \end{aligned}$$

A **dust** is a collection of non-interacting particles, a simplest example of a fluid. This means the particles in a dust are at rest with respect to each other for no interaction. The 4-velocity $u^\mu(x)$ will be constant for each particle at each point. A dust can be solely described by the energy density in its rest frame. We define the stress-energy tensor of dust to be

$$T_{\mu\nu} = \rho u_\mu u_\nu. \quad (2.15)$$

2.6.1 Perfect Fluids

The most useful fluid in general relativity is a **perfect fluid**. We should first familiarize with it in special relativity before generalizing to curved spacetime. A perfect fluid is a dust with an isotropic pressure P , no heat conduction, and no viscosity. Since the pressure is isotropic, there is no shear in the stress-energy tensor. Also, there should be no preferred direction for pressure, so the component T^{ii} should be the same for each i . In the rest frame of the fluid, the stress-energy tensor should take the form

$$T_{\mu\nu} = \begin{bmatrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{bmatrix}.$$

In a covariant form, the stress-energy tensor for a perfect fluid is

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + P\eta_{\mu\nu}. \quad (2.16)$$

One can check using $T_{\mu\nu}(\mathbf{e}_\alpha)^\mu (\mathbf{e}_\beta)^\nu$ that it produces the correct physical quantities:

$$T_{\mu\nu}u^\mu u^\nu = (\rho + P)u_\mu u_\nu u^\mu u^\nu + P\eta_{\mu\nu}u^\mu u^\nu = (\rho + P)(-1)^2 + (-P) = \rho,$$

$$T_{\mu\nu}(\mathbf{e}_i)^\mu (\mathbf{e}_i)^\nu = (\rho + P)u_\mu u_\nu (\mathbf{e}_i)^\mu (\mathbf{e}_i)^\nu + P\eta_{\mu\nu}(\mathbf{e}_i)^\mu (\mathbf{e}_i)^\nu = 0 + P = P.$$

The equation of motion for a perfect fluid is

$$\partial^\mu T_{\mu\nu} = 0. \quad (2.17)$$

Using (2.16), the equation of motion can be written as

$$\partial^\mu (\rho + P)u_\mu u_\nu + (\rho + P)(u_\nu \partial^\mu u_\mu + u_\mu \partial^\mu u_\nu) + \partial_\nu P = 0. \quad (2.18)$$

To understand the physics in this equation, we project it into the timelike basis vector u^μ and orthogonal spacelike basis vectors $(\mathbf{e}_i)^\mu$. First, we should remember a useful identity,

$$u^\nu \partial^\mu u_\nu = \frac{1}{2} \partial^\mu (u_\nu u^\nu) = \frac{1}{2} \partial^\mu (-1) = 0.$$

The timelike projection of (2.18) is a contraction with u^ν . Using $u_\nu u^\nu = -1$,

$$\begin{aligned} 0 &= \partial^\mu (\rho + P)u_\mu + (\rho + P)(-\partial^\mu u_\mu + u_\mu \cancel{u^\nu \partial^\mu u_\nu}) + u^\nu \partial_\nu P \\ &= -u^\mu \partial_\mu \rho - \cancel{u^\mu \partial_\mu P} - (\rho + P)\partial_\mu u^\mu + \cancel{u^\nu \partial_\nu P}, \end{aligned}$$

or

$$\partial_\mu (\rho u^\mu) + P \partial_\mu u^\mu = 0.$$

This is the conservation of energy for a perfect fluid. It will be clearer in the non-relativistic limit, where $u^\mu = (1, v^i)$ with $v \ll 1$. Also, the pressure is much less than the energy density, $P \ll \rho$. (Remember that ρ is the mass density multiplied by c^2 , while

the pressure comes from momentum transfer of particles in random motions with $v \ll c$.) Thus, in the non-relativistic limit, the conservation of energy becomes

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0,$$

which is the continuity equation from fluid mechanics.

Next, the spacelike projection of (2.18) requires the orthogonal projection tensor

$$P^\mu{}_\nu = \delta^\mu{}_\nu + u^\mu u_\nu. \quad (2.19)$$

It is easy to check that

$$\begin{aligned} P^\mu{}_\nu v^\nu_\perp &= v^\mu_\perp \quad \text{for } v^\nu_\perp u_\nu = 0 \text{ (orthogonal),} \\ P^\mu{}_\nu v^\nu_\parallel &= 0 \quad \text{for } v^\nu_\parallel = k u^\nu \text{ (parallel),} \end{aligned}$$

for some constant k . Applying $P^\nu{}_\alpha$ to (2.18), and noting that every term with u_ν vanishes after the contraction,

$$\begin{aligned} 0 &= (\delta^\nu{}_\alpha + u^\nu u_\alpha) [\partial^\mu (\rho + P) u_\mu u_\nu + (\rho + P) (u_\nu \partial^\mu u_\mu + u_\mu \partial^\mu u_\nu) + \partial_\nu P] \\ &= (\delta^\nu{}_\alpha + u^\nu u_\alpha) [(\rho + P) u_\mu \partial^\mu u_\nu + \partial_\nu P] \\ &= (\rho + P) u_\mu \partial^\mu u_\alpha + (\rho + P) u_\mu u_\alpha \cancel{\partial^\mu u_\nu} + \partial_\alpha P + u_\alpha u^\nu \partial_\nu P \end{aligned}$$

or

$$(\rho + P) u^\mu \partial_\mu u_\nu + \partial_\nu P + u_\nu u^\mu \partial_\mu P = 0.$$

If we care only about the spatial part ($\nu = i$), and again apply the non-relativistic limit $u^\mu = (1, v^i)$, $v \ll 1$, and $P \ll \rho$,

$$(\rho + P) [\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}] + \nabla P + \mathbf{v} (\partial_t P + \mathbf{v} \cdot \nabla P) = 0.$$

Neglecting all the second-order terms (i.e. products of $\mathbf{v}$ and P), we get

$$\rho [\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}] + \nabla P = 0,$$

which is the Euler equation from fluid mechanics.

In fact, equation (2.17) represents the conservation of the “stress-energy”, as $\partial^\mu T_{\mu\nu} = 0$ itself is in the form of a continuity equation. Therefore, it should work for all fluids, not just for perfect fluids. We will examine more in the following section.

2.6.2 Electromagnetic Fields

The stress energy tensor is not obtained by guessing. In Chapter 5, we will see that varying the metric tensor in the action gives a stress-energy tensor. In this way, we can get the stress-energy tensor for a scalar field,

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \eta_{\mu\nu} \left[\frac{1}{2} \partial_\alpha \phi \partial^\alpha \phi + V(\phi) \right], \quad (2.20)$$

and that for electromagnetic fields,

$$T_{\mu\nu} = F_{\mu\alpha} F_\nu{}^\alpha - \frac{1}{4} \eta_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}. \quad (2.21)$$

One can check that $\partial^\mu T_{\mu\nu} = 0$ in both cases. However, it is important to know that (2.21) satisfies $\partial^\mu T_{\mu\nu}$ via Maxwell’s equations only when $J^\mu = 0$. If $J^\mu \neq 0$, there should be an extra term that tells the stress-energy tensor of the charge density and currents.

We shall examine (2.21) more to see whether it make sense in its components. Here are some relevant matrices for reference:

$$F_{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{bmatrix}, \quad F_\mu{}^\nu = \eta^{\nu\alpha} F_{\mu\alpha} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{bmatrix}, \quad F^{\mu\nu} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{bmatrix}.$$

The $F_{\alpha\beta} F^{\alpha\beta}$ term gives

$$F_{\alpha\beta} F^{\alpha\beta} = -2(E_x^2 + E_y^2 + E_z^2) + 2(B_x^2 + B_y^2 + B_z^2) = -2\mathbf{E}^2 + 2\mathbf{B}^2.$$

The first term $F_{\mu\alpha}F_\nu{}^\alpha$ can be done by matrix multiplication,

$$\begin{aligned} [F_{\mu\alpha}F_\nu{}^\alpha] &= [F_{\mu\alpha}][F_\nu{}^\alpha]^T = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{bmatrix} \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{E}^2 & -E_y B_z + E_z B_y & E_x B_z - E_z B_x & -E_x B_y + E_y B_x \\ -E_y B_z + E_z B_y & -E_x^2 + B_z^2 + B_y^2 & -E_x E_y - B_x B_y & -E_x E_z - B_x B_z \\ E_x B_z - E_z B_x & -E_x E_y - B_x B_y & -E_y^2 + B_z^2 + B_x^2 & -E_y E_z - B_y B_z \\ -E_x B_y + E_y B_x & -E_x E_z - B_x B_z & -E_y E_z - B_y B_z & -E_z^2 + B_y^2 + B_x^2 \end{bmatrix}. \end{aligned}$$

It is easier to write each components in terms of $\mathbf{E}^2$, $\mathbf{B}^2$, and the Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B}$,

$$F_{\mu\alpha}F_\nu{}^\alpha = \begin{bmatrix} \mathbf{E}^2 & -S_x & -S_y & -S_z \\ -S_x & -E_x^2 + \mathbf{B}^2 - B_x^2 & -E_x E_y - B_x B_y & -E_x E_z - B_x B_z \\ -S_y & -E_x E_y - B_x B_y & -E_y^2 + \mathbf{B}^2 - B_y^2 & -E_y E_z - B_y B_z \\ -S_z & -E_x E_z - B_x B_z & -E_y E_z - B_y B_z & -E_z^2 + \mathbf{B}^2 - B_z^2 \end{bmatrix}.$$

Substituting these terms into (2.21),

$$T_{\mu\nu} = \begin{bmatrix} \mathbf{E}^2 & -S_x & -S_y & -S_z \\ -S_x & -E_x^2 + \mathbf{B}^2 - B_x^2 & -E_x E_y - B_x B_y & -E_x E_z - B_x B_z \\ -S_y & -E_x E_y - B_x B_y & -E_y^2 + \mathbf{B}^2 - B_y^2 & -E_y E_z - B_y B_z \\ -S_z & -E_x E_z - B_x B_z & -E_y E_z - B_y B_z & -E_z^2 + \mathbf{B}^2 - B_z^2 \end{bmatrix} + \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

We can write the spatial part of the result in terms of the Maxwell stress tensor σ_{ij} ,

$$T_{\mu\nu} = \begin{bmatrix} \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) & -S_x & -S_y & -S_z \\ -S_x & -\sigma_{xx} & -\sigma_{xy} & -\sigma_{xz} \\ -S_y & -\sigma_{xy} & -\sigma_{yy} & -\sigma_{yz} \\ -S_z & -\sigma_{xz} & -\sigma_{yz} & -\sigma_{zz} \end{bmatrix}, \quad \text{where} \quad \sigma_{ij} = E_i E_j + B_i B_j - \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)\delta_{ij}.$$

The components of $T_{\mu\nu}$ makes sense. The T_{00} component is exactly the energy density stored in the electric and magnetic fields derived from electromagnetism. The T_{0i} components are the those of the Poynting vector, which is proportional to the momentum density in the fields. (The Poynting vector and momentum density are in fact the same in natural units.) Moreover, It is easy to check that $T_{\mu\nu}$ of electromagnetism is traceless:

$$T^\mu{}_\mu = F_{\mu\alpha}F^{\mu\alpha} - \frac{1}{4}\delta^\mu{}_\mu F_{\alpha\beta}F^{\alpha\beta} = F_{\mu\alpha}F^{\mu\alpha} - F_{\alpha\beta}F^{\alpha\beta} = 0.$$

If we consider a “perfect fluid” of electric and magnetic fields, we have

$$T^\mu{}_\mu = \eta^{\mu\nu}T_{\mu\nu} = \rho - 3P = 0.$$

This indicates a relation between the pressure and energy density:

$$P = \frac{1}{3}\rho.$$

This is exactly the equation of state of a **photon gas**, which can be derived from statistical mechanics. We see that the stress-energy tensor of electromagnetism exhibits a perfect connection among relativity, electromagnetism, and statistical mechanics.

3 DIFFERENTIAL GEOMETRY

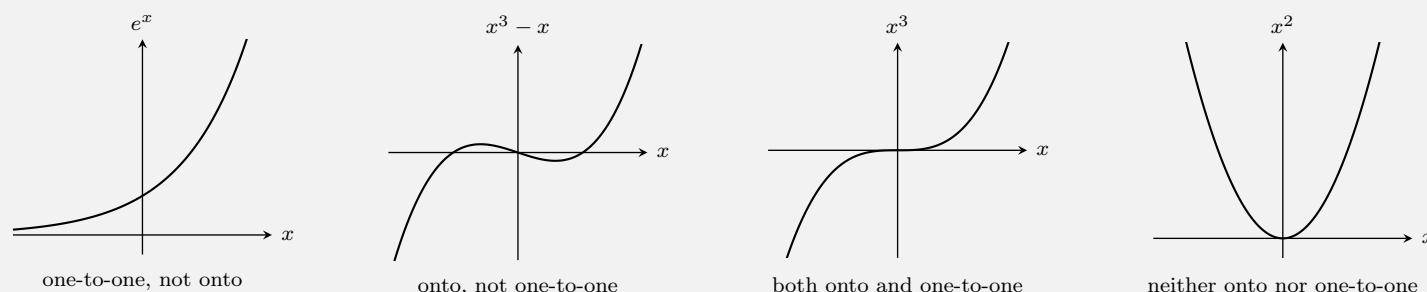
3.1 Manifolds

3.1.1 Sets and Maps

Consider two sets M and N . A **map** $\phi : M \rightarrow N$ is a relationship that assigns each element of M to exactly one element of N . For two maps $\phi : A \rightarrow B$ and $\psi : B \rightarrow C$, the **composition** $\psi \circ \phi : A \rightarrow C$ is defined by the operation $(\psi \circ \phi)(a) = \psi(\phi(a))$. If $a \in A$, then $\phi(a) \in B$ and $(\psi \circ \phi)(a) \in C$. A map is called **one-to-one** or **injective** if each element of N has at most one element of M mapped into it. A map is called **onto** or **surjective** if each element of N has at least one element of M mapped into it. The set M is called the **domain** of the map ϕ , and the set of points in N that M gets mapped into is called the **image** of ϕ .

Example 3.1. Functions

The most familiar maps would be single-variable functions, $\phi : \mathbb{R} \rightarrow \mathbb{R}$. Consider the following four functions



$\phi(x) = e^x$ is one-to-one but not onto because the image does not cover negative values of $\mathbb{R}$. $\phi(x) = x^3 - x$ is onto but not one-to-one because there are multiple x giving the same value of $\phi(x)$, for example, $\phi(-1) = \phi(1) = \phi(0) = 0$. $\phi(x) = x^3$ is both one-to-one and onto. $\phi(x) = x^2$ is neither.

A map that is both one-to-one and onto is **invertible** or **bijjective**. In this case, there exists an **inverse map** $\phi^{-1} : N \rightarrow M$ such that $(\phi^{-1} \circ \phi)(a) = a$. We can generalize these maps to $\phi : \mathbb{R}^m \rightarrow \mathbb{R}^n$, which takes an m -tuple $(x^1, x^2, \dots, x^m)$ to an n -tuple $(y^1, y^2, \dots, y^n)$. Each component y^μ is a function of $(x^1, x^2, \dots, x^m)$. Each of these functions is referred to as C^p if its p th derivative exists and is continuous. The map $\phi : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is C^p if all its component functions are at least C^p . A C^∞ map is called **smooth**.

Two sets M and N are **diffeomorphic** if there exists a C^∞ map $\phi : M \rightarrow N$ with a C^∞ inverse map $\phi^{-1} : N \rightarrow M$. This map ϕ is called a diffeomorphism.

Example 3.2. Diffeomorphism from $\mathbb{R}$ to $\mathbb{R}$

The function $y = x^3 + x$ is a diffeomorphism from $\mathbb{R}$ to $\mathbb{R}$. It is one-to-one and onto, and it is smooth. It is not easy to see its inverse in closed form, but its inverse is smooth. On the contrary, $y = x^3$ is not a diffeomorphism from $\mathbb{R}$ to $\mathbb{R}$. Though $y = x^3$ is one-to-one, onto, and smooth, its inverse $x = \sqrt[3]{y}$ is not because its derivatives at $y = 0$ is not continuous.

3.1.2 Formal Definition of a Manifold

There are several steps to define a manifold. An **open ball** is the set of all points x in $\mathbb{R}^n$ such that

$$|x - y| = \left[\sum_i (x^i - y^i)^2 \right]^{1/2} < r$$

for some fixed $y \in \mathbb{R}^n$ and $r \in \mathbb{R}$. Basically, an open ball is the interior of an n -sphere of radius r centered at y (no boundary). An **open set** in $\mathbb{R}^n$ is a set that is an arbitrary union of open balls. A set $M \subset \mathbb{R}^n$ is open if there is an open ball centered at y that is completely inside M for any y .

A **chart** (called by mathematicians) or **coordinate system** (called by physicists) is a subset U of a set M , with a one-to-one map $\phi : U \rightarrow \mathbb{R}^n$, such that $\phi(U)$ is *open* in $\mathbb{R}^n$. We say that U is an open set in M . Finally, an n -dimensional **manifold** M is a set, which is a collection of all possible charts, that satisfy the following properties:

1. The union of the charts is equal to M .
2. If two charts overlap, $U_\alpha \cap U_\beta \neq \emptyset$, then the (smooth) map $(\phi_\alpha \circ \phi_\beta^{-1})$ takes points in the set $\phi_\beta(U_\alpha \cap U_\beta) \subset \mathbb{R}^n$ *onto* an *open* set $\phi_\alpha(U_\alpha \cap U_\beta) \subset \mathbb{R}^n$. In other words, the two charts are smoothly sewn together.

For our interests, we will assume that all manifolds are C^∞ from now on. Most manifolds cannot be covered by a single chart. Let's see by some examples.

Example 3.3. The 1-sphere S^1

A 1-sphere, or S^1 , is just a circle of some radius r . A natural chart is the polar angle θ . This maps all points on S^1 to real numbers from 0 to 2π . However, by the definition of a chart, $\theta(S^1)$ should be open in $\mathbb{R}$, so it cannot include $\theta = 0$ or $\theta = 2\pi$. But excluding both 0 and 2π makes the chart not covering the entire circle. Thus, at least two charts is necessary to cover S^1 .

Example 3.4. The infinite cylinder $\mathbb{R} \times S^1$

Though a circle S^1 needs to be covered by at least two charts, the infinite cylinder $\mathbb{R} \times S^1$ can be covered by only one chart. We can describe the infinite cylinder (or radius 1) as

$$\mathbb{R} \times S^1 = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 = 1\} = \{(r, \theta, z) \in \mathbb{R}^3 | r = 1\}.$$

Again, the cylindrical coordinates does not work. The map $\phi(x, y, z) = (\theta, z)$ does not map to an open set in $\mathbb{R}^2$ because of $\theta \in [0, 2\pi)$. One possible map to an open set in $\mathbb{R}^2$ by

$$\phi(x, y, z) = (e^z \cos \theta, e^z \sin \theta),$$

where e^z work as some radius term, and θ is the angle in polar coordinates. We have $\theta \in [0, 2\pi)$ still, but the image is the entire $\mathbb{R}^2$ except the origin, so we map $\mathbb{R} \times S^1 \rightarrow \mathbb{R}^2 - (0, 0)$. The map is clearly one-to-one and smooth. This is a valid chart that covers $\mathbb{R} \times S^1$.

3.2 Vectors and Tensors

3.2.1 Directional Derivatives

In curved manifolds, especially in general relativity, we cannot compare or do operations vectors at different points to each other. Vectors are objects specified at a single point p . The space at point p in which these vectors live is called the **tangent space** T_p . It

contains all tangent vectors to the manifolds. But we have not define vectors on a manifold, so the word “tangent vector” is nonsense for now. The goal is to find an object defined at point p that is independent of coordinates and satisfy all requirements for a vector.

Consider parametric curves passing through p on an n -dimensional manifold M . Let f be a smooth functions on M that maps to $\mathbb{R}$. Each curve through p defines an operator, called the **directional derivative**, that maps $f \rightarrow df/d\lambda$ at p . We claim that the tangent space T_p can be identified with the space of directional derivatives along curves through p . There are essentially two things to prove: 1. T_p forms a vector space, and 2. it is also n -dimensional, since we want a vector that can point in n independent directions for an n -manifold.

Proof. For convenience, consider coordinates x^μ at p . Let $d/d\lambda$ and $d/d\eta$ be the derivatives along two curves $x^\mu(\lambda)$ and $x^\mu(\eta)$ passing through p , respectively. (Note that $d/d\lambda$ and $d/d\eta$ are independent of coordinates; they depend only on the parametrization λ and η .) A new operator can be obtained by addition and scalar multiplication:

$$a \frac{d}{d\lambda} + b \frac{d}{d\eta}.$$

A good derivative operator should act linearly on functions and obey Leibniz product rule. The new operator is obviously linear. It indeed obeys the Leibniz rule:

$$\left(a \frac{d}{d\lambda} + b \frac{d}{d\eta}\right)(fg) = af \frac{dg}{d\lambda} + ag \frac{df}{d\lambda} + bf \frac{dg}{d\eta} + bg \frac{df}{d\eta} = \left(a \frac{df}{d\lambda} + b \frac{df}{d\eta}\right)g + \left(a \frac{dg}{d\lambda} + b \frac{dg}{d\eta}\right)f.$$

Hence the set of directional derivatives is closed. They form a vector space. □

With coordinates x^μ (where $\mu = 1, 2, 3, \dots, n$), there is a natural set of n directional derivatives at p . They are the partial derivatives ∂_μ . This is merely the artifact of the definition of the partial derivative: it is the directional derivative along a curve defined by x^ν constant for all $\nu \neq \mu$, parametrized by x^μ . The partial derivatives $\{\partial_\mu\}$ at p form a basis for the tangent space T_p .

Proof. First, the partial derivatives $\{\partial_\mu\}$ span the tangent space. The action of directional derivative on a smooth function f can be written as

$$\frac{df}{d\lambda} = \frac{dx^\mu}{d\lambda} \frac{\partial f}{\partial x^\mu}$$

by the chain rule. This works for any smooth function, so we can decompose the directional derivative as

$$\frac{d}{d\lambda} = \frac{dx^\mu}{d\lambda} \partial_\mu,$$

where ∂_μ are the basis vectors, and $dx^\mu/d\lambda$ are components.

Second, the partial derivatives are linearly independent. Suppose the vector ∂_ν that can be expressed as a linear combination of other basis vectors ∂_μ for $\nu \neq \mu$. We already know how to write this expression: by the chain rule

$$\partial_\nu = \frac{\partial x^\mu}{\partial x^\nu} \partial_\mu = \delta^\mu_\nu \partial_\mu.$$

Because $\nu \neq \mu$, the RHS is zero but the LHS is not—contradiction. Therefore, the partial derivatives are linearly independent. In conclusion, $\{\partial_\mu\}$ form a basis for T_p . It is known as a **coordinate basis** for T_p . □

The transformation law under a change of coordinates is easy to obtain if we work with a coordinate basis. Again, by the chain rule,

$$\partial_{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu,$$

where $\partial_{\mu'}$ is the basis for a new coordinate system $x^{\mu'}$. This is the transformation of basis vectors. The transformation law of components can be obtained by setting the abstract vector $\mathbf{v} = v^\mu \partial_\mu$ unchanged:

$$v^\mu \partial_\mu = v^{\mu'} \partial_{\mu'} = v^{\mu'} \frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu.$$

We see that the components transform as

$$v^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu} v^\mu.$$

So far these are properties of vectors at a single point p . We define a **vector field** to be a map from smooth functions over the manifold to smooth functions by taking a directional derivative at each point.

3.2.2 Commutators

$$[v, w]^\mu = v^\nu \partial_\nu w^\mu - w^\nu \partial_\nu v^\mu. \quad (3.1)$$

3.2.3 Tensors

Given the notion of vectors on a manifold, it is straightforward to define dual vectors (one-forms) and higher rank tensors. The basis vectors of one-forms are often denoted as $\{dx^\mu\}$. As usual, we set the action of basis one-forms on basis vectors to be

$$dx^\mu(\partial_\nu) = \delta^\mu_\nu = \frac{\partial x^\mu}{\partial x^\nu}.$$

A one-form then can be expanded as $\omega = \omega_\mu dx^\mu$. The transformation law of basis one-forms and components are the usual ones

$$dx^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu} dx^\mu, \quad \omega_{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \omega_\mu.$$

Generalizing to rank (k, l) tensors, the transformation law is

$$T^{\mu'_1 \dots \mu'_k}_{\nu'_1 \dots \nu'_l} = \frac{\partial x^{\mu'_1}}{\partial x^{\mu_1}} \dots \frac{\partial x^{\mu'_k}}{\partial x^{\mu_k}} \frac{\partial x^{\nu_1}}{\partial x^{\nu'_1}} \dots \frac{\partial x^{\nu_l}}{\partial x^{\nu'_l}} T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l}. \quad (3.2)$$

What is so great about tensors? According to the transformation law (3.2), if the components of a tensor $T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} = 0$, it is zero in any other frames. It means a tensor equation holds in any coordinates. For example, let's take the most famous tensor equation in general relativity: the Einstein field equation,

$$G^{\mu\nu} + \Lambda g^{\mu\nu} = 8\pi T^{\mu\nu}.$$

Suppose we don't know about these terms, but is provided that both sides are tensors of rank $(2, 0)$. Evidently we can also write

$$G^{\mu\nu} + \Lambda g^{\mu\nu} - 8\pi T^{\mu\nu} = 0.$$

Now we have some tensor equal to zero. Transforming all tensors to a set of new coordinates gives

$$\frac{\partial x^{\mu'}}{\partial x^\mu} \frac{\partial x^{\nu'}}{\partial x^\nu} (G^{\mu\nu} + \Lambda g^{\mu\nu} - 8\pi T^{\mu\nu}) = G^{\mu'\nu'} + \Lambda g^{\mu'\nu'} - 8\pi T^{\mu'\nu'} = 0,$$

or

$$G^{\mu'\nu'} + \Lambda g^{\mu'\nu'} = 8\pi T^{\mu'\nu'}.$$

This may sound trivial, but it is in fact a power statement.

3.3 The Metric Tensor

3.3.1 Properties of the Metric Tensor

The line element in curved spacetime is defined by the metric tensor $g_{\mu\nu}$, as in Minkowski spacetime,

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

What becomes complicated in general relativity is that now elements of $g_{\mu\nu}$ are in general functions of spacetime coordinates. Even in Minkowski spacetime, $g_{\mu\nu}$ can be functions of coordinates. For example, if we express the 3-space using spherical coordinates, then

the line element is

$$ds^2 = -dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \iff g_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}.$$

Here $g_{\theta\theta} = r^2$ and $g_{\phi\phi} = r^2 \sin^2 \theta$ are functions of coordinates. Because ds^2 depends only on the geometry of spacetime, the coordinates we choose can have no physical meaning. Also note that $dx^\mu dx^\nu = dx^\nu dx^\mu$, so

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = g_{\mu\nu} dx^\nu dx^\mu \implies g_{\mu\nu} = g_{\nu\mu}.$$

In other words, the metric tensor is a symmetric tensor. Its matrix representation can always be seen as a symmetric matrix. Hence this 4×4 matrix will have ten independent entries (4 diagonal and 6 off-diagonal). Given that we are allowed to transform coordinates using four functions $x'^\mu(x^0, x^1, x^2, x^3)$, there will be only 6 independent degrees of freedom left without changing any physics.

It is useful to define the **inverse metric** $g^{\mu\nu}$ such that

$$g^{\mu\alpha} g_{\alpha\nu} = \delta^\mu_\nu.$$

In matrices, $g^{\mu\nu}$ and $g_{\mu\nu}$ multiplying together will give the identity matrix. For example, the inverse metric of Minkowski spacetime in spherical coordinates is

$$g^{\mu\nu} = \text{diag} \left(-1, 1, \frac{1}{r^2}, \frac{1}{r^2 \sin^2 \theta} \right).$$

For convenience, the indices of an arbitrary tensor is raised and lowered by the inverse metric and the metric, respectively. For example,

$$g^{\mu\alpha} T_{\alpha\nu} = T^\mu{}_\nu, \quad g_{\mu\alpha} T^{\alpha\nu} = T^\nu{}_\mu.$$

3.3.2 Local Inertial Frames

By the principle of equivalence, as we will see in Section 4.1, any local effects of gravity can be transformed to a free-falling frame by some coordinate transformations. Mathematically, we can always find a set of coordinates $\{x'^\mu\}$ in which the metric is Minkowski at a specific event in spacetime. We say that a curved spacetime is *locally* flat. In matrix language, $g_{\mu\nu}$ is a symmetric, real matrix. It can always be diagonalized at a point by an orthogonal matrix with columns being eigenvectors, and the transformed matrix is diagonal with entries being its eigenvalues. Both the eigenvalues and eigenvectors are real. Then rescaling the coordinate can make the diagonal entries $(-1, 1, 1, 1)$.

Proof. The transformation of the metric tensor can be written as

$$g'_{\mu\nu} = g_{\alpha\beta} \frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x'^\beta}{\partial x^\nu} = g_{\alpha\beta} \Lambda^\alpha{}_\mu \Lambda^\beta{}_\nu = \Lambda^\alpha{}_\mu g_{\alpha\beta} \Lambda^\beta{}_\nu.$$

In matrix language,

$$g' = \Lambda^T g \Lambda.$$

Since g can be diagonalized by an orthogonal matrix O , we have

$$g' = O^{-1} g O = O^T g O,$$

where $O^T = O^{-1}$ or $O^T O = I$ for an orthogonal matrix. We see that $\Lambda = O$. Now suppose the diagonal metric tensor g' has diagonal entries $\lambda_0, \lambda_1, \lambda_2, \lambda_3$, which means

$$ds^2 = \lambda_0 (dx'^0)^2 + \lambda_1 (dx'^1)^2 + \lambda_2 (dx'^2)^2 + \lambda_3 (dx'^3)^2$$

for some coordinates $\{dx'^\mu\}$. Then we can always rescale dx'^μ to make $g'_{\mu\nu} = \eta_{\mu\nu}$:

$$ds^2 = -(\sqrt{-\lambda_0} dx'^0)^2 + (\sqrt{\lambda_1} dx'^1)^2 + (\sqrt{\lambda_2} dx'^2)^2 + (\sqrt{\lambda_3} dx'^3)^2.$$

For physical coordinates, we must require one of the λ 's to be negative (in this case λ_0). Otherwise, we cannot rescale a positive eigenvalue to a negative one with dx'^μ squared. $\square$

This form of the metric is called the **canonical form**, where it becomes a diagonal matrix with elements only ± 1 or 0 . The number of $+1$, -1 , and 0 is called the **signature** of the metric. We will always deal with nondegenerate metric, which means that there is no zero eigenvalues. If all eigenvalues are positive, the metric is called positive definite or **Riemannian**. The metric with a single minus sign is called **Lorentzian** or **pseudo-Riemannian**. General relativity deals with Lorentzian metrics.

Now because there always is a local inertial frame, the causal structure at a particular point is the same as it is in special relativity,

$$ds^2 \begin{cases} < 0, & \text{timelike,} \\ = 0, & \text{null/lightlike,} \\ > 0, & \text{spacelike.} \end{cases}$$

This has significant consequence. Since ds^2 is frame invariant, the type of separation is preserved under all coordinate transformations. Then we can integrate along world line of particles to find quantities we want. First, light rays always travel between events that are null-separated: $ds^2 = 0$ along its world lines no matter how it is curved in spacetime. Next, for material particles that travel below the speed of light, its proper time is the integral

$$\tau = \int_A^B \sqrt{-g_{\mu\nu} dx^\mu dx^\nu} = \int_A^B d\lambda \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}}$$

for any **affine parameter** λ . The choice of affine parameter will not affect any of the physics. We will see that they always cancel in physical equations such as the geodesic equation.

We can do better than setting the metric to its canonical form. At each point p , we can always find **local inertial coordinates** such that the metric is Minkowski at point p and all its first derivative is zero,

$$g_{\mu\nu}|_p = \eta_{\mu\nu}, \quad \partial_\alpha g_{\mu\nu}|_p = 0.$$

However, one cannot make all second derivatives of the metric vanish by a coordinate transformation. There will be $n^2(n^2 - 1)/12$ of them cannot be set to zero for an n -dimensional manifold. *Remember this number.* We will meet it again.

Proof. Consider a coordinate transformation such that the metric transform as

$$g'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta}.$$

First, we want to set $g_{\alpha\beta}|_p = \eta_{\alpha\beta}$. The coordinate transformation equation is

$$\frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \eta_{\alpha\beta} \Big|_p = g'_{\mu\nu} \Big|_p.$$

We can count the number of equations and the number of variables. The variables are the coordinate transformations $\partial x^\alpha / \partial x'^\mu$. There are $n^2 = 16$ of them (n for α and n for μ). The number of independent equations is $n(n+1)/2 = 10$, which are 10 independent components of the symmetric $g'_{\mu\nu}$. This means we have 6 additional degrees of freedom to choose coordinates, which correspond to Lorentz transformations: 3 rotations and 3 boosts.

For first derivatives of the metric, we take the derivative of $g'_{\mu\nu}$,

$$\frac{\partial g'_{\mu\nu}}{\partial x'^\gamma} = \frac{\partial}{\partial x'^\gamma} \left(\frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta} \right) = \frac{\partial}{\partial x'^\gamma} \left(\frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \right) g_{\alpha\beta} + \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \frac{\partial x^\delta}{\partial x'^\gamma} \frac{\partial g_{\alpha\beta}}{\partial x^\delta}.$$

Suppose we want $g_{\alpha\beta}|_p = \eta_{\alpha\beta}$ and $\partial_\delta g_{\alpha\beta}|_p = 0$. Then

$$\left(\frac{\partial^2 x^\alpha}{\partial x'^\mu \partial x'^\gamma} \frac{\partial x^\beta}{\partial x'^\nu} + \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial^2 x^\beta}{\partial x'^\nu \partial x'^\gamma} \right) \Big|_p \eta_{\alpha\beta} = \frac{\partial g'_{\mu\nu}}{\partial x'^\gamma} \Big|_p.$$

Since $g_{\alpha\beta}|_p = \eta_{\alpha\beta}$ fixes the first derivatives of coordinates $\partial x^\alpha / \partial x'^\mu$ (with extra freedoms). This equation should fix the second derivatives of coordinates $\partial^2 x^\alpha / \partial x'^\mu \partial x'^\gamma$; they are the variables. Because partial derivatives commute, the number of variables is $n^2(n+1)/2 = 40$ (n for α , and $n(n+1)/2$ for symmetric μ and γ .) Meanwhile, the number of independent equations is determined by $\partial g'_{\mu\nu} / \partial x'^\gamma$, which is also $n^2(n+1)/2 = 40$ (n for γ , and $n(n+1)/2$ for symmetric μ and ν .) The number of variables matches

exactly the number of independent equations! This means we can solve for appropriate $\partial^2 x^\alpha / \partial x'^\mu \partial x'^\gamma$ such that $g_{\alpha\beta}|_p = \eta_{\alpha\beta}$ and $\partial_\delta g_{\alpha\beta}|_p = 0$ at the same time.

Now we take a look at second derivatives of the metric,

$$\frac{\partial g'_{\mu\nu}}{\partial x'^\gamma \partial x'^\delta} = \frac{\partial^2}{\partial x'^\gamma \partial x'^\delta} \left(\frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \right) g_{\alpha\beta} + (\text{first derivatives of } g_{\alpha\beta}) + \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \frac{\partial x^\rho}{\partial x'^\gamma} \frac{\partial x^\sigma}{\partial x'^\delta} \frac{\partial^2 g_{\alpha\beta}}{\partial x^\rho \partial x^\sigma}.$$

If we want $g_{\alpha\beta}|_p = \eta_{\alpha\beta}$, all first derivatives of $g_{\alpha\beta}$ to be zero, and all second derivatives of $g_{\alpha\beta}$ to be zero, we have

$$\left(\frac{\partial^3 x^\alpha}{\partial x'^\gamma \partial x'^\delta \partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} + \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial^3 x^\alpha}{\partial x'^\gamma \partial x'^\delta \partial x'^\nu} \right) \Big|_p \eta_{\alpha\beta} = \frac{\partial g'_{\mu\nu}}{\partial x'^\gamma \partial x'^\delta} \Big|_p.$$

The variables are third-derivatives of coordinates, $\partial^3 x^\alpha / \partial x'^\gamma \partial x'^\delta \partial x'^\mu$. To find the number of variables, we basically need to find the number of possible combinations of γ, δ , and μ . Classifying them into three cases: none of the indices are the same; two of the indices are the same; and all three indices are the same. If no indices are the same, the number is n choose 3, or $\binom{n}{3} = n(n-1)(n-2)/3!$. If two of the indices are the same, the number is n choose 2 multiplied by 2, or $2\binom{n}{2} = n(n-1)$. (This is because permutation matters in this case: 1, 1, 4 is not the same as 1, 4, 4.) If all three indices are the same, the number is n choose 1, or n . Adding them up and multiply an additional n for α , the number of variables is

$$\left[\frac{n(n-1)(n-2)}{6} + n(n-1) + n \right] n = \frac{n^2(n+1)(n+2)}{6}.$$

The number of equations in $\partial g'_{\mu\nu} / \partial x'^\gamma \partial x'^\delta$ is $[n(n+1)/2]^2$ for symmetric μ and ν and symmetric γ and δ . The number of equations is more than the number of variables:

$$\left[\frac{n(n+1)}{2} \right]^2 - \frac{n^2(n+1)(n+2)}{6} = \boxed{\frac{1}{12}n^2(n^2-1)}.$$

This means there are $n^2(n^2-1)/12$ second derivatives (20 in 4-dimensional spacetime) of the metric that cannot be set to zero by coordinate transformations. They represent gravity, or curvature of spacetime. $\square$

3.3.3 Tensor Densities

Recall that the Levi-Civita symbol is defined as

$$\tilde{\epsilon}_{\mu_1 \mu_2 \dots \mu_n} = \begin{cases} +1, & \text{if } \mu_1 \mu_2 \dots \mu_n \text{ is an even permutation of } 01 \dots (n-1), \\ -1, & \text{if } \mu_1 \mu_2 \dots \mu_n \text{ is an odd permutation of } 01 \dots (n-1), \\ 0, & \text{otherwise.} \end{cases}$$

Here we use $\tilde{\epsilon}$ to denote the Levi-Civita symbol. It is called a symbol because it does not obey tensor transformation rule, as the symbol is defined to be the same in every coordinate system. We can turn it into a tensor. It is known that for any $n \times n$ matrix $M^\mu{}_{\mu'}$, its determinant $\det M$ obeys (see proof in Appendix A.2.1)

$$\tilde{\epsilon}_{\mu'_1 \mu'_2 \dots \mu'_n} |M| = \tilde{\epsilon}_{\mu_1 \mu_2 \dots \mu_n} M^{\mu_1}{}_{\mu'_1} M^{\mu_2}{}_{\mu'_2} \dots M^{\mu_n}{}_{\mu'_n}.$$

By setting $M^\mu{}_{\mu'} = \partial x^\mu / \partial x^{\mu'}$, the matrix of partial derivatives,

$$\tilde{\epsilon}_{\mu'_1 \mu'_2 \dots \mu'_n} = \det \left(\frac{\partial x^{\mu'}}{\partial x^\mu} \right) \tilde{\epsilon}_{\mu_1 \mu_2 \dots \mu_n} \frac{\partial x^{\mu_1}}{\partial x^{\mu'_1}} \frac{\partial x^{\mu_2}}{\partial x^{\mu'_2}} \dots \frac{\partial x^{\mu_n}}{\partial x^{\mu'_n}}, \quad (3.3)$$

where the matrix $\partial x^{\mu'} / \partial x^\mu$ is the inverse of the matrix $\partial x^\mu / \partial x^{\mu'}$. (The determinant of partial derivatives are also known as the **Jacobian** from elementary vector calculus.) The Levi-Civita symbol transform almost as a tensor, except for the determinant factor. Objects transforming in this way are called **tensor densities**. The power of the Jacobian is called the **weight** of the tensor density, so the Levi-Civita symbol has weight 1.

Example 3.5. Determinant of the metric

The determinant of the metric, usually written as $g \equiv \det g_{\mu\nu}$, is also a tensor density:

$$g(x^{\mu'}) = \det\left(\frac{\partial x^{\mu'}}{\partial x^{\mu}}\right)^{-2} g(x^{\mu}). \quad (3.4)$$

Here is the proof:

To be written.

Of course, there is no transformation partial derivatives because g is a scalar. The weight of g is -2 .

With (3.4), we can relate the Jacobian to the determinant of a metric. Specifically, we can use it to promote a tensor density to a tensor by multiplying by $|g|^{w/2}$. The absolute sign is because $g < 0$ for Lorentzian metrics. For example, the **Levi-Civita tensor** is defined as

$$\epsilon_{\mu_1\mu_2\cdots\mu_n} = \sqrt{|g|} \tilde{\epsilon}_{\mu_1\mu_2\cdots\mu_n}, \quad (3.5)$$

where we drop the tilde symbol. One can easily show that (3.5) transform as a tensor just by combining (3.3) and (3.4). Now with the Levi-Civita tensor, we can raise and lower indices using the metric, and develop other useful results:

$$\epsilon^{\mu_1\mu_2\cdots\mu_n} = \frac{1}{\sqrt{|g|}} \tilde{\epsilon}^{\mu_1\mu_2\cdots\mu_n},$$

and the contraction

$$\epsilon^{\mu_1\mu_2\cdots\mu_p\alpha_1\cdots\alpha_{n-p}} \epsilon_{\mu_1\mu_2\cdots\mu_p\beta_1\cdots\beta_{n-p}} = (-1)^s p!(n-p)! \delta^{[\alpha_1}_{\beta_1} \cdots \delta^{\alpha_{n-p}]}_{\beta_{n-p}}. \quad (3.6)$$

3.4 Differential Forms

3.4.1 Exterior Derivatives

There is a special type of tensors called **differential forms**. They are useful when we talk about differentiation and integration on manifolds. A p -form is defined to be a $(0, p)$ tensor that is totally antisymmetric. The space of all p -forms is denoted as Λ^p ; the space of all p -form fields over a manifold M is $\Lambda^p(M)$. The number of linearly independent p -forms on an n -dimensional manifold is $n!/p!(n-p)!$, so there cannot be p -forms with $p > n$.

Directly multiplying two differential forms need not give another differential form because the resulting object may not be totally antisymmetric. Therefore, we define the **wedge product** of a p -form A and a q -form B to be

$$(A \wedge B)_{\mu_1\cdots\mu_{p+q}} = \frac{(p+q)!}{p!q!} A_{[\mu_1\cdots\mu_p} B_{\mu_{p+1}\cdots\mu_{p+q}]}. \quad (3.7)$$

For example, the wedge product of two 1-forms is

$$(A \wedge B)_{\mu\nu} = 2A_{[\mu} B_{\nu]} = A_{\mu} B_{\nu} - A_{\nu} B_{\mu}.$$

The wedge product does not commute nor anticommute in general. It obeys

$$A \wedge B = (-1)^{pq} B \wedge A.$$

Partial derivatives transform does not transform like tensors in curved spacetime, so we define other derivatives that transform like a tensor. Here we shall mention the **exterior derivative** d , which differentiates p -form fields to a $(p+1)$ -form fields:

$$(dA)_{\mu_1\cdots\mu_{p+1}} = (p+1)\partial_{[\mu_1} A_{\mu_2\cdots\mu_{p+1}]}. \quad (3.8)$$

There is a modified Leibniz rule for exterior derivatives. The exterior derivative of the product of a p -form ω and a q -form η is

$$d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^p \omega \wedge (d\eta). \quad (3.9)$$

The exterior derivative of another exterior derivative always vanishes,

$$d(dA) = 0, \quad \text{or} \quad [d(dA)]_{\mu_1 \dots \mu_{p+2}} = (p+1)(p+2) \partial_{[\mu_1} \partial_{\mu_2} A_{\mu_3 \dots \mu_{p+2}]} = 0.$$

Essentially, it is because partial derivatives commute, so antisymmetrizing them gives zero. A p -form A is called **closed** if $dA = 0$, and is **exact** if $A = dB$ for some $(p-1)$ -form B . It turns out that all closed forms are exact except for zero-forms in Minkowski spacetime.

The good thing about the exterior derivative is that it removes the nontensorial part of partial derivatives on a form. For example, the partial derivative of a 1-form transform like

$$\frac{\partial}{\partial x^{\mu'}} \omega_{\nu'} = \frac{\partial x^{\mu}}{\partial x^{\mu'}} \frac{\partial}{\partial x^{\mu}} \left(\frac{\partial x^{\nu}}{\partial x^{\nu'}} \omega_{\nu} \right) = \frac{\partial x^{\mu}}{\partial x^{\mu'}} \frac{\partial x^{\nu}}{\partial x^{\nu'}} \left(\frac{\partial}{\partial x^{\mu}} \omega_{\nu} \right) + \omega_{\nu} \frac{\partial x^{\mu}}{\partial x^{\mu'}} \frac{\partial}{\partial x^{\mu}} \frac{\partial x^{\nu}}{\partial x^{\nu'}}. \quad (3.10)$$

If $\partial_{\mu} \omega_{\nu}$ is a tensor, the first term on the RHS is a good tensor transformation, but the second term should not be there. In Minkowski spacetime where $\partial x^{\nu} / \partial x^{\nu'}$ does not depend on x^{μ} , the second term vanishes, but it is not true in curved spacetime. Note that the nontensorial term can be written as

$$\omega_{\nu} \frac{\partial x^{\mu}}{\partial x^{\mu'}} \frac{\partial}{\partial x^{\mu}} \frac{\partial x^{\nu}}{\partial x^{\nu'}} = \omega_{\nu} \frac{\partial^2 x^{\nu}}{\partial x^{\mu'} \partial x^{\nu'}}.$$

Since partial derivatives commute, this expression is symmetric in μ' and ν' . The exterior derivative (3.8) antisymmetrize commuting partial derivatives, so the result is zero. In other words, the exterior derivative remove this nontensorial term, so it transform like a tensor.

Finally, we define the **Hodge star operator** on an n -dimensional manifold as a map from p -forms to $(n-p)$ -forms,

$$(*A)_{\mu_1 \dots \mu_{n-p}} = \frac{1}{p!} \epsilon^{\nu_1 \dots \nu_p}_{\mu_1 \dots \mu_{n-p}} A_{\nu_1 \dots \nu_p}. \quad (3.11)$$

We call $*A$ the **Hodge dual** of A . Applying twice the Hodge star operator twice gives

$$*(*A) = (-1)^{s+p(n-p)} A,$$

where s is the number of minus signs in the eigenvalues of the metric.

Example 3.6. Maxwell's equations

By the definition of the exterior derivative (3.8), the electromagnetic field tensor can be written as

$$F_{\mu\nu} = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu} = dA.$$

The field tensor is antisymmetric, so it can be seen as a two-form. Clearly, F is closed, because

$$dF = d(dA) = 0.$$

This is just $\nabla \cdot \mathbf{B} = 0$ and Faraday's law. Gauss's law and Maxwell-Ampère's law can be written in terms of the Hodge dual,

$$d(*F) = *J.$$

To be written.

3.4.2 Integration

It turns out that differential forms are perfect objects for integration on manifolds. In ordinary vector calculus in $\mathbb{R}^n$, we know that changing coordinates picks up a factor of Jacobian in the volume element,

$$d^n x' = \det \left(\frac{\partial x^{\mu'}}{\partial x^\mu} \right) d^n x.$$

In other words, the volume element $d^n x$ transform like a tensor density. To integrate on a general n -dimensional manifold, we write

$$d^n x = dx^0 \wedge \cdots \wedge dx^{n-1}.$$

The wedge product is defined to product totally antisymmetric objects, so we can write it in terms of the Levi-Civita symbol,

$$d^n x = dx^0 \wedge \cdots \wedge dx^{n-1} = \frac{1}{n!} \tilde{\epsilon}_{\mu_1 \cdots \mu_n} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_n}.$$

Under a coordinate transformation, the Levi-Civita symbol is invariant, so

$$\tilde{\epsilon}_{\mu_1 \cdots \mu_n} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_n} = \tilde{\epsilon}_{\mu_1 \cdots \mu_n} \frac{\partial x^{\mu_1}}{\partial x^{\mu'_1}} \cdots \frac{\partial x^{\mu_n}}{\partial x^{\mu'_n}} dx^{\mu'_1} \wedge \cdots \wedge dx^{\mu'_n} = \det \left(\frac{\partial x^\mu}{\partial x^{\mu'}} \right) \tilde{\epsilon}_{\mu'_1 \cdots \mu'_n} dx^{\mu'_1} \wedge \cdots \wedge dx^{\mu'_n}.$$

In the second equality, we used (3.3). Multiplying by the Jacobian $\det(\partial x^{\mu'}/\partial x^\mu)$ and dividing by $n!$ on both sides, we have

$$\frac{1}{n!} \det \left(\frac{\partial x^{\mu'}}{\partial x^\mu} \right) \tilde{\epsilon}_{\mu_1 \cdots \mu_n} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_n} = \frac{1}{n!} \tilde{\epsilon}_{\mu'_1 \cdots \mu'_n} dx^{\mu'_1} \wedge \cdots \wedge dx^{\mu'_n}.$$

This recovers the usual volume element transformation rule,

$$\det \left(\frac{\partial x^{\mu'}}{\partial x^\mu} \right) d^n x = d^n x'.$$

To make the volume element transform like a tensor, we multiply both sides by $\sqrt{|g|}$ in individual coordinates because $\sqrt{|g|}$ transform according to (3.4),

$$\sqrt{|g'|} d^n x' = \sqrt{|g|} d^n x.$$

In conclusion, the correct volume element of integration on a manifold is $\sqrt{|g|} d^n x$. The metric determinant The integral of a scalar function $\phi(x^\mu)$ on a manifold is written as

$$I = \int \phi(x^\mu) \sqrt{|g|} d^n x. \quad (3.12)$$

Example 3.7. Diagonal metric

For simplicity, consider a diagonal metric, ($\{e_\mu\}$ are orthogonal basis vectors),

$$ds^2 = -g_{tt} dt^2 + g_{xx} dx^2 + g_{yy} dy^2 + g_{zz} dz^2,$$

with all of $g_{tt}, g_{xx}, g_{yy}, g_{zz} > 0$. A two-dimensional surface of spacetime can be described by the set of events with two constant coordinates. For example, let $dt = dx = 0$. Because we are using a orthogonal coordinate system and spacetime is locally flat, the infinitesimal area dA can be seen as a small rectangle of side lengths $d\ell^y$ and $d\ell^z$,

$$dA = d\ell^y d\ell^z \quad \text{where} \quad d\ell^y = ds|_{\text{fixed } t, x, z} = \sqrt{g_{yy}} dy, \quad d\ell^z = ds|_{\text{fixed } t, x, y} = \sqrt{g_{zz}} dz.$$

Thus, $dA = \sqrt{g_{yy}g_{zz}} dy dz$. The 3-volume element is similar. For example, events with the same t constitutes a volume of

$$dV = \sqrt{g_{xx}g_{yy}g_{zz}} dx dy dz,$$

and finally, the 4-volume element

$$dV_4 = \sqrt{g_{tt}g_{xx}g_{yy}g_{zz}} dt dx dy dz.$$

Since the metric is diagonal, the metric determinant is just the products of diagonal entries, which means $\sqrt{|g|} = \sqrt{g_{tt}g_{xx}g_{yy}g_{zz}}$. In this example, we clearly see the geometric interpretation of the factor $\sqrt{|g|}$ in the volume element.

Example 3.8. Spatial geometry in Schwarzschild spacetime

The Schwarzschild metric describes the spacetime *outside* a static, spherically symmetric object of mass M (like a star or a black hole) in vacuum. The metric is given by

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

We will derive this metric from Einstein's field equation in Chapter 7. For now we shall use it directly as an example. Here we require $r > 2GM$. Consider events that occur at the same coordinate time t . A sphere of fixed r -coordinate has surface area

$$A(r) = \int \sqrt{g_{\theta\theta}} \sqrt{g_{\phi\phi}} d\theta d\phi = \int \sqrt{r^2} \sqrt{r^2 \sin^2 \theta} d\theta d\phi = r^2 \int_0^\pi d\theta \sin \theta \int_0^{2\pi} d\phi = 4\pi r^2.$$

This is consistent with Euclidean geometry, but we will see that the spatial geometry is not Euclidean. The problem is with the radial distance separating two spheres at $r = r_1$ and $r = r_2 > r_1$,

$$\Delta R = \int_{r_1}^{r_2} \sqrt{g_{rr}} dr = \int_{r_1}^{r_2} \left(1 - \frac{2GM}{r}\right)^{-1/2} dr > \int_{r_1}^{r_2} dr = r_2 - r_1$$

for $r_1, r_2 > 2GM$ (outside the Schwarzschild radius). We see that even though the surface areas of the two spheres are $4\pi r_1^2$ and $4\pi r_2^2$, the radial distance between them is greater than $r_2 - r_1$. This indicates a nontrivial curved spacetime.

4 CURVATURE

4.1 On the Relation of Gravitation to Curvature

The difference between special relativity and general relativity is that the latter takes gravitational fields into account. We already know that in special relativity, things happen differently in frames with relative velocity, such as time dilation. This is the case where we neglect gravitational fields. With a non-uniform gravitational field in one frame, clocks do not run at the same rate either.

4.1.1 The Principle of Equivalence

Gravity is the special one among all forces. Particles with the same initial conditions (position and velocity) follow the same trajectory in a gravitational field, regardless of their internal composition. Other forces do not have this property: electromagnetism affects only charged particles and their trajectory depends on the charge-mass ratio. Strong and weak interactions are similar—different particles act differently. Only gravity acts on all particles. Alternatively, we can think of *mass* as the “charge” of gravity so the trajectory depends on the “mass-mass” ratio, which obviously does not affect the trajectory of an object.

In general relativity, a freely falling frame in a gravitational field is actually the “true” inertial frame. Consider two frames, each of which has an identical particle in it:

1. A frame with a uniform downward gravitational field g , e.g. a lab on the surface of the Earth.
2. A frame with a uniform upward acceleration with magnitude $a = g$ with no gravity, e.g. a rocket in free space accelerating.

The second frame is a “non-inertial” frame in the sense of special relativity. If we stick to this frame, we will see the particle accelerating downward, as if it is in a gravitational field. An observer fixed relative to the frame, with measurements all solely based on experiments in this frame, cannot tell whether the frame is accelerated or is in a gravitational field. This leads to the **Principle of Equivalence**:

The Principle of Equivalence

There is no experiment that locally can distinguish a uniform gravitational field from a laboratory which undergoes uniform acceleration.

Now consider the same two frames, but now there is a laser beam on the left wall that emits light horizontally. If you are in the second frame, light bends downward because the frame is accelerating upward. By the equivalence principle, the light beam in the frame with gravity should also bend downward. Then we arrive at a very astonishing conclusion: light in a gravitational field follows a curved path, even if it has no mass. Because light travels along null geodesics, and those geodesics appear curved, we say that gravity curves spacetime. Minkowski space is a flat space with Euclid’s parallelism axiom but it is not a Euclidean space. For example, photons with straight world lines have zero proper length. Thus, special relativity has a flat, non-Euclidean geometry.

In a nonuniform gravitational field, parallel world lines of two particles do not generally remain parallel, so a spacetime with gravity is not flat. Einstein proposed that objects follow geodesics that appear to be locally straight parallel lines but not globally. Hence the theory of gravity uses a curved spacetime to represent the effects of gravity on particles’ trajectory. We need to first study the mathematics that describes **curvature**.

4.1.2 The Geodesic Equation

We begin with a physical view of geodesics. The world lines of free particles in a gravitational field are **geodesics** of the spacetime. To find the trajectory of a free particle from events A and B , we extremize its proper time:

$$\tau = \int_A^B d\lambda \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}}.$$

The Euler-Lagrange equation reads,

$$\frac{\partial L}{\partial x^\mu} - \frac{d}{d\lambda} \frac{\partial L}{\partial (dx^\mu/d\lambda)} = 0, \quad \text{where} \quad L = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} = \frac{d\tau}{d\lambda}$$

can be seen as the Lagrangian. Before we go on, we should look at a simple example.

Example 4.1. Geodesic equation in cylindrical polar coordinates

Consider a flat spacetime in cylindrical polar coordinates:

$$d\tau^2 = dt^2 - dr^2 - r^2 d\phi^2 - dz^2.$$

The Lagrangian is

$$L = \frac{d\tau}{d\lambda} = \left[\left(\frac{dt}{d\lambda} \right)^2 - \left(\frac{dr}{d\lambda} \right)^2 - r^2 \left(\frac{d\phi}{d\lambda} \right)^2 - \left(\frac{dz}{d\lambda} \right)^2 \right]^{1/2}.$$

Then calculate the derivatives needed in the Euler-Lagrange equation. For example,

$$\frac{\partial L}{\partial (dt/d\lambda)} = \frac{1}{2L} \frac{2 dt}{d\lambda} = \frac{d\lambda}{d\tau} \frac{dt}{d\lambda} = \frac{dt}{d\tau}.$$

Other derivatives are similar,

$$\begin{aligned} \frac{\partial L}{\partial (dr/d\lambda)} &= -\frac{dr}{d\tau}, & \frac{\partial L}{\partial (d\phi/d\lambda)} &= -r^2 \frac{d\phi}{d\tau}, & \frac{\partial L}{\partial (dz/d\lambda)} &= -\frac{dz}{d\tau}. \\ \frac{\partial L}{\partial t} &= \frac{\partial L}{\partial \phi} = \frac{\partial L}{\partial z} = 0, & \frac{\partial L}{\partial r} &= -\frac{r}{L} \left(\frac{d\phi}{d\lambda} \right)^2 = -r \frac{d\lambda}{d\tau} \left(\frac{d\phi}{d\lambda} \right)^2. \end{aligned}$$

We can immediately read the Euler-Lagrange equation for t ,

$$0 = \frac{d}{d\lambda} \left(\frac{dt}{d\lambda} \right) \implies \frac{d\lambda}{d\tau} \frac{d}{d\lambda} \left(\frac{dt}{d\lambda} \right) = 0 \implies \boxed{\frac{d^2 t}{d\tau^2} = 0}.$$

The ϕ -, z -, and r - equations are similar,

$$\begin{aligned} 0 = \frac{d}{d\lambda} \left(-r^2 \frac{d\phi}{d\lambda} \right) &\implies \boxed{r \frac{d^2 \phi}{d\tau^2} + 2 \frac{dr}{d\tau} \frac{d\phi}{d\tau} = 0}, & \text{and} & & \frac{d}{d\lambda} \left(-\frac{dz}{d\lambda} \right) = 0 &\implies \boxed{\frac{d^2 z}{d\tau^2} = 0}, \\ -r \frac{d\lambda}{d\tau} \left(\frac{d\phi}{d\lambda} \right)^2 - \frac{d}{d\lambda} \left(-\frac{dr}{d\lambda} \right) &= 0 &\implies & & \boxed{\frac{d^2 r}{d\tau^2} - r \left(\frac{d\phi}{d\tau} \right)^2 = 0}. \end{aligned}$$

The r -, ϕ -, and z -equations are the usual acceleration components in cylindrical polar coordinates in Newtonian mechanics, with dt replaced by $d\tau$,

$$\mathbf{a} = (\ddot{r} - r\dot{\phi}^2)\hat{\mathbf{r}} + (r\ddot{\phi} + 2\dot{r}\dot{\phi})\hat{\boldsymbol{\phi}} + \ddot{z}\hat{\mathbf{z}} = 0.$$

The Lagrangian helps find geodesic equations quickly even in curved spacetime. The general **geodesic equation** for material particles is written as

$$\boxed{\frac{d^2 x^\alpha}{d\tau^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0 \iff \frac{du^\alpha}{d\tau} + \Gamma^\alpha_{\mu\nu} u^\mu u^\nu = 0.} \quad (4.1)$$

where $\Gamma^\alpha_{\mu\nu}$ are known as **Christoffel symbols**. They are related to the *first derivatives* of the metric tensor $g_{\mu\nu}$ by

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2}g^{\alpha\beta}(\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}). \quad (4.2)$$

See Appendix A.3 for proof (from the Euler-Lagrange equation). From (4.2), we can see that $\Gamma^\alpha_{\mu\nu} = \Gamma^\alpha_{\nu\mu}$: exchanging μ and ν does nothing to the sum of first two terms, and $g_{\mu\nu} = g_{\nu\mu}$ for the third term. Hence instead of $4 \times 4 \times 4 = 64$ components in $\Gamma^\alpha_{\mu\nu}$, there are only 4×10 independent components (4 from α , and 10 from symmetric μ, ν). The Christoffel symbols play crucial roles in general relativity, not just in geodesics. Their mathematical meaning in geometry will appear in the next section. We will also derive the geodesic equation using *parallel transport* in Section 4.3 after we have enough mathematical tools.

Example 4.2. Christoffel symbols in cylindrical polar coordinates

Consider the metric

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = -dt^2 + dr^2 + r^2d\phi^2 + dz^2, \quad g_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad g^{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/r^2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

We won't calculate all 40 of the Christoffel symbols because most of them are zero in cylindrical coordinates with many symmetries. There are only three nonzero Christoffel symbols:

$$\begin{aligned} \Gamma^r_{\phi\phi} &= \frac{1}{2}g^{r\beta} \left(\frac{\partial g_{\beta\phi}}{\partial\phi} + \frac{\partial g_{\beta\phi}}{\partial\phi} - \frac{\partial g_{\phi\phi}}{\partial x^\beta} \right) = \frac{1}{2}g^{rr} \left(-\frac{\partial g_{\phi\phi}}{\partial r} \right) = -r. \\ \Gamma^\phi_{\phi r} &= \Gamma^\phi_{r\phi} = \frac{1}{2}g^{\phi\beta} \left(\frac{\partial g_{\beta\phi}}{\partial r} + \frac{\partial g_{\beta r}}{\partial\phi} - \frac{\partial g_{\phi r}}{\partial x^\beta} \right) = \frac{1}{2}g^{\phi\phi} \frac{\partial g_{\phi\phi}}{\partial r} = \frac{1}{r}. \end{aligned}$$

The geodesic equation is

$$\frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Check that the Christoffel symbols produce the geodesic equations for cylindrical coordinates:

$$\begin{aligned} \frac{d^2t}{d\tau^2} &= 0 \iff \Gamma^t_{\mu\nu} = 0, \\ \frac{d^2r}{d\tau^2} - r \left(\frac{d\phi}{d\tau} \right)^2 &= 0 \iff \Gamma^r_{\phi\phi} = -r, \text{ others vanish} \\ \frac{d^2\phi}{d\tau^2} + \frac{2}{r} \frac{dr}{d\tau} \frac{d\phi}{d\tau} &= 0 \iff \Gamma^\phi_{\phi r} = \Gamma^\phi_{r\phi} = \frac{1}{r}, \text{ others vanish} \\ \frac{d^2z}{d\tau^2} &= 0 \iff \Gamma^z_{\mu\nu} = 0. \end{aligned}$$

(Note the factor of 2 in the third equation because $\Gamma^\phi_{\phi r} = \Gamma^\phi_{r\phi}$.)

In general, we don't calculate the Christoffel symbols from derivatives of metric using (4.2). It is easier to use the Euler-Lagrange equation to find the geodesic equations, and then read off the Christoffel symbols. This works especially quickly in high dimensions, where there are $n^2(n+1)/2$ (too many) independent Christoffel symbols. If we are interested in geodesics of massless particles such as a photon, the proper time τ in (4.1) should be replaced with an affine parameter λ because τ is undefined,

$$\frac{d^2x^\alpha}{d\lambda^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0.$$

This also works for material particles, but the equation is more physically meaningful with derivatives with respect to τ .

4.2 The Covariant Derivative*

In this section, we introduce the most important derivative in general relativity. It is well-known that partial derivatives of a tensor

does not transform like a tensor in general. We would like to find a derivative operator, ∇_μ , such that when it acts on any (k, l) tensor, the resulting object is a $(k, l + 1)$ tensor.

4.2.1 Connection Coefficients

Since the goal is some kind of derivative operator, it should obey linearity and Leibniz rule. We also introduce three more conditions that ∇_μ should satisfy, so in total there are five conditions:

1. Linearity: for all (k, l) tensor A and B and $a, b \in \mathbb{R}$,

$$\nabla_\mu(aA^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_l} + bB^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_k}) = a\nabla_\mu A^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_l} + b\nabla_\mu B^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_k}.$$

2. Leibniz rule: for all (k, l) tensor A and (k', l') tensor B ,

$$\nabla_\mu(A^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_l} B^{\gamma_1 \dots \gamma_{k'}}_{\delta_1 \dots \delta_{l'}}) = (\nabla_\mu A^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_l}) B^{\gamma_1 \dots \gamma_{k'}}_{\delta_1 \dots \delta_{l'}} + A^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_l} (\nabla_\mu B^{\gamma_1 \dots \gamma_{k'}}_{\delta_1 \dots \delta_{l'}}).$$

3. Commutativity with contraction: it does not matter whether you take derivative first or contraction first,

$$\nabla_\mu(A^{\alpha_1 \dots \gamma \dots \alpha_k}_{\beta_1 \dots \gamma \dots \beta_l}) = \nabla_\mu A^{\alpha_1 \dots \gamma \dots \alpha_k}_{\beta_1 \dots \gamma \dots \beta_l}.$$

In other words, we can write

$$\nabla_\mu(\delta^\gamma_\alpha A^{\alpha_1 \dots \alpha \dots \alpha_k}_{\beta_1 \dots \gamma \dots \beta_l}) = \delta^\gamma_\alpha \nabla_\mu A^{\alpha_1 \dots \alpha \dots \alpha_k}_{\beta_1 \dots \gamma \dots \beta_l}.$$

So condition is equivalent to saying $\nabla_\mu \delta^\gamma_\alpha$, which is of course a reasonable thing to ask for.

4. Consistency with the notion of tangent vectors as directional derivatives on scalar functions:

$$\xi(f) = \xi^\mu \nabla_\mu f \implies \nabla_\mu f = \partial_\mu f.$$

This means ∇_μ is the same as partial derivatives ∂_μ on any scalar function f .

5. Torsion free: for any scalar function f , the derivative operator commutes,

$$\nabla_\mu \nabla_\nu f = \nabla_\nu \nabla_\mu f.$$

There are certain conditions where this condition is dropped, but we will assume this to be true in this notes. (Note that this condition is not the same as condition 4 because $\nabla_\mu \nabla_\nu f = \nabla_\mu(\partial_\nu f) \neq \partial_\mu \partial_\nu f$. When ∇_ν act on f , the resulting object $\partial_\nu f$ is no longer a scalar function.)

It is easy to show that there exist an operator that satisfy all these conditions—the partial derivative ∂_μ is one example. However, these conditions do not define a unique derivative operator.

Consider two derivative operators ∇_μ and $\tilde{\nabla}_\mu$. Condition 4 infers that they agree in their action on scalar field f . We want to find the disagreements when acting on general tensors. First, let ω_ν be a dual vector field. We compute $(\tilde{\nabla}_\mu - \nabla_\mu)(f\omega_\nu)$,

$$\tilde{\nabla}_\mu(f\omega_\nu) - \nabla_\mu(f\omega_\nu) = (\tilde{\nabla}_\mu f)\omega_\nu + f\tilde{\nabla}_\mu \omega_\nu - (\nabla_\mu f)\omega_\nu - f\nabla_\mu \omega_\nu = f(\tilde{\nabla}_\mu - \nabla_\mu)\omega_\nu,$$

where we used condition 4 so that $\tilde{\nabla}_\mu f = \nabla_\mu f$. Note that $(\tilde{\nabla}_\mu - \nabla_\mu)$ is not only linear in numbers, but also in functions. We define the **connection coefficients** $C^\alpha_{\mu\nu}$:

$$(\tilde{\nabla}_\mu - \nabla_\mu)\omega_\nu = C^\alpha_{\mu\nu} \omega_\alpha. \quad (4.3)$$

We will use this definition and conditions on ∇_μ to find the effect of $\tilde{\nabla}_\mu - \nabla_\mu$ on tensors of other ranks. First, the connection coefficient is symmetric in the two lower indices. Let $\omega_\nu = \nabla_\nu f = \tilde{\nabla}_\nu f$, then by (4.3),

$$\tilde{\nabla}_\mu \tilde{\nabla}_\nu f - \nabla_\mu \nabla_\nu f = C^\alpha_{\mu\nu} \nabla_\alpha f.$$

By condition 5, the LHS is symmetric in μ and ν , so $C^\alpha_{\mu\nu}$ must satisfy

$$C^\alpha_{\mu\nu} = C^\alpha_{\nu\mu}.$$

Next, let $\tilde{\nabla}_\mu - \nabla_\mu$ act on $\omega_\nu v^\nu$ for some vector field v^ν and dual vector field ω_ν . By Leibniz rule,

$$(\tilde{\nabla}_\mu - \nabla_\mu)(\omega_\nu v^\nu) = (C^\alpha_{\mu\nu} \omega_\alpha) v^\nu + \omega_\nu (\tilde{\nabla}_\mu - \nabla_\mu) v^\nu.$$

By condition 4, the LHS is zero because $\omega_\nu v^\nu$ is a scalar field. Notice that both α and ν are dummy indices in the first term on the RHS, so we are free to exchange them,

$$0 = (C^\nu_{\mu\alpha} \omega_\nu) v^\alpha + \omega_\nu (\tilde{\nabla}_\mu - \nabla_\mu) v^\nu = \omega_\nu [(\tilde{\nabla}_\mu - \nabla_\mu) v^\nu + C^\nu_{\mu\alpha} v^\alpha] = 0.$$

Since ω_ν is arbitrary, we must have

$$(\tilde{\nabla}_\mu - \nabla_\mu) v^\nu = -C^\nu_{\mu\alpha} v^\alpha. \quad (4.4)$$

This is how $\tilde{\nabla}_\mu - \nabla_\mu$ act on vectors. To be concrete, let us work on one more example.

Example 4.3. $\tilde{\nabla}_\mu - \nabla_\mu$ acting on a $(1,1)$ tensor T^α_β

We would like to find the effects of $\tilde{\nabla}_\mu - \nabla_\mu$ on T^α_β in terms of connection coefficients. Let $\tilde{\nabla}_\mu - \nabla_\mu$ act on a scalar function $T^\alpha_\beta \omega_\alpha v^\beta$. By Leibniz rule, and using both (4.3) and (4.4),

$$(\tilde{\nabla}_\mu - \nabla_\mu)(T^\alpha_\beta \omega_\alpha v^\beta) = \omega_\alpha v^\beta (\tilde{\nabla}_\mu - \nabla_\mu) T^\alpha_\beta + (C^\nu_{\mu\alpha} \omega_\nu) T^\alpha_\beta v^\beta - (C^\beta_{\mu\nu} v^\nu) T^\alpha_\beta \omega_\alpha.$$

On the RHS, we want each term to have a factor of $\omega_\alpha v^\beta$, so we exchange some dummy indices for each term to get

$$\begin{aligned} (\tilde{\nabla}_\mu - \nabla_\mu)(T^\alpha_\beta \omega_\alpha v^\beta) &= \omega_\alpha v^\beta (\tilde{\nabla}_\mu - \nabla_\mu) T^\alpha_\beta + C^\alpha_{\mu\nu} \omega_\alpha T^\nu_\beta v^\beta - C^\nu_{\mu\beta} v^\beta T^\alpha_\nu \omega_\alpha \\ &= \omega_\alpha v^\beta [(\tilde{\nabla}_\mu - \nabla_\mu) T^\alpha_\beta + C^\alpha_{\mu\nu} T^\nu_\beta - C^\nu_{\mu\beta} T^\alpha_\nu]. \end{aligned}$$

The LHS is zero because of condition 4, so

$$(\tilde{\nabla}_\mu - \nabla_\mu) T^\alpha_\beta = -C^\alpha_{\mu\nu} T^\nu_\beta + C^\nu_{\mu\beta} T^\alpha_\nu.$$

The pattern may be obvious already. In $(\tilde{\nabla}_\mu - \nabla_\mu)T$, each upper index of T is summed over with a lower index of the connection coefficient C with a minus sign, and each lower index of T is summed over with the upper index of C with a plus sign. The rule for a (k,l) tensor can be generalized to the following equation,

$$(\tilde{\nabla}_\mu - \nabla_\mu) T^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_l} = \sum_{i=1}^l C^\nu_{\mu\beta_i} T^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \nu \dots \beta_l} - \sum_{j=1}^k C^{\alpha_j}_{\mu\nu} T^{\alpha_1 \dots \nu \dots \alpha_k}_{\beta_1 \dots \beta_l}. \quad (4.5)$$

We can see the importance of the connection coefficients as every information in the difference of derivative operators acting on a tensor is stored in $C^\alpha_{\mu\nu}$. Recall that ∇_μ and $\tilde{\nabla}_\mu$ are any derivative operators that satisfy the five conditions listed on the previous page. Evidently, it is most convenient to choose $\tilde{\nabla}_\mu$ to be the familiar partial derivative ∂_μ . In this case, the connection coefficients $C^\alpha_{\mu\nu}$ are denoted as $\Gamma^\alpha_{\mu\nu}$, the **Christoffel symbols**. We will let $\tilde{\nabla}_\mu = \partial_\mu$ in the rest of this notes. Here are some useful results here for future reference:

$$\begin{aligned} \nabla_\mu \omega_\nu &= \partial_\mu \omega_\nu - \Gamma^\alpha_{\mu\nu} \omega_\alpha, \\ \nabla_\mu v^\nu &= \partial_\mu v^\nu + \Gamma^\nu_{\mu\alpha} v^\alpha, \\ \nabla_\mu T_{\alpha\beta} &= \partial_\mu T_{\alpha\beta} - \Gamma^\nu_{\mu\alpha} T_{\nu\beta} - \Gamma^\nu_{\mu\beta} T_{\alpha\nu}, \\ \nabla_\mu T^\alpha_\beta &= \partial_\mu T^\alpha_\beta + \Gamma^\alpha_{\mu\nu} T^\nu_\beta - \Gamma^\nu_{\mu\beta} T^\alpha_\nu, \\ \nabla_\mu T^{\alpha\beta} &= \partial_\mu T^{\alpha\beta} + \Gamma^\alpha_{\mu\nu} T^{\nu\beta} + \Gamma^\beta_{\mu\nu} T^{\alpha\nu}. \end{aligned} \quad (4.6)$$

These can be obtained from (4.5). Note the sign of each term, because we have moved $\tilde{\nabla}_\mu = \partial_\mu$ to the RHS.

4.2.2 The Covariant Derivative

We just asserted that $C^\alpha_{\mu\nu}$ are the Christoffel symbols when we set $\tilde{\nabla}_\mu = \partial_\mu$, but so far we know nothing about how to compute $C^\alpha_{\mu\nu}$. In general relativity, it is most useful to give an additional requirement on the derivative operator ∇_μ . We now prove that

given a metric $g_{\alpha\beta}$, there exist a unique derivative operator ∇_μ such that

$$\boxed{\nabla_\mu g_{\alpha\beta} = 0.} \quad (4.7)$$

In this case, we say that ∇_μ is **metric compatible**.

Proof. The metric tensor is a tensor, so for any derivative operator $\tilde{\nabla}_\mu$, it should satisfy (4.5),

$$\nabla_\mu g_{\alpha\beta} = \tilde{\nabla}_\mu g_{\alpha\beta} - C^\nu{}_{\mu\alpha} g_{\nu\beta} - C^\nu{}_{\mu\beta} g_{\alpha\nu} = \tilde{\nabla}_\mu g_{\alpha\beta} - C_{\beta\mu\alpha} - C_{\alpha\mu\beta},$$

where we lowered the indices of the connection coefficients. Because $\nabla_\mu g_{\alpha\beta} = 0$, this means

$$C_{\beta\mu\alpha} + C_{\alpha\mu\beta} = \tilde{\nabla}_\mu g_{\alpha\beta}.$$

By permuting the indices on both sides, we can get two more equations,

$$C_{\beta\alpha\mu} + C_{\mu\alpha\beta} = \tilde{\nabla}_\alpha g_{\mu\beta},$$

$$C_{\alpha\beta\mu} + C_{\mu\beta\alpha} = \tilde{\nabla}_\beta g_{\mu\alpha}.$$

For the three equations above, adding the first two and subtracting the third gives,

$$C_{\beta\mu\alpha} + C_{\alpha\mu\beta} + C_{\beta\alpha\mu} + C_{\mu\alpha\beta} - C_{\alpha\beta\mu} - C_{\mu\beta\alpha} = \tilde{\nabla}_\mu g_{\alpha\beta} + \tilde{\nabla}_\alpha g_{\mu\beta} - \tilde{\nabla}_\beta g_{\mu\alpha}.$$

Using the fact that C is symmetric in the last two indices, this can be simplified to

$$2C_{\beta\mu\alpha} = \tilde{\nabla}_\mu g_{\alpha\beta} + \tilde{\nabla}_\alpha g_{\mu\beta} - \tilde{\nabla}_\beta g_{\mu\alpha},$$

or, raising back the index using the metric,

$$C^\beta{}_{\mu\alpha} = \frac{1}{2} g^{\beta\nu} (\tilde{\nabla}_\mu g_{\alpha\nu} + \tilde{\nabla}_\alpha g_{\mu\nu} - \tilde{\nabla}_\nu g_{\mu\alpha}). \quad (4.8)$$

The choice $C^\beta{}_{\mu\alpha}$, expressed solely by the metric and its derivative, solves the equation $\nabla_\mu g_{\alpha\beta} = 0$. Suppose $\tilde{\nabla}_\mu$ also satisfies $\tilde{\nabla}_\mu g_{\alpha\beta} = 0$. Then all the terms in the connection coefficient $C^\beta{}_{\mu\alpha}$ are just zero—there is no connection coefficients. In other words, $(\tilde{\nabla}_\mu - \nabla_\mu)T = 0$ for any tensor, which means $\tilde{\nabla}_\mu = \nabla_\mu$. The derivative operator ∇_μ is unique. $\square$

The connection coefficients (4.8) in terms of the metric tensor should look similar. If we replace $\tilde{\nabla}_\mu$ with ∂_μ , with some index re-labeling $\nu \rightarrow \beta$, $\beta \rightarrow \alpha$, $\alpha \rightarrow \nu$,

$$\boxed{\Gamma^\alpha{}_{\mu\nu} = \frac{1}{2} g^{\alpha\beta} (\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}).} \quad (4.9)$$

This is exactly (4.2), the Christoffel symbol derived from physical geodesic equations. Finally, this unique derivative operator that gives $\nabla_\alpha g_{\mu\nu}$ is known as the **covariant derivative**. Every metric defines its own covariant derivative, with connection coefficients as Christoffel symbols given by (4.9). There are many advantages of the metric compatibility, $\nabla_\alpha g_{\mu\nu} = 0$. First, the covariant derivative commutes with raising and lowering indices,

$$\nabla_\alpha v_\mu = \nabla_\alpha (g_{\mu\nu} v^\nu) = g_{\mu\nu} \nabla_\alpha v^\nu.$$

Second, the covariant derivative of the inverse metric is zero,

$$0 = \nabla_\alpha \delta^\mu{}_\nu = \nabla_\alpha (g^{\mu\beta} g_{\beta\nu}) = g_{\beta\nu} \nabla_\alpha g^{\mu\beta} \implies 0 = g^{\gamma\nu} g_{\beta\nu} \nabla_\alpha g^{\mu\beta} = \delta^\gamma{}_\beta \nabla_\alpha g^{\mu\beta} = \nabla_\alpha g^{\mu\gamma}.$$

Third, the covariant derivative of the Levi-Civita tensor is zero. This can be proved by the definition of the Levi-Civita tensor $\epsilon_{\mu\nu\alpha\beta} = \sqrt{|g|} \tilde{\epsilon}_{\mu\nu\alpha\beta}$. The Levi-Civita symbol $\tilde{\epsilon}_{\mu\nu\alpha\beta}$ are just constants. The metric determinant g is some combinations of the components, $g_{\mu\nu}$ and $g^{\mu\nu}$, which all vanish under the covariant derivative. Hence $\nabla_\gamma \epsilon_{\mu\nu\alpha\beta} = 0$.

There is a useful connection between the covariant derivative and the exterior derivative. For arbitrary tensor with only lower indices, $T_{\beta_1 \dots \beta_l}$, consider the following expression:

$$\nabla_{[\mu} T_{\beta_1 \dots \beta_l]}.$$

That is, we antisymmetrize all the lower indices. According to (4.5), the covariant derivative of $T_{\beta_1 \dots \beta_l}$ is

$$\nabla_\mu T_{\beta_1 \dots \beta_l} = \partial_\mu T_{\beta_1 \dots \beta_l} - \sum_{i=1}^l \Gamma^\nu_{\mu \beta_i} T_{\beta_1 \dots \nu \dots \beta_l}.$$

Now antisymmetrizing all lower indices $\mu, \beta_1, \dots, \beta_l$,

$$\nabla_{[\mu} T_{\beta_1 \dots \beta_l]} = \partial_{[\mu} T_{\beta_1 \dots \beta_l]} - \sum_{i=1}^l \Gamma^\nu_{[\mu \beta_i} T_{\beta_1 \dots \nu \dots \beta_l]}.$$

(The $[\nu]$ means that ν is not involved in the antisymmetrization because it does not appear in $\nabla_{[\mu} T_{\beta_1 \dots \beta_l]}$.) But $\Gamma^\nu_{\mu \beta_i}$ is symmetric in ν and β_i , so antisymmetrizing it gives zero. Every Christoffel symbols disappear in the above expression, so

$$\nabla_{[\mu} T_{\beta_1 \dots \beta_l]} = \partial_{[\mu} T_{\beta_1 \dots \beta_l]} = \frac{1}{l+1} (dT)_{\mu \beta_1 \dots \beta_l}, \quad (4.10)$$

where in the second equality we used the definition of the exterior derivative (3.8).

4.2.3 Coordinate Transformations

From the beginning, we want the covariant derivative of a tensor to also be a tensor. Here we shall check the transformation law of it. First, the Christoffel symbols transform as

$$\Gamma^{\alpha'}_{\mu' \nu'} = \frac{\partial x^{\alpha'}}{\partial x^\alpha} \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \Gamma^\alpha_{\mu \nu} + \frac{\partial x^{\alpha'}}{\partial x^\alpha} \frac{\partial^2 x^\alpha}{\partial x^{\mu'} \partial x^{\nu'}}. \quad (4.11)$$

See proof in Appendix A.2.2. The first term is like a tensorial transformation, but the second is not. This means the Christoffel symbols are not components of a tensor.

Dual Vectors. Now consider the transformed covariant derivative on a dual vector (referring to Equations 4.6),

$$\nabla_{\mu'} \omega_{\nu'} = \partial_{\mu'} \omega_{\nu'} - \Gamma^{\alpha'}_{\mu' \nu'} \omega_{\alpha'}.$$

The partial derivative term, according to (3.10), transforms like

$$\partial_{\mu'} \omega_{\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \partial_\mu \omega_\nu + \omega_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}}.$$

The Christoffel symbol term transforms like

$$\Gamma^{\alpha'}_{\mu' \nu'} \omega_{\alpha'} = \left(\frac{\partial x^{\alpha'}}{\partial x^\alpha} \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \Gamma^\alpha_{\mu \nu} + \frac{\partial x^{\alpha'}}{\partial x^\alpha} \frac{\partial^2 x^\alpha}{\partial x^{\mu'} \partial x^{\nu'}} \right) \left(\frac{\partial x^\alpha}{\partial x^{\alpha'}} \omega_\alpha \right)$$

Well, it is not good to have more than two α 's to be summed over. It is better to change $(\partial x^\alpha / \partial x^{\alpha'}) \omega_\alpha \rightarrow (\partial x^\beta / \partial x^{\alpha'}) \omega_\beta$,

$$\begin{aligned} \Gamma^{\alpha'}_{\mu' \nu'} \omega_{\alpha'} &= \left(\frac{\partial x^{\alpha'}}{\partial x^\alpha} \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \Gamma^\alpha_{\mu \nu} + \frac{\partial x^{\alpha'}}{\partial x^\alpha} \frac{\partial^2 x^\alpha}{\partial x^{\mu'} \partial x^{\nu'}} \right) \left(\frac{\partial x^\beta}{\partial x^{\alpha'}} \omega_\beta \right) \\ &= \delta^\beta_{\alpha'} \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \Gamma^\alpha_{\mu \nu} \omega_\beta + \delta^\beta_{\alpha'} \omega_\beta \frac{\partial^2 x^\alpha}{\partial x^{\mu'} \partial x^{\nu'}} \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \Gamma^\alpha_{\mu \nu} \omega_\alpha + \omega_\alpha \frac{\partial^2 x^\alpha}{\partial x^{\mu'} \partial x^{\nu'}}. \end{aligned}$$

Combining them, the covariant derivative of a dual vector transforms as

$$\begin{aligned} \nabla_{\mu'} \omega_{\nu'} &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \partial_\mu \omega_\nu + \omega_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}} - \left(\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \Gamma^\alpha_{\mu \nu} \omega_\alpha + \omega_\alpha \frac{\partial^2 x^\alpha}{\partial x^{\mu'} \partial x^{\nu'}} \right) \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} (\partial_\mu \omega_\nu - \Gamma^\alpha_{\mu \nu} \omega_\alpha) \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \nabla_\mu \omega_\nu. \end{aligned}$$

In the second equality, the nontensorial terms cancel (ν and α are dummy indices). The covariant derivative of a dual vector transforms like a tensor, as desired.

Vectors. The transformed covariant derivative on a vector is

$$\nabla_{\mu'} v^{\nu'} = \partial_{\mu'} v^{\nu'} + \Gamma^{\nu'}_{\mu' \alpha'} v^{\alpha'}.$$

The partial derivative term transforms as

$$\partial_{\mu'} v^{\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu \left(\frac{\partial x^{\nu'}}{\partial x^\nu} x^\nu \right) = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \partial_\mu v^\nu + v^\nu \frac{\partial^2 x^{\nu'}}{\partial x^{\mu'} \partial x^\nu}.$$

The Christoffel symbol term is a little trickier. Using the transformation law (4.11)

$$\begin{aligned} \Gamma^{\nu'}_{\mu' \alpha'} v^{\alpha'} &= \left(\frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\alpha'}} \Gamma^\alpha_{\mu \nu} + \frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial^2 x^\alpha}{\partial x^{\mu'} \partial x^{\alpha'}} \right) \left(\frac{\partial x^{\alpha'}}{\partial x^\beta} v^\beta \right) \\ &= \frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial x^\mu}{\partial x^{\mu'}} \delta^\nu_\beta \Gamma^\alpha_{\mu \nu} v^\beta + v^\beta \frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial x^{\alpha'}}{\partial x^\beta} \frac{\partial}{\partial x^{\mu'}} \frac{\partial x^\alpha}{\partial x^{\alpha'}} \\ &= \frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial x^\mu}{\partial x^{\mu'}} \Gamma^\alpha_{\mu \nu} v^\nu + v^\nu \frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial x^{\alpha'}}{\partial x^\nu} \frac{\partial}{\partial x^{\mu'}} \frac{\partial x^\alpha}{\partial x^{\alpha'}}. \end{aligned}$$

The second term does not look nice. Also, it looks like there is no minus sign for us to cancel the nontensorial part in the partial derivative term. To acquire a minus sign, we use the product rule,

$$\begin{aligned} \Gamma^{\nu'}_{\mu' \alpha'} v^{\alpha'} &= \frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial x^\mu}{\partial x^{\mu'}} \Gamma^\alpha_{\mu \nu} v^\nu + v^\nu \frac{\partial x^{\nu'}}{\partial x^\alpha} \left[\frac{\partial}{\partial x^{\mu'}} \left(\frac{\partial x^{\alpha'}}{\partial x^\nu} \frac{\partial x^\alpha}{\partial x^{\alpha'}} \right) - \frac{\partial x^\alpha}{\partial x^{\alpha'}} \frac{\partial}{\partial x^{\mu'}} \frac{\partial x^{\alpha'}}{\partial x^\nu} \right] \\ &= \frac{\partial x^{\nu'}}{\partial x^\nu} \frac{\partial x^\mu}{\partial x^{\mu'}} \Gamma^\nu_{\mu \alpha} v^\alpha + v^\nu \frac{\partial x^{\nu'}}{\partial x^\alpha} \left[\cancel{\frac{\partial}{\partial x^{\mu'}} (\delta^\alpha_\nu)} - \frac{\partial x^\alpha}{\partial x^{\alpha'}} \frac{\partial^2 x^{\alpha'}}{\partial x^{\mu'} \partial x^\nu} \right] \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma^\nu_{\mu \alpha} v^\alpha - v^\nu \frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial x^\alpha}{\partial x^{\alpha'}} \frac{\partial^2 x^{\alpha'}}{\partial x^{\mu'} \partial x^\nu} \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma^\nu_{\mu \alpha} v^\alpha - v^\nu \delta^{\nu'}_{\alpha'} \frac{\partial^2 x^{\alpha'}}{\partial x^{\mu'} \partial x^\nu} \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma^\nu_{\mu \alpha} v^\alpha - v^\nu \frac{\partial^2 x^{\nu'}}{\partial x^{\mu'} \partial x^\nu}. \end{aligned}$$

Now combining $\partial_{\mu'} v^{\nu'}$ and $\Gamma^{\nu'}_{\mu' \alpha'} v^{\alpha'}$, the covariant derivative of a vector transforms as

$$\begin{aligned} \nabla_{\mu'} v^{\nu'} &= \partial_{\mu'} v^{\nu'} + \Gamma^{\nu'}_{\mu' \alpha'} v^{\alpha'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \partial_\mu v^\nu + v^\nu \frac{\partial^2 x^{\nu'}}{\partial x^{\mu'} \partial x^\nu} + \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma^\nu_{\mu \alpha} v^\alpha - v^\nu \frac{\partial^2 x^{\nu'}}{\partial x^{\mu'} \partial x^\nu} \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} (\partial_\mu v^\nu + \Gamma^\nu_{\mu \alpha} v^\alpha) \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \nabla_\mu v^\nu. \end{aligned}$$

Thus, the covariant derivative of a vector also transforms like a tensor. We know that components of any higher rank tensors can be written as products of components of vectors and dual vectors. Then by Leibniz rule, the covariant derivative of higher rank tensors is also a tensor,

$$\nabla_{\alpha'} T^{\mu'_1 \dots \mu'_k}_{\nu'_1 \dots \nu'_l} = \frac{\partial x^{\alpha'}}{\partial x^\alpha} \frac{\partial x^{\mu'_1}}{\partial x^{\mu_1}} \dots \frac{\partial x^{\mu'_k}}{\partial x^{\mu_k}} \frac{\partial x^{\nu_1}}{\partial x^{\nu'_1}} \dots \frac{\partial x^{\nu_l}}{\partial x^{\nu'_l}} \nabla_\alpha T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l}.$$

4.2.4 Commutator of Vector Fields

To be written.

$$[\xi, \eta]^\mu = \xi^\nu \nabla_\nu \eta^\mu - \eta^\nu \nabla_\nu \xi^\mu. \quad (4.12)$$

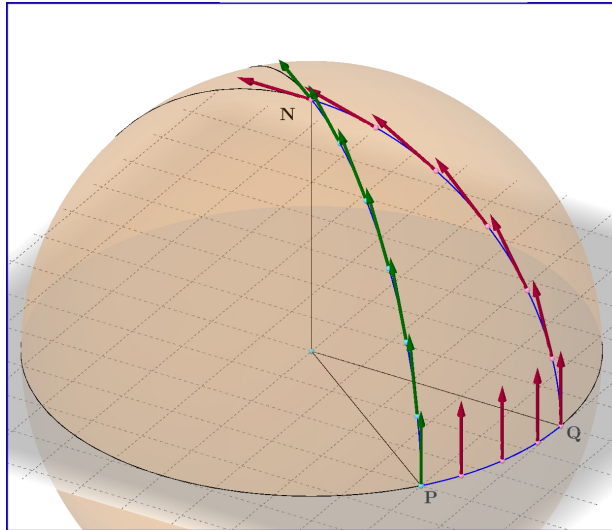


Figure 4.1: An example of parallel transport. On a 2-sphere, a vector is parallel transported from the north pole to the equator, then along the equator for some angle ϕ , and back to the north pole. The final vector is different from the initial one.

4.2.5 Stokes's Theorem on Curved Manifolds

For a vector field v^μ over a “volume” V . The integral of the covariant divergence of this vector over V can be turned into a surface integral over the boundary ∂V .

$$\int_V \nabla_\mu v^\mu \sqrt{|g|} d^n x = \int_{\partial V} n_\mu v^\mu \sqrt{|\gamma|} d^{n-1} x, \quad (4.13)$$

where n_μ is the unit normal to the boundary “surface” ∂V , and γ_{ij} is called the **induced metric** on ∂V .

Proof. To be written. □

4.3 Parallel-Transport and Geodesics

4.3.1 The Equation of Parallel Transport

In flat spacetime, we are used to compare vectors at different points and do some operations on them (dot product, cross product, etc.). Mathematically, we rely on the fact that vectors are parallel transported to the same point. By then we can compare them. The concept of parallel transport generalizes to curved spacetime. The difference between parallel transport in flat and curved spacetime is that the result of parallel transporting vector from one point to another will depend on the path taken between two points. A parallel transport along a closed loop may not result in the original vector (see Figure 4.1 for example).

The **parallel transport** of a vector is defined as moving it along a specified path while keeping it constant. Generalizing to tensors, given a curve $x^\mu(\lambda)$ parametrized by λ , we require the component of a tensor to be constant along the curve. In flat spacetime, it means

$$\frac{d}{d\lambda} T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} = \xi^\mu \frac{\partial}{\partial x^\mu} T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} = 0,$$

where $\xi^\mu = dx^\mu/d\lambda$ are components of the tangent vector along the curve. Generalizing to curved spacetime, we just replace the partial derivative with the covariant derivative:

$$\frac{dx^\mu}{d\lambda} \nabla_\mu T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} = 0. \quad (4.14)$$

This is the **equation of parallel transport**. It is a valid tensor equation because both the tangent vector $dx^\mu/d\lambda$ and the covariant derivative ∇T are tensors. For a vector in its component form, it reduces to

$$\xi^\mu \nabla_\mu v^\alpha = 0, \quad \text{or} \quad \frac{d}{d\lambda} v^\alpha + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\lambda} v^\nu = 0. \quad (4.15)$$

A good thing about parallel transport of two vectors is that their inner product is preserved. For two vectors v^μ and w^ν satisfying

the equation of parallel transport (4.15),

$$\xi^\alpha \nabla_\alpha (g_{\mu\nu} v^\mu w^\nu) = g_{\mu\nu} w^\nu \xi^\alpha \nabla_\alpha v^\mu + g_{\mu\nu} v^\mu \xi^\alpha \nabla_\alpha w^\nu = 0.$$

This is essentially a consequence of $\nabla_\alpha g_{\mu\nu} = 0$.

4.3.2 The Geodesic Equation

Physically, a geodesic is the path of a freely falling particle, a path which extremizes the proper time of the particle. There is an equally valid mathematical definition for a geodesic. A **geodesic** is a path that parallel transports its own tangent vector. Equation (4.15) tells us how to parallel transport a vector. We just replace v^α with the tangent vector along the path $\xi^\alpha = dx^\alpha/d\lambda$:

$$\xi^\mu \nabla_\mu \xi^\alpha = 0, \quad \text{or} \quad \frac{d^2 x^\alpha}{d\lambda^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0. \quad (4.16)$$

This is the same **geodesic equation** as the one derived from extremizing the proper time. It consists of four second-order nonlinear differential equations. If the initial conditions $x^\mu(\lambda_0)$ and $dx^\mu/d\lambda|_{\lambda_0}$ are specified, there is a unique solution to x^μ . Notice that the geodesic equation (4.1) has $\Gamma^\alpha_{\mu\nu}$ arises from extremizing the path length. The Christoffel symbol emerges before we introduce the covariant derivative and all sorts of connection coefficients. This means the definition of geodesics from “parallel-transport view” and “extremized path view” are equivalent only when the connection coefficients are Christoffel symbols.

The parameter λ that satisfy the geodesic equation (4.16) is called an **affine parameter**. The proper time τ is an affine parameter. It is clear from the geodesic equation that if we change $\tau \rightarrow \lambda = a\tau + b$, the equation still holds. Of course, we can parametrize the equation with parameters that are not affine. Then we will get an extra term on the RHS of the geodesic equation:

$$\frac{d^2 x^\alpha}{d\beta^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\beta} \frac{dx^\nu}{d\beta} = f(\beta) \frac{dx^\alpha}{d\beta}$$

for some parameter $\beta(\lambda)$. We can find $f(\beta)$ by relating this geodesic equation with the affinely parametrized one:

$$\begin{aligned} \frac{d^2 x^\alpha}{d\beta^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\beta} \frac{dx^\nu}{d\beta} &= \frac{d}{d\beta} \left(\frac{d\lambda}{d\beta} \frac{dx^\alpha}{d\lambda} \right) + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{d\lambda}{d\beta} \frac{dx^\nu}{d\lambda} \frac{d\lambda}{d\beta} \\ &= \frac{d^2 \lambda}{d\beta^2} \frac{dx^\alpha}{d\lambda} + \frac{d\lambda}{d\beta} \frac{d}{d\beta} \frac{dx^\alpha}{d\lambda} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \left(\frac{d\lambda}{d\beta} \right)^2 \\ &= \frac{d^2 \lambda}{d\beta^2} \frac{d\beta}{d\lambda} \frac{dx^\alpha}{d\beta} + \left(\frac{d\lambda}{d\beta} \right)^2 \left(\frac{d^2 x^\alpha}{d\lambda^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right). \end{aligned}$$

The second parenthesis is the LHS of the affinely parametrized geodesic equation, so it is zero. We find that

$$f(\beta) = \frac{d^2 \lambda}{d\beta^2} \frac{d\beta}{d\lambda}.$$

4.4 Curvature*

There are many ways that can infer about the intrinsic curvature of a manifold. We already know that parallel transporting a vector along a loop on a curved manifold does not give back the original vector in general. (For example, the parallel transport of a vector on a 2-sphere.) We will also see that the covariant derivative of a tensor field fails to commute if there is curvature. Euclid’s axioms also fails on a curved manifold—parallel lines at one point do not remain parallel at other points. This is generalized to the geodesic deviation, another indication of curvature.

4.4.1 The Riemann Curvature Tensor

Consider two covariant derivatives ∇_μ and ∇_ν . Let ω_α be a dual vector field and f be a scalar function. Then

$$\nabla_\mu \nabla_\nu (f \omega_\alpha) = \nabla_\mu (\omega_\alpha \nabla_\nu f + f \nabla_\mu \omega_\alpha) = (\nabla_\mu \nabla_\nu f) \omega_\alpha + \nabla_\nu f \nabla_\mu \omega_\alpha + \nabla_\mu f \nabla_\nu \omega_\alpha + f \nabla_\mu \nabla_\nu \omega_\alpha.$$

On the other hand, if we first act ∇_μ then ∇_ν ,

$$\nabla_\nu \nabla_\mu (f\omega_\alpha) = (\nabla_\nu \nabla_\mu f)\omega_\alpha + \nabla_\mu f \nabla_\nu \omega_\alpha + \nabla_\nu f \nabla_\mu \omega_\alpha + f \nabla_\nu \nabla_\mu \omega_\alpha.$$

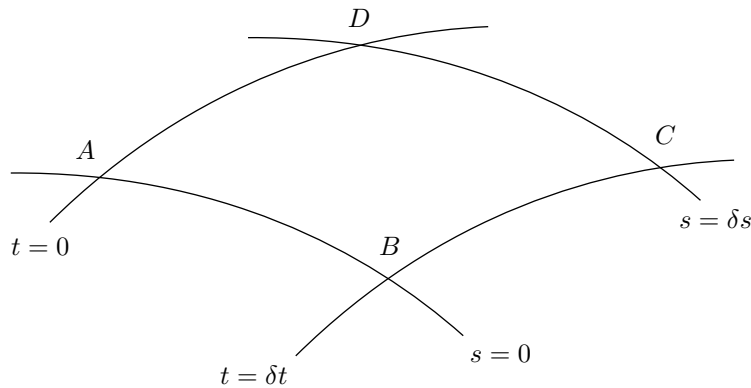
Computing the commutator $[\nabla_\mu, \nabla_\nu]$ acting on $f\omega_\alpha$ gives

$$[\nabla_\mu, \nabla_\nu](f\omega_\alpha) = \nabla_\mu \nabla_\nu (f\omega_\alpha) - \nabla_\nu \nabla_\mu (f\omega_\alpha) = f[\nabla_\mu, \nabla_\nu]\omega_\alpha,$$

where we used condition 5 of the covariant derivative ($\nabla_\mu \nabla_\nu f = \nabla_\nu \nabla_\mu f$). Thus, the commutator $[\nabla_\mu, \nabla_\nu]$ is functionally linear. We define a tensor associated with it, called the **Riemann curvature tensor** $R^\alpha{}_{\beta\mu\nu}$, such that

$$[\nabla_\mu, \nabla_\nu]\omega_\beta = -R^\alpha{}_{\beta\mu\nu}\omega_\alpha. \quad (4.17)$$

(Textbooks define the Riemann curvature tensor differently. See Appendix B for reference.) The Riemann curvature tensor is closely related to parallel transporting a vector along a closed loop. This can be used to construct a notion of curvature if we take the loop to be infinitesimally small on the manifold. Consider the following loop, assumed to be small. For simplicity, we choose A to be the origin. The loop is formed by moving δt along the $s = 0$ curve, then moving δs along the $t = \delta t$ curve, then moving $-\delta t$ along the $s = \delta s$ curve, and finally moving back to the origin along the $t = 0$ curve.



Let $\xi^\mu = dx^\mu/ds$ be the components of the tangent vector $\xi = \partial/\partial s$ along constant t lines, and $\eta^\mu = dx^\mu/dt$ be the components of the tangent vector $\eta = \partial/\partial t$ along constant s lines. We want to parallel transport a vector v^α along the loop. According to the equation of parallel transport (4.15), we have $\xi^\mu \nabla_\mu v^\alpha = \eta^\mu \nabla_\mu v^\alpha = 0$. Let ω_α be an arbitrary dual vector. Instead of computing how v^α changes, it is easier to compute how $v^\alpha \omega_\alpha$ changes along the curve.

From A to B , the change in $v^\alpha \omega_\alpha$ is

$$\delta_{AB} \simeq \delta t \frac{\partial}{\partial t} (v^\alpha \omega_\alpha) \Big|_{(0, \delta t/2)} = \delta t \frac{dx^\mu}{dt} \partial_\mu (v^\alpha \omega_\alpha) \Big|_{(0, \delta t/2)} = \delta t \xi^\mu \nabla_\mu (v^\alpha \omega_\alpha) \Big|_{(0, \delta t/2)}.$$

In the last equality, we used condition 4 of the covariant derivative ($\nabla_\mu f = \partial_\mu f$). Using the Leibniz rule and the fact that v^α is parallel transported ($\xi^\mu \nabla_\mu v^\alpha = 0$),

$$\delta_{AB} \simeq \delta t \xi^\mu (v^\alpha \nabla_\mu \omega_\alpha + \omega_\alpha \nabla_\mu v^\alpha) \Big|_{(0, \delta t/2)} = \delta t v^\alpha \xi^\mu \nabla_\mu \omega_\alpha \Big|_{(0, \delta t/2)}.$$

We can do the same for other paths:

$$\begin{aligned} \delta_{BC} &\simeq \delta s v^\alpha \eta^\mu \nabla_\mu \omega_\alpha \Big|_{(\delta s/2, \delta t)}, \\ \delta_{CD} &\simeq -\delta t v^\alpha \xi^\mu \nabla_\mu \omega_\alpha \Big|_{(\delta s, \delta t/2)}, \\ \delta_{DA} &\simeq -\delta s v^\alpha \eta^\mu \nabla_\mu \omega_\alpha \Big|_{(\delta s/2, 0)}. \end{aligned}$$

The net effect on $v^\alpha \omega_\alpha$ is

$$\delta_{AB} + \delta_{BC} + \delta_{CD} + \delta_{DA} \simeq \delta t \left[v^\alpha \xi^\mu \nabla_\mu \omega_\alpha \Big|_{(0, \delta t/2)} - v^\alpha \xi^\mu \nabla_\mu \omega_\alpha \Big|_{(\delta s, \delta t/2)} \right] + \delta s \left[v^\alpha \eta^\mu \nabla_\mu \omega_\alpha \Big|_{(\delta s/2, \delta t)} - v^\alpha \eta^\mu \nabla_\mu \omega_\alpha \Big|_{(\delta s/2, 0)} \right].$$

Note that terms inside the bracket is a difference of a quantity evaluated at points separated by δt (or δs), but there is an additional factor of δs (or δt) outside the bracket. Hence the net effect is at least in second order. In other words, curvature is related to the *second-order derivatives*. Indeed, the first bracket is the difference of $v^\alpha \xi^\mu \nabla_\mu \omega_\alpha$ evaluated at $(0, \delta t/2)$ from that evaluated at

$(\delta s, \delta t/2)$. The change of $v^\alpha \xi^\mu \nabla_\mu \omega_\alpha$, to first order in δs , is

$$\left[v^\alpha \xi^\mu \nabla_\mu \omega_\alpha \Big|_{(0, \delta t/2)} - v^\alpha \xi^\mu \nabla_\mu \omega_\alpha \Big|_{(\delta s, \delta t/2)} \right] \simeq -\delta s \eta^\nu \partial_\nu (v^\alpha \xi^\mu \nabla_\mu \omega_\alpha) = -\delta s \eta^\nu \nabla_\nu (v^\alpha \xi^\mu \nabla_\mu \omega_\alpha).$$

We again used condition 4 of the covariant derivative in the last equality. By Leibniz rule and the equation of parallel transport ($\eta^\nu \nabla_\nu v^\alpha = 0$),

$$\delta_{AB} + \delta_{CD} \simeq -\delta t \delta s v^\alpha \eta^\nu \nabla_\nu (\xi^\mu \nabla_\mu \omega_\alpha).$$

Similarly, the sum of the other two paths is

$$\delta_{BC} + \delta_{DA} \simeq \delta s \left[v^\alpha \eta^\mu \nabla_\mu \omega_\alpha \Big|_{(\delta s/2, \delta t)} - v^\alpha \eta^\mu \nabla_\mu \omega_\alpha \Big|_{(\delta s/2, 0)} \right] = +\delta s \delta t v^\alpha \xi^\mu \nabla_\nu (\eta^\mu \nabla_\mu \omega_\alpha).$$

The net effect on $v^\alpha \omega_\alpha$ becomes

$$\begin{aligned} \delta(v^\alpha \omega_\alpha) &\simeq \delta s \delta t v^\alpha [\xi^\nu \nabla_\nu (\eta^\mu \nabla_\mu \omega_\alpha) - \eta^\nu \nabla_\nu (\xi^\mu \nabla_\mu \omega_\alpha)] \\ &= \delta s \delta t v^\alpha (\xi^\nu \nabla_\nu \eta^\mu \nabla_\mu \omega_\alpha + \xi^\nu \eta^\mu \nabla_\nu \nabla_\mu \omega_\alpha - \eta^\nu \nabla_\nu \xi^\mu \nabla_\mu \omega_\alpha - \eta^\nu \xi^\mu \nabla_\nu \nabla_\mu \omega_\alpha) \\ &= \delta s \delta t v^\alpha (\xi^\nu \nabla_\nu \eta^\mu \nabla_\mu \omega_\alpha + \xi^\mu \eta^\nu \nabla_\mu \nabla_\nu \omega_\alpha - \eta^\nu \nabla_\nu \xi^\mu \nabla_\mu \omega_\alpha - \eta^\nu \xi^\mu \nabla_\nu \nabla_\mu \omega_\alpha) \\ &= \delta s \delta t v^\alpha ([\xi, \eta]^\mu \nabla_\mu \omega_\alpha + \xi^\mu \eta^\nu [\nabla_\mu, \nabla_\nu] \omega_\alpha). \end{aligned}$$

In the last step, we identify the commutator $[\xi, \eta]^\mu = \xi^\nu \nabla_\nu \eta^\mu - \eta^\nu \nabla_\nu \xi^\mu$ using (4.12). Because the tangent vectors $\xi = \partial/\partial s$ and $\eta = \partial/\partial t$ are just partial derivatives, they commute, $[\xi, \eta]^\mu = 0$. Using the definition of the Riemann curvature tensor (4.17), the result is

$$\delta(v^\alpha \omega_\alpha) \simeq -\delta s \delta t v^\alpha \xi^\mu \eta^\nu R^\beta_{\alpha\mu\nu} \omega_\beta.$$

The RHS is evaluated at A , valid to second order. The approximate sign becomes an equal sign if we take the loop to be infinitesimally small. Re-labeling $\alpha \leftrightarrow \beta$, and expanding the LHS,

$$\omega_\alpha \delta v^\alpha + v^\alpha \delta \omega_\alpha = -\delta s \delta t v^\beta \xi^\mu \eta^\nu R^\alpha_{\beta\mu\nu} \omega_\alpha.$$

We only parallel transport the vector v^α . The dual vector ω_α is not transported, so its value at A is fixed, $\delta \omega_\alpha = 0$. Since ω_α is arbitrary, we can cancel it on both sides, giving

$$\delta v^\alpha = -\delta s \delta t \xi^\mu \eta^\nu R^\alpha_{\beta\mu\nu} v^\beta.$$

This is how the Riemann curvature tensor connects to the parallel transport of a vector around a closed loop.

Analogous to how we find the action of the covariant derivative on vectors and arbitrary tensors and express them in terms of Christoffel symbols, here we shall find how $[\nabla_\mu, \nabla_\nu]$ acts on tensors and express them in terms of the Riemann curvature tensor. Consider a vector v^α and a dual vector ω_α . Covariant derivatives commute on scalar functions, so

$$0 = [\nabla_\mu, \nabla_\nu](v^\alpha \omega_\alpha) = \omega_\alpha [\nabla_\mu, \nabla_\nu] v^\alpha + v^\alpha [\nabla_\mu, \nabla_\nu] \omega_\alpha = \omega_\alpha [\nabla_\mu, \nabla_\nu] v^\alpha - v^\alpha R^\beta_{\alpha\mu\nu} \omega_\beta.$$

Interchanging $\alpha \leftrightarrow \beta$, and removing ω_α , we have

$$[\nabla_\mu, \nabla_\nu] v^\alpha = R^\alpha_{\beta\mu\nu} v^\beta. \quad (4.18)$$

Generalizing to higher rank tensors:

$$[\nabla_\mu, \nabla_\nu] T^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_l} = \sum_{i=1}^k R^{\alpha_i}_{\gamma\mu\nu} T^{\alpha_1 \dots \gamma \dots \alpha_k}_{\beta_1 \dots \beta_l} - \sum_{j=1}^l R^\gamma_{\beta_j\mu\nu} T^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \gamma \dots \beta_l}. \quad (4.19)$$

4.4.2 Properties of the Curvature Tensor

The Riemann curvature tensor has many symmetries in its indices and satisfies many algebraic and differential identities. It is more convenient to discuss the symmetry properties using the tensor $R_{\alpha\mu\nu\beta} = g_{\alpha\gamma} R^\gamma_{\mu\nu\beta}$. Here is a list of properties of the Riemann curvature tensor:

1. It is antisymmetric in its last two indices,

$$R^\alpha_{\beta\mu\nu} = -R^\alpha_{\beta\nu\mu}, \quad \text{or} \quad R^\alpha_{\beta[\mu\nu]} = R^\alpha_{\beta\mu\nu}. \quad (4.20)$$

This follows from the definition of the Riemann curvature tensor (4.17).

2. The total antisymmetrization of its last three indices is zero,

$$R^\alpha_{[\beta\mu\nu]} = 0. \quad (4.21)$$

Proof. consider an arbitrary dual vector ω_β . Then

$$[\nabla_\mu, \nabla_\nu]\omega_\beta = 2\nabla_{[\mu}\nabla_{\nu]}\omega_\beta = -R^\alpha_{\beta[\mu\nu]}\omega_\alpha.$$

Incorporating β in the antisymmetrization,

$$-R^\alpha_{[\beta\mu\nu]}\omega_\alpha = 2\nabla_{[\mu}\nabla_{\nu]}\omega_\beta = 2\partial_{[\mu}\partial_{\nu]}\omega_\beta = 0.$$

In the second equality, we invoked (4.10) twice, and the last equality is because partial derivatives commute. This work for any ω_β , so $R^\alpha_{[\beta\mu\nu]} = 0$. $\square$

Using property 1, this is equivalent to

$$R_{\alpha\beta\mu\nu} + R_{\alpha\mu\nu\beta} + R_{\alpha\nu\beta\mu} = 0, \quad (4.22)$$

which is known as the **first Bianchi identity** or **algebraic Bianchi identity**.

3. It is antisymmetric in its first two indices,

$$R_{\alpha\beta\mu\nu} = -R_{\beta\alpha\mu\nu}, \quad \text{or} \quad R_{[\alpha\beta]\mu\nu} = R_{\alpha\beta\mu\nu}. \quad (4.23)$$

This can be proved by applying $[\nabla_\mu, \nabla_\nu]$ to the metric tensor using (4.19),

$$[\nabla_\mu, \nabla_\nu]g_{\alpha\beta} = -R^\gamma_{\alpha\mu\nu}g_{\gamma\beta} - R^\gamma_{\beta\mu\nu}g_{\alpha\gamma} = -R_{\beta\alpha\mu\nu} - R_{\alpha\beta\mu\nu}.$$

But the covariant derivative of the metric tensor is always zero, so

$$R_{\beta\alpha\mu\nu} + R_{\alpha\beta\mu\nu} = 0.$$

4. It is symmetric in the first pair and last pair of its indices:

$$R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}. \quad (4.24)$$

Proof. This is the result of all of properties 1, 2, and 3. We start with two algebraic Bianchi identity, with first indices α and μ respectively:

$$\begin{aligned} 0 = R_{\alpha\beta\mu\nu} + R_{\alpha\mu\nu\beta} + R_{\alpha\nu\beta\mu} &\implies R_{\alpha\beta\mu\nu} = -R_{\alpha\mu\nu\beta} - R_{\alpha\nu\beta\mu}, \\ 0 = R_{\mu\nu\alpha\beta} + R_{\mu\alpha\beta\nu} + R_{\mu\beta\nu\alpha} &\implies R_{\mu\nu\alpha\beta} = -R_{\mu\alpha\beta\nu} - R_{\mu\beta\nu\alpha}. \end{aligned}$$

Subtract the second from the first, and using both property 1 and 3,

$$R_{\alpha\beta\mu\nu} - R_{\mu\nu\alpha\beta} = -\cancel{R_{\alpha\mu\nu\beta}} + \cancel{R_{\mu\alpha\beta\nu}} - R_{\alpha\nu\beta\mu} + R_{\mu\beta\nu\alpha} = R_{\nu\alpha\beta\mu} - R_{\beta\mu\nu\alpha}.$$

Now write out the other two algebraic Bianchi identity, with first indices β and ν respectively:

$$\begin{aligned} 0 = R_{\beta\mu\nu\alpha} + R_{\beta\nu\alpha\mu} + R_{\beta\alpha\mu\nu} &\implies -R_{\beta\mu\nu\alpha} = R_{\beta\nu\alpha\mu} + R_{\beta\alpha\mu\nu}, \\ 0 = R_{\nu\alpha\beta\mu} + R_{\nu\mu\beta\alpha} + R_{\nu\beta\mu\alpha} &\implies R_{\nu\alpha\beta\mu} = -R_{\nu\mu\beta\alpha} - R_{\nu\beta\mu\alpha}. \end{aligned}$$

Substituting them into $R_{\alpha\beta\mu\nu} - R_{\mu\nu\alpha\beta}$:

$$R_{\alpha\beta\mu\nu} - R_{\mu\nu\alpha\beta} = -\cancel{R_{\nu\beta\mu\alpha}} - R_{\nu\mu\beta\alpha} + \cancel{R_{\beta\nu\alpha\mu}} + R_{\beta\alpha\mu\nu}.$$

Moving the terms to the right,

$$0 = -R_{\alpha\beta\mu\nu} + R_{\beta\alpha\mu\nu} + R_{\mu\nu\alpha\beta} - R_{\nu\mu\alpha\beta} = 2(R_{\mu\nu\alpha\beta} - R_{\alpha\beta\mu\nu}) \implies R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}.$$

$\square$

5. It satisfies the **second Bianchi identity** or **differential Bianchi identity** (or simply just **Bianchi identity**):

$$\nabla_{[\gamma} R_{\alpha\beta]\mu\nu} = 0, \quad \text{or} \quad \nabla_{\gamma} R_{\alpha\beta\mu\nu} + \nabla_{\alpha} R_{\beta\gamma\mu\nu} + \nabla_{\beta} R_{\gamma\alpha\mu\nu} = 0. \quad (4.25)$$

Proof. Consider the following two equations for any dual vector ω_{β} :

$$\begin{aligned} [\nabla_{\mu}, \nabla_{\nu}] \nabla_{\alpha} \omega_{\beta} &= 2 \nabla_{[\mu} \nabla_{\nu]} \nabla_{\alpha} \omega_{\beta} = -R^{\gamma}_{\alpha\mu\nu} \nabla_{\gamma} \omega_{\beta} - R^{\gamma}_{\beta\mu\nu} \nabla_{\alpha} \omega_{\gamma}, \\ \nabla_{\alpha} [\nabla_{\mu}, \nabla_{\nu}] \omega_{\beta} &= 2 \nabla_{\alpha} \nabla_{[\mu} \nabla_{\nu]} \omega_{\beta} = \nabla_{\alpha} (-R^{\gamma}_{\beta\mu\nu} \omega_{\gamma}) = -\omega_{\gamma} \nabla_{\alpha} R^{\gamma}_{\beta\mu\nu} - R^{\gamma}_{\beta\mu\nu} \nabla_{\alpha} \omega_{\gamma}. \end{aligned}$$

Here we used (4.19) to express the commutator in terms of the Riemann curvature tensor. The LHS of the two equations are the same if we incorporate α into antisymmetrization, as $\nabla_{[\mu} \nabla_{\nu]} \nabla_{\alpha]} = \nabla_{[\alpha} \nabla_{\mu} \nabla_{\nu]}$. (The indices are related by even permutations.) Thus, the RHS of the two equations should also be the same if we antisymmetrize α, μ, ν :

$$R^{\gamma}_{[\alpha\mu\nu]} \nabla_{\gamma} \omega_{\beta} + R^{\gamma}_{\beta[\mu\nu]} \nabla_{\alpha]} \omega_{\gamma} = \omega_{\gamma} \nabla_{[\alpha} R^{\gamma}_{\beta|\mu\nu]} + R^{\gamma}_{\beta[\mu\nu]} \nabla_{\alpha]} \omega_{\gamma}.$$

The second term on both sides cancels. The first term on the LHS vanishes because of (4.21). We are left with

$$\omega_{\gamma} \nabla_{[\alpha} R^{\gamma}_{\beta|\mu\nu]} = 0 \quad \implies \quad \nabla_{[\alpha} R^{\gamma}_{\beta|\mu\nu]} = 0$$

because this works for any ω_{γ} . Lowering the γ index gives $\nabla_{[\alpha} R_{\gamma\beta|\mu\nu]} = 0$. Because $R_{\gamma\beta\mu\nu} = R_{\mu\nu\gamma\beta}$ (property 4),

$$\nabla_{[\alpha} R_{\mu\nu]\gamma\beta} = \nabla_{[\alpha} R_{\gamma\beta|\mu\nu]} = 0.$$

Re-labeling the indices gives the differential Bianchi identity (4.25). □

We can see that different properties arise from different conditions on covariant derivatives. Property 1 always hold, because it is just saying that the commutator is antisymmetric. Property 2 requires the covariant derivative to be torsion free (condition 5), or equivalently, $\Gamma^{\alpha}_{\mu\nu} = \Gamma^{\alpha}_{\nu\mu}$. Property 3 requires metric compatibility of the covariant derivative, $\nabla_{\mu} g_{\alpha\beta} = 0$. These three properties are independent of each other. Other properties are all based on the first three.

In an n -dimensional space, the Riemann curvature tensor has n^4 components. Luckily, because it has many symmetries, the number of independent components in the Riemann curvature tensor is $\frac{1}{12}n^2(n^2 - 1)$, much less than n^4 .

Proof. There are many ways to prove this. Many of them involves the algebraic Bianchi identity (4.22). Here we will use only (4.20), (4.21), and (4.23) because, as mentioned above, they are independent of each other. First, we should know that the number of independent terms in an antisymmetric part of a tensor with l indices is

$$\binom{n}{l} = \frac{n!}{l!(n-l)!} = \frac{1}{l!} n(n-1)(n-2) \cdots (n-l+1).$$

For example, the number of independent terms in $A_{\mu\nu\alpha} = A_{[\mu\nu\alpha]}$ is $n(n-1)(n-2)/3!$. There are n possible choices for μ ; there are $n-1$ possible choices for ν because if $\nu = \mu$ then $A_{[\mu\mu\alpha]} = 0$ trivially; there are $n-2$ possible choices for $\alpha \neq \mu, \nu$ by the same reasoning. The $3!$ accounts for the possible permutations in μ, ν, α . This is because if we know $A_{\mu\nu\alpha}$, other components can be known by permutation as A is totally antisymmetric.

Now we analyze the properties one by one. Property 1: $R_{\alpha\beta\mu\nu} = R_{\alpha\beta[\mu\nu]}$ says that the number of independent components in μ, ν part is $n(n-1)/2!$. Property 3: $R_{\alpha\beta\mu\nu} = R_{[\alpha\beta]\mu\nu}$ says that the number of independent components in α, β part is also $n(n-1)/2!$. Based on these two properties, the number of independent components in $R_{\alpha\beta\mu\nu} = R_{[\alpha\beta][\mu\nu]}$ is

$$\left[\frac{n(n-1)}{2} \right]^2 = \frac{1}{4} n^2 (n-1)^2.$$

Property 2: $R_{\alpha[\beta\mu\nu]} = 0$ are constraint equations. These constraint equations reduces the number of independent components by the number of equations. (This is just the algebraic property of a system of linear equations. For example, if there are three variables satisfying two linear equations, then only one variable is independent, or free, and the other ones can be expressed in terms of it.) So the goal is to count how many (independent) equations are there in $R_{\alpha[\beta\mu\nu]} = 0$. The antisymmetric part $[\beta\mu\nu]$ contributes

$n(n-1)(n-2)/3!$, and α is free, so it contributes a factor of n . The number of independent equations in $R_{\alpha[\beta\mu\nu]} = 0$ is

$$n \cdot \frac{1}{3!} n(n-1)(n-2) = \frac{1}{6} n^2(n-1)(n-2).$$

Putting things all together, the number of independent components in the Riemann curvature tensor is

$$N = \frac{1}{4} n^2(n-1)^2 - \frac{1}{6} n^2(n-1)(n-2) = \frac{1}{12} n^2(n-1)[3(n-1) - 2(n-2)] = \frac{1}{12} n^2(n-1)(n+1) = \frac{1}{12} n^2(n^2-1).$$

□

For example, in 4-dimensional spacetime, the number of independent components is 20, much smaller than $4^4 = 256$. They correspond to the degrees of freedom in the *second derivatives* of the metric that cannot be set to zero by a coordinate transformation.

4.4.3 Coordinate Component Representation

The Riemann curvature tensor is significant to the governing equation, Einstein's equation, in general relativity. We should know how to calculate it in terms of coordinate components. This is simple. We just express $[\nabla_\mu, \nabla_\nu]\omega_\beta$ in terms of partial derivatives and Christoffel symbols using equations in (4.6). By explicit calculation,

$$\nabla_\mu \nabla_\nu \omega_\beta = \nabla_\mu (\partial_\nu \omega_\beta - \Gamma^\alpha_{\nu\beta} \omega_\alpha) = \partial_\mu (\partial_\nu \omega_\beta - \Gamma^\alpha_{\nu\beta} \omega_\alpha) - \Gamma^\lambda_{\mu\nu} (\partial_\lambda \omega_\beta - \Gamma^\alpha_{\lambda\beta} \omega_\alpha) - \Gamma^\lambda_{\mu\beta} (\partial_\nu \omega_\lambda - \Gamma^\alpha_{\nu\lambda} \omega_\alpha).$$

(Note that the covariant derivative does not care about the dummy index α by condition 3.) Now antisymmetrize μ and ν ,

$$\nabla_{[\mu} \nabla_{\nu]} \omega_\beta = \partial_{[\mu} \partial_{\nu]} \omega_\beta - \omega_\alpha \partial_{[\mu} \Gamma^\alpha_{\nu]\beta} - \partial_{[\mu} \omega_\alpha \Gamma^\alpha_{\nu]\beta} - \Gamma^\lambda_{[\mu\nu]} (\partial_\lambda \omega_\beta - \Gamma^\alpha_{\lambda\beta} \omega_\alpha) - \Gamma^\lambda_{\beta[\mu} \partial_{\nu]} \omega_\lambda + \Gamma^\lambda_{\beta[\mu} \Gamma^\alpha_{\nu]\lambda} \omega_\alpha.$$

Partial derivatives commute, and $\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\nu\mu}$, so the first and the fourth term vanishes. The third term can be written as

$$-\partial_{[\mu} \omega_\alpha \Gamma^\alpha_{\nu]\beta} = -\Gamma^\alpha_{\beta[\nu} \partial_{\mu]} \omega_\alpha = +\Gamma^\alpha_{\beta[\mu} \partial_{\nu]} \omega_\alpha.$$

There are only two terms left:

$$\nabla_{[\mu} \nabla_{\nu]} \omega_\beta = -\partial_{[\mu} \Gamma^\alpha_{\nu]\beta} \omega_\alpha + \Gamma^\lambda_{\beta[\mu} \Gamma^\alpha_{\nu]\lambda} \omega_\alpha = -\frac{1}{2} R^\alpha_{\beta\mu\nu} \omega_\alpha.$$

The dual vector ω_α is arbitrary, so we get

$$R^\alpha_{\beta\mu\nu} = \partial_\mu \Gamma^\alpha_{\nu\beta} - \partial_\nu \Gamma^\alpha_{\mu\beta} + \Gamma^\alpha_{\mu\lambda} \Gamma^\lambda_{\nu\beta} - \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\mu\beta} = 2\partial_{[\mu} \Gamma^\alpha_{\nu]\beta} + 2\Gamma^\alpha_{\lambda[\mu} \Gamma^\lambda_{\nu]\beta}. \quad (4.26)$$

4.4.4 Geodesic Deviation

The final indication of curvature that we shall discuss is the geodesic deviation. Basically, it means parallel lines does not remain parallel on a curved manifold. Instead, they accelerate towards or away from each other.

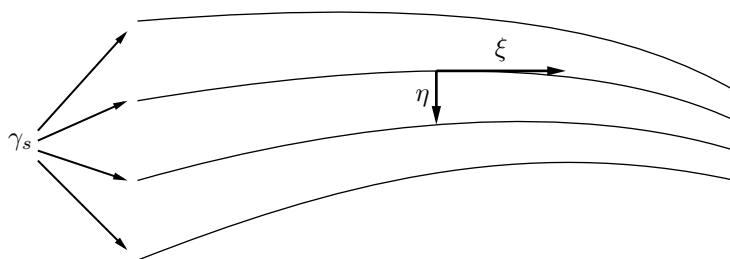


Figure 4.2: A set of geodesics γ_s , with tangent vector ξ and deviation vector η .

Let $\gamma_s(\lambda)$ be a set of geodesics, where each geodesic is labeled by a smooth parameter $s \in \mathbb{R}$. The parameter λ of the geodesics is assumed to be an affine parameter. We consider the space spanned by the curves $\gamma_s(\lambda)$. Let $\xi = \partial/\partial\lambda$ be the tangent vector along the geodesics, and $\xi^\mu = dx^\mu/d\lambda$ its components. We also define the **deviation vector** $\eta = \partial/\partial s$. Its components $\eta^\mu = dx^\mu/ds$ is related

to the displacement to an infinitesimally nearby geodesic. Again, the vectors ξ and η commute as they are just partial derivatives,

$$[\xi, \eta]^\mu = 0 \implies \xi^\nu \nabla_\nu \eta^\mu = \eta^\nu \nabla_\nu \xi^\mu.$$

Because η^μ is the infinitesimal separation between nearby geodesics, we want to study how it changes as a function of λ as we move along geodesics. The “relative velocity” of the separation between nearby geodesics is

$$v^\alpha \equiv \frac{d}{d\lambda} \eta^\alpha = \frac{dx^\mu}{d\lambda} \nabla_\mu \eta^\alpha = \xi^\mu \nabla_\mu \eta^\alpha.$$

One step further, the “relative acceleration” of the separation is

$$a^\alpha \equiv \frac{d}{d\lambda} v^\alpha = \xi^\nu \nabla_\nu (\xi^\mu \nabla_\mu \eta^\alpha).$$

It turns out that a^α (again, a second derivative) is related to the Riemann curvature tensor. We start with

$$a^\alpha = \xi^\nu \nabla_\nu (\xi^\mu \nabla_\mu \eta^\alpha) = \xi^\nu \nabla_\nu (\eta^\mu \nabla_\mu \xi^\alpha),$$

where we used the commutator $[\xi, \eta]^\mu = 0$. By the Leibniz rule,

$$a^\alpha = (\xi^\nu \nabla_\nu \eta^\mu) (\nabla_\mu \xi^\alpha) + \eta^\mu \xi^\nu \nabla_\nu \nabla_\mu \xi^\alpha = (\eta^\nu \nabla_\nu \xi^\mu) (\nabla_\mu \xi^\alpha) + \eta^\mu \xi^\nu \nabla_\nu \nabla_\mu \xi^\alpha.$$

Now bring in the Riemann curvature tensor by swapping the covariant derivatives using $[\nabla_\mu, \nabla_\nu] \xi^\alpha = R^\alpha_{\beta\mu\nu} \xi^\beta$,

$$a^\alpha = (\eta^\nu \nabla_\nu \xi^\mu) (\nabla_\mu \xi^\alpha) + \eta^\mu \xi^\nu \nabla_\mu \nabla_\nu \xi^\alpha - R^\alpha_{\beta\mu\nu} \eta^\mu \xi^\nu \xi^\beta.$$

Exchanging the dummy indices $\mu \leftrightarrow \nu$ for the second term and use the Leibniz rule again,

$$a^\alpha = \eta^\nu \nabla_\nu (\xi^\mu \nabla_\mu \xi^\alpha) - R^\alpha_{\beta\mu\nu} \eta^\mu \xi^\nu \xi^\beta = \eta^\nu \nabla_\nu (\xi^\mu \nabla_\mu \xi^\alpha) + R^\alpha_{\beta\mu\nu} \xi^\beta \xi^\mu \eta^\nu.$$

But the first term is zero: ξ is the tangent vector along the geodesics by definition, so it must satisfy the geodesic equation $\xi^\mu \nabla_\mu \xi^\nu = 0$. In conclusion, we get

$$a^\alpha = R^\alpha_{\beta\mu\nu} \xi^\beta \xi^\mu \eta^\nu. \quad (4.27)$$

This is known as the **equation of geodesic deviation**. The acceleration vector $a^\mu = 0$ for all geodesics if and only if $R^\alpha_{\beta\mu\nu} = 0$.

4.4.5 The Ricci Tensor, Ricci Scalar, Weyl Tensor, and Einstein Tensor

We see that the Riemann curvature tensor is a wonderful tensor that characterizes curvature. It appears in both the parallel transport of a vector along a loop and the geodesic deviation. It is often useful to decompose it into a trace part and a trace-free part. Because of the symmetry properties 1 and 3, the non-vanishing (and independent) trace of $R^\alpha_{\beta\mu\nu}$ is the contraction of the first and the third indices:

$$R_{\mu\nu} \equiv R^\alpha_{\mu\alpha\nu} = R_{\nu\mu}. \quad (4.28)$$

This tensor is called the **Ricci tensor**. (Of course, we can contract the first and the fourth indices. That will give minus the Ricci tensor as $R^\alpha_{\beta\mu\nu} = -R^\alpha_{\beta\nu\mu}$, which is not independent.) The Ricci tensor is a symmetric tensor because $R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}$. We can find the trace of the Ricci tensor, and that is called the **Ricci scalar** (or sometimes scalar curvature),

$$R \equiv R^\mu_{\mu}. \quad (4.29)$$

The trace-free part of the Riemann curvature tensor is called the **Weyl tensor**, $C_{\alpha\beta\mu\nu}$, defined by

$$C_{\alpha\beta\mu\nu} \equiv R_{\mu\nu\alpha\beta} - \frac{2}{(n-2)} (g_{\alpha[\mu} R_{\nu]\beta} - g_{\beta[\mu} R_{\nu]\alpha}) + \frac{2}{(n-1)(n-2)} g_{\alpha[\mu} g_{\nu]\beta} R. \quad (4.30)$$

The Weyl tensor is traceless in contraction in any pair of its indices, and it works only in $n \geq 3$ dimensions. It also retains all the symmetries of the Riemann curvature tensor. The Weyl tensor is not as useful as the Ricci tensor and Ricci scalar as it does not appear in Einstein’s equations.

Finally, a certain combination of the Ricci tensor and Ricci scalar appears to be important. Recall that the Bianchi identity (4.25) is

$$0 = \nabla_\gamma R_{\alpha\beta\mu\nu} + \nabla_\alpha R_{\beta\gamma\mu\nu} + \nabla_\beta R_{\gamma\alpha\mu\nu} = \nabla_\gamma R_{\alpha\beta\mu\nu} - \nabla_\alpha R_{\beta\gamma\mu\nu} - \nabla_\beta R_{\alpha\gamma\mu\nu}.$$

We can raise α and contract it with μ ,

$$0 = \nabla_\gamma R_{\beta\nu} - \nabla^\alpha R_{\beta\gamma\nu\alpha} - \nabla_\beta R_{\gamma\nu},$$

and then raise β and contract it with ν ,

$$0 = \nabla_\gamma R - \nabla^\alpha R_{\gamma\alpha} - \nabla^\beta R_{\gamma\beta} \implies 0 = 2\nabla^\mu R_{\mu\nu} - g_{\mu\nu} \nabla^\nu R = 2\nabla^\mu R_{\mu\nu} - \nabla^\nu (Rg_{\mu\nu}).$$

(Notice that we can raise the indices of the covariant derivative and move the metric tensor around it because it is metric compatible, $\nabla_\mu g_{\alpha\beta} = 0$.) This is the so-called “twice contracted Bianchi identity”, which relates the divergence of the Ricci tensor to the Ricci scalar,

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu R. \quad (4.31)$$

The twice contracted Bianchi identity suggests us to define a new tensor

$$\boxed{G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} Rg_{\mu\nu}.} \quad (4.32)$$

In this case, $G_{\mu\nu}$ will be divergence-free,

$$\nabla^\mu G_{\mu\nu} = \nabla_\mu G^{\mu\nu} = 0.$$

This famous tensor $G_{\mu\nu}$ is called the **Einstein tensor**, which will appear in Einstein’s equation. Because the divergence-free property often comes with *conserved quantities*, $\nabla^\mu G_{\mu\nu} = 0$ is crucial in the motivation of Einstein’s equation that connects matter to curvature.

4.5 Symmetries and Conserved Quantities

4.5.1 The Lie Derivative

Define a vector field $v^\mu(x)$ on a manifold. Given this vector field, we can define the **integral curves** of the vector field to be the curves $x^\mu(\lambda)$ that solve

$$\frac{dx^\mu}{d\lambda} = v^\mu.$$

This is just the components of a tangent vector $d/d\lambda$ along the curve, with coordinate basis vectors $\partial/\partial x^\mu$.

The **Lie derivative** is a derivative operator that measures the change of a tensor T along the vector field v , denoted as $\mathcal{L}_v$. It is a map from (k, l) tensor fields to (k, l) tensor fields, so it is coordinate independent. As any other derivative operators, the Lie derivative is linear and obeys the Leibniz rule. We also require that the Lie derivative acting on a scalar function reduces to the ordinary directional derivative,

$$\mathcal{L}_v f = v(f) = v^\mu \partial_\mu f.$$

Even though the Lie derivative is coordinate independent, it is convenient to define coordinates $x^\mu = (x^1, \dots, x^n)$ on a manifold. Let $\lambda = x^1$ be the parameter along the integral curves, then the components $v^\mu = dx^\mu/d\lambda$ becomes just

$$v^\mu = \frac{dx^\mu}{dx^1} = \delta^\mu_1 \implies v^\mu = (1, 0, 0, \dots, 0) \quad \text{or} \quad v = \frac{\partial}{\partial x^1}.$$

The Lie derivative of a rank (k, l) tensor along this vector field becomes simply

$$\mathcal{L}_v T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} = \frac{\partial}{\partial x^1} T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l}. \quad (4.33)$$

This does not seem coordinate independent due to x^1 , but we will see that it can be expressed in terms of coordinate-independent quantities. Consider another vector field $w^\mu(x)$. Recall that the commutator of two vector fields is given by (3.1),

$$[v, w]^\mu = v^\nu \partial_\nu w^\mu - w^\nu \partial_\nu v^\mu.$$

But $v^\mu = (1, 0, 0, \dots, 0)$ are constants, so $\partial_\nu v^\mu = 0$, and the commutator is

$$[v, w]^\mu = v^\nu \partial_\nu w^\mu = v^1 \partial_1 w^\mu = \frac{\partial w^\mu}{\partial x^1}.$$

This is the same as the Lie derivative of w^μ along v , so we arrive at

$$\mathcal{L}_v w^\mu = [v, w]^\mu = v^\nu \partial_\nu w^\mu - (\partial_\nu v^\mu) w^\nu. \quad (4.34)$$

(Sometimes the commutator is called the [Lie bracket](#).) Equation (4.34) is now covariant. This will be the coordinate-independent form of the Lie derivative (at least on a vector field for now). To generalize to other tensors, consider the Lie derivative on the scalar $\omega_\mu u^\mu$ for arbitrary vector field u^μ and dual vector field ω_μ . The Lie derivative on a scalar is just the directional derivative,

$$\mathcal{L}_v(\omega_\mu u^\mu) = v(\omega_\mu u^\mu) = v^\nu \partial_\nu(\omega_\mu u^\mu) = v^\nu (\partial_\nu \omega_\mu) u^\mu + v^\nu \omega_\mu (\partial_\nu u^\mu).$$

On the other hand, by the Leibniz rule,

$$\mathcal{L}_v(\omega_\mu u^\mu) = (\mathcal{L}_v \omega_\mu) u^\mu + \omega_\mu (\mathcal{L}_v u^\mu) = (\mathcal{L}_v \omega_\mu) u^\mu + \omega_\mu [v, u]^\mu = (\mathcal{L}_v \omega_\mu) u^\mu + \omega_\mu v^\nu \partial_\nu u^\mu - \omega_\mu u^\nu \partial_\nu v^\mu.$$

Setting them equal and canceling some terms,

$$(\mathcal{L}_v \omega_\mu) u^\mu = u^\mu v^\nu \partial_\nu \omega_\mu + \omega_\mu u^\nu \partial_\nu v^\mu = u^\mu v^\nu \partial_\nu \omega_\mu + \omega_\nu u^\mu \partial_\mu v^\nu.$$

Since u^μ is arbitrary, we can remove it from both sides and get the covariant form of $\mathcal{L}_v \omega_\mu$:

$$\mathcal{L}_v \omega_\mu = v^\nu \partial_\nu \omega_\mu + (\partial_\mu v^\nu) \omega_\nu.$$

It is straightforward to generalize the Lie derivative on higher rank tensors using this equation and (4.34):

$$\mathcal{L}_v T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} = v^\gamma \partial_\gamma T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} - \sum_{i=1}^k (\partial_\gamma v^{\mu_i}) T^{\mu_1 \dots \gamma \dots \mu_k}_{\nu_1 \dots \nu_l} + \sum_{j=1}^l (\partial_{\nu_j} v^\gamma) T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \gamma \dots \nu_l}. \quad (4.35)$$

Note that equation (4.35) is derived only from the commutator and the fact that $\mathcal{L}_v f = v(f)$. Recall that in Section 4.2, any derivative operator ∇_μ satisfying the “five conditions” can represent the commutator,

$$[\xi, \eta]^\mu = \xi^\nu \nabla_\nu \eta^\mu - \eta^\nu \nabla_\nu \xi^\mu.$$

Also, any ∇_μ acting on functions satisfy $\xi(f) = \xi^\mu \nabla_\mu f$, just like the partial derivative $v(f) = v^\mu \partial_\mu f$. Therefore, we are free to replace all ∂ with ∇ in (4.35) and get

$$\mathcal{L}_\xi T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} = \xi^\gamma \nabla_\gamma T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} - \sum_{i=1}^k (\nabla_\gamma \xi^{\mu_i}) T^{\mu_1 \dots \gamma \dots \mu_k}_{\nu_1 \dots \nu_l} + \sum_{j=1}^l (\nabla_{\nu_j} \xi^\gamma) T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \gamma \dots \nu_l}. \quad (4.36)$$

So far we haven’t introduce any metric yet. All we need is a manifold and a vector field defined on it. If we introduce a metric $g_{\mu\nu}$, (4.36) will be very useful when we choose ∇_μ to be the covariant derivative, with $\nabla_\mu g_{\alpha\beta} = 0$. In particular, the Lie derivative of the metric tensor, according to (4.36), is

$$\mathcal{L}_\xi g_{\mu\nu} = \xi^\alpha \nabla_\alpha g_{\mu\nu} + g_{\alpha\nu} \nabla_\mu \xi^\alpha + g_{\mu\alpha} \nabla_\nu \xi^\alpha = \xi^\alpha \nabla_\alpha g_{\mu\nu} + \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu.$$

The first term vanishes for the covariant derivative, leaving

$$\mathcal{L}_\xi g_{\mu\nu} = 2\nabla_{(\mu} \xi_{\nu)}. \quad (4.37)$$

4.5.2 Killing Vectors

General relativity is a difficult theory due to its nonlinearity. A spacetime with lots of symmetries in it will make problems easier. In particular, if the metric tensor is independent of a coordinate everywhere, we will get conserved quantities that help simplify the equations of motion. The Lie derivative of the metric tells us how to do this. Let ξ be a vector field with components $dx^\mu/d\lambda$ along

the integral curves. If we choose a coordinate system x^μ with $x^1 \in \{x^\mu\}$ being the parameter along the curve, then

$$\xi = \frac{\partial}{\partial x^1}, \quad \text{and} \quad \mathcal{L}_\xi g_{\mu\nu} = \frac{\partial}{\partial x^1} g_{\mu\nu} = 2\nabla_{(\mu} \xi_{\nu)},$$

where we used (4.37). This means if all components of the metric tensor is independent of the coordinate x^1 ,

$$\nabla_{(\mu} \xi_{\nu)} = 0. \quad (4.38)$$

This equation is known as **Killing's equation** (after Wilhelm Killing). The vector ξ^μ that satisfy this equation is called a **Killing vector**. Symmetries in general relativity are represented by Killing vector fields. Notice that Killing's equation is a linear equation. This means if ξ^μ and η^μ are both Killing vectors, so is any linear combination of them.

Killing vectors are related to conserved quantities of a particle along geodesics. If ξ^μ are components of a Killing vector, then its inner product with the 4-velocity of the particle, $\xi \cdot u = \xi_\mu u^\mu$, is a conserved quantity along geodesics.

Proof. We know that the 4-velocity of a particle $u^\mu = dx^\mu/d\tau$ satisfies the geodesic equation,

$$u^\mu \nabla_\mu u^\alpha = 0 \quad \Longleftrightarrow \quad \frac{d^2 x^\mu}{d\tau^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

The change of $\xi_\mu u^\mu$ along the geodesic is

$$\frac{d}{d\lambda} (\xi_\mu u^\mu) = u^\nu \nabla_\nu (\xi_\mu u^\mu) = u^\mu u^\nu \nabla_\nu \xi_\mu + \xi_\mu u^\nu \nabla_\nu u^\mu.$$

The second term becomes zero because u^μ satisfies the geodesic equation. The first term can be written as

$$u^\mu u^\nu \nabla_\nu \xi_\mu = \frac{1}{2} (u^\mu u^\nu \nabla_\nu \xi_\mu + u^\nu u^\mu \nabla_\mu \xi_\nu) = u^\mu u^\nu \nabla_{(\mu} \xi_{\nu)} = 0,$$

where we made a copy of $u^\mu u^\nu \nabla_\nu \xi_\mu = u^\nu u^\mu \nabla_\mu \xi_\nu$ by switching the dummy indices in the first equality. This is also zero because ξ^μ satisfies Killing's equation (4.38). Hence $d(\xi_\mu u^\mu)/d\tau = 0$, the quantity $\xi_\mu u^\mu$ is a conserved quantity along geodesics. $\square$

Killing's equation can be derived directly from the geodesic equation as well, but the ideal is more or less the same as the Lie derivative. Start from the geodesic equation

$$u^\mu \nabla_\mu u^\nu = 0.$$

We are free to lower the index of u^ν because $\nabla_\mu g_{\alpha\beta} = 0$. Referencing (4.6),

$$u^\mu \nabla_\mu u_\nu = 0 \quad \Longrightarrow \quad u^\mu \partial_\mu u_\nu - \Gamma^\alpha_{\mu\nu} u^\mu u_\alpha = 0.$$

Since the 4-velocity is defined as $u^\mu = dx^\mu/d\tau$, the first term will be

$$u^\mu \partial_\mu u_\nu = \frac{dx^\mu}{d\tau} \frac{\partial}{\partial x^\mu} u_\nu = \frac{du_\nu}{d\tau}.$$

The second term, in coordinate representation, is

$$\Gamma^\alpha_{\mu\nu} u^\mu u_\alpha = \frac{1}{2} g^{\alpha\beta} (\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}) u^\mu u_\alpha = \frac{1}{2} (\partial_\nu g_{\mu\beta} + 2\partial_{[\mu} g_{\beta]\nu}) u^\mu u^\beta = \frac{1}{2} (\partial_\nu g_{\mu\beta}) u^\mu u^\beta.$$

Therefore, the geodesic equation can be equivalently written as

$$\frac{du_\nu}{d\tau} = \frac{1}{2} (\partial_\nu g_{\mu\beta}) u^\mu u^\beta.$$

From this form of the geodesic equation, it is clear that if all components of the metric tensor is independent of a coordinate x^1 ,

$$\frac{\partial}{\partial x^1} g_{\mu\nu} = 0 \quad \Longrightarrow \quad \frac{du_1}{d\tau} = 0.$$

This will make more sense physically if change u_1 to p_1 by multiplying a mass m , this component of the 4-momentum is conserved. In a more covariant way, we introduce a vector $\xi = \partial/\partial x^1$, with components $\xi^\mu = (\partial_1)^\mu = \delta^\mu_1$. Then $p_1 = \xi^\mu p_\mu = \xi_\mu p^\mu$. Conservation

of p_1 says that

$$\frac{d}{d\tau}(\xi_\mu p^\mu) = 0.$$

Now $\xi_\mu p^\mu$ is just a function, we can express this change in the covariant derivative form,

$$0 = p^\mu \nabla_\mu (\xi_\nu p^\nu) = p^\mu \xi_\nu \nabla_\mu p^\nu + p^\mu p^\nu \nabla_\mu \xi_\nu = p^\mu p^\nu \nabla_\mu \xi_\nu = p^\mu p^\nu \nabla_{(\mu} \xi_{\nu)}.$$

For arbitrary 4-velocity or 4-momentum, this is equivalent to Killing's equation

$$\nabla_{(\mu} \xi_{\nu)} = 0.$$

4.5.3 Killing Tensors

To be written.

4.5.4 Relation to Curvature

There is a useful relation between Killing vectors and the Riemann curvature tensor:

$$\nabla_\mu \nabla_\nu \xi^\alpha = R^\alpha{}_{\nu\mu\beta} \xi^\beta. \quad (4.39)$$

Proof. We start with the definition of the Riemann curvature tensor:

$$\nabla_\mu \nabla_\nu \xi_\beta - \nabla_\nu \nabla_\mu \xi_\beta = -R^\alpha{}_{\beta\mu\nu} \xi_\alpha.$$

If ξ_β satisfies Killing's equation (4.38), then $\nabla_\mu \xi_\beta = -\nabla_\beta \xi_\mu$, and

$$\begin{aligned} \nabla_\mu \nabla_\nu \xi_\beta + \nabla_\nu \nabla_\beta \xi_\mu &= -R^\alpha{}_{\beta\mu\nu} \xi_\alpha, \\ \nabla_\nu \nabla_\beta \xi_\mu + \nabla_\beta \nabla_\mu \xi_\nu &= -R^\alpha{}_{\mu\nu\beta} \xi_\alpha, \\ \nabla_\beta \nabla_\mu \xi_\nu + \nabla_\mu \nabla_\nu \xi_\beta &= -R^\alpha{}_{\nu\beta\mu} \xi_\alpha, \end{aligned}$$

where the last two equations are just index permutations of the first. Adding the first two equations and subtracting the third,

$$\nabla_\mu \nabla_\nu \xi_\beta + \nabla_\nu \nabla_\beta \xi_\mu + \nabla_\nu \nabla_\beta \xi_\mu + \nabla_\beta \nabla_\mu \xi_\nu - \nabla_\beta \nabla_\mu \xi_\nu - \nabla_\mu \nabla_\nu \xi_\beta = -(R^\alpha{}_{\beta\mu\nu} + R^\alpha{}_{\mu\nu\beta} - R^\alpha{}_{\nu\beta\mu}) \xi_\alpha.$$

We find that many terms cancel. The LHS is $2\nabla_\nu \nabla_\beta \xi_\mu$, while the RHS can be simplified using the algebraic Bianchi identity (4.22),

$$-(R^\alpha{}_{\beta\mu\nu} + R^\alpha{}_{\mu\nu\beta} - R^\alpha{}_{\nu\beta\mu}) \xi_\alpha = -(R^\alpha{}_{\beta\mu\nu} + R^\alpha{}_{\mu\nu\beta} + R^\alpha{}_{\nu\beta\mu} - 2R^\alpha{}_{\nu\beta\mu}) \xi_\alpha = 2R^\alpha{}_{\nu\beta\mu} \xi_\alpha.$$

Thus, the Killing vector and the Riemann curvature tensor is related by

$$\nabla_\nu \nabla_\beta \xi_\mu = R^\alpha{}_{\nu\beta\mu} \xi_\alpha.$$

To raise the index of ξ_μ , we do the following using various symmetries of the Riemann curvature tensor:

$$\nabla_\nu \nabla_\beta \xi_\mu = R_{\alpha\nu\beta\mu} \xi^\alpha = R_{\beta\mu\alpha\nu} \xi^\alpha = R_{\mu\beta\nu\alpha} \xi^\alpha.$$

Then raise the index μ to get (4.39),

$$\nabla_\nu \nabla_\beta \xi^\mu = R^\mu{}_{\nu\beta\alpha} \xi^\alpha.$$

□

By (4.39), we can contract α with μ to related Killing vectors to the Ricci tensor,

$$\nabla_\mu \nabla_\nu \xi^\mu = R_{\mu\nu} \xi^\mu. \quad (4.40)$$

Then there is another important result of Killing vectors, which says that the Ricci scalar is constant along a Killing vector field:

$$\xi^\mu \nabla_\mu R = 0. \quad (4.41)$$

This is another way to say that the geometry is not changing along a Killing vector field.

Proof. To prove this, we need the twice contracted Bianchi identity (4.31) and Killing's equation (4.38),

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu R, \quad \nabla_\mu \nabla_\nu \xi^\mu = R_{\mu\nu} \xi^\mu, \quad \text{and} \quad \nabla_{(\mu} \xi_{\nu)} = 0.$$

Consider the quantity $\nabla^\mu (R_{\mu\nu} \xi^\nu)$. By the Leibniz rule,

$$\nabla^\mu (R_{\mu\nu} \xi^\nu) = \xi^\nu \nabla^\mu R_{\mu\nu} + R_{\mu\nu} \nabla^\mu \xi^\nu.$$

The second term vanishes because $R_{\mu\nu}$ is symmetric in its indices but $\nabla^\mu \xi^\nu$ is antisymmetric by Killing's equation. Meanwhile, using (4.40),

$$\nabla^\mu (R_{\mu\nu} \xi^\nu) = \nabla^\mu \nabla_\nu \nabla_\mu \xi^\nu = \nabla_\mu \nabla_\nu \nabla^\mu \xi^\nu.$$

This time we cannot use Killing's equation directly because covariant derivatives do not commute in general, but we can use (4.19) to relate the commutator of covariant derivatives to the Riemann curvature tensor:

$$[\nabla_\mu, \nabla_\nu](\nabla^\alpha \xi^\beta) = R^\alpha{}_{\gamma\mu\nu} \nabla^\gamma \xi^\beta + R^\beta{}_{\gamma\mu\nu} \nabla^\alpha \xi^\gamma.$$

Contracting α with μ and β with ν ,

$$[\nabla_\mu, \nabla_\nu](\nabla^\mu \xi^\nu) = R^\mu{}_{\alpha\mu\nu} \nabla^\alpha \xi^\nu + R^\nu{}_{\alpha\mu\nu} \nabla^\mu \xi^\alpha = R_{\alpha\nu} \nabla^\alpha \xi^\nu - R_{\alpha\mu} \nabla^\mu \xi^\alpha = 0.$$

So ∇_μ and ∇_ν do commute (and is symmetric) when they act on and contract with $\nabla^\mu \xi^\nu$. Then Killing's equation infers that $\nabla_\mu \nabla_\nu \nabla^\mu \xi^\nu = 0$. Therefore, by the twice contracted Bianchi identity.

$$0 = \xi^\nu \nabla^\mu R_{\mu\nu} = \frac{1}{2} \xi^\nu \nabla_\nu R$$

□

5 GRAVITATION

5.1 Transition from Special Relativity to General Relativity

The transition of laws of physics from special relativity to general relativity is simple. There is a principle called the **minimal-coupling principle**, which says that the transition of laws of physics obeys following steps:

1. Take a law of physics that is valid in inertial coordinates in flat (Minkowski) spacetime.
2. Write it in a relativistic covariant (tensorial) form.
3. Assert that the resulting law to be true in curved spacetime. A more practical solution to do this is to take write $\eta_{\mu\nu} \rightarrow g_{\mu\nu}$ and $\partial_\mu \rightarrow \nabla_\mu$.

5.1.1 From Newton's Second Law to Geodesic Equation

In special relativity, Newton's second law for a free particle says that the second derivative of it vanishes along a path,

$$\frac{d^2 x^\mu}{d\lambda^2} = 0.$$

This is not a tensorial equation, because only the first derivative $dx^\mu/d\lambda$ is a well-defined tensor, while the second derivative is not. It does not work even in polar coordinates. Using the chain rule,

$$\frac{dx^\nu}{d\lambda} \partial_\nu \frac{dx^\mu}{d\lambda} = 0.$$

This will be a valid tensorial equation in flat spacetime. In curved spacetime, we follow the recipe $\eta_{\mu\nu} \rightarrow g_{\mu\nu}$ and $\partial_\mu \rightarrow \nabla_\mu$,

$$\frac{dx^\nu}{d\lambda} \partial_\nu \frac{dx^\mu}{d\lambda} = 0 \quad \longrightarrow \quad \frac{dx^\nu}{d\lambda} \nabla_\nu \frac{dx^\mu}{d\lambda} = 0,$$

or, writing out the covariant derivative,

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0.$$

This is just the geodesic equation. Thus, in general relativity, free particles move along geodesics.

The geodesic equation should produce Newtonian equations of motion in the Newtonian limit. This limit will give some information about the metric. The Newtonian limit is defined with three requirements:

1. The particles are moving slowly compared to the speed of light. In other words,

$$\frac{dx^i}{d\tau} \ll \frac{dt}{d\tau}.$$

2. The gravitational field is weak, so the metric can be written as the Minkowski metric $\eta_{\mu\nu}$ plus a small perturbation $|h_{\mu\nu}| \ll 1$,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad g^{\mu\nu} = (g_{\mu\nu})^{-1} \simeq \eta^{\mu\nu} - h^{\mu\nu}.$$

3. The field is static, or unchanging with time, so any (partial) time-derivative of the metric is zero.

By requirement 1, the only important terms in the geodesic equation are

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{tt} \left(\frac{dt}{d\tau} \right)^2 = 0.$$

With static field, $\partial_t g_{\mu\nu} = 0$, the Christoffel symbols Γ^μ_{tt} are

$$\Gamma^\mu_{tt} = \frac{1}{2} g^{\mu\nu} (\partial_t g_{\nu t} + \partial_t g_{t\nu} - \partial_\nu g_{tt}) = -\frac{1}{2} g^{\mu\nu} \partial_\nu g_{tt} \simeq -\frac{1}{2} \eta^{\mu\nu} \partial_\nu h_{tt}.$$

The geodesic equation then becomes

$$\frac{d^2 x^\mu}{d\tau^2} = \frac{1}{2} \eta^{\mu\nu} \partial_\nu h_{tt} \left(\frac{dt}{d\tau} \right)^2.$$

Again because the field is static, $\partial_t h_{tt}$, the $\mu = 0$ component of the equation is

$$\frac{d^2 t}{d\tau^2} = 0 \quad \Rightarrow \quad \frac{dt}{d\tau} = \text{const.}$$

The spatial components are

$$\frac{d^2 x^i}{d\tau^2} = \frac{1}{2} \left(\frac{dt}{d\tau} \right)^2 \partial^i h_{tt} \quad \Rightarrow \quad \frac{d^2 x^i}{dt^2} = \frac{1}{2} \partial^i h_{tt}.$$

If we compare this to Newton's second law, $\ddot{\mathbf{x}} = -\nabla\Phi$, we find that the Newtonian limit takes

$$h_{tt} = -2\Phi, \quad \text{or} \quad g_{tt} = -(1 + 2\Phi). \quad (5.1)$$

5.1.2 The Stress-Energy Tensor

In flat spacetime, we know that the stress-energy tensor conservation is written as

$$\partial^\mu T_{\mu\nu} = 0.$$

This is already in a covariant form. Generalizing it to curved spacetime gives

$$\nabla^\mu T_{\mu\nu} = 0.$$

The stress-energy tensor of the perfect fluid is useful in general relativity, so we should generalize it as well. From (2.16), we change $\eta_{\mu\nu} \rightarrow g_{\mu\nu}$,

$$T_{\mu\nu} = (\rho + P) u_\mu u_\nu + P g_{\mu\nu}. \quad (5.2)$$

5.1.3 Geometrized Units

Since general relativity is a theory of relativity + gravity, it should involve two fundamental constants: the speed of light c and the gravitational constant G . We will adopt **geometrized units** $c = 1$ and $G = 1$ in general relativity. As a consequence, quantities in general relativity are measured in *lengths*.

5.2 Einstein's Equation

The general theory of relativity is the study of how matter curves spacetime and how spacetime tells matter to move. The latter is already known—freely falling particles follow geodesics in curved spacetime, so the governing equation is the geodesic equation. The former is Einstein's field equation, the fundamental equation of general relativity. Just as Newton's second law, Maxwell's equations, and other fundamental equations, Einstein's equation cannot be “derived” from first principles. It was postulated by Einstein, and it needed to withstand testing experiments because physics is an empirical science. Nonetheless, Einstein's equation does not come from nowhere. It can be motivated from Newtonian gravity with some extra modifications. Once Einstein's equation is motivated, the goal is to find the action and Lagrangian that give rise to this equation.

5.2.1 From Poisson's Equation to Einstein's Equation

In Newtonian gravity, the gravitational potential Φ tells how matter should move subject to gravity. The gravitational potential can be determined by Poisson's equation

$$\nabla^2 \Phi = 4\pi\rho,$$

where ∇^2 is the Laplacian and ρ is the mass density. We want to generalize Poisson's equation to some tensorial equation, with some analogous quantities replacing $\nabla^2\Phi$ and ρ . The tensor generalization of the mass density should be the stress-energy tensor $T_{\mu\nu}$. The generalization of Φ should be the metric, because the metric is related to the curvature and it determines the geodesic equation that tells how matter should move. Thus, some second derivatives of $g_{\mu\nu}$ with respect to spacetime should replace $\nabla^2\Phi$. In conclusion, we look for something like

$$[\nabla^2 g]_{\mu\nu} \sim T_{\mu\nu}.$$

Guess 1. Since $T_{\mu\nu}$ is symmetric, the LHS tensor should also be a symmetric tensor. The simplest guess might be a straightforward generalization of $\nabla^2 = \partial_i \partial^i$ to $\nabla_\mu \nabla^\mu$. However, the covariant derivative annihilates the metric tensor automatically, $\nabla_\mu g_{\alpha\beta} = 0$, this will not work for nontrivial $T_{\mu\nu}$.

Guess 2. There is some known tensor that characterize curvature and involves second-order derivatives of $g_{\mu\nu}$. One possibility is the Riemann curvature tensor $R^\alpha{}_{\beta\mu\nu}$. But it has too many indices, so we shall try the Ricci tensor next,

$$R_{\mu\nu} = \kappa T_{\mu\nu}$$

for some constant κ . The Ricci tensor is symmetric, so it seems to be a good guess. However, the conservation law $\nabla^\mu T_{\mu\nu} = 0$ suggests that $\nabla^\mu R_{\mu\nu} = 0$. This is not always true.

Guess 3. With the previous two guesses, we put more constraints on the tensor on the LHS in $[\nabla^2 g]_{\mu\nu} \sim T_{\mu\nu}$. It should be a symmetric tensor that involves second-order derivatives in the metric, and is divergence-free like $T_{\mu\nu}$. Recall that the Einstein tensor (4.32) satisfies all the requirements,

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}, \quad G_{\mu\nu} = G_{\nu\mu}, \quad \nabla^\mu G_{\mu\nu} = 0.$$

Therefore, we propose that

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \kappa T_{\mu\nu}. \quad (5.3)$$

The job now is to fix the constant κ . In 4-dimensional spacetime, taking the trace on both sides of the equation leads to

$$R - \frac{1}{2}R\delta^\mu{}_\mu = \kappa T^\mu{}_\mu \quad \implies \quad R = -\kappa T,$$

where $T \equiv T^\mu{}_\mu$, so we can rewrite (5.3) as

$$R_{\mu\nu} = \kappa \left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu} \right). \quad (5.4)$$

For general relativity to be a valid theory, we expect it to reduce to Poisson's equation when the field is weak, time-independent, and particles are moving slowly compared to the speed of light. Consider a perfect fluid with

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + Pg_{\mu\nu}.$$

In the low-speed limit, the pressure P is negligible compared to the energy density ρ (which will also be small in the weak-field limit), so we are in fact interested in a dust:

$$T_{\mu\nu} = \rho u_\mu u_\nu,$$

where u^μ is the 4-velocity of the fluid. In the rest frame of the fluid, $u^\mu = (u^t, 0, 0, 0)$, and it always satisfies $g_{\mu\nu}u^\mu u^\nu = -1$. In the following calculations, we will keep only first-order terms in perturbation $h_{\mu\nu}$ and ρ , for $h_{\mu\nu}$ defined by

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad g^{\mu\nu} \simeq \eta^{\mu\nu} - h^{\mu\nu}.$$

Computing $g_{\mu\nu}u^\mu u^\nu$ and setting it equal to -1 , we have

$$-1 = g_{tt}u^t u^t = (-1 + h_{tt})(u^t)^2 \quad \implies \quad u^t = (1 - h_{tt})^{-1/2} \simeq 1 + \frac{1}{2}h_{tt}.$$

In the low-speed limit, the only important term is the T_{tt} component,

$$T_{tt} = \rho(u_t)^2 \simeq \rho \left(-1 - \frac{1}{2}h_{tt} \right)^2 \simeq \rho.$$

The trace of $T_{\mu\nu}$ to first order is

$$T = g^{tt}T_{tt} \simeq -T_{tt} = -\rho.$$

Substituting T_{tt} and T into the time-time component of (5.4),

$$R_{tt} = \frac{1}{2}\kappa\rho.$$

The goal now is to explicitly express $R_{tt} = R^\mu{}_{t\mu t}$ in terms of the metric. Due to the symmetry property (antisymmetric in the last two indices) of the Riemann curvature tensor, $R^t{}_{ttt} = 0$, so $R_{tt} = R^i{}_{tit}$. By the coordinate representation (4.26),

$$R^i{}_{tjt} = \partial_j\Gamma^i{}_{tt} - \partial_t\Gamma^i{}_{jt} + \Gamma^i{}_{j\mu}\Gamma^\mu{}_{tt} - \Gamma^i{}_{t\mu}\Gamma^\mu{}_{jt}.$$

Let's analyze terms one by one: the second term is a time-derivative of the Christoffel symbol, so it vanishes for static fields. Also, an important trick in perturbation theory is that all Christoffel symbols are at least first-order in perturbation $h_{\mu\nu}$,

$$\Gamma^\alpha{}_{\mu\nu} = \frac{1}{2}g^{\alpha\beta}(\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}) = \frac{1}{2}g^{\alpha\beta}(\partial_\mu h_{\beta\nu} + \partial_\nu h_{\beta\mu} - \partial_\beta h_{\mu\nu}).$$

Therefore, all $(\Gamma)^2$ terms are second-order terms in the Riemann curvature tensor. The leading-order term is only

$$R^i{}_{tjt} = \partial_j\Gamma^i{}_{tt}.$$

Contracting i and j to get the Ricci tensor,

$$R_{tt} = R^i{}_{tit} = \partial_i \left[\frac{1}{2}g^{i\mu}(\partial_t g_{\mu t} + \partial_t g_{t\mu} - \partial_\mu g_{tt}) \right] = -\frac{1}{2}\partial_i(\eta^{ij} - h^{ij})\partial_j h_{tt} \simeq -\frac{1}{2}\delta^{ij}\partial_i\partial_j h_{tt} = -\frac{1}{2}\nabla^2 h_{tt}.$$

Hence in the Newtonian limit,

$$-\frac{1}{2}\nabla^2 h_{tt} = \frac{1}{2}\kappa\rho.$$

This looks very similar to Poisson's equation. In the previous section, we already know $h_{tt} = -2\Phi$ from the geodesic equation in the Newtonian limit, so $\nabla^2\Phi = \frac{1}{2}\kappa\rho$. Comparing this with Poisson's equation, $\nabla^2\Phi = 4\pi\rho$, we find that $\kappa = 8\pi$. Putting κ back into the proposal (5.3), we arrive at **Einstein's field equation** for general relativity:

$$\boxed{R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi T_{\mu\nu}.} \quad (5.5)$$

This form of Einstein's equation is useful when there is nonzero stress-energy tensor. On the other hand, as we will see in later chapters, many solutions are already interesting for $T_{\mu\nu} = 0$ (called vacuum solutions). If $T_{\mu\nu} = 0$, it is more convenient to use the form of (5.4),

$$R_{\mu\nu} = 8\pi \left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu} \right). \quad (5.6)$$

This reduces to $R_{\mu\nu} = 0$ when $T_{\mu\nu} = 0$.

5.2.2 Properties of Einstein's Equation

1. Mathematically, Einstein's equation is a set of coupled second-order nonlinear partial differential equations. Both sides of Einstein's equation are symmetric tensors, so there are ten equations. With four constraint equations $\nabla^\mu G_{\mu\nu} = 0$, there are six independent equations left. This is consistent with the number of physical degrees of freedom in $g_{\mu\nu}$. The metric tensor $g_{\mu\nu}$ is symmetric with 10 independent components, but there are 4 unphysical degrees of freedom by coordinate transformation $x^{\mu'}(x^\mu)$.

2. The LHS of the equation has the Ricci tensor and Ricci scalar, which are contractions of the Riemann curvature tensor. The Riemann curvature tensor consists of derivatives and products of Christoffel symbols. The Christoffel symbols involve the inverse metric contracted with partial derivatives of the metric. Also, the stress-energy tensor can depend on the metric either explicitly or implicitly as well. Therefore, Einstein's equation is *highly nonlinear*. There is no principle of superposition in general relativity. Even the vacuum solution ($T_{\mu\nu} = 0$) are very difficult to solve. There are only a few exact solutions known, and many of them require lots of symmetry to simplify the problem.

3. Given a metric $g_{\mu\nu}$, it will always be a solution to Einstein's equation. Just compute the Einstein tensor $G_{\mu\nu}$ and then set it equal to $8\pi T_{\mu\nu}$. There will always be a stress-energy tensor associated with it. However, some arbitrary $g_{\mu\nu}$ cannot guarantee a stress-energy tensor that are physical (e.g. the energy density might be negative). The goal of Einstein's equation is to solve for the metric $g_{\mu\nu}$ with some form of the stress-energy tensor $T_{\mu\nu}$. The stress-energy tensor generally depend on the metric. For example, a perfect fluid has $T_{\mu\nu} = (\rho + P)u_\mu u_\nu + Pg_{\mu\nu}$, which has an explicit dependence on $g_{\mu\nu}$. Even for a dust, $T_{\mu\nu} = \rho u_\mu u_\nu$, the metric is hidden in the stress-energy tensor. This is because you need a metric to specify a valid 4-velocity u^μ , since it needs to be a unit vector, $g_{\mu\nu}u^\mu u^\nu = -1$. Thus, in general relativity, we need to simultaneously solve for the metric and the stress-energy tensor.

4. If $g_{\mu\nu}$ is a vacuum solution, then $\lambda^2 g_{\mu\nu}$ is also a valid solution for some constant λ . First, the Christoffel symbols $\Gamma \sim g^{-1}\partial g$ are unchanged by this re-scaling. As a result, the Riemann curvature tensor $R^\alpha{}_{\beta\mu\nu} \sim 2\partial\Gamma + 2(\Gamma)^2$ is unchanged, and so is the Ricci tensor $R_{\mu\nu} = R^\alpha{}_{\mu\alpha\nu}$. Then according to (5.6), a vacuum solution satisfies $R_{\mu\nu} = 0$, re-scaling the solution by a constant has no effect on the equation.

5. In fact, even though Einstein's equation and the geodesic equation are often discussed separately, the conservation law $\nabla^\mu T_{\mu\nu} = 0$ contains the geodesic equation already. This is reasonable, because $T_{\mu\nu}$ can have time-dependence as it contains the metric. Moreover, any sufficiently small bodies that constitutes the fluid should follow geodesics. This is consistent with $\nabla^\mu T_{\mu\nu} = 0$, but we will not prove this statement in this notes.

6. There is no quantization in Einstein's equation. General relativity is a classical field theory (no $\hbar$). A precise theory of quantum gravity is yet unknown. This does not mean that we cannot do quantum physics in general relativity. In fact, there are examples of quantum field theory in curved spacetime. A good example would be Hawking radiation of a black hole, which deals with quantum fluctuations near the surface of a black hole. Another example would be the Early universe, where quantum fluctuations lead to non-uniform matter density that evolves into large-scale structures like galaxies we see today.

5.3 The Lagrangian Formulation

5.3.1 The Einstein-Hilbert Action

The laws of physics are transitioned from special relativity to general relativity with the prescription $\eta_{\mu\nu} \rightarrow g_{\mu\nu}$ and $\partial_\mu \rightarrow \nabla_\mu$. In an n -dimensional manifold, the action S for a field ϕ^i is

$$S = \int \mathcal{L}(\phi^i, \nabla_\mu \phi^i) d^n x = \int \tilde{\mathcal{L}}(\phi^i, \nabla_\mu \phi^i) \sqrt{-g} d^n x.$$

Note that the volume element carries a $\sqrt{-g}$ for a Lorentzian metric, where $g = \det g_{\mu\nu}$. Sometimes it is more useful to use $\mathcal{L} = \tilde{\mathcal{L}}\sqrt{-g}$ as the Lagrangian, and sometimes it is more useful to drop the $\sqrt{-g}$ and use $\tilde{\mathcal{L}}$ directly. The Euler-Lagrange equations is simply obtained by replacing $\partial_\mu \rightarrow \nabla_\mu$ in (2.11),

$$\frac{\partial \tilde{\mathcal{L}}}{\partial \phi^i} - \nabla_\mu \left(\frac{\partial \tilde{\mathcal{L}}}{\partial (\nabla_\mu \phi^i)} \right) = 0, \quad (5.7)$$

In general relativity, the dynamical fields are components of the metric tensor $g_{\mu\nu}$, while the equations of motion for $g_{\mu\nu}$ are Einstein's equations. We want to find an action for general relativity such that extremizing the action $\delta S = 0$ with respect to $g_{\mu\nu}$ gives Einstein's equation. This sounds complicated, because Einstein's equation involves $g_{\mu\nu}$ implicitly with contractions and nonlinearity. Also, we cannot reversely derive the Lagrangian from the Euler-Lagrange equation (5.7) because the second term is automatically zero with $\nabla_\mu g_{\alpha\beta} = 0$. Nevertheless, there is still some hint from the Lagrangian formulation itself and the nature of gravity:

1. The Lagrangian must be a *scalar*.
2. Since we can always set up local inertial coordinates such that the metric is locally Minkowski and that its first derivatives vanish, the Lagrangian must involve at least second derivatives.
3. The Lagrangian should have something to do with curvature.

In 1915, David Hilbert proposed that the action should be

$$S = \int R \sqrt{-g} d^n x. \quad (5.8)$$

This action is known as the **Einstein-Hilbert action**. The Lagrangian inferred from this action is probably the simplest possible scalar in general relativity that meets those hints above—it is just the Ricci scalar. It turns out that this simple action is indeed the correct action to derive Einstein's equation.

5.3.2 Einstein's Equation in Vacuum

As mentioned above, the Euler-Lagrange's equation is useless for $g_{\mu\nu}$ as dynamical variables because its covariant derivative is automatically zero. We need to extremize the Einstein-Hilbert action with respect to $g_{\mu\nu}$ from scratch. It is more convenient to vary with respect to $g^{\mu\nu}$. The variation of the metric $\delta g_{\mu\nu}$ and that of the inverse metric $\delta g^{\mu\nu}$ are related by

$$0 = \delta(\delta^\mu{}_\nu) = \delta(g^{\mu\alpha} g_{\alpha\nu}) = \delta g^{\mu\alpha} g_{\alpha\nu} + g^{\mu\alpha} \delta g_{\alpha\nu} \implies \delta g^{\mu\alpha} g_{\alpha\nu} = -g^{\mu\alpha} \delta g_{\alpha\nu}.$$

Multiplying both sides by $g^{\nu\beta}$ gives

$$\delta g^{\mu\beta} = -g^{\mu\alpha} g^{\nu\beta} \delta g_{\alpha\nu}.$$

Relabeling some indices, we can get

$$\delta g_{\mu\nu} = -g_{\mu\alpha} g_{\nu\beta} \delta g^{\alpha\beta} \iff \delta g^{\alpha\beta} = -g^{\mu\alpha} g^{\nu\beta} \delta g_{\mu\nu}. \quad (5.9)$$

Varying $g_{\mu\nu}$ is equivalent to varying $g^{\mu\nu}$; they are interchangeable via (5.9). Now we vary the Einstein-Hilbert action (5.8). First write the Ricci scalar as $R = g^{\mu\nu} R_{\mu\nu}$. The action can then be divided into three parts: $g^{\mu\nu} \delta R_{\mu\nu}$, $R_{\mu\nu} \delta g^{\mu\nu}$, and $\delta \sqrt{-g}$,

$$\delta S = \int d^n x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} + \int d^n x \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + \int d^n x R \delta \sqrt{-g}. \quad (5.10)$$

The second part is already in the form of $\delta g^{\mu\nu}$. We should examine the other two with care. First, the Ricci tensor is the trace of the Riemann curvature tensor, and it further depends on the Christoffel symbol. Hence the tasks are the following: find $\delta \Gamma^\alpha{}_{\mu\nu}$, then $\delta R^\alpha{}_{\beta\mu\nu}$, then $\delta R_{\mu\nu}$.

Variation of the Christoffel symbol $\Gamma^\alpha{}_{\mu\nu}$. The coordinate representation of the Christoffel symbol is

$$\Gamma^\alpha{}_{\mu\nu} = \frac{1}{2} g^{\alpha\beta} (\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}).$$

With the variation $g_{\mu\nu} \rightarrow g_{\mu\nu} + \delta g_{\mu\nu}$, $g^{\mu\nu} \rightarrow g^{\mu\nu} + \delta g^{\mu\nu}$, the Christoffel symbols varies as

$$\delta \Gamma^\alpha{}_{\mu\nu} = \frac{1}{2} (g^{\alpha\beta} + \delta g^{\alpha\beta}) [\partial_\mu (g_{\beta\nu} + \delta g_{\beta\nu}) + \partial_\nu (g_{\beta\mu} + \delta g_{\beta\mu}) - \partial_\beta (g_{\mu\nu} + \delta g_{\mu\nu})] - \frac{1}{2} g^{\alpha\beta} (\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}).$$

The zeroth-order terms cancel as usual, we are left with

$$\delta \Gamma^\alpha{}_{\mu\nu} = \frac{1}{2} \delta g^{\alpha\beta} (\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}) + \frac{1}{2} g^{\alpha\beta} (\partial_\mu \delta g_{\beta\nu} + \partial_\nu \delta g_{\beta\mu} - \partial_\beta \delta g_{\mu\nu}) + \mathcal{O}(\delta g^2).$$

Using (5.9), we can write $\delta g^{\alpha\beta} = -\delta g_{\rho\sigma} g^{\rho\alpha} g^{\sigma\beta}$, and

$$\begin{aligned} \delta \Gamma^\alpha{}_{\mu\nu} &= -\frac{1}{2} \delta g_{\rho\sigma} g^{\rho\alpha} g^{\sigma\beta} (\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}) + \frac{1}{2} g^{\alpha\beta} (\partial_\mu \delta g_{\beta\nu} + \partial_\nu \delta g_{\beta\mu} - \partial_\beta \delta g_{\mu\nu}) \\ &= -\delta g_{\beta\sigma} g^{\alpha\beta} \Gamma^\sigma{}_{\mu\nu} + \frac{1}{2} g^{\alpha\beta} (\partial_\mu \delta g_{\beta\nu} + \partial_\nu \delta g_{\beta\mu} - \partial_\beta \delta g_{\mu\nu}), \end{aligned}$$

where we recombined the first parenthesis into a Christoffel symbol and then replaced the dummy index ρ with β . Before moving on, observe that the first term has a Christoffel symbol with a variation $\delta g_{\beta\sigma}$ but no derivatives. The second term has derivatives of a variation but no Christoffel symbols. Meanwhile, all terms have the same $g^{\alpha\beta}$ factor. If we combine all the terms, there should be some covariant derivative appearing.

The trick is to rewrite the partial derivatives back into the covariant derivative, and see what happens:

$$\begin{aligned}\partial_\mu \delta g_{\beta\nu} &= \nabla_\mu \delta g_{\beta\nu} + \Gamma^\sigma_{\mu\beta} \delta g_{\sigma\nu} + \Gamma^\sigma_{\mu\nu} \delta g_{\beta\sigma}, \\ \partial_\nu \delta g_{\beta\mu} &= \nabla_\nu \delta g_{\beta\mu} + \Gamma^\sigma_{\nu\beta} \delta g_{\sigma\mu} + \Gamma^\sigma_{\nu\mu} \delta g_{\beta\sigma}, \\ \partial_\beta \delta g_{\mu\nu} &= \nabla_\beta \delta g_{\mu\nu} + \Gamma^\sigma_{\beta\mu} \delta g_{\sigma\nu} + \Gamma^\sigma_{\beta\nu} \delta g_{\sigma\mu}. \\ \partial_\mu \delta g_{\beta\nu} + \partial_\nu \delta g_{\beta\mu} - \partial_\beta \delta g_{\mu\nu} &= \nabla_\mu \delta g_{\beta\nu} + \nabla_\nu \delta g_{\beta\mu} - \nabla_\beta \delta g_{\mu\nu} + 2\Gamma^\sigma_{\mu\nu} \delta g_{\beta\sigma}.\end{aligned}$$

Substituting this result into $\delta\Gamma^\alpha_{\mu\nu}$:

$$\delta\Gamma^\alpha_{\mu\nu} = -\delta g_{\beta\sigma} g^{\alpha\beta} \Gamma^\sigma_{\mu\nu} + \frac{1}{2} g^{\alpha\beta} (\nabla_\mu \delta g_{\beta\nu} + \nabla_\nu \delta g_{\beta\mu} - \nabla_\beta \delta g_{\mu\nu}) + g^{\alpha\beta} \Gamma^\sigma_{\mu\nu} \delta g_{\beta\sigma}.$$

The first and the last term cancel, so the variation $\delta\Gamma^\alpha_{\mu\nu}$ takes the same form as $\Gamma^\alpha_{\mu\nu}$, but with $\partial \rightarrow \nabla$ and $g_{\mu\nu} \rightarrow \delta g_{\mu\nu}$:

$$\boxed{\delta\Gamma^\alpha_{\mu\nu} = \frac{1}{2} g^{\alpha\beta} (\nabla_\mu \delta g_{\beta\nu} + \nabla_\nu \delta g_{\beta\mu} - \nabla_\beta \delta g_{\mu\nu}).} \quad (5.11)$$

The presence of covariant derivatives instead of partial derivatives means that the variation of the Christoffel symbol is actually a tensor, unlike the Christoffel symbol itself. Since $\delta\Gamma^\alpha_{\mu\nu}$ is a tensor, we are allowed to take its covariant derivative. This fact turns out to be useful when we vary the Riemann curvature tensor.

Variation of the Riemann Curvature Tensor $R^\alpha_{\beta\mu\nu}$. The coordinate representation of the Riemann curvature tensor is

$$R^\alpha_{\beta\mu\nu} = \partial_\mu \Gamma^\alpha_{\nu\beta} - \partial_\nu \Gamma^\alpha_{\mu\beta} + \Gamma^\alpha_{\mu\lambda} \Gamma^\lambda_{\beta\nu} - \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\beta\mu}.$$

Its variation in terms of $\delta\Gamma^\alpha_{\mu\nu}$ is

$$\begin{aligned}\delta R^\alpha_{\beta\mu\nu} &= \partial_\mu (\Gamma^\alpha_{\nu\beta} + \delta\Gamma^\alpha_{\nu\beta}) - \partial_\nu (\Gamma^\alpha_{\mu\beta} + \delta\Gamma^\alpha_{\mu\beta}) + (\Gamma^\alpha_{\mu\lambda} + \delta\Gamma^\alpha_{\mu\lambda})(\Gamma^\lambda_{\beta\nu} + \delta\Gamma^\lambda_{\beta\nu}) - (\Gamma^\alpha_{\nu\lambda} + \delta\Gamma^\alpha_{\nu\lambda})(\Gamma^\lambda_{\beta\mu} + \delta\Gamma^\lambda_{\beta\mu}) - R^\alpha_{\beta\mu\nu} \\ &= \partial_\mu \delta\Gamma^\alpha_{\nu\beta} - \partial_\nu \delta\Gamma^\alpha_{\mu\beta} + \Gamma^\alpha_{\mu\lambda} \delta\Gamma^\lambda_{\beta\nu} + \delta\Gamma^\alpha_{\mu\lambda} \Gamma^\lambda_{\beta\nu} - \Gamma^\alpha_{\nu\lambda} \delta\Gamma^\lambda_{\beta\mu} - \delta\Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\beta\mu} + \mathcal{O}(\delta\Gamma^2).\end{aligned}$$

Again, there are lots of Christoffel symbols with partial derivatives, so we hope to turn them into covariant derivatives. Fortunately, $\delta\Gamma^\alpha_{\mu\nu}$ is a tensor. We can take its covariant derivative: following the rule (4.5),

$$\begin{aligned}\nabla_\mu \delta\Gamma^\alpha_{\nu\beta} &= \partial_\mu \delta\Gamma^\alpha_{\nu\beta} + \Gamma^\alpha_{\mu\lambda} \delta\Gamma^\lambda_{\nu\beta} - \Gamma^\lambda_{\mu\nu} \delta\Gamma^\alpha_{\lambda\beta} - \Gamma^\lambda_{\mu\beta} \delta\Gamma^\alpha_{\nu\lambda}, \\ \nabla_\nu \delta\Gamma^\alpha_{\mu\beta} &= \partial_\nu \delta\Gamma^\alpha_{\mu\beta} + \Gamma^\alpha_{\nu\lambda} \delta\Gamma^\lambda_{\mu\beta} - \Gamma^\lambda_{\nu\mu} \delta\Gamma^\alpha_{\lambda\beta} - \Gamma^\lambda_{\nu\beta} \delta\Gamma^\alpha_{\mu\lambda}.\end{aligned}$$

Substituting the partial derivatives in $\delta R^\alpha_{\beta\mu\nu}$ with these equations,

$$\begin{aligned}\delta R^\alpha_{\beta\mu\nu} &= (\nabla_\mu \delta\Gamma^\alpha_{\nu\beta} - \Gamma^\alpha_{\mu\lambda} \delta\Gamma^\lambda_{\nu\beta} + \Gamma^\lambda_{\mu\nu} \delta\Gamma^\alpha_{\lambda\beta} + \Gamma^\lambda_{\mu\beta} \delta\Gamma^\alpha_{\nu\lambda}) - (\nabla_\nu \delta\Gamma^\alpha_{\mu\beta} - \Gamma^\alpha_{\nu\lambda} \delta\Gamma^\lambda_{\mu\beta} + \Gamma^\lambda_{\nu\mu} \delta\Gamma^\alpha_{\lambda\beta} + \Gamma^\lambda_{\nu\beta} \delta\Gamma^\alpha_{\mu\lambda}) \\ &\quad + \Gamma^\alpha_{\mu\lambda} \delta\Gamma^\lambda_{\beta\nu} + \delta\Gamma^\alpha_{\mu\lambda} \Gamma^\lambda_{\beta\nu} - \Gamma^\alpha_{\nu\lambda} \delta\Gamma^\lambda_{\beta\mu} - \delta\Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\beta\mu}.\end{aligned}$$

We will find that all terms cancel except the covariant derivatives, so the variation of the Riemann curvature tensor also takes a simple form:

$$\boxed{\delta R^\alpha_{\beta\mu\nu} = \nabla_\mu \delta\Gamma^\alpha_{\nu\beta} - \nabla_\nu \delta\Gamma^\alpha_{\mu\beta}.} \quad (5.12)$$

One can express this in terms of $\delta g_{\mu\nu}$, but that is not necessary for now.

Variation of the Ricci Tensor $R_{\mu\nu}$. There is nothing special about the variation of the Ricci tensor. It is just the contraction

$$\delta R_{\beta\nu} = \delta R^\alpha_{\beta\alpha\nu} = \nabla_\alpha \delta\Gamma^\alpha_{\nu\beta} - \nabla_\nu \delta\Gamma^\alpha_{\alpha\beta}, \quad (5.13)$$

which is known as the **Palatini identity**. What we are interested is $g^{\mu\nu} \delta R_{\mu\nu}$ because this appears in δS . It turns out that this quantity can be written as a covariant divergence:

$$g^{\mu\nu} \delta R_{\mu\nu} = g^{\mu\nu} \nabla_\alpha \delta\Gamma^\alpha_{\mu\nu} - g^{\mu\nu} \nabla_\nu \delta\Gamma^\beta_{\mu\beta} = \nabla_\alpha (g^{\mu\nu} \delta\Gamma^\alpha_{\mu\nu} - g^{\alpha\mu} \delta\Gamma^\nu_{\mu\nu}),$$

where we used the metric compatibility $\nabla_\alpha g_{\mu\nu} = 0$ to move the metric around the derivatives. In fact, this divergence is all we need, but we can proceed further to express $g^{\mu\nu}\delta R_{\mu\nu}$ in terms of $\delta g_{\mu\nu}$ in a simple form. Using (5.11),

$$g^{\mu\nu}\delta R_{\mu\nu} = \nabla_\alpha \left[\frac{1}{2} g^{\mu\nu} g^{\alpha\beta} (\nabla_\mu \delta g_{\beta\nu} + \nabla_\nu \delta g_{\beta\mu} - \nabla_\beta \delta g_{\mu\nu}) - \frac{1}{2} g^{\alpha\mu} g^{\nu\beta} (\nabla_\mu \delta g_{\beta\nu} + \nabla_\nu \delta g_{\beta\mu} - \nabla_\beta \delta g_{\mu\nu}) \right].$$

The product of two inverse metrics and a metric variation suggests us to turn $\delta g_{\mu\nu}$ into $\delta g^{\mu\nu}$ using (5.9),

$$g^{\mu\nu}\delta R_{\mu\nu} = \frac{1}{2} \nabla_\alpha [(-\nabla_\mu \delta g^{\mu\alpha} - \nabla_\nu \delta g^{\nu\alpha} - g^{\mu\nu} g^{\alpha\beta} \nabla_\beta \delta g_{\mu\nu}) - (g^{\alpha\mu} g^{\nu\beta} \nabla_\mu \delta g_{\beta\nu} - \nabla_\nu \delta g^{\nu\alpha} + \nabla_\beta \delta g^{\alpha\beta})].$$

Observe that the third and fourth terms are the same, the first and last terms are the same, and the second and fifth terms cancel. This leads to

$$g^{\mu\nu}\delta R_{\mu\nu} = \nabla_\alpha (g^{\alpha\beta} g^{\mu\nu} \nabla_\beta \delta g_{\mu\nu} - \nabla_\mu \delta g^{\mu\alpha}). \quad (5.14)$$

Again, this is a covariant divergence of some vector related to the metric variation. When we integrate it,

$$\int d^n x \sqrt{-g} \nabla_\alpha (g^{\alpha\beta} g^{\mu\nu} \nabla_\beta \delta g_{\mu\nu} - \nabla_\mu \delta g^{\mu\alpha}) = (\text{some surface integral evaluated at infinity}),$$

This volume integral is turned into a surface integral by Stokes's theorem (4.13). The integral is over whole space, so the boundary of this volume is at infinity. We always set the variation to be zero at infinity in classical field theories. (This assumption is subtle, as it may affect some of the advanced physics, but we do not worry about them in this subject.) Because now $\delta g_{\mu\nu} = 0$ at infinity, the surface integral vanishes. Everything we calculated above contributes *nothing*.

Variation of the Determinant $\sqrt{-g}$. Finally, we come to the variation of the determinant, $\delta\sqrt{-g}$. To do this, we need to use the identity

$$\ln(\det M) = \text{Tr}(\ln M),$$

where the natural log of a matrix M is defined such that $e^{\ln M} = M$. The variation of this identity gives

$$\frac{1}{\det M} \delta(\det M) = \text{Tr}(M^{-1} \delta M).$$

This is known as **Jacobi's formula** in matrix calculus. In general, M^{-1} may not commute with δM . Fortunately, the trace is cyclic, $\text{Tr}(AB) = \text{Tr}(BA)$, so the order of M^{-1} and δM does not matter in the trace. Set M to be the metric $g_{\mu\nu}$, and $\det M = g$. The variation of g is

$$\delta g = g \text{Tr}(g^{\mu\nu} \delta g_{\nu\alpha}) = g(g^{\mu\nu} \delta g_{\nu\mu}) = -g(g_{\mu\nu} \delta g^{\mu\nu}).$$

The second equality is the definition of the trace. In the last equality, we used (5.9),

$$g^{\mu\nu} \delta g_{\mu\nu} = -g^{\mu\nu} g_{\mu\alpha} g_{\nu\beta} \delta g^{\alpha\beta} = -\delta^\nu_\alpha g_{\nu\beta} \delta g^{\alpha\beta} = -g_{\alpha\beta} \delta g^{\alpha\beta}.$$

With δg , the variation of $\sqrt{-g}$ is straightforward,

$$\delta\sqrt{-g} = -\frac{1}{2\sqrt{-g}} \delta g = \frac{g}{2\sqrt{-g}} g_{\mu\nu} \delta g^{\mu\nu} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}.$$

Back into the variation of the action in (5.10), with the first term not contributing,

$$\delta S = \int d^n x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} + \int d^n x \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + \int d^n x R \delta\sqrt{-g} = \int d^n x \sqrt{-g} \left[R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right] \delta g^{\mu\nu}.$$

Now apply Hamilton's principle: the physical fields $g^{\mu\nu}$ satisfies $\delta S = 0$ for any small variation $\delta g^{\mu\nu}$. This means the integrand in δS must be zero, or

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \frac{1}{\sqrt{-g}} \frac{\delta \mathcal{L}}{\delta g^{\mu\nu}} = 0.$$

This is Einstein's equation in vacuum, derived from extremizing the Einstein-Hilbert action.

5.3.3 Einstein's Equation with Matter

Einstein's equation from the Einstein-Hilbert action alone are “free fields” because there is no matter. To recover the full Einstein's equation with a stress-energy tensor, we need to include an action for matter. Consider the action

$$S = \frac{1}{16\pi} S_H + S_M,$$

where S_H is the Einstein-Hilbert action and S_M is the matter action. In the last section, extremizing the action gives the result

$$\frac{1}{\sqrt{-g}} \frac{\delta \mathcal{L}_H}{\delta g_{\mu\nu}} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0.$$

Following this method,

$$0 = \frac{1}{\sqrt{-g}} \frac{\delta \mathcal{L}}{\delta g^{\mu\nu}} = \frac{1}{16\pi} \frac{1}{\sqrt{-g}} \frac{\delta \mathcal{L}_H}{\delta g^{\mu\nu}} + \frac{1}{\sqrt{-g}} \frac{\delta \mathcal{L}_M}{\delta g^{\mu\nu}} = \frac{1}{16\pi} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) + \frac{1}{\sqrt{-g}} \frac{\delta \mathcal{L}_M}{\delta g^{\mu\nu}}.$$

Define the stress-energy tensor to be

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta \mathcal{L}_M}{\delta g^{\mu\nu}}. \quad (5.15)$$

Extremizing the action with respect to $\delta g^{\mu\nu}$ recovers the full Einstein's equation:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi T_{\mu\nu}.$$

Example 5.1. $T_{\mu\nu}$ for a scalar field

In special relativity, the Lagrangian of a scalar field ϕ is given by

$$\mathcal{L}_\phi = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi).$$

In general relativity, this becomes

$$\mathcal{L}_\phi = \sqrt{-g} \left[-\frac{1}{2} \nabla_\mu \phi \nabla^\mu \phi - V(\phi) \right], \quad S = \int d^n x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) \right].$$

Varying the action with respect to $\delta g^{\mu\nu}$ (not ϕ),

$$\begin{aligned} \delta S &= \int d^n x \delta \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) \right] + \int d^n x \sqrt{-g} \left[-\frac{1}{2} \delta g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi \right] \\ &= \int d^n x \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \right) \left[-\frac{1}{2} g^{\alpha\beta} \nabla_\alpha \phi \nabla_\beta \phi - V(\phi) \right] + \int d^n x \sqrt{-g} \left[-\frac{1}{2} \delta g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi \right] \\ &= \int d^n x \sqrt{-g} \delta g^{\mu\nu} \left[\frac{1}{4} g_{\mu\nu} g^{\alpha\beta} \nabla_\alpha \phi \nabla_\beta \phi + \frac{1}{2} g_{\mu\nu} V(\phi) - \frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi \right]. \end{aligned}$$

According to (5.15), the stress-energy tensor for the scalar field is

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta \mathcal{L}_\phi}{\delta g^{\mu\nu}} = \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \nabla_\alpha \phi \nabla^\alpha \phi - g_{\mu\nu} V(\phi).$$

This is consistent to the one in special relativity (2.20), where $\nabla_\mu \rightarrow \partial_\mu$ and $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$.

Example 5.2. $T_{\mu\nu}$ for electromagnetic fields

In special relativity, the Lagrangian of electromagnetism is (2.14),

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + A_\mu J^\mu.$$

In general relativity (assuming no charge density and current, $J^\mu = 0$), this Lagrangian becomes

$$\mathcal{L}_{\text{EM}} = \sqrt{-g} \left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \right), \quad S = \int d^n x \sqrt{-g} \left(-\frac{1}{4}g^{\alpha\mu}g^{\beta\nu}F_{\mu\nu}F_{\alpha\beta} \right).$$

Varying the action with respect to $\delta g^{\mu\nu}$,

$$\delta S = \int d^n x \delta \sqrt{-g} \left(-\frac{1}{4}g^{\alpha\mu}g^{\beta\nu}F_{\mu\nu}F_{\alpha\beta} \right) + \int d^n x \sqrt{-g} \left(-\frac{1}{4}\delta g^{\alpha\mu}g^{\beta\nu}F_{\mu\nu}F_{\alpha\beta} - \frac{1}{4}g^{\alpha\mu}\delta g^{\beta\nu}F_{\mu\nu}F_{\alpha\beta} \right).$$

Let's look at the second integral. The two terms in the integrand are in fact the same. We can show this by switching the dummy indices $\alpha \leftrightarrow \beta$ and $\mu \leftrightarrow \nu$,

$$\delta g^{\alpha\mu}g^{\beta\nu}F_{\mu\nu}F_{\alpha\beta} = \delta g^{\beta\nu}g^{\alpha\mu}F_{\nu\mu}F_{\beta\alpha} = g^{\alpha\mu}\delta g^{\beta\nu}(-F_{\mu\nu})(-F_{\alpha\beta}) = g^{\alpha\mu}\delta g^{\beta\nu}F_{\mu\nu}F_{\alpha\beta}.$$

So the action variation simplifies to

$$\delta S = \int d^n x \left(-\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu} \right) \left(-\frac{1}{4}g^{\alpha\rho}g^{\beta\sigma}F_{\rho\sigma}F_{\alpha\beta} \right) + \int d^n x \sqrt{-g} \left(-\frac{1}{2}\delta g^{\alpha\mu}g^{\beta\nu}F_{\mu\nu}F_{\alpha\beta} \right).$$

To factor out $\delta g^{\mu\nu}$, we relabel $\alpha \leftrightarrow \nu$ for the second term,

$$\begin{aligned} \delta S &= \int d^n x \sqrt{-g} \delta g^{\mu\nu} \left(\frac{1}{8}g^{\alpha\rho}g^{\beta\sigma}F_{\rho\sigma}F_{\alpha\beta} \right) - \int d^n x \sqrt{-g} \left(\frac{1}{2}\delta g^{\nu\mu}g^{\beta\alpha}F_{\mu\alpha}F_{\nu\beta} \right) \\ &= \int d^n x \sqrt{-g} \delta g^{\mu\nu} \left(\frac{1}{8}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} - \frac{1}{2}F_{\mu\alpha}F_{\nu}{}^{\alpha} \right). \end{aligned}$$

Using (5.15), the stress-energy tensor for electromagnetic fields is

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta \mathcal{L}_{\text{EM}}}{\delta g^{\mu\nu}} = F_{\mu\alpha}F_{\nu}{}^{\alpha} - \frac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}.$$

This is consistent to (2.21). We already discussed its implication there.

5.4 The Cosmological Constant

In non-gravitational physics, we talk only about changes in energy. For example, the trajectory of a particle in a potential $V(x)$ is exactly the same as the one in a potential $V(x) + V_0$ for any constant V_0 . Oppositely, gravity cares about everything in the stress-energy tensor. Adding anything to a stress-energy tensor causes the curvature to change. It is possible that there exists some **vacuum energy**, ρ_{vac} , an energy density associated with empty space.

Evidently, the vacuum energy should be a constant because vacuum is homogeneous and isotropic. The stress-energy tensor associated with vacuum energy is

$$T_{\mu\nu}^{\text{vac}} = -\rho_{\text{vac}}g_{\mu\nu}.$$

(Note that there is a minus sign. It has to do with the expansion of the universe, in contrast to contraction, which gravity usually does.) Comparing this stress-energy tensor to the perfect-fluid one, $T_{\mu\nu} = (\rho + P)u_\mu u_\nu + Pg_{\mu\nu}$, we find that

$$P_{\text{vac}} = -\rho_{\text{vac}}.$$

Usually, the vacuum stress-energy tensor is separated from matter stress-energy tensor. If we add the vacuum piece to Einstein's equation (5.5),

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi(T_{\mu\nu} - \rho_{\text{vac}}g_{\mu\nu}),$$

the new Einstein's equation can be written as

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}. \quad (5.16)$$

The new term $\Lambda = 8\pi\rho_{\text{vac}}$ is called the **cosmological constant**. The cosmological constant is believed to be a fundamental constant of nature, so it cannot be calculated from first principles. In fact, the vacuum energy calculated from quantum field theory is over 120 orders of magnitude larger than the observed one. This problem is known as the **cosmological constant problem**. The observed value of the cosmological constant is about 10^{-52} m^{-2} , so it is often neglected in most of general relativity. In cosmology, however, it plays a crucial role in the accelerating expansion of the universe, which occurs in the very-early universe, now, and in the future.

6 WEAK FIELDS AND GRAVITATIONAL RADIATION

Before solving the full nonlinear Einstein's equation, it is better to discuss Einstein's equation in the weak-field limit in terms of linearized gravity. The weak-field limit is less restrictive than the Newtonian limit. In the weak-field limit, gravitational fields can vary with time, and test particles can be relativistic. These two situations correspond to two important topics in this chapter: gravitational waves and deflection of light in weak fields.

6.1 Linearized Gravity

In linearized gravity, we decompose the metric $g_{\mu\nu}$ into the Minkowski metric $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ plus a small perturbation $h_{\mu\nu}$,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad g^{\mu\nu} \simeq \eta^{\mu\nu} - h^{\mu\nu}, \quad |h_{\mu\nu}| \ll 1.$$

Again, we keep only first-order terms in $h_{\mu\nu}$ and $h^{\mu\nu}$. As a consequence, indices in first-order terms are raised and lowered using the Minkowski metric. (Sometimes it is necessary to choose other metric instead of the Minkowski metric as the background. For example, in cosmological perturbation theory, the background metric is often taken to be the FRW metric discussed in Chapter 9.)

6.1.1 Weak-Field Einstein's Equation

We need to find Einstein's equation for linearized gravity before solving it. In particular, we need to calculate the Einstein tensor to first order in perturbation. The coordinate representations of the Christoffel symbols, Riemann curvature tensor, Ricci tensor, Ricci scalar, and Einstein tensor are

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2}g^{\alpha\beta}(\partial_\mu g_{\beta\nu} + \partial_\nu g_{\beta\mu} + \partial_\beta g_{\mu\nu}), \quad R^\alpha_{\beta\mu\nu} = \partial_\mu \Gamma^\alpha_{\nu\beta} - \partial_\nu \Gamma^\alpha_{\mu\beta} + \Gamma^\alpha_{\mu\lambda}\Gamma^\lambda_{\beta\nu} - \Gamma^\alpha_{\nu\lambda}\Gamma^\lambda_{\beta\mu},$$

$$R_{\mu\nu} = R^\alpha_{\mu\alpha\nu}, \quad R = R^\mu_{\mu}, \quad G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}.$$

We linearize them to first-order perturbation one by one.

- Christoffel symbols: the Minkowski metric are constants, so the derivative of $g_{\mu\nu}$ is the derivative of the perturbation $h_{\mu\nu}$. This means the terms in the parenthesis of the Christoffel symbols are first-order terms, so their indices are raised by $\eta^{\alpha\beta}$:

$$\Gamma^\alpha_{\mu\nu} \simeq \frac{1}{2}\eta^{\alpha\beta}(\partial_\mu h_{\beta\nu} + \partial_\nu h_{\beta\mu} - \partial_\beta h_{\mu\nu}). \quad (6.1)$$

- Riemann curvature tensor: the Christoffel symbols are all first-order terms, so the $(\Gamma)^2$ terms in the Riemann curvature tensor are second order. We are left with

$$\begin{aligned} R^\alpha_{\beta\mu\nu} &\simeq \partial_\mu \Gamma^\alpha_{\nu\beta} - \partial_\nu \Gamma^\alpha_{\mu\beta} = \frac{1}{2}\eta^{\alpha\gamma}(\partial_\mu(\partial_\beta h_{\gamma\nu} + \partial_\nu h_{\gamma\beta} - \partial_\gamma h_{\beta\nu}) - \partial_\nu(\partial_\beta h_{\gamma\mu} + \partial_\mu h_{\gamma\beta} - \partial_\gamma h_{\beta\mu})) \\ &= \frac{1}{2}\eta^{\alpha\gamma}(\partial_\mu\partial_\beta h_{\gamma\nu} - \partial_\mu\partial_\gamma h_{\beta\nu} - \partial_\nu\partial_\beta h_{\gamma\mu} + \partial_\nu\partial_\gamma h_{\beta\mu}). \end{aligned} \quad (6.2)$$

- Ricci tensor: contracting α and μ in the Riemann curvature tensor to get the Ricci tensor,

$$\begin{aligned} R_{\beta\nu} &= R^\alpha_{\beta\alpha\nu} \simeq \frac{1}{2}\eta^{\alpha\gamma}(\partial_\alpha\partial_\beta h_{\gamma\nu} - \partial_\alpha\partial_\gamma h_{\beta\nu} - \partial_\nu\partial_\beta h_{\gamma\alpha} + \partial_\nu\partial_\gamma h_{\beta\alpha}) \\ &= \frac{1}{2}(\partial_\alpha\partial_\beta h^\alpha_{\nu} - \partial^\gamma\partial_\gamma h_{\beta\nu} - \partial_\nu\partial_\beta h^\alpha_{\alpha} + \partial_\nu\partial_\gamma h^\gamma_{\beta}). \end{aligned}$$

Define $h \equiv h^\mu_{\mu}$ to be the trace of the metric perturbation and $\square \equiv \partial_\mu\partial^\mu$ to be the **d'Alembertian** (or wave operator). The

Ricci tensor can then be written as

$$R_{\mu\nu} \simeq \frac{1}{2}(\partial_\mu \partial_\alpha h^\alpha{}_\nu + \partial_\nu \partial_\alpha h^\alpha{}_\mu - \partial_\mu \partial_\nu h - \square h_{\mu\nu}). \quad (6.3)$$

- Ricci scalar: the Ricci tensor is also first order in perturbation, so raising its index by $\eta^{\mu\nu}$ and contracting gives the Ricci scalar,

$$\begin{aligned} R &= g^{\mu\nu} R_{\mu\nu} \simeq \frac{1}{2} \eta^{\mu\nu} (\partial_\mu \partial_\alpha h^\alpha{}_\nu + \partial_\nu \partial_\alpha h^\alpha{}_\mu - \partial_\mu \partial_\nu h - \square h_{\mu\nu}) \\ &= \frac{1}{2} (\partial_\mu \partial_\alpha h^{\alpha\mu} + \partial_\nu \partial_\alpha h^{\alpha\nu} - \partial^\nu \partial_\nu h - \square h^\nu{}_\nu) \\ &= \partial_\mu \partial_\nu h^{\mu\nu} - \square h. \end{aligned} \quad (6.4)$$

- Finally, the Einstein tensor is

$$G_{\mu\nu} \simeq R_{\mu\nu} - \frac{1}{2} R \eta_{\mu\nu} \simeq \frac{1}{2} (\partial_\mu \partial_\alpha h^\alpha{}_\nu + \partial_\nu \partial_\alpha h^\alpha{}_\mu - \partial_\mu \partial_\nu h - \square h_{\mu\nu} - \eta_{\mu\nu} \partial_\alpha \partial_\beta h^{\alpha\beta} + \eta_{\mu\nu} \square h). \quad (6.5)$$

Even expressed only to first-order perturbation, this Einstein tensor is somewhat complicated. There are several steps to simplify it. First, define the **trace-reverse** perturbation tensor $\bar{h}_{\mu\nu}$,

$$\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \quad \Longleftrightarrow \quad h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} \bar{h}. \quad (6.6)$$

It is called “trace-reverse” because the trace of $\bar{h}_{\mu\nu}$ is $\bar{h} \equiv \bar{h}^\mu{}_\mu = -h$. Expressing the Einstein tensor in terms of $\bar{h}_{\mu\nu}$ can get rid of the trace terms:

$$\begin{aligned} G_{\mu\nu} &= \frac{1}{2} (\partial_\mu \partial_\alpha \eta^{\alpha\beta} h_{\beta\nu} + \partial_\nu \partial_\alpha \eta^{\alpha\beta} h_{\beta\mu} - \partial_\mu \partial_\nu h - \square h_{\mu\nu} - \eta_{\mu\nu} \partial_\alpha \partial_\beta h^{\alpha\beta} + \eta_{\mu\nu} \square h) \\ &= \frac{1}{2} \left[\partial_\mu \partial_\alpha \eta^{\alpha\beta} \left(\bar{h}_{\beta\nu} - \frac{1}{2} \eta_{\beta\nu} \bar{h} \right) + \partial_\nu \partial_\alpha \eta^{\alpha\beta} \left(\bar{h}_{\beta\mu} - \frac{1}{2} \eta_{\beta\mu} \bar{h} \right) + \partial_\mu \partial_\nu \bar{h} - \square \left(\bar{h}_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} \bar{h} \right) - \eta_{\mu\nu} \partial_\alpha \partial_\beta \left(\bar{h}^{\alpha\beta} - \frac{1}{2} \eta^{\alpha\beta} \bar{h} \right) - \eta_{\mu\nu} \square \bar{h} \right] \\ &= \frac{1}{2} \left(\partial_\mu \partial_\alpha \bar{h}^\alpha{}_\nu - \frac{1}{2} \partial_\mu \partial_\alpha \delta^\alpha{}_\nu \bar{h} + \partial_\nu \partial_\alpha \bar{h}^\alpha{}_\mu - \frac{1}{2} \partial_\nu \partial_\alpha \delta^\alpha{}_\mu \bar{h} + \partial_\mu \partial_\nu \bar{h} - \square \bar{h}_{\mu\nu} + \frac{1}{2} \cancel{\eta_{\mu\nu} \square \bar{h}} - \eta_{\mu\nu} \partial_\alpha \partial_\beta \bar{h}^{\alpha\beta} + \frac{1}{2} \cancel{\eta_{\mu\nu} \square \bar{h}} - \cancel{\eta_{\mu\nu} \square \bar{h}} \right) \\ &= \frac{1}{2} \left(\partial_\mu \partial_\alpha \bar{h}^\alpha{}_\nu - \cancel{\frac{1}{2} \partial_\mu \partial_\nu \bar{h}} + \partial_\nu \partial_\alpha \bar{h}^\alpha{}_\mu - \cancel{\frac{1}{2} \partial_\nu \partial_\mu \bar{h}} + \partial_\mu \partial_\nu \bar{h} - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \partial_\alpha \partial_\beta \bar{h}^{\alpha\beta} \right) \\ &= \frac{1}{2} (\partial_\mu \partial_\alpha \bar{h}^\alpha{}_\nu + \partial_\nu \partial_\alpha \bar{h}^\alpha{}_\mu - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \partial_\alpha \partial_\beta \bar{h}^{\alpha\beta}). \end{aligned}$$

Setting this to the stress-energy tensor $T_{\mu\nu}$, we obtain Einstein’s equation for linearized gravity:

$$\frac{1}{2} (\partial_\mu \partial_\alpha \bar{h}^\alpha{}_\nu + \partial_\nu \partial_\alpha \bar{h}^\alpha{}_\mu - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \partial_\alpha \partial_\beta \bar{h}^{\alpha\beta}) = 8\pi T_{\mu\nu}.$$

This is still hard to solve given the number of partial derivatives. Notice that all terms but $\square \bar{h}_{\mu\nu}$ on the LHS involves the divergence of $\bar{h}_{\mu\nu}$ in some way. This equation will be greatly simplified if we set the divergence

$$\partial_\mu \bar{h}^{\mu\nu} = 0. \quad (6.7)$$

With this condition, we arrive at the weak-field Einstein’s equation:

$$\boxed{\square \bar{h}_{\mu\nu} = -16\pi T_{\mu\nu}.} \quad (6.8)$$

The minus sign on the RHS is important: it means the weak-field Einstein’s equation is a set of *wave equations*. The wave equation is inhomogeneous if there is matter ($T_{\mu\nu} \neq 0$), and is homogeneous in vacuum ($T_{\mu\nu} = 0$). There is no covariant derivative/Christoffel symbols, and the equation is linear in $\bar{h}_{\mu\nu}$, so it describes **linearized gravity**. Now the question is: do we always have the freedom to set $\partial_\mu \bar{h}^{\mu\nu} = 0$? The answer turns out to be yes.

6.1.2 Gauge Transformations

In electromagnetism, the physical quantities are components of $F_{\mu\nu}$ while the fields are components of the 4-potential A_μ . Under a gauge transformation, $A_\mu \rightarrow A_\mu + \partial_\mu \lambda$ for arbitrary function λ , the field tensor $F_{\mu\nu}$ is invariant. In general relativity, not all degrees of freedom in the metric tensor are physical. There are some **gauge freedoms** that can simplify the problem. Below is a

correspondence between gauge transformations in electromagnetism and linearized gravity in general relativity:

Electromagnetism

- Dynamical fields: the 4-potential A_μ .
- Physical Quantities: the field tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.
- Source of fields: the 4-current J^μ .
- Equations of motion: Maxwell's equations $\partial_\mu F^{\mu\nu} = J^\nu$.
- Gauge freedom: gauge transformations $A_\mu \rightarrow A_\mu + \partial_\mu \lambda$.

Linearized Gravity

- Dynamical fields: the metric perturbation $h_{\mu\nu}$.
- Physical Quantities: the curvature $G_{\mu\nu}$.
- Source of fields: the stress-energy tensor $T_{\mu\nu}$.
- Equations of motion: Einstein's equations $G_{\mu\nu} = 8\pi T_{\mu\nu}$.
- Gauge freedom: coordinate transformations $h_{\mu\nu} \rightarrow h'_{\mu\nu}$.

Gauge transformations in general relativity are just coordinate transformations. A wise choice of coordinates makes equations a lot easier. In linearized gravity, consider a small change of coordinates by a vector ξ^μ ,

$$x'^\mu = x^\mu + \xi^\mu(x^\nu),$$

where $|\xi^\mu| \ll 1$ and all its derivatives are small. Then the coordinate transformation matrix has elements

$$\frac{\partial x'^\mu}{\partial x^\nu} = \delta^\mu_\nu + \partial_\nu \xi^\mu, \quad \frac{\partial x^\mu}{\partial x'^\nu} \simeq \delta^\mu_\nu - \partial_\nu \xi^\mu.$$

(We write x'^μ instead of $x^{\mu'}$ for index correspondence, but they mean the same.) The indices of ξ^μ are raised and lowered by the Minkowski metric because they are small. Using the transformation law (3.2), the transformed metric tensor is

$$\begin{aligned} g'_{\mu\nu} &= \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta} = g_{\alpha\beta} (\delta^\alpha_\mu - \partial_\mu \xi^\alpha) (\delta^\beta_\nu - \partial_\nu \xi^\beta) \\ &= (\eta_{\alpha\beta} + h_{\alpha\beta}) (\delta^\alpha_\mu \delta^\beta_\nu - \delta^\alpha_\mu \partial_\nu \xi^\beta - \delta^\beta_\nu \partial_\mu \xi^\alpha) \\ &\simeq \eta_{\mu\nu} - \eta_{\beta\mu} \partial_\nu \xi^\beta - \eta_{\alpha\nu} \partial_\mu \xi^\alpha + h_{\mu\nu} \\ &= \eta_{\mu\nu} + (h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu). \end{aligned}$$

This means the metric perturbation $h_{\mu\nu}$ transforms as

$$h'_{\mu\nu} = h_{\mu\nu} - 2\partial_{(\mu} \xi_{\nu)}. \quad (6.9)$$

This new perturbation is still a valid perturbation, because we assume ξ^μ and its partial derivatives to be small. Now we want to show that we can always do a coordinate transformation such that $\partial_\mu \bar{h}'^{\mu\nu} = 0$. This gauge is often called the **Lorenz gauge** (not Lorentz), analogous to the Lorenz gauge in electromagnetism $\partial_\mu A^\mu = 0$. To do this, suppose $\bar{h}_{\mu\nu}$ is not divergence-free. First find $\bar{h}'_{\mu\nu}$ in terms of $\bar{h}_{\mu\nu}$ and ξ^μ :

$$\bar{h}'_{\mu\nu} = h'_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h' = h_{\mu\nu} - 2\partial_{(\mu} \xi_{\nu)} - \frac{1}{2} \eta_{\mu\nu} (h - 2\partial_\alpha \xi^\alpha) = \bar{h}_{\mu\nu} - 2\partial_{(\mu} \xi_{\nu)} + \eta_{\mu\nu} \partial_\alpha \xi^\alpha.$$

Its divergence is

$$\partial_\mu \bar{h}'^{\mu\nu} = \partial_\mu \bar{h}^{\mu\nu} - \partial_\mu \partial^\mu \xi^\nu - \cancel{\partial_\mu \partial^\nu \xi^\mu} + \cancel{\eta^{\mu\nu} \partial_\mu \partial_\alpha \xi^\alpha} = \partial_\mu \bar{h}^{\mu\nu} - \square \xi^\nu.$$

We want $\partial_\mu \bar{h}'^{\mu\nu}$ to vanish, so the transformation vector ξ^ν is determined by

$$\square \xi^\nu = \partial_\mu \bar{h}^{\mu\nu}, \quad (6.10)$$

which is just an inhomogeneous wave equation and it always has a solution for a well-behaved $\bar{h}^{\mu\nu}$. This is all we need: we don't care about the exact form of ξ^μ , all we need to know is that they exist. In conclusion, it is always possible to find a coordinate transformation such that $\partial_\mu \bar{h}'^{\mu\nu} = 0$ and meanwhile sets $\bar{h}'^{\mu\nu}$ to be small (so that it is a perturbation).

There is an important property of the gauge transformation (6.9) is that the linearized Riemann curvature tensor (6.2) is invariant

such a transformation. It can be proved by explicit calculation:

$$\begin{aligned}
R'^{\alpha}_{\beta\mu\nu} &= \frac{1}{2}\eta^{\alpha\gamma}(\partial_{\mu}\partial_{\beta}h'_{\gamma\nu} - \partial_{\mu}\partial_{\gamma}h'_{\beta\nu} - \partial_{\nu}\partial_{\beta}h'_{\gamma\mu} + \partial_{\nu}\partial_{\gamma}h'_{\beta\mu}) \\
&= \frac{1}{2}\eta^{\alpha\gamma}[\partial_{\mu}\partial_{\beta}(h_{\gamma\nu} - 2\partial_{(\gamma}\xi_{\nu)}) - \partial_{\mu}\partial_{\gamma}(h_{\beta\nu} - \partial_{(\beta}\xi_{\nu)}) - \partial_{\nu}\partial_{\beta}(h_{\gamma\mu} - \partial_{(\gamma}\xi_{\mu)}) + \partial_{\nu}\partial_{\gamma}(h_{\beta\mu} - \partial_{(\beta}\xi_{\mu)})] \\
&= R^{\alpha}_{\beta\mu\nu} + \frac{1}{2}\eta^{\alpha\gamma}[-2\partial_{\mu}\partial_{\beta}\partial_{(\gamma}\xi_{\nu)} + 2\partial_{\mu}\partial_{\gamma}\partial_{(\beta}\xi_{\nu)} + 2\partial_{\nu}\partial_{\beta}\partial_{(\gamma}\xi_{\mu)} - 2\partial_{\nu}\partial_{\gamma}\partial_{(\beta}\xi_{\mu)}] \\
&= R^{\alpha}_{\beta\mu\nu} + \frac{1}{2}\eta^{\alpha\gamma}(-\partial_{\mu}\partial_{\beta}\partial_{\gamma}\xi_{\nu} - \partial_{\mu}\partial_{\beta}\partial_{\nu}\xi_{\gamma} + \partial_{\mu}\partial_{\gamma}\partial_{\beta}\xi_{\nu} + \partial_{\mu}\partial_{\gamma}\partial_{\nu}\xi_{\beta} + \partial_{\nu}\partial_{\beta}\partial_{\gamma}\xi_{\mu} + \partial_{\nu}\partial_{\beta}\partial_{\mu}\xi_{\gamma} - \partial_{\nu}\partial_{\gamma}\partial_{\beta}\xi_{\mu} - \partial_{\nu}\partial_{\gamma}\partial_{\mu}\xi_{\beta}) \\
&= R^{\alpha}_{\beta\mu\nu}.
\end{aligned}$$

If the Riemann curvature tensor is invariant under this gauge transformation, then so is the Ricci tensor, Ricci scalar, and Einstein tensor. We can calculate them in a convenient gauge and use the same result in other gauges.

6.2 The Newtonian Limit

6.2.1 The Newtonian Metric

We are in the place of deriving the metric tensor for the Newtonian limit. In the weak-field limit, assume that both the field and the speed of particles are small: $|h_{\mu\nu}| \ll 1$ and $v \ll 1$. As a result, the dominant term in the stress-energy tensor is the energy density, $|T_{tt}| \gg |T_{ti}| \gg |T_{ij}|$. The weak-field Einstein's equation is (6.8).

$$\left(-\frac{\partial^2}{\partial t^2} + \nabla^2\right)\bar{h}_{tt} = -16\pi\rho.$$

For a particle moving with speed $v = |d\mathbf{x}/dt|$ in the gravitational field, the partial time-derivative can be written as

$$\frac{\partial}{\partial t} = \frac{dx}{dt}\frac{\partial}{\partial x} + \frac{dy}{dt}\frac{\partial}{\partial y} + \frac{dz}{dt}\frac{\partial}{\partial z} = \mathbf{v} \cdot \nabla \ll \nabla$$

because $v \ll 1$. Hence the weak-field Einstein's equation in the Newtonian limit is

$$\nabla^2\bar{h}_{tt} = -16\pi\rho.$$

Comparing this to Poisson's equation $\nabla^2\Phi = 4\pi\rho$, we find that

$$\bar{h}_{tt} = -4\Phi.$$

If the oscillations of $\bar{h}_{\mu\nu}$ is also small, $\square\bar{h}_{\mu\nu} \ll 1$, the inequality of the stress-energy tensor should carry over to the perturbation, $|\bar{h}_{tt}| \gg |\bar{h}_{ti}| \gg |\bar{h}_{ij}|$. In other words, $\bar{h}_{\mu\nu} = -4\Phi\delta^t_{\mu}\delta^t_{\nu}$. The trace of $h_{\mu\nu}$ is then

$$h = -\bar{h} \simeq \bar{h}_{tt} = -4\Phi.$$

Using (6.6) to give back the original metric perturbation,

$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h} = -4\Phi\delta^t_{\mu}\delta^t_{\nu} + 2\Phi\eta_{\mu\nu},$$

or $h_{tt} = h_{xx} = h_{yy} = h_{zz} = -2\Phi$. The metric for the Newtonian limit is

$$\boxed{ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Phi)(dx^2 + dy^2 + dz^2).} \quad (6.11)$$

The quantity Φ will be the usual Newtonian gravitational potential, e.g. $\Phi(r) = -M/r$ for a spherical object with mass M . The metric (6.11) cannot serve as the metric far from a relativistic source (such as a neutron star) even though the field is weak. This is because we assume that the field is weak *everywhere*, including inside the source, so that we can use Poisson's equation. Moreover, the Newtonian gravitational potential Φ can no longer be defined when the source is relativistic. Therefore, (6.11) only works for regular non-relativistic objects like the Earth or the Sun.

6.2.2 Gravitational Lensing I: Newtonian Solution

To be written.

6.3 The Gravitational Wave Solution

In the weak-field limit, there is another family of solutions in which we do not assume the field to be static. These solutions are known as gravitational waves, a very hot subject in nowadays astrophysics.

6.3.1 General Solutions

We start from the weak-field Einstein's equation (6.8) in vacuum,

$$\left(-\frac{\partial^2}{\partial t^2} + \nabla^2\right) \bar{h}_{\mu\nu} = 0 \quad \text{or} \quad \partial_\alpha \partial^\alpha \bar{h}_{\mu\nu} = 0.$$

This time we keep the time-derivative, so the equation is a homogeneous wave equation. Its solution is given by

$$\bar{h}_{\mu\nu} = A_{\mu\nu} e^{ik_\alpha x^\alpha} \quad (6.12)$$

for some real dual vector k_α and complex constant tensor $A_{\mu\nu}$. Substituting this solution into the wave equation, we get

$$k_\alpha k^\alpha \bar{h}_{\mu\nu} = 0 \quad \implies \quad k_\alpha k^\alpha = 0.$$

This means the 4-vector k^α is a null vector: it is tangent to the world line of a photon. Suppose the components of this 4-vector are $k^\alpha = (\omega, \mathbf{k})$, where ω is the frequency of the wave and $\mathbf{k}$ is the wave vector. The condition $k_\alpha k^\alpha = 0$ gives the dispersion relation

$$\omega^2 = \mathbf{k}^2.$$

The phase velocity ω/k (which is the same as the group velocity $d\omega/dk$ in this case) of the wave is $v_{\text{phase}} = v_{\text{group}} = 1$, or the speed of light. The real part of the solution (6.12) are now known as **gravitational waves**. They are waves traveling through vacuum at the speed of light, generated by accelerating masses. The Lorenz condition $\partial^\mu \bar{h}_{\mu\nu}$ implies that

$$k^\mu A_{\mu\nu} = 0. \quad (6.13)$$

This means $A_{\mu\nu}$ must be orthogonal to k^μ , a property that will be useful in the following section.

6.3.2 The Transverse-Traceless Gauge

Recall that our weak-field Einstein's equation (6.8) is based on the Lorenz gauge condition (6.9) which requires (6.10):

$$\square \bar{h}'_{\mu\nu} = -16\pi T_{\mu\nu} \quad \text{holds when} \quad h'_{\mu\nu} = h_{\mu\nu} - 2\partial_{(\mu} \xi_{\nu)}, \quad \text{where} \quad \square \xi_\nu = \partial^\mu \bar{h}_{\mu\nu}.$$

From this we see that the Lorenz gauge is a class of gauges because $\square \xi_\nu = \partial^\mu \bar{h}_{\mu\nu}$ is an inhomogeneous wave equation. We have not used all gauge freedoms yet. We can always add ξ'_ν to the solution to the inhomogeneous wave equation if ξ'_ν solves the homogeneous one, $\square \xi'_\nu = 0$. Let's choose $\xi'_\nu = B_\nu e^{ik_\alpha x^\alpha}$, where B_ν is a complex constant and k^α is still the wave vector. Though adding ξ'_ν does not alter the Lorenz gauge, it changes the metric perturbation. Suppose $h_{\mu\nu}$ already satisfies the Lorenz gauge, $\xi_\nu \rightarrow \xi_\nu + \xi'_\nu$ changes $h_{\mu\nu}$ according to (6.9),

$$h'_{\mu\nu} = h_{\mu\nu} - 2\partial_{(\mu} \xi'_{\nu)},$$

and $\bar{h}_{\mu\nu}$ changes by

$$\bar{h}'_{\mu\nu} = h'_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h' = \bar{h}_{\mu\nu} - 2\partial_{(\mu} \xi_{\nu)} + \eta_{\mu\nu} \partial_\alpha \xi^\alpha.$$

Using the gravitational wave solution $\bar{h}_{\mu\nu} = A_{\mu\nu} e^{ik_\alpha x^\alpha}$, we can compute the derivatives on ξ_μ and cancel the exponential factor,

$$A'_{\mu\nu} = A_{\mu\nu} - 2iB_{(\mu} k_{\nu)} + i\eta_{\mu\nu} B^\alpha k_\alpha. \quad (6.14)$$

Note that we can choose B_μ whatever we want to make $A'_{\mu\nu}$ easy. It can be chosen such that

$$A'^\mu{}_\mu = 0 \quad \text{and} \quad A'_{\mu\nu}u^\nu = 0 \quad (6.15)$$

for some 4-velocity u . After then, there will be no gauge freedoms left. The gauge (6.15) is often called the **transverse-traceless gauge** (or TT gauge).

Proof. Refer to (6.14). First, the “traceless” condition $A'^\mu{}_\mu = 0$ provides one constraint on B_μ :

$$0 = A^\mu{}_\mu - iB^\mu k_\mu - iB_\mu k^\mu + i\delta^\mu{}_\mu B^\alpha k_\alpha = A^\mu{}_\mu - 2iB^\mu k_\mu + 4iB^\alpha k_\alpha = A^\mu{}_\mu + 2iB^\mu k_\mu.$$

Setting this to zero relates $B^\mu k_\mu$ to the trace of $A^\mu{}_\mu$,

$$B^\mu k_\mu = -\frac{1}{2i}A^\mu{}_\mu. \quad (6.16)$$

Second, the “transverse” condition $A_{\mu\nu}u^\nu = 0$ requires

$$0 = A_{\mu\nu}u^\nu - iB_\mu k_\nu u^\nu - iB_\nu k_\mu u^\nu + iu_\mu B^\alpha k_\alpha. \quad (6.17)$$

It seems like four constraints for $\mu = 0, 1, 2, 3$, but there are actually only three. This is because $k^\mu A'_{\mu\nu}u^\nu = 0$ for any B_μ ,

$$k^\mu A'_{\mu\nu}u^\nu = k^\mu A_{\mu\nu}u^\nu - ik^\mu B_\mu k_\nu u^\nu - ik^\mu B_\nu k_\mu u^\nu + ik^\mu u_\mu B^\alpha k_\alpha = 0,$$

where we used $k^\mu A_{\mu\nu} = 0$, $k_\mu k^\mu = 0$, and the second and fourth terms cancel. Hence there are only three independent constraint equations, as one of the μ is related to the other three by $k^\mu A_{\mu\nu}u^\nu = 0$.

With one constraint from traceless condition and three from transverse condition, we look for four components B_μ in terms of $A_{\mu\nu}$, u^μ , and k_μ such that both transverse and traceless conditions hold. Starting with the transverse condition (6.17),

$$iB_\mu k_\nu u^\nu + iB_\nu k_\mu u^\nu = A_{\mu\nu}u^\nu + iu_\mu B^\alpha k_\alpha.$$

Using the traceless condition (6.16),

$$iB_\mu k_\nu u^\nu + iB_\nu k_\mu u^\nu = A_{\mu\nu}u^\nu - \frac{1}{2}u_\mu A^\alpha{}_\alpha. \quad (6.18)$$

The first term on the LHS contains B_μ alone, but the second term contains its contraction with u^ν , which is annoying. To eliminate $B_\nu u^\nu$, we multiply both sides by u^μ ,

$$iB_\mu k_\nu u^\mu u^\nu + iB_\nu k_\mu u^\nu u^\mu = A_{\mu\nu}u^\mu u^\nu + \frac{1}{2}A^\alpha{}_\alpha u^\mu u_\mu.$$

The two terms on the LHS are the same, so

$$2iB_\nu k_\mu u^\nu u^\mu = A_{\mu\nu}u^\mu u^\nu + \frac{1}{2}A^\alpha{}_\alpha u^\mu u_\mu \implies B_\nu u^\nu = \frac{1}{2ik_\mu u^\mu} \left(A_{\mu\nu}u^\mu u^\nu + \frac{1}{2}A^\alpha{}_\alpha u^\mu u_\mu \right).$$

Plugging this back into (6.18) and solving for B_μ ,

$$B_\mu = \frac{1}{ik_\nu u^\nu} \left[-\frac{k_\mu}{2k_\nu u^\nu} \left(A_{\alpha\beta}u^\alpha u^\beta + \frac{1}{2}A^\alpha{}_\alpha \right) + A_{\mu\nu}u^\nu - \frac{1}{2}u_\mu A^\alpha{}_\alpha \right].$$

□

Though B_μ looks ugly, we successfully express it in terms of known quantities $A_{\mu\nu}$, u^μ , and k_μ . Substituting B_μ into the gauge transformation (6.14) gives the transverse-traceless $A'_{\mu\nu}$. Note that the TT gauge is valid only when we are working with the vacuum weak-field Einstein’s equation $\square \bar{h}_{\mu\nu} = 0$ because the condition $k^\mu A_{\mu\nu} = 0$ is necessary, as shown in the proof.

By now we assume that we are working in the TT gauge, $A^\mu{}_\mu = 0$, $A_{\mu\nu}u^\mu = 0$, and $A_{\mu\nu}k^\mu = 0$. Since all gauge freedoms are used, the remaining components of $A_{\mu\nu}$ must be physically important. As the metric perturbation becomes traceless, there is no difference between $\bar{h}_{\mu\nu}$ and $h_{\mu\nu}$. Now consider a frame with the 4-velocity being the timelike basis vector, $u^\mu = (1, 0, 0, 0) = \delta^\mu{}_t$. (It requires only background Lorentz transformation to this frame, so it is a separate gauge transformation from the metric perturbation.) The

gauge condition $A_{\mu\nu}u^\mu = 0$ implies

$$A_{\mu t} = 0.$$

Let's orient the axis such that the gravitational wave travel in the z -direction, so the wave vector becomes $k^\mu = (\omega, 0, 0, \omega)$. With $A_{\mu t} = A_{t\mu} = 0$, the gauge condition $0 = A_{\mu\nu}k^\mu = A_{t\nu}k^t + A_{z\nu}k^z$ implies

$$A_{\mu z} = 0.$$

(In fact, this is the origin of the word “transverse”—there is no wave component along the direction of propagation.) Now we are only left with A_{xx} , A_{xy} , A_{yx} , and A_{yy} . The final gauge condition $A^\mu{}_\mu = 0$ forces $A_{xx} = -A_{yy}$, and $A_{\mu\nu}$ being symmetric forces $A_{xy} = A_{yx}$. The constant tensor $A_{\mu\nu}$ thus has the simple form

$$A_{\mu\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

where we renamed $A_{xx} \rightarrow h_+$ and $A_{xy} \rightarrow h_\times$ for reasons that will later be obvious. There are only two physical degrees of freedom in $A_{\mu\nu}$; they correspond to two independent *polarizations* of gravitational waves.

6.3.3 Effects on Free Particles

Consider a free particle at rest with a 4-velocity $u^\mu = (1, 0, 0, 0)$ in the lab frame. What is the effect of a gravitational wave on the particle? We know that a free particle obeys the geodesic equation (4.16),

$$\frac{du^\alpha}{d\tau} + \Gamma^\alpha{}_{\mu\nu}u^\mu u^\nu = 0.$$

Since only $u^t = 1$ is nonzero initially, the geodesic equation reduces to

$$\frac{du^\alpha}{d\tau} + \Gamma^\alpha{}_{tt} = 0.$$

However, $h_{\mu t} = 0$ in the TT gauge means that

$$\Gamma^\alpha{}_{tt} = -\frac{1}{2}\eta^{\alpha\beta}(\partial_t h_{\beta t} + \partial_t h_{\beta t} - \partial_\beta h_{tt}) = 0$$

for all α , so the initial acceleration is $du^\alpha/d\tau = 0$ and $u^\mu = (1, 0, 0, 0)$ after $d\tau$. By induction, the acceleration vanishes for all time later. The free particle that encounters the gravitational wave seems to be “at rest” always. This is not what happens physically, because $du^\alpha/d\tau = 0$ only means that the particle is at constant *coordinate* position, not geometric position. There are two ways to see the physical effect: the simple way is to compute the proper distance between free particles, another way is to use the equation of geodesic deviation, which is more general.

Proper Distance. Consider two particles close to each other, one at the origin and the other at $x = \epsilon$ and $y = z = 0$. Assume they are initially at rest and the wave comes from the z -direction. The physical distance is given by the proper distance between the particles,

$$\Delta l \equiv \int \sqrt{ds^2} = \int \sqrt{g_{\mu\nu}dx^\mu dx^\nu} = \int_0^\epsilon \sqrt{g_{xx}} dx.$$

If ϵ is small, this integral is approximately

$$\Delta l \simeq |g_{xx}(x=0)|^{1/2}\epsilon = [1 + h_{xx}(x=0)]^{1/2} \simeq \left[1 + \frac{1}{2}h_{xx}(x=0)\right]\epsilon.$$

In general h_{xx} is nonzero. If it is oscillating, then the separation between the two nearby particles is oscillating at the same frequency as the wave. The proper distance is a measurable quantity. In fact, this is what modern gravitational wave detectors measure. Note that the effect of the wave is linear in ϵ , so larger detector means larger change in separation. The amplitude of the wave (such as A_{xx}) is often of the order 10^{-21} , so linearized gravity is almost a perfect approximation.

Geodesic Deviation. Consider again two nearby free particles A and B in the xy -plane. This time we do not set up equations in the TT gauge. Instead, we move to the local inertial frame of particle A . By the definition of the local inertial frame, the metric is $g_{\mu\nu} = \eta_{\mu\nu}$ and all first derivatives of $g_{\mu\nu}$ vanish at particle A 's position. Let s^μ be the deviation vector between the two particles as measured from particle A . The goal is to compute how s^μ changes as both particles move along their individual geodesics.

By the equation of geodesic deviation (4.27), the acceleration vector a^α of the deviation satisfies

$$a^\alpha = R^\alpha{}_{\beta\mu\nu} u^\beta u^\mu s^\nu, \quad \text{where} \quad a^\alpha = \frac{d^2 s^\alpha}{d\tau^2} = u^\nu \nabla_\nu (u^\mu \nabla_\mu s^\alpha).$$

All first derivatives vanish in the local inertial frame, so do all Christoffel symbols. The covariant derivative reduces to the partial derivative, $a^\alpha = u^\nu \partial_\nu (u^\mu \partial_\mu s^\alpha)$. Meanwhile, if we assume the velocity of the particles are small, $dx^i/d\tau \ll dt/d\tau$, the 4-velocity field is given by $u^\mu = dx^\mu/d\tau \simeq \delta^\mu_t + \mathcal{O}(v^2)$ for geodesics near the particle A . (The $\mathcal{O}(v^2)$ is because the Lorentz factor γ is second order in v from 1.) Keeping only to first order in small quantities, the acceleration vector becomes $a^\alpha = \partial_t^2 s^\alpha$. The RHS of the geodesic deviation equation is now $R^\alpha{}_{tt\nu} s^\nu$. Now we use the fact that the Riemann curvature tensor is gauge invariant in linearized gravity proved in the previous section. Thus, we can compute it in the TT gauge and use in our local inertial frame. Referring to linearized Christoffel symbols (6.1) and linearized Riemann curvature tensor (6.2),

$$R^\alpha{}_{tt\nu} = \partial_t \Gamma^\alpha{}_{t\nu} - \cancel{\partial_\nu \Gamma^\alpha{}_{tt}} = \frac{1}{2} \eta^{\alpha\beta} \partial_t (\partial_t h_{\beta\nu} + \partial_\nu h_{\beta t} - \partial_\beta h_{t\nu}) = \frac{1}{2} \eta^{\alpha\mu} \partial_t^2 h_{\mu\nu},$$

where we used $\Gamma^\alpha{}_{tt} = 0$ and $h_{\mu t} = \bar{h}_{\mu t} = 0$ in the TT gauge. In the local inertial frame, the geodesic deviation equation is now

$$\frac{\partial^2}{\partial t^2} s^\alpha = \frac{1}{2} s^\nu \eta^{\alpha\mu} \frac{\partial^2}{\partial t^2} h_{\mu\nu}. \quad (6.19)$$

Suppose the gravitational wave come from the z -direction and we measure the deviation at the same coordinate time so that $s^\mu = (0, s^x, s^y, 0)$. Recall that $h_{\mu\nu} = A_{\mu\nu} e^{ik_\alpha x^\alpha}$ has amplitude $A_{\mu\nu}$ specified by two numbers, $A_{xx} = -A_{yy} = h_+$ and $A_{xy} = A_{yx} = h_\times$. If $h_\times = 0$ but $h_+ \neq 0$, then

$$\frac{\partial^2}{\partial t^2} s^x = \frac{1}{2} s^x \frac{\partial^2}{\partial t^2} (h_+ e^{ik_\alpha x^\alpha}) \quad \text{and} \quad \frac{\partial^2}{\partial t^2} s^y = -\frac{1}{2} s^y \frac{\partial^2}{\partial t^2} (h_+ e^{ik_\alpha x^\alpha}).$$

To first order, the solution to these equations are

$$s^x = \left(1 + \frac{1}{2} h_+ e^{ik_\alpha x^\alpha}\right) s^x(0) \quad \text{and} \quad s^y = \left(1 - \frac{1}{2} h_+ e^{ik_\alpha x^\alpha}\right) s^y(0) \quad (6.20)$$

for initial separation $s^x(0)$ and $s^y(0)$. If $h_+ = 0$ but $h_\times \neq 0$, the geodesic deviation (6.19) reads

$$\frac{\partial^2}{\partial t^2} s^x = \frac{1}{2} s^y \frac{\partial^2}{\partial t^2} (h_\times e^{ik_\alpha x^\alpha}) \quad \text{and} \quad \frac{\partial^2}{\partial t^2} s^y = \frac{1}{2} s^x \frac{\partial^2}{\partial t^2} (h_\times e^{ik_\alpha x^\alpha}).$$

To first order, the solutions are

$$s^x = s^x(0) + \frac{1}{2} h_\times e^{ik_\alpha x^\alpha} s^y(0) \quad \text{and} \quad s^y = s^y(0) + \frac{1}{2} h_\times e^{ik_\alpha x^\alpha} s^x(0). \quad (6.21)$$

6.3.4 Polarization

Let's interpret these gravitational wave solutions. The solution (6.20) with $h_\times = 0$ but $h_+ \neq 0$ would distort the circle of test particles into a “+” pattern with time. Figure 6.1 shows this pattern. This type of gravitational waves has **+ polarization**.

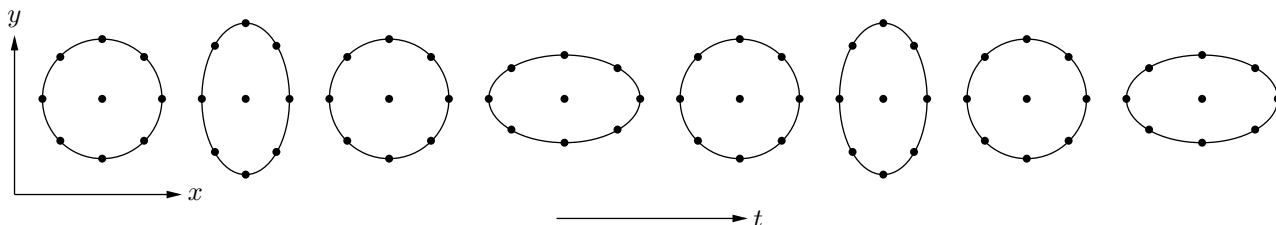


Figure 6.1: The effect of a gravitational wave with + polarization to a circle of test particles in to ellipses.

On the other hand, if $h_+ = 0$ but $h_\times$, the solution (6.21) predicts that the gravitational wave will distort the circle of test particles into a “ $\times$ ” pattern with time. Figure 6.2 shows this pattern. This type of gravitational waves has $\times$ polarization.

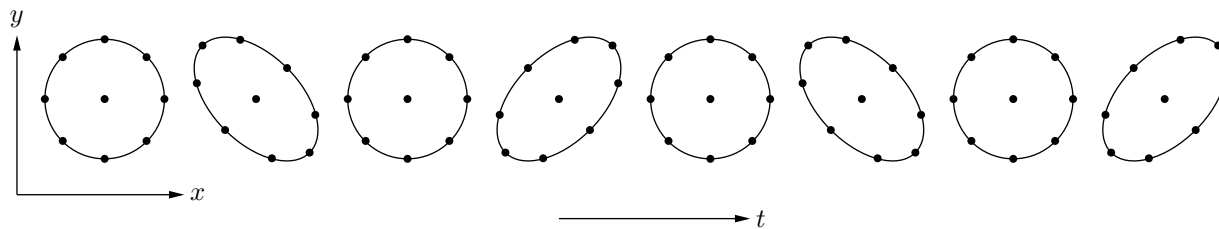


Figure 6.2: The effect of a gravitational wave with $\times$ polarization to a circle of test particles in to ellipses.

Just like electromagnetic waves, gravitational waves also have two independent polarizations. The difference is the two polarization states of an electromagnetic wave are at 90° to each other, while those of a gravitational wave are at 45° to each other. We can also have analogous right- and left-handed circularly polarized gravitational waves,

$$h_R = \frac{1}{\sqrt{2}}(h_+ + ih_\times), \quad h_L = \frac{1}{\sqrt{2}}(h_+ - ih_\times).$$

They are discussed in Appendix A.4.1 (To be written.)

6.4 Generation of Gravitational Waves

6.4.1 The Quadrupole Moment Tensor

To study the generation of gravitational waves, we need to use the weak-field Einstein's equation (6.8) that couples to matter ($T_{\mu\nu} \neq 0$),

$$\square \bar{h}_{\mu\nu} = -16\pi T_{\mu\nu}.$$

The following steps will be very similar to the ones in deriving electromagnetic radiation. The weak-field Einstein's equation is an inhomogeneous wave equation, which has been studied over a hundred of years. The usual method is to find the Green's $G(x - x')$ for the wave operator $\square$:

$$\square G(x - x') = \delta^4(x - x').$$

(The wave operator $\square$ differentiates with respect to x^μ , not x'^μ .) Once we find the Green's function, the solution $\bar{h}_{\mu\nu}(x)$ can be obtained by

$$\bar{h}_{\mu\nu}(x) = -16\pi \int G(x - x') T_{\mu\nu}(x') d^4x',$$

which can be verified easily by taking $\square \bar{h}_{\mu\nu}$. We will just quote the Green's function to the wave operator here:

$$G(x - x') = -\frac{1}{4\pi|\mathbf{x} - \mathbf{x}'|} \delta[|\mathbf{x} - \mathbf{x}'| - (t - t')] \theta(t - t'). \quad (6.22)$$

There are several things to note in this Green's function. First, the $\mathbf{x} = (x^1, x^2, x^3)$ and $\mathbf{x}' = (x'^1, x'^2, x'^3)$ are spatial vectors. The norm is given by $|\mathbf{x} - \mathbf{x}'| = [\delta_{ij}(x^i - x'^i)(x^j - x'^j)]^{1/2}$. The θ is the Heaviside θ -function, which is 1 if $t > t'$ and zero otherwise. The solution $\bar{h}_{\mu\nu}(t, \mathbf{x})$ is then

$$\bar{h}_{\mu\nu}(x) = 4 \int \frac{1}{|\mathbf{x} - \mathbf{x}'|} \delta[|\mathbf{x} - \mathbf{x}'| - (t - t')] \theta(t - t') T_{\mu\nu}(x') d^4x'.$$

This integral makes sense physically. For a source $T_{\mu\nu}(t', \mathbf{x}')$, the θ function means that we care only about its effect on the metric perturbation $h_{\mu\nu}(t, \mathbf{x})$ when t is to the future of t' . The source just can only affect metric in its future light cone; the metric is only affected by the source in its past light cone. The delta function means that the effect travels at the speed of light $c = 1$ (as matched by gravitational waves). In other words, the metric at $(t, \mathbf{x})$ is only affected by the source at distance $|\mathbf{x} - \mathbf{x}'| = (t - t')$ away. Doing the delta function integral over t' ,

$$h(t, \mathbf{x}) = 4 \int \frac{1}{|\mathbf{x} - \mathbf{x}'|} T_{\mu\nu}(t_r, \mathbf{x}') d^3x', \quad (6.23)$$

where $t_r = t - |\mathbf{x} - \mathbf{x}'|$ is called the **retarded time**. The goal is to find $\bar{h}_{ij}$. After that, the Lorenz gauge condition $\partial^\mu \bar{h}_{\mu\nu} = 0$ will help us find $\bar{h}_{tj}$, and using the gauge condition again helps find $\bar{h}_{tt}$ from $\bar{h}_{tj}$. Assume that we are interested in $h_{\mu\nu}$ far from the source region so that $r = |\mathbf{x}| \gg r' = |\mathbf{x}'|$. Then we can approximate

$$\begin{aligned} |\mathbf{x} - \mathbf{x}'| &= (r^2 + r'^2 - 2\mathbf{x} \cdot \mathbf{x}')^{1/2} = r \left(1 - \frac{2\mathbf{x} \cdot \mathbf{x}'}{r^2} + \frac{r'^2}{r^2} \right)^{1/2} \simeq r - \frac{\mathbf{x} \cdot \mathbf{x}'}{r}, \\ \frac{1}{|\mathbf{x} - \mathbf{x}'|} &= \frac{1}{r} \left(1 - \frac{2\mathbf{x} \cdot \mathbf{x}'}{r^2} + \frac{r'^2}{r^2} \right)^{-1/2} \simeq \frac{1}{r} + \frac{\mathbf{x} \cdot \mathbf{x}'}{r^3}, \\ T_{\mu\nu}(t_r, \mathbf{x}') &= T_{\mu\nu}(t - r + \mathbf{x} \cdot \mathbf{x}'/r + \mathcal{O}(r^{-2}), \mathbf{x}') \simeq T_{\mu\nu}(t - r, \mathbf{x}') + \partial_t T_{\mu\nu}(t - r, \mathbf{x}') \frac{\mathbf{x} \cdot \mathbf{x}'}{r}. \end{aligned}$$

Making these approximation in (6.23) and keeping leading order of $1/r$,

$$\bar{h}_{\mu\nu} \simeq 4 \int \left(\frac{1}{r} - \frac{\mathbf{x} \cdot \mathbf{x}'}{r^3} \right) \left[T_{\mu\nu}(t - r, \mathbf{x}') + \partial_t T_{\mu\nu}(t - r, \mathbf{x}') \frac{\mathbf{x} \cdot \mathbf{x}'}{r} \right] d^3 x' \simeq \frac{4}{r} \int T_{\mu\nu}(t - r, \mathbf{x}') d^3 x'.$$

Now we need to find the integral of $T_{\mu\nu}$ over the source region. For notation simplicity, we drop the primes when we do the integral (later it will be added back). The ij components of the stress-energy tensor can be written as

$$T_{ij} = T_{ik} \partial^k x_j = \partial^k (T_{ik} x_j) - \partial^k T_{ik} x_j = \partial^k (T_{ik} x_j) + \partial^t T_{it} x_j.$$

In the first equality, we used the fact that $\partial^k x_j = \delta^k_j$. In the last equality, we used the Lorenz gauge condition ($\partial^\mu \bar{h}_{\mu\nu} = 0$ implies that $\partial^\mu T_{\mu\nu} = \square \partial^\mu \bar{h}_{\mu\nu} = 0$). The term $\partial^k (T_{ik} x_j)$ is a divergence term. When integrated over space, it is turned to a surface integral evaluated at infinity where $T_{\mu\nu} = 0$. Therefore,

$$\int T_{ij} d^3 x = \int \partial^k (T_{ik} x_j) d^3 x + \partial^t \int T_{it} x_j d^3 x = \partial^t \int T_{it} x_j d^3 x.$$

Since T_{ij} is symmetric, this resultant integral should be symmetric in i and j , i.e. the integral of $T_{it} x_j$ and $T_{jt} x_i$ are the same. Use the above trick again,

$$\int \partial^k (T_{kt} x_i x_j) d^3 x = \int (\partial^k T_{kt} x_i x_j + T_{it} x_j + T_{jt} x_i) d^3 x = \int (-\partial^t T_{tt} x_i x_j + 2T_{it} x_j) d^3 x.$$

The LHS is a divergence term, so its integral over all space is again zero, and

$$\int T_{it} x_j d^3 x = \frac{1}{2} \partial^t \int T_{tt} x_i x_j d^3 x \implies \int T_{ij} d^3 x = \frac{1}{2} \partial^t \partial^t \int T_{tt} x_i x_j d^3 x.$$

Finally, substituting the integral of T_{ij} into $\bar{h}_{ij}$, we obtain the **quadrupole formula**:

$$\boxed{\bar{h}_{ij}(t, \mathbf{x}) = \frac{2}{r} \frac{\partial^2 q_{ij}}{\partial t^2} \Big|_{t_r}}, \quad (6.24)$$

where q_{ij} is called the **quadrupole moment tensor**

$$q_{ij} = \int T_{tt}(t_r, \mathbf{x}') x'_i x'_j d^3 x' \quad \text{and} \quad t_r = t - r. \quad (6.25)$$

6.4.2 Energy Loss

To be written.

$$P = \frac{1}{5} \left| \frac{d^3 Q_{ij}}{dt^3} \right|^2. \quad (6.26)$$

6.4.3 Exact Solutions

To be written.

7 THE SCHWARZSCHILD SOLUTION

The Schwarzschild solution is one of the most important solutions of Einstein's equation. It was solved by Karl Schwarzschild just a little more than one month after Einstein published his field equations and was the first nontrivial *exact* solution. Physically, the Schwarzschild solution describes the exterior gravitational field of a static, spherically symmetric body. It consequently provides an excellent approximation to many astrophysical objects such as the Sun and the planets in the solar system. Many tests of general relativity were based on the Schwarzschild solution. Some classic tests include the precession of Mercury, the bending of light, and the gravitational redshift. They will be discussed in this chapter. Moreover, the Schwarzschild solution predicts one of the most interesting types of objects in the universe—black holes, which will be discussed in the next chapter.

7.1 The Schwarzschild Metric

7.1.1 Conditions on the Metric

The goal of this section is to find a metric that describes a static gravitational field outside a spherically symmetric object. The term **static** means that the metric should be time-independent and invariant under time-reversal $t \rightarrow -t$. The metric without the time-reversal symmetry but time-independent in its components is called **stationary**. For example, a uniform sphere with constant angular velocity is stationary because time-reversal can reverse its spin direction; a non-spinning uniform sphere is static.

Let's impose some conditions on the metric $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$ to reduce it to the simplest possible form for a static, spherically symmetric object:

1. First, the static condition immediately implies that there are no time-space cross terms, $dt dx^i$, or otherwise changing $t \rightarrow -t$ will reverse these terms. The dt^2 and $dx^i dx^j$ terms are unaffected. This simplifies the metric to

$$ds^2 = -g_{tt}dt^2 + g_{ij}dx^i dx^j.$$

2. Since we are interested in spherically symmetric objects, it is useful to adopt spherical coordinates (t, r, θ, ϕ) . The angular metric should be proportional to that of a two-sphere, $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$. Spherical symmetry also rules out the radial-angular cross terms, $dr d\theta$ and $dr d\phi$, with reasoning analogous to time-reversal symmetry. Then we expect the metric to look like

$$ds^2 = -g_{tt}dt^2 + g_{rr}dr^2 + r^2 d\Omega^2,$$

where g_{tt} and g_{rr} are positive functions of r only. This is just some generalization of the Minkowski metric $ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2$, which is obviously spherically symmetric. Of course, we are allowed to multiply a factor g_{Ω^2} before $r^2 d\Omega^2$, but that is not necessary because we can just redefine the radial coordinate scaled by some function so that the above metric holds.

3. To fix the signature of the metric, we let $g_{tt} = e^{2\alpha(r)}$ and $g_{rr} = e^{2\beta(r)}$ for some unknown functions $\alpha(r)$ and $\beta(r)$. We conclude that the metric should have the form

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2. \quad (7.1)$$

7.1.2 The Schwarzschild Metric

With the metric (7.1), it is straightforward to compute the Christoffel symbols using their coordinate representation (4.9). Here is a list of them:

$$\begin{aligned}\Gamma^t_{tr} &= \partial_r \alpha \\ \Gamma^r_{tt} &= e^{2(\alpha-\beta)} \partial_r \alpha & \Gamma^r_{rr} &= \partial_r \beta & \Gamma^r_{\theta\theta} &= -re^{-2\beta} & \Gamma^r_{\phi\phi} &= -re^{-2\beta} \sin^2 \theta \\ \Gamma^\theta_{r\theta} &= \frac{1}{r} & \Gamma^\theta_{\phi\phi} &= -\sin \theta \cos \theta \\ \Gamma^\phi_{r\phi} &= \frac{1}{r} & \Gamma^\phi_{\theta\phi} &= \cot \theta.\end{aligned}$$

Other Christoffel symbols are either related to these by symmetry or are zero. The Riemann curvature tensor can then be calculated by its coordinate representation (4.26),

$$\begin{aligned}R^t_{rtr} &= -\partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta & R^t_{\theta t\theta} &= -re^{-2\beta} \partial_r \alpha & R^t_{\phi t\phi} &= -re^{-2\beta} \sin^2 \theta \partial_r \alpha \\ R^r_{trt} &= e^{2(\alpha-\beta)} [\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta] & R^r_{\theta r\theta} &= re^{-2\beta} \partial_r \beta & R^r_{\phi r\phi} &= re^{-2\beta} \sin^2 \theta \partial_r \beta \\ R^\theta_{t\theta t} &= \frac{1}{r} e^{2(\alpha-\beta)} \partial_r \alpha & R^\theta_{r\theta r} &= \frac{1}{r} \partial_r \beta & R^\theta_{\phi\theta\phi} &= (1 - e^{-2\beta}) \sin^2 \theta \\ R^\phi_{t\phi t} &= \frac{1}{r} e^{2(\alpha-\beta)} \partial_r \alpha & R^\phi_{r\phi r} &= \frac{1}{r} \partial_r \beta & R^\phi_{\theta\phi\theta} &= 1 - e^{-2\beta}.\end{aligned}$$

Some of them are related by symmetry when indices are lowered, but it is clearer to list them out because we will compute the Ricci tensor using them. Other unlisted components are either related to above ones by antisymmetry or are zero.

The Ricci tensor $R_{\mu\nu} = R^\alpha_{\mu\alpha\nu}$ is diagonal, with components

$$\begin{aligned}R_{tt} &= e^{2(\alpha-\beta)} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \alpha \right] \\ R_{rr} &= -\partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \beta \\ R_{\theta\theta} &= e^{-2\beta} [r(-\partial_r \alpha + \partial_r \beta) - 1] + 1 \\ R_{\phi\phi} &= R_{\theta\theta} \sin^2 \theta,\end{aligned}\tag{7.2}$$

and the Ricci scalar $R = R^\mu_{\mu}$ (not necessary for now) is

$$R = -2e^{-2\beta} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r} (\partial_r \alpha - \partial_r \beta) + \frac{1}{r^2} (1 - e^{2\beta}) \right].\tag{7.3}$$

We are dealing with the geometry outside the spherical symmetric object, so the stress-energy tensor is zero. With this condition, it is more convenient to use Einstein's equation in the form (5.6),

$$R_{\mu\nu} = 8\pi \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right) \implies R_{\mu\nu} = 0.$$

In other words, to solve for α and β (or $e^{2\alpha}$ and $e^{2\beta}$), we set the Ricci tensor to zero. Consider the combination of R_{tt} and R_{rr} ,

$$0 = e^{-2(\alpha-\beta)} R_{tt} + R_{rr} = \frac{2}{r} (\partial_r \alpha + \partial_r \beta).$$

This implies $\alpha = -\beta + C$ for some constant C . Actually, this constant C can be set to zero because we can always rescale $dt \rightarrow e^{-C} dt$. The form of the metric (7.1) becomes

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{-2\alpha(r)} dr^2 + r^2 d\Omega^2.$$

The equation $R_{\theta\theta} = 0$ with $\alpha = -\beta$ reduces to

$$e^{2\alpha} (2r \partial_r \alpha + 1) = 1 \implies \frac{\partial}{\partial r} (re^{2\alpha}) = 1.$$

The solution is

$$e^{2\alpha} = 1 - \frac{R_S}{r},$$

where R_S , with the dimension of length, is called the **Schwarzschild radius**. The final form of the metric is

$$ds^2 = -\left(1 - \frac{R_S}{r}\right) dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2 d\Omega^2.$$

Since we used the combination of R_{tt} and R_{rr} to solve for $e^{2\alpha}$, it is not guaranteed that $R_{tt} = 0$ and $R_{rr} = 0$ separately, but one can check that this metric indeed satisfies all $R_{\mu\nu} = 0$. Now we want to relate this Schwarzschild radius to some physical quantities. The expectation is that the metric at large r reduces to the Newtonian limit (6.11). For a spherically symmetric mass with $\Phi = M/r$, the component tt component of the Newtonian metric is $g_{tt} = -(1 - 2M/r)$. But this is already in the same form as the metric above, so $R_S = 2M$.

In conclusion, the exterior metric of a static, spherically symmetric object is given by

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (7.4)$$

This metric is known as the **Schwarzschild metric**. The coordinates (t, r, θ, ϕ) are **Schwarzschild coordinates**. They *do not* have the same physical meaning as the Minkowski, spherical coordinates. By observation, there are two interesting radius r that blows up the metric: $r = 0$ and $r = 2M$. We will analyze them in Chapter 8. For now, let's focus on the exterior geometry of the Schwarzschild metric.

There is one parameter M that we identify with the mass of the central object. To justify this physical meaning, we shall show that at large r , we recover the Newtonian weak-field spacetime metric,

$$ds^2 = -(1 - 2\Phi)dt^2 + (1 + 2\Phi)(dr^2 + r^2 d\Omega^2).$$

Note that the Schwarzschild metric already resembles the weak-field metric if we take $r \rightarrow \infty$, but the angular part is not matching. Consider a coordinate transformation from Schwarzschild coordinates (t, r, θ, ϕ) to **isotropic coordinates** $(t, \bar{r}, \theta, \phi)$, with $\bar{r}$ defined by

$$r = \bar{r} \left(1 + \frac{M}{2\bar{r}}\right)^2 = \bar{r} + M + \frac{M^2}{4\bar{r}}, \quad dr = \left(1 - \frac{M^2}{4\bar{r}^2}\right) d\bar{r} = \left(1 + \frac{M}{2\bar{r}}\right) \left(1 - \frac{M}{2\bar{r}}\right) d\bar{r}.$$

Then

$$1 - \frac{2M}{r} = 1 - \frac{2M}{\bar{r}(1 + M/2\bar{r})^2} = \frac{(1 + M/2\bar{r})^2 - 2M/\bar{r}}{(1 + M/2\bar{r})^2} = \frac{(1 - M/2\bar{r})^2}{(1 + M/2\bar{r})^2}.$$

The Schwarzschild metric in isotropic coordinates is

$$ds^2 = -\left(\frac{1 - M/2\bar{r}}{1 + M/2\bar{r}}\right)^2 dt^2 + \left(1 + \frac{M}{2\bar{r}}\right)^4 (d\bar{r}^2 + \bar{r}^2 d\Omega^2).$$

Now the spatial part looks like Euclidean space in spherical coordinates. So far the metric is still *exact*. Now as $r \rightarrow \infty$, or $M/\bar{r} \rightarrow 0$, we can Taylor expand the metric using $(1 + \epsilon)^n \simeq 1 + n\epsilon$:

$$\left(\frac{1 - M/2\bar{r}}{1 + M/2\bar{r}}\right)^2 \simeq \left(1 - \frac{M}{2\bar{r}}\right)^4 \simeq 1 - \frac{2M}{\bar{r}}, \quad \left(1 + \frac{M}{2\bar{r}}\right)^4 \simeq 1 + \frac{2M}{\bar{r}}.$$

This gives the Newtonian weak-field metric,

$$ds^2 \simeq (1 + 2\Phi)dt^2 + (1 - 2\Phi)(d\bar{r}^2 + \bar{r}^2 d\Omega^2),$$

where $\Phi = -GM/\bar{r}$ (expressing in ordinary units for physical meaning).

7.1.3 Gravitational Redshift

There are many ways to obtain the gravitational redshift in Schwarzschild spacetime.

By Time Dilation. We may also check this using gravitational time dilation directly. Since a photon's wave period is itself a clock, the gravitational time dilation therefore affects the frequency. We know that a static observer measures their proper time by

$$d\tau = \left(1 - \frac{2M}{r}\right)^{1/2} dt$$

in Schwarzschild spacetime. Suppose there is a static light source emitting two wave crests of a photon within a proper time period $\delta\tau = (1 - 2M/r)\delta t$, so the emitted frequency is $\nu_r = 1/\delta\tau$. A static observer at $r \rightarrow \infty$ measures the proper time interval between two wave crests to be δt , or a frequency $\nu_\infty = 1/\delta t$. Compare these frequencies,

$$\frac{\nu_\infty}{\nu_r} = \frac{\delta\tau}{\delta t} = \left(1 - \frac{2M}{r}\right)^{1/2} \implies z = \frac{\lambda_\infty}{\lambda_r} - 1 = \frac{\nu_r}{\nu_\infty} - 1 = \left(1 - \frac{2M}{r}\right)^{-1/2} - 1.$$

By Killing Vectors. We can check gravitational redshift using Killing vectors for conserved quantities. For a static observer at radius r , its 4-velocity satisfy $u \cdot u = -1$, and its spatial component is zero. In Schwarzschild metric, it means

$$g_{\mu\nu}u^\mu u^\nu = g_{tt}(u^t)^2 = -1 \implies u^t = \left(1 - \frac{2M}{r}\right)^{-1/2}.$$

Suppose there is an orthonormal basis in the observer's frame and u is its timelike basis vector. Then the energy measured by the observer will satisfy $\eta_{\mu\nu}p^\mu = p \cdot u$, or

$$E = -p \cdot u = -g_{\mu\nu}p^\mu u^\nu = -g_{tt}p^t u^t = \left(1 - \frac{2M}{r}\right)p^t \left(1 - \frac{2M}{r}\right)^{-1/2} = \left(1 - \frac{2M}{r}\right)^{1/2} p^t.$$

In other words,

$$h\nu_r = \left(1 - \frac{2M}{r}\right)^{1/2} p^t \Big|_r, \quad h\nu_\infty = p^t \Big|_\infty.$$

The goal now is to relate $p^t|_r$ to $p^t|_\infty$. It is achieved by realizing that there exists a timelike Killing vector ξ since the metric tensor does not depend on time. Then $\xi \cdot p = g_{\mu\nu}\xi^\mu p^\nu = g_{tt}(1)p^t$ is conserved. Hence

$$-\left(1 - \frac{2M}{r}\right)p^t \Big|_r = -p^t \Big|_\infty \implies \left(1 - \frac{2M}{r}\right)^{1/2} \nu_r = \nu_\infty,$$

which is the desired gravitational redshift.

7.2 Birkhoff's Theorem

Theorem 7.1. Birkhoff's Theorem

The spacetime outside any spherically symmetric, electrically neutral body is given by Schwarzschild geometry,

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \left(1 - \frac{2M}{r}\right)^{-1}dr^2 + r^2d\Omega^2.$$

In other words, if we drop the assumption that the spacetime is static, Birkhoff's theorem says that the spherically symmetric spacetime *leads to* the static solution (the Schwarzschild solution).

Proof. To be written. □

7.3 Orbits in Schwarzschild Spacetime

In Newtonian mechanics, we know that a spherically symmetric object gives rise to a $-1/r$ potential, and the orbits are conic sections

(circles, ellipses, parabolas, and hyperbolas). In general relativity, the orbits of free particles are geodesics in spacetime. We will see that many quantities in Schwarzschild spacetime resemble those in Newtonian gravity.

7.3.1 The Effective Potential

We don't need to solve the geodesic equations. Instead, we can invoke conserved quantities using Killing vectors. Since the metric tensor is independent of t and ϕ , there are two conserved quantities,

$$e \equiv -\xi \cdot u \quad \text{and} \quad L \equiv \eta \cdot u,$$

where $\xi^\mu = \delta^\mu_t$ and $\eta^\mu = \delta^\mu_\phi$. These inner products (in Schwarzschild geometry) are

$$e = \left(1 - \frac{2M}{r}\right) \frac{dt}{d\tau} \quad \text{and} \quad L = r^2 \sin^2 \theta \frac{d\phi}{d\tau}.$$

They are like conserved energy and conserved angular momentum per unit mass, and are integrals of motion for us. We argue that the particle should move in a plane, just like in Newtonian mechanics. It is always possible to orient the coordinates such that $\phi = 0$ and $u^\phi = d\phi/d\tau = 0$ at some instant. Then because $L = r^2 \sin^2 \theta u^\phi = 0$ is conserved, the particle will continue to move along $\phi = 0$ with $d\phi/d\tau = 0$. Thus, without loss of generality, we can choose a convenient coordinates such that the particle always move in the $\theta = \pi/2$ plane with $u^\theta = 0$. Now every $\sin \theta$ becomes a 1. In particular, $L = r^2 d\phi/d\tau$. The last integral of motion is $u \cdot u = -1$, which is true in all geometries,

$$-\left(1 - \frac{2M}{r}\right) \left(\frac{dt}{d\tau}\right)^2 + \left(1 - \frac{2M}{r}\right)^{-1} \left(\frac{dr}{d\tau}\right)^2 + r^2 \left(\frac{d\phi}{d\tau}\right)^2 = -1.$$

Multiplying both sides by $(1 - 2M/r)$ and plugging in the conserved quantities, we get an equation for $r(\tau)$,

$$-e^2 + \left(\frac{dr}{d\tau}\right)^2 + \left(1 + \frac{L^2}{r^2}\right) \left(1 - \frac{2M}{r}\right) = 0.$$

Multiplying both sides by $1/2$ and then minus $1/2$,

$$\frac{e^2 - 1}{2} = \frac{1}{2} \left(\frac{dr}{d\tau}\right)^2 + \frac{1}{2} \left(1 + \frac{L^2}{r^2}\right) \left(1 - \frac{2M}{r}\right) - \frac{1}{2} = \frac{1}{2} \left(\frac{dr}{d\tau}\right)^2 + \frac{L^2}{2r^2} - \frac{M}{r} - \frac{L^2 M}{r^3}.$$

Now the equation looks more familiar if we write

$$\boxed{\frac{1}{2} \left(\frac{dr}{d\tau}\right)^2 + V_{\text{eff}}(r) = \mathcal{E} \quad \text{where} \quad V_{\text{eff}}(r) = -\frac{M}{r} + \frac{L^2}{2r^2} - \frac{L^2 M}{r^3}, \quad \mathcal{E} \equiv \frac{e^2 - 1}{2}.} \quad (7.5)$$

The quantity $\mathcal{E}$ is not the total energy. If we write $e = 1 + E$, then E is the energy per unit mass without the rest mass energy, and

$$\mathcal{E} = \frac{e^2 - 1}{2} = \frac{E^2 - 2E}{2} = E + \frac{1}{2}E^2 \simeq E$$

for $E \ll 1$. This means if the total energy of the particle is much less than its rest mass energy, the quantity $\mathcal{E}$ is like the Newtonian-limit energy E_{Newt} . Observing the effective potential V_{eff} : the first term is the gravitational potential, while the second is the centrifugal barrier. If we were in Newtonian gravity, these two terms will produce orbits in conic sections. If the particle is too close to $r = 0$, the centrifugal barrier will prevent it from falling into the central object. One can find that there exists a stable equilibrium of $V_{\text{eff,Newton}}$ at $r = L^2/M$. However, in GR, there is a third, *negative* term that dominates at small r . It enhance the gravitational attraction of the central mass M . This means that if the particle is too close to the central object, it will definitely falls into it. Another difference of this equation of motion in GR from Newtonian mechanics is that the time-derivative is with respect to τ , but not t . Moreover, the coordinate radius r should not be taken as the real radial distance from the center to a sphere of radius r . Figure 7.1 contains V_{eff} in Newtonian and GR pictures for different parameters M and L that we shall analyze below.

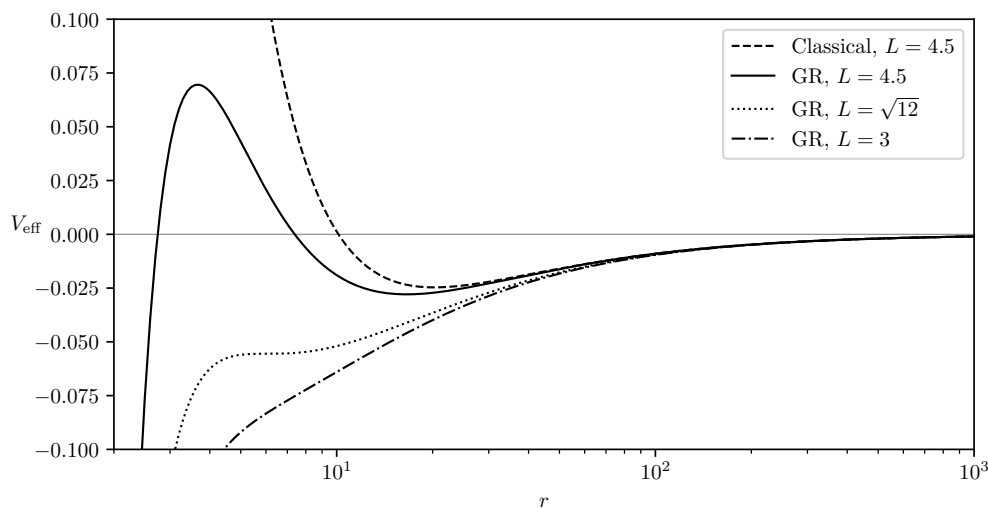


Figure 7.1: Newtonian and GR effective potential with $M = 1$. In Newtonian gravity, there always exist a stable equilibrium (minimum in V_{eff}) at which the orbit is circular. There is a centrifugal barrier. In GR, the shape of V_{eff} depends on the relative size of M and L . For $L^2 > 12M^2$, there exists both a minimum and a maximum. For $L^2 = 12M^2$, there is an innermost stable circular orbit at $r = 6M$. For $L^2 < 12M^2$, there are no stable circular orbits.

The Schwarzschild radius $r = 2M$ is not surprisingly also a special point for V_{eff} . At $r = 2M$,

$$V_{\text{eff}} = -\frac{M}{2M} + \frac{L^2}{8M^2} - \frac{ML^2}{8M^3} = -\frac{1}{2},$$

independent of L . Next, the maximum and minimum of V_{eff} are at

$$r_{\min} = \frac{L^2}{2M} \left[1 + \sqrt{1 - 12 \left(\frac{M}{L} \right)^2} \right] \quad \text{and} \quad r_{\max} = \frac{L^2}{2M} \left[1 - \sqrt{1 - 12 \left(\frac{M}{L} \right)^2} \right].$$

Note that $r_{\max}$ occurs closer to the central object. Because of the square root in $r_{\min}$ and $r_{\max}$, there are three cases of V_{eff} .

- $L^2 < 12M^2$: V_{eff} has no extrema. There is no circular orbits.
- $L^2 = 12M^2$: for this particular angular momentum, $r_{\min} = r_{\max} = 6M$. At this point, V_{eff} has a point of inflection, so $r = 6M$ is usually referred to as the **ISCO**, or **innermost stable circular orbit**.
- $L^2 > 12M^2$: $V_{\text{eff}}(r)$ has both a minimum and a maximum, with the maximum closer to the origin. If $L^2 = 16M^2$, then $r_{\max} = 4M$ and $V_{\text{eff}}(4M) = 0$. It turns out that for $L^2 < 16M^2$, the effective potential is negative everywhere. Even if a particle starts from rest at infinity ($e = 1$ or $\mathcal{E} = 0$), it will reach the central object. Since $L^2 = 12M^2$ sets the boundary of $r_{\min} = r_{\max} = 6M$, any $L^2 \geq 12M^2$ will have $r_{\min} \geq 6M$ and $r_{\max} \leq 6M$. As $L \rightarrow \infty$,

$$r_{\max} = \frac{L^2}{2M} \left[1 - \sqrt{1 - 12 \left(\frac{M}{L} \right)^2} \right] = \frac{L^2}{2M} \left[1 - 1 + 6 \left(\frac{M}{L} \right)^2 \right] = 3M.$$

Hence there is a lower bound for $r_{\max}$ at $3M$. The range of $r_{\max}$ is $3M < r_{\max} \leq 6M$.

7.3.2 Radial Plunge Orbits

The simplest “orbits” are orbits with zero angular momentum and the particle starts from rest at infinity. This implies $L = 0$, $e = 1$, and $\mathcal{E} = 0$. We would like to know how long does it take the particle to reach $r = 2M$. The energy equation (7.5) reads

$$\frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 - \frac{M}{r} = 0 \quad \Rightarrow \quad \frac{dr}{d\tau} = \pm \sqrt{\frac{2M}{r}}.$$

This looks like a particle in a Newtonian gravitational field, but note that t is replaced by τ . Since we are interested in inward-falling particles, we shall choose $u^r = dr/d\tau = -\sqrt{2M/r}$. The solution is

$$\tau(r) = \tau_* - \frac{2}{3} \sqrt{\frac{r^3}{2M}} \iff r(\tau) = \left(\frac{9M}{2}\right)^{1/3} (\tau_* - \tau)^{2/3}, \quad (7.6)$$

where τ_* is some integration constant. It takes infinite proper time for the particle to reach coordinate radius r if it starts at infinity because $\tau \rightarrow -\infty$ as $r \rightarrow \infty$. If the particle moves from a finite r_i to another finite r_f , the proper time elapsed is finite. This is important because it holds true even when $r_f = 2M$.

For coordinate time t , things get more complicated. From the definition of the conserved quantity e (which we define to be 1 in Plunge orbits),

$$e = 1 = \left(1 - \frac{2M}{r}\right) \frac{dt}{d\tau} = \left(1 - \frac{2M}{r}\right) \frac{dt}{dr} \frac{dr}{d\tau} = -\sqrt{\frac{2M}{r}} \left(1 - \frac{2M}{r}\right) \frac{dt}{dr}.$$

The resulting differential equation is

$$\frac{dt}{dr} = -\left(\frac{2M}{r}\right)^{-1/2} \left(1 - \frac{2M}{r}\right)^{-1}.$$

The solution is

$$t = t_* + 2M \left[-\frac{2}{3} \left(\frac{r}{2M}\right)^{3/2} - 2 \left(\frac{r}{2M}\right)^{1/2} + \ln \left| \frac{(r/2M)^{1/2} + 1}{(r/2M)^{1/2} - 1} \right| \right], \quad (7.7)$$

where t_* is another integration constant. Again, it takes infinite coordinate time for the particle to reach r if it starts at infinity, $t \rightarrow -\infty$ as $r \rightarrow \infty$. However, it also takes infinite coordinate time for the particle to reach $r = 2M$ from any finite radius because the logarithm blows up at $r \rightarrow 2M$. This is another special property of the Schwarzschild radius.

Example 7.1. Speed of a particle measured by a static observer

We know the particle's 4-velocity in a radial plunge orbit,

$$u = \left(\frac{dt}{d\tau}, \frac{dr}{d\tau}, 0, 0 \right) = \left(\left[1 - \frac{2M}{r} \right]^{-1}, -\sqrt{\frac{2M}{r}}, 0, 0 \right), \quad u \cdot u = -1.$$

For the observer, construct their orthonormal basis $\{e'_\mu\}$. The timelike basis vector is the 4-velocity of the observer, u_{ob} . The only nonzero component is u_{ob}^t ,

$$g_{tt}(u_{\text{ob}}^t)^2 = -1 \implies u_{\text{ob}}^t = \left(1 - \frac{2M}{r}\right)^{-1/2} = (e'_t)^t.$$

(The notation $(e'_t)^t$ means the time component of the observer's timelike basis vector e'_t in the coordinate basis.) We can construct other orthonormal basis. The four orthonormal basis vectors are

$$e'_t = \left(1 - \frac{2M}{r}\right)^{-1/2} e_t, \quad e'_r = \left(1 - \frac{2M}{r}\right)^{1/2} e_r, \quad e'_\theta = \frac{1}{r} e_\theta, \quad e'_\phi = \frac{1}{r \sin \theta} e_\phi. \quad (7.8)$$

It is straightforward to check that $e'_\mu \cdot e'_\nu = \eta_{\mu\nu}$. The velocity of the particle measured by the observer is then

$$v = \frac{dr'}{dt'} = \frac{dr'/d\tau'}{dt'/d\tau'} = \frac{u \cdot e'_r}{-u \cdot e'_t} = \frac{g_{rr} u^r (e'_r)^r}{-g_{tt} u^t (e'_t)^t} = \frac{(1 - 2M/r)^{-1} (-\sqrt{2M/r}) \sqrt{1 - 2M/r}}{-(2M/r - 1)(1 - 2M/r)^{-1}(1 - 2M/r)^{-1/2}} = -\sqrt{\frac{2M}{r}}.$$

Here the primed quantities $dr'/d\tau'$ belong to the particle, but measured in the observer's frame. The radial velocity is negative because it is falling inward. In fact, recall that the 4-velocity of a particle in an orthonormal basis is just $u = (\gamma, \gamma v, 0, 0)$. Thus, the Lorentz factor is $\gamma = -u \cdot e'_t = (1 - 2M/r)^{-1/2} = 1/\sqrt{1 - v^2}$. These are all consistent in a locally Minkowski spacetime (the observer's frame). To perceive this velocity, we turn the particle to an outgoing particle. Because the particle reaches infinity with zero velocity, $v = \sqrt{2M/r}$ is the *escape velocity*. Note that this is identical to the Newtonian limit, but with real radius replaced by the Schwarzschild coordinate radius r .

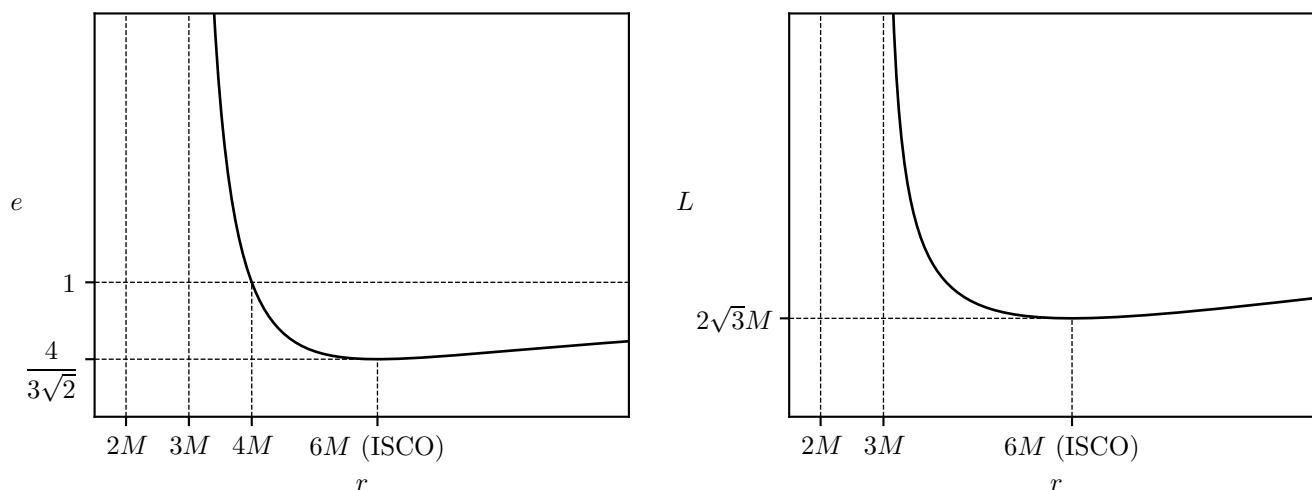


Figure 7.2: A plot of $e(r)$ and $L(r)$. There are several important radii on each graph.

For a particle at infinity starting with a generic e , its radial velocity measured by a stationary observer at coordinate radius r is

$$v = -\frac{1}{e} \sqrt{e^2 - 1 + \frac{2M}{r}}.$$

This reduces to the result above if we plug in $e = 1$.

7.3.3 Circular Orbits

Circular orbits occur at $dV_{\text{eff}}/dr = 0$ only when $L^2 \geq 12M^2$, which gives an equation and the solution for the angular momentum L :

$$Mr^2 - L^2r + 3ML^2 = 0 \implies L = \pm r \sqrt{\frac{M}{r - 3M}}.$$

We will take the positive solution; the particle is moving in counterclockwise direction.

By the energy equation (7.5), we can calculate the energy e for a circular orbit ($dr/d\tau = 0$):

$$\frac{e^2 - 1}{2} = \frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 + \left(-\frac{M}{r} + \frac{L^2}{2r^2} - \frac{ML^2}{r^3} \right) \implies e = \frac{r - 2M}{\sqrt{r(r - 3M)}}.$$

From Figure 7.2, there are two possible circular orbits for every $L > 2\sqrt{3}M$. The unstable orbit is inside $r = 6M$, while the stable one is outside $r = 6M$. There is no circular orbits inside $r = 3M$, no matter how large the angular momentum is. We will see why in later sections. From the graph of $e(r)$, we see that the ISCO at $r = 6M$ is the most tightly bound orbit because e is at minimum. The energy (per unit mass) is $e = 4/3\sqrt{2} \approx 0.9428$.

Example 7.2. Brightest source in the universe

In black hole physics, the ISCO is an important radius. It is around where the accretion disk is formed. The energy per unit mass $e \approx 0.9428$ means that if an object starts at rest and come to the ISCO, it will release an per unit mass of $1 - e \approx 0.057$. Here 1 (or mc^2/m) can be seen as the rest mass energy per unit mass. Thus, an object reaching the ISCO releases the equivalent of 5.7% of its rest mass energy before falling into the black hole since there is no stable circular orbits inward. 5.7% does not sound much, but it is an enormous amount of energy. The nuclear fusion rate of hydrogen into helium (the current power source of the Sun) only releases 0.7% of the rest mass energy of the reactants. Moreover, if the black hole is a Kerr black hole (a spinning one), the object will release as high as 42% of its rest mass energy. Therefore, the accretion disk around a black hole

is one of the most luminous source in the universe.

Example 7.3. Period of a circular orbit

For a circular orbit,

$$e = \left(1 - \frac{2M}{r}\right) \frac{dt}{d\tau} = \frac{(r-2M)}{\sqrt{r(r-3M)}}, \quad L = r^2 \frac{d\phi}{d\tau} = r \sqrt{\frac{M}{r-3M}}. \quad (7.9)$$

We would like to know the period of a particle revolving around the central object at radius r . There are at least three interpretation of this period: the period in coordinate time measured by a static observer at infinity, the period in the proper time of the particle, and the period measured by a static observer at radius r in the gravitational field. We first consider the coordinate time period T_t . The coordinate angular velocity (the angular velocity measured by a static observer at infinity) and the coordinate time period are

$$\omega_t = \frac{d\phi}{dt} = \frac{d\phi/d\tau}{dt/d\tau} = \frac{L/r^2}{e/(1-2M/r)} = \sqrt{\frac{M}{r^3}} \implies T_t = 2\pi r \sqrt{\frac{r}{M}}.$$

The period measured by the particle gives the proper time period,

$$T_\tau = \frac{2\pi}{\omega_\tau} = \frac{2\pi}{d\phi/d\tau} = \frac{2\pi}{L/r^2} = 2\pi r \sqrt{\frac{r-3M}{M}}.$$

The period measured by a static observer at r can be calculated from the coordinate time period by gravitational time dilation because a circular orbit with fixed speed is a perfect clock:

$$d\tau_{\text{ob}} = \sqrt{g_{tt}} dt \implies T_{\text{ob}} = \sqrt{g_{tt}} T_t = 2\pi r \sqrt{\frac{r-2M}{M}}.$$

Alternatively, we can compute the speed of the particle in the observer's frame. The 4-velocity of the particle in its circular orbit is

$$u = \left(\frac{dt}{d\tau}, 0, 0, \frac{d\phi}{d\tau}\right) = \left(\frac{e}{1-2M/r}, 0, 0, \frac{L}{r^2}\right).$$

Using the same approach as in Example 7.1, the period of the particle measured by the observer is

$$v = \left| \frac{u \cdot e'_\phi}{-u \cdot e'_t} \right| = \left| \frac{g_{\phi\phi}(d\phi/d\tau)(g_{\phi\phi})^{-1/2}}{-g_{tt}(dt/d\tau)(-g_{tt})^{-1/2}} \right| = \sqrt{\frac{M}{r-2M}} \implies T_{\text{ob}} = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{r-2M}{M}}.$$

In fact, we can compute the Lorentz factor and find the desired relation between T_{ob} and T_τ :

$$\gamma = \frac{1}{\sqrt{1-v^2}} = \sqrt{\frac{r-2M}{r-3M}}, \quad T_{\text{ob}} = \gamma T_\tau.$$

The hierarchy of the three periods is $T_t > T_{\text{ob}} > T_\tau$. The first inequality comes from gravitational time dilation, while the second comes from special relativistic time dilation. The particle's proper time for one revolution is the extremal (in this case minimum) period because it is a free particle in Schwarzschild spacetime.

7.3.4 Elliptical Orbits and Precession

There are only two potentials for which all bound orbits are closed: Newtonian $-1/r$ potential and the harmonic oscillator potential r^2 . These bound orbits are ellipses. In relativity, the $1/r$ potential no longer give rise to a closed orbit in general. In Schwarzschild spacetime, it is mainly because of the $-ML^2/r^3$ term in the effective potential. Hence a particle following an elliptical orbit does not have a closed track, but instead precesses in the same direction as the orbital direction. This is known as **apsidal precession**, or **precession of the perihelion** for planets. The apsidal angle $\delta\phi$ is defined as $\Delta\phi - 2\pi$, where $\Delta\phi$ is the angular increment of the particle over one revolution. (If there is no precession, then $\Delta\phi = 2\pi$, the definition of one revolution.) To find $\delta\phi$, we compute

$$\frac{d\phi}{dr} = \frac{d\phi/d\tau}{dr/d\tau} = \frac{L/r^2}{\pm\sqrt{e^2 - 1 - 2V_{\text{eff}}}} = \pm \frac{L}{r^2} \left(e^2 - 1 + \frac{2M}{r} - \frac{L^2}{r^2} + \frac{2ML^2}{r^3} \right)^{-1/2},$$

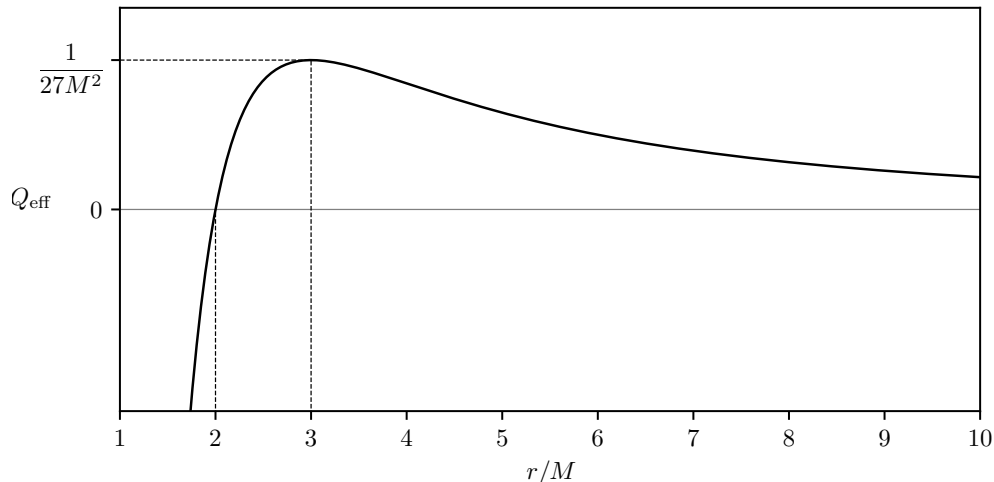


Figure 7.3: Q_{eff} as a function of r . There is only one unstable circular orbit at $3M$.

where we replace $dr/d\tau$ by (7.5). Then over one revolution,

$$\Delta\phi = 2 \int_{r_1}^{r_2} \frac{d\phi}{dr} dr = 2L \int_{r_1}^{r_2} \frac{dr}{r^2} \left(e^2 - 1 + \frac{2M}{r} - \frac{L^2}{r^2} + \frac{2ML^2}{r^3} \right)^{-1/2}.$$

Here r_1 and r_2 are periapsis and apoapsis where $dr/d\tau = 0$. To lowest order of relativistic effects, we will find (see derivation in Appendix A.5.1)

$$\Delta\phi = 2\pi + \delta\phi, \quad \delta\phi \simeq \frac{6\pi M^2}{L^2} = \frac{6\pi M}{a(1-\epsilon^2)}, \quad (7.10)$$

where ϵ is the eccentricity of the ellipse, defined as $\epsilon = \sqrt{1 - b^2/a^2}$ where b is the semiminor axis and a is the semimajor axis. The apsidal angle is large for small semimajor axis because the particle experience stronger gravity. It is also large for large eccentricity because the periapsis is in region with stronger gravity.

7.3.5 Photon Orbits

For photon orbits, the conserved quantities are similar:

$$e = -\xi \cdot u = \left(1 - \frac{2M}{r}\right) \frac{dt}{d\lambda}, \quad L = \eta \cdot u = r^2 \frac{d\phi}{d\lambda}, \quad u \cdot u = 0,$$

where again we are considering photon orbit in a plane, so $\sin\theta = 1$. Note that all τ is replaced by the affine parameter λ , and $u \cdot u$ is null instead of -1 . The equation $u \cdot u = 0$ gives

$$0 = -\left(1 - \frac{2M}{r}\right) \left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{2M}{r}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + r^2 \left(\frac{d\phi}{d\lambda}\right)^2 = -\left(1 - \frac{2M}{r}\right)^{-1} e^2 + \left(1 - \frac{2M}{r}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + \frac{L^2}{r^2},$$

or

$$\boxed{\frac{1}{L^2} \left(\frac{dr}{d\lambda}\right)^2 + Q_{\text{eff}}(r) = \frac{1}{b^2} \quad \text{where} \quad b = \frac{L}{e}, \quad Q_{\text{eff}}(r) = \frac{1}{r^2} \left(1 - \frac{2M}{r}\right).} \quad (7.11)$$

Here Q_{eff} is like an effective potential. The shape of Q_{eff} is shown in Figure 7.3. Note that Q_{eff} approaches 0 from above as $r \rightarrow \infty$. It is zero at $r = 2M$, and approaches $-\infty$ as $r \rightarrow 0$. Recall that for material particles, we arrived at the conclusion that there is no circular orbit at or below $r = 3M$ no matter how large their angular momentum is. This is because the circular orbit at $r = 3M$ is for photons. A material particle must travel faster than the speed of light to have circular orbit at $r < 3M$, since,

$$v = \sqrt{\frac{M}{r-2M}} \rightarrow 1 \quad \text{as} \quad r \rightarrow 3M.$$

Now consider a photon approaching the central object from infinity. If $1/b^2 < Q_{\text{eff}}(3M) = 1/27M^2$, then it will reach a minimum radius and travel to infinity. If $1/b^2 > Q_{\text{eff}}(3M)$, then it will overcome the centrifugal barrier and fall into the object. The quantity

b has a more physical meaning. At infinity, the spacetime is flat, so

$$b = \frac{L}{e} = r^2 \left(1 - \frac{2M}{r}\right)^{-1} \frac{d\phi}{dt} \rightarrow r^2 \frac{d\phi}{dt}.$$

In the xy -plane ($\theta = \pi/2$ plane), $y = r \sin \phi \simeq r\phi$ as $r \rightarrow \infty$ or $\phi \rightarrow 0$, where y is some small distance compared to r .

$$\phi = \frac{y}{r}, \quad \frac{d\phi}{dr} = -\frac{y}{r^2} \implies b = r^2 \left(-\frac{y}{r^2}\right) \frac{dr}{dt} = -y \frac{dr}{dt}.$$

But again, at infinity, $dr/dt = -1$, which means $b = y$ is the **impact parameter** of the photon (see Figure 7.4).

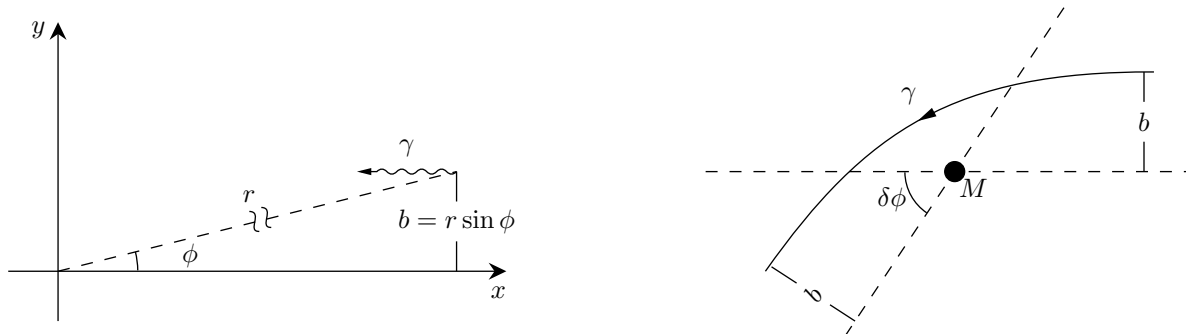


Figure 7.4: Interpretation of b as the impact parameter for a photon from infinity (left) and gravitational light bending (right).

Because any photon with the impact parameter $b_{\text{capture}} < 3\sqrt{3}M$ will be captured by the central mass, there is a capture cross-section:

$$\sigma = \pi b_{\text{capture}}^2 = 27\pi M^2. \quad (7.12)$$

7.3.6 Gravitational Lensing II: Schwarzschild Solution

Consider a photon coming from infinity with an impact parameter $b \gg 3\sqrt{3}M$ so that its trajectory is weakly bent. We want to calculate the deflection angle $\delta\phi$ (see Figure 7.4, right). From (7.11),

$$\frac{1}{b^2} = \frac{1}{L^2} \left(\frac{dr}{d\lambda}\right)^2 + \frac{1}{r^2} \left(1 - \frac{2M}{r}\right) = \frac{1}{r^4} \left(\frac{dr}{d\phi}\right)^2 + \frac{1}{r^2} \left(1 - \frac{2M}{r}\right).$$

Let $u \equiv M/r$, $dr = -M du/u^2$,

$$\frac{1}{b^2} = \frac{u^4}{M^4} \frac{M^2}{u^4} \left(\frac{du}{d\phi}\right)^2 + \frac{u^2}{M^2} (1 - 2u) \implies \left(\frac{du}{d\phi}\right)^2 + u^2 - \frac{M^2}{b^2} = 2u^3.$$

Note that if $b \gg 3\sqrt{3}M$, the closest approach of the photon to the central mass is $r_1 \sim b$, or $u_{\text{max}} \sim M/b \ll 1$ at the point of closest approach. The term $2u^3$ on the RHS is third order in u , which is much smaller than terms on the LHS. Neglecting $2u^3$, we have a harmonic oscillator equation,

$$\left(\frac{du}{d\phi}\right)^2 + u^2 \simeq \frac{M^2}{b^2}.$$

The general solution is $u = A \cos \phi + B \sin \phi$. For a finite impact parameter b , we have $u = M/r \rightarrow 0$ as $r \rightarrow \infty$ and $\phi \rightarrow 0$. This forces $A = 0$. Substituting $u = B \sin \phi$ back into the equation gives $B = M/b$. It turns out that we will have the trivial (or classical) result:

$$u = \frac{M}{r} = \frac{M}{b} \sin \phi \implies b = r \sin \phi = \text{const.}$$

This makes sense because the term we neglected $2u^3 \propto 1/r^3$ is the one arising from general relativity. Neglecting it means returning back to classical orbital mechanics. Now we should include $2u^3$, but keeping it small as a perturbation:

$$\left(\frac{du}{d\phi}\right)^2 + u^2 - \frac{M^2}{b^2} = 2u^3. \quad (7.13)$$

We expect the solution be perturbed by a small amount $u_1 \ll M/b$ so that $u = (M/b) \sin \phi + u_1$. Plugging this into (7.13), and keeping terms up to linear order in u_1 on the LHS but zeroth order on the RHS because $2u^3$ is already small,

$$\frac{M^2}{b^2} \cos^2 \phi + \frac{2M}{b} \cos \phi \frac{du_1}{d\phi} + \frac{M^2}{b^2} \sin^2 \phi + u_1 \frac{2M}{b} \sin \phi - \frac{M^2}{b^2} = 2 \frac{M^3}{b^3} \sin^3 \phi \implies \cos \phi \frac{du_1}{d\phi} + u_1 \sin \phi = \frac{M^2}{b^2} \sin^3 \phi.$$

To solve this equation, we write the LHS as a total derivative,

$$\cos^2 \phi \frac{d}{d\phi} \left(\frac{u_1}{\cos \phi} \right) = \frac{M^2}{b^2} \sin^3 \phi \implies \frac{u_1}{\cos \phi} = \frac{M^2}{b^2} \int \tan^2 \phi \sin \phi d\phi = \frac{M^2}{b^2} \left(\frac{1}{\cos \phi} + \cos \phi \right) + C.$$

The initial condition $u_1 = 0$ as $r \rightarrow \infty$ and $\phi \rightarrow 0$ requires $C = -2M^2/b^2$. The final solution of u is

$$u = \frac{M}{b} \sin \phi + \frac{M^2}{b^2} (1 - \cos \phi)^2 + \mathcal{O} \left(\frac{M^3}{b^3} \right). \quad (7.14)$$

Now, $u \rightarrow 0$ ($r \rightarrow \infty$) should also be satisfied as $\phi \rightarrow \pi + \delta\phi$ (see Figure 7.4, right) after it bypasses the mass. In the weak bending limit $\delta\phi \ll 1$,

$$0 = \frac{M}{b} \sin(\pi + \delta\phi) + \frac{M^2}{b^2} [1 - \cos(\pi + \delta\phi)]^2 \simeq -\frac{M}{b} \delta\phi + \frac{4M^2}{b^2},$$

or

$$\boxed{\delta\phi = \frac{4M}{b}}. \quad (7.15)$$

The classical (exact) deflection angle of a *material* particle passing through a $1/r$ potential such as in Rutherford scattering or in a gravitational potential $\Phi = -GM/r$,

$$\delta\phi = 2 \tan^{-1} \left(\frac{GM}{v_0^2 b} \right) \rightarrow \frac{2GM}{v_0^2 b}$$

for large b , where v_0 is the speed at infinity. It seems reasonable to put $v_0 = c$ for photon, but it is only half the value obtained in general relativity. Thus, gravitational light bending is an excellent test of GR. In fact, it was a test of gravitational lensing in 1919 that made Einstein and his theory of general relativity famous.

Example 7.4. Gravitational lensing: a Schwarzschild lens

Consider the **Schwarzschild lens**, where a background point source at large impact parameter b is lensed by an isolated mass M . The lensing event always has the source and lens close together on the sky. From Figure (7.5), there are many angles related:

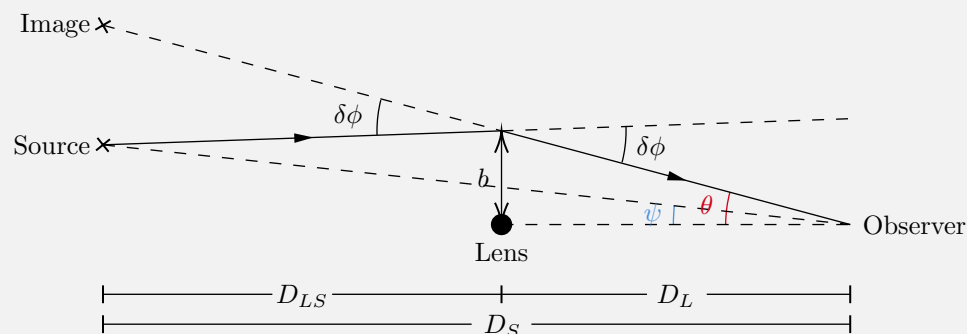


Figure 7.5: An illustration of gravitational lensing. The trajectory of photon can be approximated as straight lines because the size of the lens is small compared to D_S , just like in geometric optics. Note that θ is not necessarily equal to $\delta\phi$.

The angle ψ is the true angular separation, while θ is the observed angular separation between the source and the lens. The bending angle $\delta\phi$ are related to ψ and θ . In small angle approximation, the source-image distance in the source plane is $\delta\phi D_{LS} = (\theta - \psi) D_S$. This provides a relation between ψ , θ , and $\delta\phi$,

$$\psi = \theta - \frac{D_{LS}}{D_S} \delta\phi = \theta - \frac{D_{LS}}{D_S D_L} \frac{4M}{\theta},$$

where we used the fact that $b = D_L \theta$ and (7.15). We can rewrite this equation as

$$\theta^2 - \psi\theta - \theta_E^2 = 0, \quad \text{where} \quad \theta_E \equiv \sqrt{\frac{4MD_{LS}}{D_S D_L}}.$$

This θ_E is known as the **Einstein radius**. Its physical meaning is the following: if a source is *directly* behind the lens so that $\psi = 0$, the source will be lensed into a ring around the lens with angular radius $\theta = \theta_E$. This is known as an **Einstein ring**, one of the most spectacular observation and test of GR. If $\psi \neq 0$, we need to solve the quadratic equation above,

$$\theta_{\pm} = \frac{1}{2} \left(\psi \pm \sqrt{\psi^2 + 4\theta_E^2} \right).$$

Both θ and θ_E are observable quantities, so the true angular separation can be found from the solution $\theta_{\pm}$.

7.3.7 The Shapiro Delay

Returning to (7.11),

$$\frac{1}{b^2} = \frac{1}{L^2} \left(\frac{dr}{d\lambda} \right)^2 + Q_{\text{eff}}(r), \quad \text{where} \quad Q_{\text{eff}}(r) = \frac{1}{r^2} \left(1 - \frac{2M}{r} \right).$$

This time we change $dr/d\lambda$ to dr/dt using the definition of e for photon,

$$\frac{1}{b^2} = \frac{e^2}{L^2} \left(1 - \frac{2M}{r} \right)^{-2} \left(\frac{dr}{dt} \right)^2 + Q_{\text{eff}}(r) = \frac{1}{b^2} \left(1 - \frac{2M}{r} \right)^{-2} \left(\frac{dr}{dt} \right)^2 + Q_{\text{eff}}(r).$$

Solving for dt/dr ,

$$\frac{dt}{dr} = \pm \frac{1}{b} \left(1 - \frac{2M}{r} \right)^{-1} \left(\frac{1}{b^2} - Q_{\text{eff}} \right)^{-1/2} \begin{cases} +, & \text{for outward motion,} \\ -, & \text{for inward motion.} \end{cases}$$

Let r_1 be the radius of closest approach, so

$$\frac{1}{b^2} = Q_{\text{eff}}(r_1) = \frac{1}{r_1^2} \left(1 - \frac{2M}{r_1} \right).$$

When the photon travels from r_1 to $r > r_1$, the Schwarzschild coordinate time elapsed is

$$t(r, r_1) = \int_{r_1}^r dr' \frac{1}{b} \left(1 - \frac{2M}{r'} \right)^{-1} \left[\frac{1}{b^2} - Q_{\text{eff}}(r') \right]^{-1/2}.$$

To first order in M/r , this integral (see derivation in Appendix A.5.2) becomes

$$t(r, r_1) = \sqrt{r^2 - r_1^2} + 2M \ln \left(\frac{r + \sqrt{r^2 - r_1^2}}{r_1} \right) + M \left(\frac{r - r_1}{r + r_1} \right)^{1/2}. \quad (7.16)$$

The first term is the classical time elapsed (with $c = 1$) by the Pythagorean theorem. The other two terms are all positive, meaning that the actual travel time of the photon is longer than classical prediction. This effect is called the **Shapiro Delay**.

7.4 Static, Spherically Symmetric Stars

In our universe, a star is a ball of hot gas supported by very stable nuclear fusion. The lifetime of a main-sequence star (a stage where the hydrogen is fused into helium) varies from millions to billions of years. Thus, during the main-sequence and most of other stages, a star is stable and can be approximated as a static, spherically symmetric perfect fluid. The exterior solution of such a star is described by the Schwarzschild spacetime. In this section, we will investigate the interior solutions of a static, spherically symmetric star. Then we will discuss gravitational collapses and the fate of stars depending on their masses.

7.4.1 Interior Solutions

The general static, spherically symmetric metric is given by (7.1),

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2.$$

We want to solve for $\alpha(r)$ and $\beta(r)$ using non-vacuum Einstein's equation:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi T_{\mu\nu}, \quad \text{with} \quad T_{\mu\nu} = (\rho + P) u_\mu u_\nu + P g_{\mu\nu},$$

where ρ is the energy density and P is the pressure inside the star. Both of them should depend only on the radial coordinate r . The Ricci tensor and Ricci scalar are known, referring to (7.2) and (7.3). We can compute the Einstein tensor,

$$\begin{aligned} G_{tt} &= \frac{1}{r^2} e^{2(\alpha-\beta)} (2r\partial_r\beta - 1 + e^{2\beta}) \\ G_{rr} &= \frac{1}{r^2} (2r\partial_r\alpha + 1 - e^{2\beta}) \\ G_{\theta\theta} &= r^2 e^{-2\beta} \left[\partial_r^2 \alpha + (\partial_r\alpha)^2 - \partial_r\alpha\partial_r\beta + \frac{1}{r}(\partial_r\alpha - \partial_r\beta) \right] \\ G_{\phi\phi} &= G_{\theta\theta} \sin^2\theta. \end{aligned} \tag{7.17}$$

Since the star is static, the 4-velocity u^μ is expected to have only the time component. Being a unit vector, $g_{\mu\nu}u^\mu u^\nu = -1$, its component should be $u^\mu = (e^\alpha, 0, 0, 0)$. With this 4-velocity, the stress-energy tensor takes the form

$$T_{\mu\nu} = \begin{bmatrix} \rho e^{2\alpha} & & & \\ & P e^{2\beta} & & \\ & & P r^2 & \\ & & & P r^2 \sin^2\theta \end{bmatrix}. \tag{7.18}$$

Then the tt -, rr -, and $\theta\theta$ -components of Einstein's equation are

$$\begin{cases} tt\text{-component} : & \frac{1}{r^2} e^{-2\beta} (2r\partial_r\beta - 1 + e^{2\beta}) = 8\pi\rho, \\ rr\text{-component} : & \frac{1}{r^2} e^{-2\beta} (2r\partial_r\alpha + 1 - e^{2\beta}) = 8\pi P, \\ \theta\theta\text{-component} : & e^{-2\beta} \left[\partial_r^2 \alpha + (\partial_r\alpha)^2 - \partial_r\alpha\partial_r\beta + \frac{1}{r}(\partial_r\alpha - \partial_r\beta) \right] = 8\pi P. \end{cases}$$

The $\phi\phi$ -equation is the same as the $\theta\theta$ -equation, so there are in total three equations. The tt -equation can be solved directly as it involves only ρ and β . Introduce a new function $m(r)$ that is related to $\beta(r)$ by

$$m(r) = \frac{r}{2} (1 - e^{-2\beta}), \quad e^{2\beta} = \left(1 - \frac{2m}{r}\right)^{-1}, \quad \partial_r\beta = \frac{1}{r} \left(1 - \frac{2m}{r}\right)^{-1} \left(\frac{dm}{dr} - \frac{2m}{r}\right).$$

From $e^{2\beta} = (1 - 2m/r)^{-1}$, we see that this is just a generalization of the Schwarzschild metric, with $m(r)$ now as a function of r . The tt -equation in terms of m is simply

$$\frac{dm}{dr} = 4\pi r^2 \rho \quad \implies \quad m(r) = 4\pi \int_0^r \rho(r') r'^2 dr',$$

If we wish to describe the exterior geometry of a star of radius R as Schwarzschild, the Schwarzschild mass should be

$$M = m(R) = 4\pi \int_0^R \rho(r) r^2 dr.$$

This should look familiar at a first glance—it seems to be just the Newtonian definition of a mass inside a spherically symmetric object. However, note that ρ here is the energy density rather than mass density. More subtly, even if ρ is the energy density, the Schwarzschild mass is still not the total energy density inside the star. This is because in general relativity, the proper volume is $\sqrt{-g} d^4x$ instead of d^4x . If we evaluate the total energy density at a specific time slice, then the spatial metric is

$$\gamma_{ij} dx^i dx^j = e^{2\beta} dr^2 + r^2 d\Omega^2,$$

which gives the proper volume element $\sqrt{\gamma} d^3x = e^\beta r^2 dr d\Omega$. The total energy density is then

$$\bar{M} = 4\pi \int_0^R \rho(r) r^2 e^{\beta(r)} dr = 4\pi \int_0^R \rho(r) r^2 \left[1 - \frac{2m(r)}{r} \right]^{-1/2} dr.$$

Note that $M < \bar{M}$ as $e^{\beta(r)} > 1$. The Schwarzschild mass M is the correct result, not the total energy density $\bar{M}$. An interpretation of this is that the Schwarzschild mass accounts for the gravitational binding energy $E_{\text{bind}} = M - \bar{M}$ within the stars, which is negative (since gravity is attractive). When we observe a star far away, its total mass appears to be less than the sum of the mass of its constituents because of this binding energy.

Next, the rr -equation in terms of $m = r(1 - e^{-2\beta})/2$ is

$$\frac{d\alpha}{dr} = \frac{m + 4\pi r^3 P}{r(r - 2m)}.$$

The $\theta\theta$ -equation from Einstein's equation is complicated, so it is easier to use the conservation of stress-energy tensor instead, $\nabla_\mu T^{\mu\nu} = 0$. Appendix A.5.3 shows that only the $\nu = r$ equation is nontrivial, which is

$$(\rho + P) \frac{d\alpha}{dr} = -\frac{dP}{dr}.$$

Combining this with the rr -equation gives

$$\frac{dP}{dr} = -\frac{(\rho + P)(m + 4\pi r^3 P)}{r(r - 2m)}. \quad (7.19)$$

This equation is known as the **Tolman-Oppenheimer-Volkoff (TOV) equation**. In the Newtonian limit, $P \ll \rho$ (low speed) and $r \gg 2m$ (not compact objects), the TOV equation reduces to

$$\frac{dP}{dr} = -\frac{\rho m}{r^2},$$

the equation of hydrostatic equilibrium in ordinary astrophysics. The TOV equation relates the pressure P to the energy density ρ . To solve it, we need another equation that relates the two: the equation of state. The equations of state vary from models to models. The simplest example is an incompressible star with uniform density ρ_* .

Example 7.5. Schwarzschild's uniform density star

Consider a star of radius R with uniform density ρ_* ,

$$\rho(r) = \begin{cases} \rho_*, & r < R, \\ 0, & r > R, \end{cases} \quad \text{and} \quad m(r) = \begin{cases} \frac{4}{3}\pi r^3 \rho_*, & r < R, \\ \frac{4}{3}\pi R^3 \rho_* \equiv M, & r > R. \end{cases}$$

Plugging in the mass $m(r)$ and density $\rho(r)$ into the TOV equation (7.19) and solving for the pressure, we obtain

$$P(r) = \rho_* \left[\frac{R(R - 2M)^{1/2} - (R^3 - 2Mr^2)^{1/2}}{(R^3 - 2Mr^2)^{1/2} - 3R(R - 2M)^{1/2}} \right].$$

With the pressure and density, we can solve for $\alpha(r)$ and thus $g_{tt} = -e^{2\alpha(r)}$ using the rr -equation,

$$e^{\alpha(r)} = \frac{3}{2} \left(1 - \frac{2M}{R} \right)^{1/2} - \frac{1}{2} \left(1 - \frac{2Mr^2}{R^3} \right)^{1/2}, \quad [r < R].$$

The pressure $P(r)$ is monotonically decreasing as r is decreasing, which is expected for stars. Note that the pressure at the center $P(0)$ blows up for a star of radius R if the total mass is large enough. The maximum mass for such a star to reach hydrostatic equilibrium is

$$M_{\text{max}} = \frac{4}{9}R.$$

If the mass of the star exceeds this value, the pressure at the center needs to be greater than infinity to support the gravitational collapse. Then the star is not static anymore. Historically, this limit (with the assumption of constant density) is called **Schwarzschild's limit**. Later, the limit is generalized in **Buchdahl's theorem**.

7.4.2 Buchdahl's Theorem

Theorem 7.2. Buchdahl's Theorem

Any static, spherically symmetric perfect fluid of mass M , radius R , and a non-increasing density $\rho'(r) \leq 0$, must satisfy $M < 4R/9$, regardless of the equation of state.

Proof. To be written. □

7.4.3 Realistic Stars

Stars have masses ranging from $0.08 M_\odot$ to $\sim 10^2 M_\odot$. The fate of a star depends on the mass that is left over after nuclear burning. When the nuclear fuel is used up, the temperature of the star decreases and the pressure cannot resist gravity.

A typical star with a mass comparable to $M_\odot$ will collapse to an object called a **white dwarf**. A white dwarf is supported by the **electron degeneracy pressure**. According to the Pauli exclusion principle in quantum mechanics, electrons (which are fermions) cannot occupy the same state (which is called degenerate). When they are compressed so close together that they need to occupy the same state, the degeneracy pressure will prevent the collapse. If the electrons in the white dwarf is non-relativistic, the degeneracy pressure scales as $\rho^{5/3}$. If the electrons are relativistic, the degeneracy pressure scales as $\rho^{4/3}$. A typical white dwarf has a size comparable to $R_\oplus$, the size of the Earth, but with a mass of $\sim M_\odot$, so white dwarfs are very dense objects.

There is an upper limit of the mass of a white dwarf called the **Chandrasekhar limit**, $M_{\text{Ch}} \approx 1.4 M_\odot$. If a white dwarf has a mass higher than M_{Ch} , the electron degeneracy pressure cannot resist gravity again. Then the gravitational collapse will combine protons and electrons to make neutrons and neutrinos through inverse beta decay,

$$p + e^- \rightarrow n + \nu_e.$$

The resulting object is called a **neutron star**. Its supporting mechanism is analogous to that of a white dwarf. An equilibrium is established by the **neutron degeneracy pressure** and gravity. A neutron star has a typical radius of 10 km and a mass below $3 M_\odot$, so it is even denser than a white dwarf and is the most compact object known in our universe (except black holes). Because a neutron star has a radius of the same order as its mass, $R \gtrsim 2M$, it is in the strong-field limit of general relativity. The precise equation of state and the mass limit of a neutron star are yet unknown because of the interplay of quantum mechanics and general relativistic effect.

It is believed that if a neutron star exceeds its maximum allowed mass limit, the gravitational collapse will be inevitable and will result in a stellar-mass black hole. There are other types of black holes in our universe. It is observed that there are **supermassive black holes** ($M \sim 10^6$ - $10^9 M_\odot$) at the centers of galaxies. Their formation is not well understood. Another type of black holes is called **primordial black holes**, usually with a mass of $\sim 10^3 M_\odot$. They are formed in the very early universe, but are not yet observed today. Black holes (especially the small ones) are in the strong-field limit of general relativity. They are discussed in detail in the next chapter.

8 BLACK HOLES

In the late 1700's, John Michell and Pierre Laplace have already thought of a star with escape velocity larger than the speed of the light. According to the conservation of energy, the escape velocity of a particle of mass m is related to the mass M and the radius R of the star by

$$\frac{1}{2}mv^2 = \frac{GMm}{R}.$$

Setting $v = c$ gives the radius of the “invisible star”:

$$R = \frac{2GM}{c^2}.$$

It is surprising that this result is exactly the formula for the radius of a Schwarzschild black hole in GR. However, the interpretations of Newtonian black holes and GR black holes are different. Newtonian one says that the light will still leave the star but will finally be pulled back by gravity. In GR, everything will never leave the surface of the black hole once it is inside $R = 2M$. The surface itself is also just empty space, but not the edge of a massive body.

8.1 Schwarzschild Black Holes

Consider the Schwarzschild metric

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \left(1 - \frac{2M}{r}\right)^{-1}dr^2 + r^2d\Omega^2.$$

At $r = 2M$ and $r = 0$, some of the metric components blow up. They are called **singularities**. In general relativity, there are two kinds of singularities. A **curvature singularity** is a singularity where the curvature of spacetime goes to infinity. A **coordinate singularity** is just an artifact of bad coordinates; an appropriate coordinate transformation can eliminate the coordinate singularity.

We need to find a measure of curvature that tells us which singularities do $r = 2M$ and $r = 0$ correspond to. Since the Schwarzschild solution is a vacuum solution, the Ricci scalar gives no information (because it is zero). A useful curvature scalar would be the **Kretschmann scalar** $K = R_{\alpha\beta\mu\nu}R^{\alpha\beta\mu\nu}$, the “square” of the Riemann curvature tensor. For the Schwarzschild metric,

$$K = \frac{48M^2}{r^6}.$$

Evidently, the curvature blows up at $r = 0$ but not at $r = 2M$, suggesting that $r = 0$ is a curvature singularity while $r = 2M$ is a coordinate singularity.

Another interesting property of the Schwarzschild metric is that at $r = 2M$, when we cross this surface, g_{rr} gets negative and g_{tt} gets positive. The coordinate radius becomes timelike, while the coordinate time becomes spacelike. Now, $r = 2M$ is merely a coordinate singularity, so it can be removed by choosing another appropriate coordinate system. In this section, we will talk about three systems of coordinates that remove Schwarzschild coordinate singularity and then study the interior of a black hole.

8.1.1 Eddington-Finkelstein Coordinates

In theory, there are infinitely many coordinate transformations, but we don't know which of them remove the coordinate singularity at $r = 2M$. Hence a good set of coordinates is usually motivated by physics. We are interested in the trajectory of particles near $r = 2M$, so one way to find good coordinates is to study the light cones. In Section 7.3.2, we see that it takes infinite coordinate time t for a particle to reach the surface $r = 2M$. This works for photons as well: consider a radial light ray ($d\theta = d\phi = 0$) traveling inward. It travels along null geodesics, so

$$-\left(1 - \frac{2M}{r}\right)dt^2 + \left(1 - \frac{2M}{r}\right)^{-1}dr^2 = 0 \quad \implies \quad \frac{dt}{dr} = \pm \left(1 - \frac{2M}{r}\right)^{-1}.$$

Define a new radial coordinate r^* (known as the **tortoise coordinate**):

$$\frac{dr^*}{dr} = \left(1 - \frac{2M}{r}\right)^{-1} \implies r^* = r + 2M \ln |r - 2M| + C.$$

It is clear that $r^* \rightarrow -\infty$ as $r \rightarrow 2M$. We choose the integration constant to be $C = -2M \ln 2M$ so that

$$r^* = r + 2M \ln \left| \frac{r}{2M} - 1 \right|. \quad (8.1)$$

Since $t(r)$ for radial light rays and $r^*(r)$ follows the same differential equation, we have $t = \pm r^* + \text{const.}$ Then define another coordinate $v \equiv t + r^*$, $dv = dt + dr^*$, and $dt = dv - dr(1 - 2M/r)^{-1}$. The ingoing null geodesics are now characterized by $v = \text{const.}$ The tortoise coordinate r^* now acts like an intermediate coordinate. Computing the metric, we find that the coordinate singularity at $r = 2M$ goes away

$$\begin{aligned} ds^2 &= -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \\ &= -\left(1 - \frac{2M}{r}\right) \left[dv^2 - 2\left(1 - \frac{2M}{r}\right)^{-1} dv dr + \left(1 - \frac{2M}{r}\right)^{-2} dr^2 \right] + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \end{aligned}$$

This set of coordinates (v, r, θ, ϕ) is known as **Eddington-Finkelstein coordinates**, first proposed by Eddington in 1924 and later by Finkelstein in 1958. The metric in Eddington-Finkelstein coordinates (v, r, θ, ϕ) is

$$\boxed{ds^2 = -\left(1 - \frac{2M}{r}\right) dv^2 + 2 dv dr + r^2 d\Omega^2.} \quad (8.2)$$

Note that Eddington-Finkelstein metric is not diagonal. Because of this property, the inverse metric also does not have a singularity at $r = 2M$. We will find the shape of light cones in Eddington-Finkelstein coordinates to study the causal structure of spacetime around $r = 2M$. In Eddington-Finkelstein coordinates, the radial null geodesics follow

$$0 = -\left(1 - \frac{2M}{r}\right) dv^2 + 2 dv dr.$$

Solving this equation, either $dv = 0$ ($v = \text{const.}$) or

$$\frac{dv}{dr} = \frac{2r}{r - 2M} = 2 \frac{dr^*}{dr} \implies v = 2r^* + \text{const.} = 2r + 4M \ln \left| \frac{r}{2M} - 1 \right| + \text{const.}$$

We often define $u \equiv v - 2r^* = t - r^*$. Back to the definition $t = \pm r^* + \text{const.}$, we see that $u = \text{const.}$ represent outgoing null geodesics. In conclusion, $v = t + r^* = \text{const.}$ and $u = t - r^* = \text{const.}$ are equations for light cones. Figure 8.1 shows light cones on a spacetime diagram of v vs. r . From Figure 8.1, we see that material particles following timelike world lines must following trajectories with decreasing r once they are inside $r = 2M$. Eventually, they will reach the $r = 0$ singularity. The $r = 2M$ surface is a null surface: an outward propagating radial light ray stays at this surface. Anything information inside $r = 2M$ thus cannot reach the outside world, which means it is impossible to probe the interior of a black hole from outside, hence the name **black hole**. The Schwarzschild radius $r = 2M$ is an **event horizon** for a Schwarzschild black hole.

8.1.2 Kruskal-Szekeres Coordinates

Though Eddington-Finkelstein coordinates solves the coordinate singularity at $r = 2M$, they are still something unresolved. We know that the Schwarzschild metric is not only time-independent, but also has time-reversal symmetry. The most significant problem with Eddington-Finkelstein coordinates is that there is no time-reversal symmetry anymore. It is clear from (8.2) that v behaves like the time-coordinate, but changing $dv \rightarrow -dv$ alters the metric because of the cross-term $2 dv dr$. It seems like there is a preferred time direction in this coordinate system. This is expected, because we started with the ingoing light rays and defined the coordinate $v = t + r^*$. In fact, reversing $dv \rightarrow -dv$ is the same as using the coordinate $u = t - r^*$ (outgoing light rays) instead of v . The outgoing Eddington-Finkelstein metric can be found,

$$ds^2 = -\left(1 - \frac{2M}{r}\right) du^2 - 2 du dr + r^2 d\Omega^2.$$

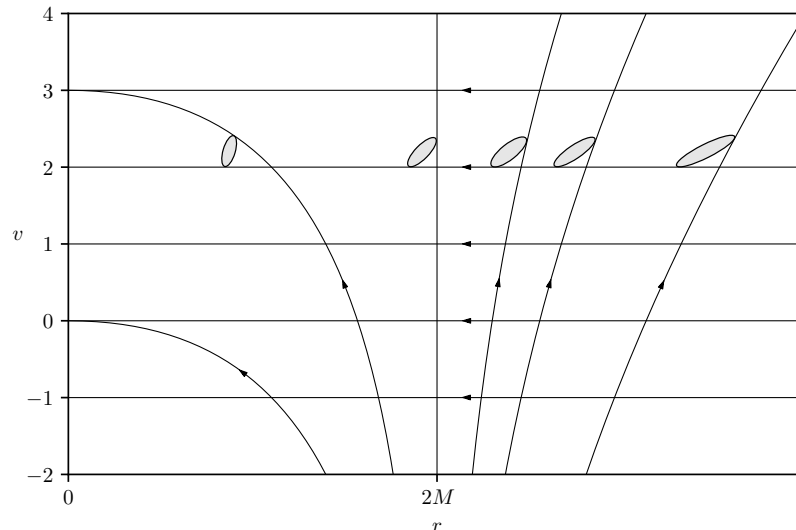


Figure 8.1: Photon world lines in (v, r) spacetime diagram. Several future light cones are drawn. The arrows indicates the direction of world lines in increasing t : the horizontal lines represent $v = t + r^* = \text{const.}$, where increasing t means decreasing r^* or r ; the curved lines represent the outgoing null geodesics $u = v - 2r^* = \text{const.}$ Increasing t means increasing r^* , or increasing r for $r > 2M$, and decreasing r for $r < 2M$ (so the “outgoing” photon is traveling *inward* towards $r = 0$ inside $r = 2M$).

Therefore, either ingoing or outgoing Eddington-Finkelstein coordinates does not describe the full Schwarzschild metric. The question now is can we use both u and v ? The answer is partially yes. The metric is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) du dv + r^2 d\Omega^2.$$

However, note that the metric now is degenerate at $r = 2M$. This is expected because v and u are also not very good coordinates at $r = 2M$ for radial null geodesics. For outgoing null geodesics, $r = 2M$ is infinitely far away because $v \rightarrow -\infty$ as $r \rightarrow 2M$. Similarly, for ingoing null geodesics, $u \rightarrow \infty$ as $r \rightarrow 2M$.

A good set of coordinates that describe the whole spacetime is known as **Kruskal-Szekeres coordinates**, discovered in 1960. The coordinates are defined as

$$u = \left(\frac{r}{2M} - 1\right)^{1/2} e^{r/4M} \cosh\left(\frac{t}{4M}\right), \quad v = \left(\frac{r}{2M} - 1\right)^{1/2} e^{r/4M} \sinh\left(\frac{t}{4M}\right),$$

for $r > 2M$ and

$$u = \left(1 - \frac{r}{2M}\right)^{1/2} e^{r/4M} \sinh\left(\frac{t}{4M}\right), \quad v = \left(1 - \frac{r}{2M}\right)^{1/2} e^{r/4M} \cosh\left(\frac{t}{4M}\right),$$

for $r < 2M$. The θ and ϕ coordinates are the same as in Schwarzschild coordinates. The metric in Kruskal-Szekeres coordinates is

$$ds^2 = -\frac{32M^3}{r} e^{-r/2M} (dv^2 - du^2) + r^2 d\Omega^2, \quad (8.3)$$

where r and t are now function of u and v , defined implicitly by

$$\left(\frac{r}{2M} - 1\right) e^{r/2M} = u^2 - v^2, \quad \tanh\left(\frac{t}{4M}\right) = \begin{cases} v/u, & r > 2M, \\ u/v, & r < 2M. \end{cases} \quad (8.4)$$

The derivation of (8.3) is in Appendix A.6.1. There is no singularity at $r = 2M$ in this metric but there is one at $r = 0$ as it should be. The Kruskal-Szekeres metric is diagonal. From (8.4), constant r lines are hyperbolae $u^2 - v^2 = \text{const.}$, while constant t lines are straight lines passing through the origin $(u, v) = (0, 0)$. Radial null lines ($d\theta = d\phi = ds = 0$) satisfy $dv = \pm du$, or $v = \pm u + \text{const.}$, which looks like null lines in Minkowski spacetime. Material particles have $g_{\mu\nu} u^\mu u^\nu = -1$,

$$\frac{32M^3}{r} e^{-r/2M} \left[-\left(\frac{dv}{d\tau}\right)^2 + \left(\frac{du}{d\tau}\right)^2 \right] = -1.$$

In order for the LHS to be negative, we must have $(dv/d\tau)^2 > (du/d\tau)^2$, or $|dv/du| > 1$, again, similar to material particle world lines in Minkowski spacetime. Figure 8.2 (sometimes called the **Kruskal diagram**) shows the constant r , constant t surfaces, and the singularity at $r = 0$.

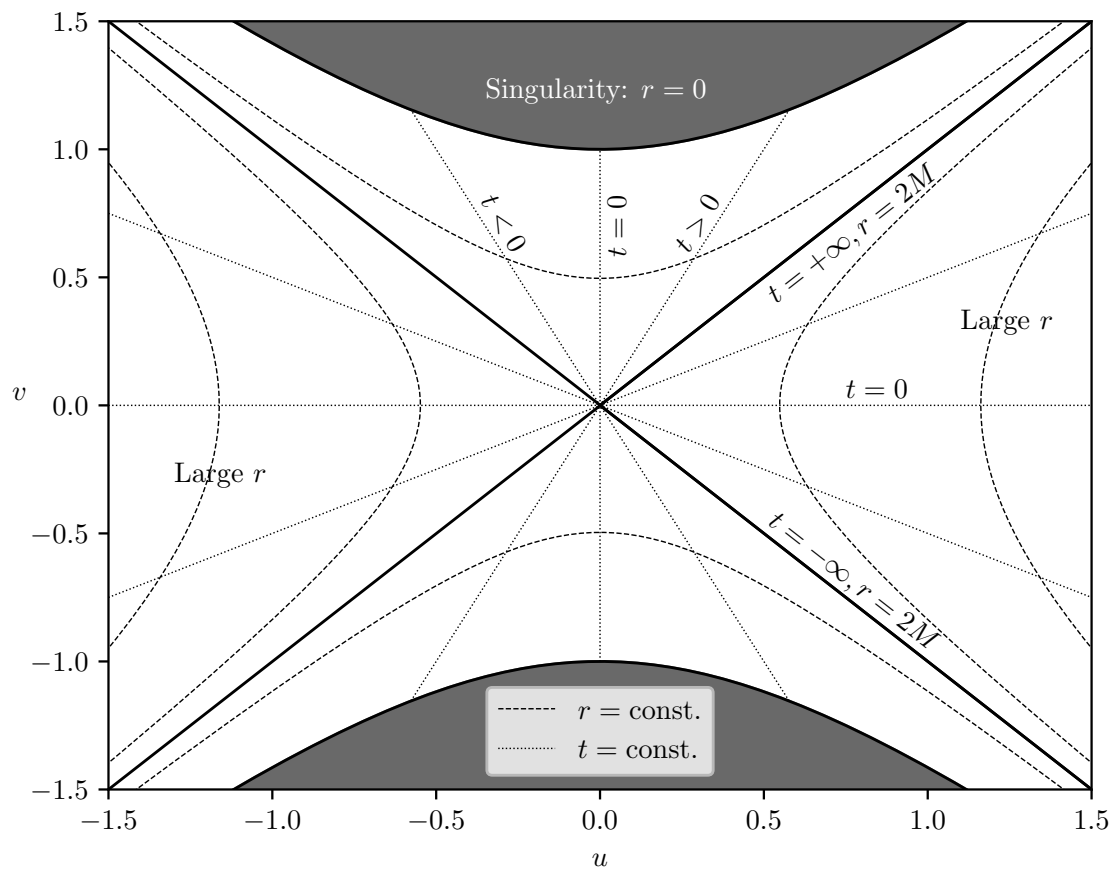
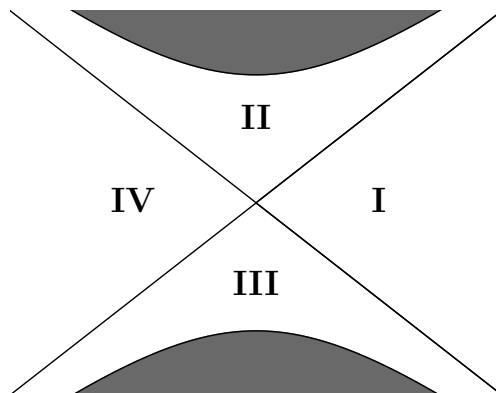


Figure 8.2: Constant r and constant t surfaces, and $r = 0$ singularity in (u, v) coordinates.

From Figure 8.2, clearly $r = 2M$ is a null surface because it obeys $v = \pm u$. It is also clear that any photon emitted inside $r = 2M$ always stays inside $r = 2M$, and eventually runs into the singularity, because $r = 0$ hyperbola is asymptotically the same as $u = \pm v$. Interestingly, the $r = 0$ singularity is a spacelike surface. It is to be interpreted as a moment in time in language of relativity.

More interesting things can be inferred from the Kruskal diagram if we divide it into 4 regions according to $r = 2M$ null surfaces:



Using this diagram, we can appreciate the full glory of the Schwarzschild geometry: the **maximally extended Schwarzschild solution**. Here, region I represent all the region outside a Schwarzschild black hole. It can be described by the original Schwarzschild coordinates (t, r, θ, ϕ) well. Using ingoing Eddington-Finkelstein coordinates (v, r, θ, ϕ) , we extended the Schwarzschild geometry to region II, which represents the interior of a black hole. The full ingoing Eddington-Finkelstein coordinates can describe both region I and II well without coordinate singularities. However, we see that the Schwarzschild geometry can also be described by outgoing Edding-Finkelstein coordinates (u, r, θ, ϕ) . Recall that light cones in the Kruskal diagram is the same as those in the Minkowski spacetime diagram. If we follow the future light cones inside region III, we find that all particles must travel away from the singularity $r = 0$ in region III. This means region III is a **white hole**, a region where nothing can enter—a time-reversal of a black hole. Finally, region IV is another asymptotically flat Schwarzschild spacetime, a mirror image of our universe. However, nothing from region IV is causally connected from region I. The only way for me to meet another person from the mirror universe is that we both jump into the black hole (region II) before we are crushed by the singularity.

There are some interesting things regarding black holes are discussed in Appendix [A.6.2](#).

8.1.3 Other Coordinates

To be written.

8.1.4 Schwarzschild Solution with $M < 0$

To be written.

8.2 General Black Holes

8.2.1 Black Hole Theorems

There are not many stationary black-hole solutions. We are interested in stationary solutions because any kinds of oscillations will be damped as energy is radiated away by gravitational waves. One can imagine that when a black hole is formed, everything is crunched into a singularity. A black hole may have a complicated history, such as a gas with complex gas motions, shock waves, magnetic fields, nucleosynthesis, and all kinds of other complications. When the black hole settles down and becomes stationary, it requires only a few parameters to be fully described:

- The simplest black hole is a Schwarzschild black hole. It is a static, spherically symmetric vacuum solution, described by only one parameter, its mass M .
- There can be a rotating black hole, or a **Kerr black hole**. Compared to a Schwarzschild black hole, a Kerr black hole loses both spherical symmetry and time-reversal symmetry. It is axially symmetric and stationary, but not static. A Kerr black hole is characterized by two parameters, its mass M and angular momentum L .
- Both Schwarzschild black holes and Kerr black holes are vacuum solutions. A common non-vacuum solution is a charged black hole, or a **Reissner-Nordström black hole**. It is a non-vacuum solution because the stress-energy tensor will involve electromagnetic fields. A Reissner-Nordström black hole is described by its mass M and charge Q .

Below are some theorems regarding these types of black holes.

Theorem 8.1. A no-hair theorem

Stationary, asymptotically flat black hole solutions to general relativity coupled to electromagnetism that are nonsingular outside the event horizon are fully characterized by mass, electric and magnetic charge, and angular momentum.

Proof. To be written. □

Theorem 8.2. The black hole uniqueness theorem (I)

The only static, asymptotically flat, vacuum solution with a horizon is the Schwarzschild solution.

Proof. To be written. □

Theorem 8.3. The black hole uniqueness theorem (II)

The only static asymptotically flat solution to Einstein-Maxwell equations with a *connected* horizon is the Reissner-Nordström solution.

Proof. To be written. □

Theorem 8.4. The black hole uniqueness theorem (III)

The only stationary, asymptotically flat vacuum solution with a horizon is the Kerr solution.

Proof. To be written. □

8.2.2 Penrose Diagrams

In general relativity, we usually want to study the causal structure of a spacetime globally. This is often done by drawing spacetime diagrams with past and future light cones. However, all the spacetime diagrams we introduced extend to infinities even for Minkowski spacetime, which is not so satisfactory. It turns out that it is possible to view an infinite spacetime in a finite region. This is fulfilled by **Penrose diagrams** (or **conformal diagrams**). Aside from shrinking an infinite spacetime to finite, another goal of Penrose diagrams is to have a “timelike” coordinate and a “radial” coordinate such that light cones remain 45° just as in Minkowski spacetime.

First consider the Minkowski metric,

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2.$$

The light cones are at 45° everywhere, with $dt/dr = \pm 1$. In general, there will be a singularity at $r = 0$, but we don’t have to worry about it now because we are in Minkowski spacetime. As usual, to study the light cones, we switch to null coordinates

$$u = t - r, \quad v = t + r.$$

The corresponding ranges are $u \in (-\infty, \infty)$, $v \in (-\infty, \infty)$, and $u \leq v$. The Minkowski metric turns into

$$ds^2 = -du dv + \frac{1}{4}(v - u)^2 d\Omega^2.$$

To bring the infinite ranges of u and v to finite, consider the following coordinate transformations,

$$U = \arctan u, \quad V = \arctan v.$$

The ranges are now $U \in (-\pi/2, \pi/2)$, $V \in (-\pi/2, \pi/2)$, and $U \leq V$. With this coordinate transformation,

$$\begin{aligned} du dv &= \frac{dU dV}{\cos^2 U \cos^2 V}, \\ (v - u)^2 &= (\tan V - \tan U)^2 = \frac{(\sin V \cos U - \cos V \sin U)^2}{\cos^2 U \cos^2 V} = \frac{\sin^2(V - U)}{\cos^2 U \cos^2 V}. \end{aligned}$$

So in (U, V, θ, ϕ) coordinates, the Minkowski metric looks like

$$ds^2 = \frac{1}{4 \cos^2 U \cos^2 V} [-4 dU dV + \sin^2(V - U) d\Omega^2].$$

The last step is to transform back to a timelike coordinate T and a radial coordinate R by

$$T = V + U, \quad R = V - U,$$

with $0 \leq R < \pi$, $|T| + R < \pi$, and the metric being

$$ds^2 = \omega^{-2}(T, R)(-dT^2 + dR^2 + \sin^2 R d\Omega^2), \quad \text{where} \quad \omega = 2 \cos U \cos V = \cos T + \cos R. \quad (8.5)$$

The Penrose diagram drawn from (8.5) is shown in Figure 8.3.

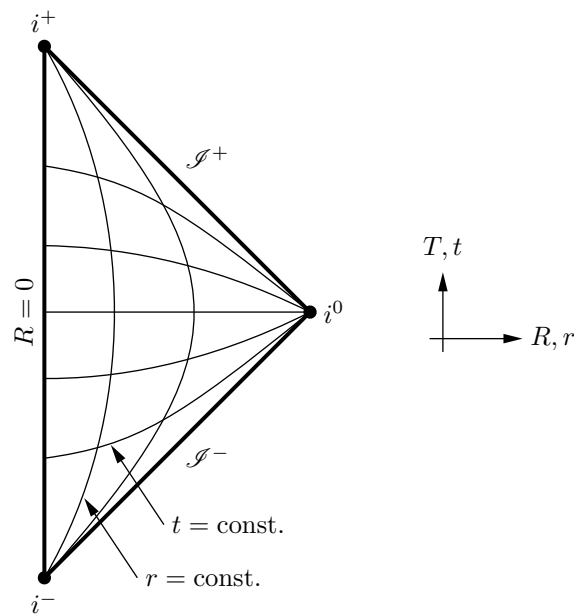


Figure 8.3: The Penrose diagram of Minkowski spacetime. Each point represents a 2-sphere. Light cones are at $\pm 45^\circ$.

Let's analyze this Penrose diagram region by region:

1. There are three important points: i^+ is the future timelike infinity ($T = \pi$ and $R = 0$, or $t = \infty$); i^- is the past timelike infinity ($T = -\pi$ and $R = 0$, or $t = -\infty$); and i^0 is the spatial infinity ($T = 0$, $R = \pi$, or $r = \infty$).
2. There are two important lines: $\mathcal{I}^+$ is the future null infinity ($T = \pi - R$ and $0 < R < \pi$); $\mathcal{I}^-$ is the past null infinity ($T = -\pi + R$ and $0 < R < \pi$). (The letter $\mathcal{I}$ is script-I, or "scri". $\mathcal{I}^+$ and $\mathcal{I}^-$ are pronounced as "scri-plus" and "scri-minus", respectively.)
3. All timelike geodesics begin at i^- and end at i^+ ; all null geodesics begin at $\mathcal{I}^-$ and end at $\mathcal{I}^+$. There can be non-geodesic timelike curves that begin at $\mathcal{I}^-$ and end at $\mathcal{I}^+$. Just draw a straight vertical line on the Penrose diagram. This will be a perfect timelike curve but not a geodesic. In fact, such a curve is the worldline of a particle with constant acceleration in one direction. Its worldline in the original Minkowski spacetime diagram would be a timelike hyperbola, which is asymptotically null (as it keeps accelerating).

The Penrose diagram of Minkowski spacetime tells us nothing new. Nevertheless, they will be particularly useful when we study black holes. As a preparation, let's consider the Penrose diagram of Schwarzschild spacetime.

To be written.

Figure 8.4

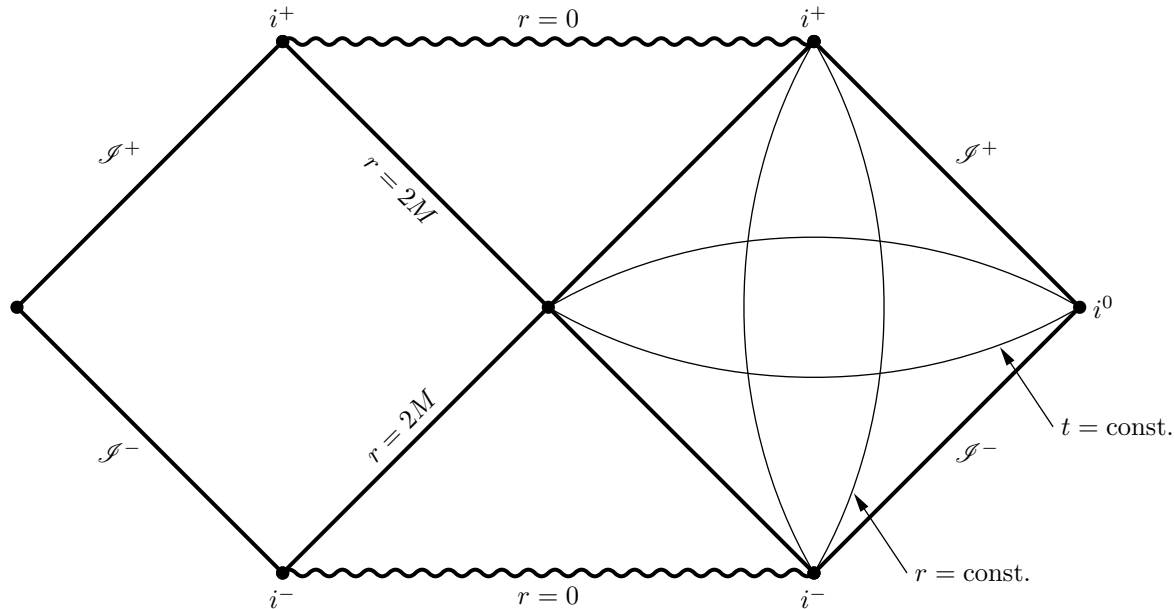


Figure 8.4: The Penrose diagram of Schwarzschild spacetime.

To be written.

8.2.3 Mass, Charge, and Angular Momentum

To be written.

8.3 Charged (Reissner-Nordström) Black Holes

8.3.1 The Reissner-Nordström Metric

Several years after the Schwarzschild solution, another exact solution of black hole was discovered. The assumptions of static, spherical symmetry and asymptotic flatness are still kept, but electromagnetic fields are allowed to present. The stress-energy tensor is therefore nonzero in Einstein's equation,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi T_{\mu\nu}.$$

Specifically, if there are no external charges present, the stress-energy tensor of electromagnetic fields is

$$T_{\mu\nu} = F_{\mu\alpha}F_{\nu}^{\alpha} + \frac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta},$$

where $F_{\mu\nu}$ is the electromagnetic field tensor (or the Maxwell tensor). Besides Einstein's equation, the field tensor also needs to satisfy Maxwell's equations,

$$g^{\mu\nu}\nabla_{\mu}F_{\nu\alpha} = 0 \quad \text{and} \quad \nabla_{[\mu}F_{\nu\alpha]} = 0.$$

Einstein's equation and Maxwell's equations are coupled because the metric solution depends on $F_{\mu\nu}$, but it meanwhile enters Maxwell's equations. We must solve them simultaneously. With some fair assumptions, i.e. static, spherical symmetry, only $F_{tr} = -F_{rt} = -E_r$ being the nonzero components, the solution is

$$ds^2 = -\left(1 - \frac{2M}{r} + \frac{Q^2}{r^2}\right)dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2}\right)^{-1}dr^2 + r^2d\Omega^2. \quad (8.6)$$

This is known as the **Reissner-Nordström** metric. The derivation of the Reissner-Nordström metric is given in Appendix A.6.3. The Reissner-Nordström metric is characterized by two independent parameters, M and Q . Here M is interpreted as the total mass of the black hole, and Q is the total electric charge. Note that physically Q can take negative values, but only Q^2 enters the metric.

The radial electric field E_r produced by the black hole can be solved from Maxwell's equation in the Reissner-Nordström metric background. The solution is $E_r = F_{rt} = Q/r^2$, the usual Coulomb's law. It is clear that if the black hole is neutral ($Q = 0$) then the solution reduces to the Schwarzschild solution.

Like the Schwarzschild metric, the Reissner-Nordström metric has a curvature singularity at $r = 0$. To study the spacetime structure near the black hole, we locate the event horizons,

$$1 - \frac{2M}{r} + \frac{Q^2}{r^2} = 0.$$

This is a quadratic equation, with solution

$$r_{\pm} = M \pm \sqrt{M^2 - Q^2}.$$

Evidently, the number of real solutions depends on the relative value of M and Q . If $Q^2 < M^2$, both horizons $r_{\pm}$ exist. If $Q^2 = M^2$, the two solutions coincide, $r_+ = r_-$, giving only one horizon. If $Q^2 > M^2$, there is no horizon, so the solution is in fact not a black hole. These three cases give rise to very different spacetime structures, so we shall study them one-by-one.

8.3.2 $Q^2 > M^2$ Solution

First, we should know what does it mean by comparing M and Q , since they usually appear to have different dimensions. The geometric units have $c = 1$, $G = 1$, and $4\pi\epsilon_0 = 1$, where ϵ_0 is the permittivity of free space. In general relativity, each quantity is expressed as a *length*. In particular, the mass and charge in geometric units are related to those in SI units by

$$M_{\text{geom}} = \frac{G}{c^2} M_{\text{SI}} \quad \text{and} \quad Q_{\text{geom}} = \sqrt{\frac{G}{4\pi\epsilon_0 c^4}} Q_{\text{SI}}.$$

Consider the case where $M_{\text{geom}} = Q_{\text{geom}}$, we can calculate the corresponding charge-mass ratio in SI units:

$$\frac{Q_{\text{SI}}}{M_{\text{SI}}} = \sqrt{\frac{4\pi\epsilon_0 c^4}{G}} \frac{G}{c^2} = \sqrt{4\pi\epsilon_0 G} \approx 8.61 \times 10^{-11} \text{ C/kg}.$$

This is not a large charge-mass ratio compared to the electron charge-mass ratio, $e/m_e \approx 1.76 \times 10^{11} \text{ C/kg}$. However, for an astrophysical object with $M \sim M_{\odot} \approx 2 \times 10^{30} \text{ kg}$, an excess charge of $Q \sim 10^{20}$ is nearly impossible. Such an object will attract opposite charges around it and become neutral quickly.

Mathematically, a black hole with $Q^2 > M^2$ is certainly a solution. In this case, $1 - 2M/r + Q^2/r^2$ is always positive and has no real roots. The coordinate t is always timelike, and r is always spacelike. There is no coordinate singularity or event horizon, so this solution is not a black hole. Hence curvature singularity $r = 0$ is a naked singularity. An observer can in principle travel arbitrarily near the black hole and return with their observation. In fact, the singularity is repulsive: timelike geodesics can never reach $r = 0$. (Null geodesics and nongeodesic timelike curves can reach the singularity.) This is very similar to the $M < 0$ Schwarzschild solution discussed in Section 8.1.4.

The Penrose diagram for $Q^2 > M^2$ Reissner-Nordström solution is shown in Figure 8.5. The structure is almost the same as Minkowski spacetime, except that $r = 0$ is replaced by the naked singularity.

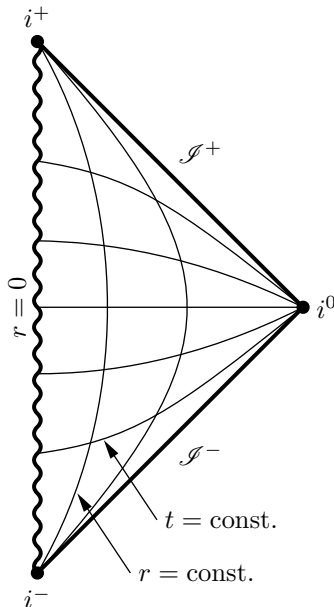


Figure 8.5: Penrose diagram for the Reissner-Nordström solution with $Q^2 > M^2$.

The $Q^2 > M^2$ Reissner-Nordström solution is usually considered unphysical. This is because M is roughly the total energy of the object, but $|Q| > M$ says that the electromagnetic energy alone is larger than the total energy.

8.3.3 $Q^2 < M^2$ Solution

The $Q^2 < M^2$ Reissner-Nordström solution is more realistic. The expression $1 - 2M/r + Q^2/r^2$ has two real roots, $r_{\pm} = M \pm \sqrt{M^2 - Q^2}$. The Reissner-Nordström metric (8.6) can be written in terms of $r_{\pm}$,

$$ds^2 = -\frac{(r - r_+)(r - r_-)}{r^2} dt^2 + \frac{r^2}{(r - r_+)(r - r_-)} dr^2 + r^2 d\Omega^2. \quad (8.7)$$

To remove the coordinate singularities at $r = r_{\pm}$, we consider radial null coordinates similar to Eddington-Finkelstein coordinates for the Schwarzschild solution. Define the tortoise coordinate as

$$r^* = \int \frac{r^2 dr}{(r - r_+)(r - r_-)} = r + \frac{r_+^2}{r_+ - r_-} \ln(r - r_+) - \frac{r_-^2}{r_+ - r_-} \ln(r - r_-) + C.$$

Let $v = t + r^*$, where $v = \text{const.}$ represents ingoing radial null geodesics. In terms of the ingoing radial null coordinates (v, r, θ, ϕ) , the Reissner-Nordström metric takes a familiar form

$$ds^2 = -\frac{(r - r_+)(r - r_-)}{r^2} dv^2 + 2 dv dr + r^2 d\Omega^2. \quad (8.8)$$

The metric (8.8) infers that the event horizons $r = r_{\pm}$ are null surface. The singularity $r = 0$ is timelike.

Back to the metric in the form (8.7), we see that when $r > r_+$, the coordinate r is spacelike. Between the horizons, $r_- < r < r_+$, r is timelike, which means a particle falling into the black hole from r_+ must decrease r . After passing the other event horizon at r_- , r becomes spacelike again. You can choose to either continue to the singularity $r = 0$ or go back to r_- . If you choose the latter, you will inevitably move in the *increasing* r direction once you pass r_- . Eventually, you will get out of r_+ , just like entering another mirror universe through a white hole. Moreover, you are then allowed to find another Reissner-Nordström black hole in the mirror universe and continue this process as many times as you want. Refer to the Penrose diagram in Figure 8.6 for example.

To be drawn.

Figure 8.6: Penrose diagram of the Reissner-Nordström solution with $Q^2 < M^2$.

8.3.4 $Q^2 = M^2$ Solution

The $Q^2 = M^2$ solution is the **extreme** Reissner-Nordström solution. The event horizon is located at $r_+ = r_- = M$. Notice that r is a null coordinate at $r = M$, and a spacelike coordinate anywhere else; it is nowhere timelike. Hence one still can avoid hitting the singularity in an extremal black hole.

One remarkable property of the $Q^2 = M^2$ solution is that we can superpose many extremal black holes with charges of the same sign and the solution is still static. In other words, the gravitational attraction and electric repulsion balance each other. This is an exact effect in general relativity and no approximation is needed. To show this, we first write the extreme Reissner-Nordström metric as the following using $M^2 = Q^2$.

$$ds^2 = -\left(1 - \frac{M}{r}\right)^2 dt^2 + \left(1 - \frac{M}{r}\right)^{-2} dr^2 + r^2 d\Omega^2. \quad (8.9)$$

Define a shifted radial coordinate $\rho = r - M$. Then the metric takes the form

$$ds^2 = -h^{-2}(\rho)dt^2 + h^2(\rho)(d\rho^2 + \rho^2 d\Omega^2), \quad \text{where} \quad h(\rho) = 1 + \frac{M}{\rho}.$$

The spatial part of the metric is isotropic. We know the electric field generated by the black hole is $E_r = Q/r^2$. It can be expressed as $E_r = \partial_r A_t$, where A_t is the usual scalar potential in electrostatics (suppose $Q > 0$),

$$A_t = -\frac{Q}{r} = -\frac{M}{\rho + M} \implies A_t = h^{-1} - 1.$$

To be written.

The entire Einstein-Maxwell equations reduce to

$$\nabla^2 h = 0,$$

where ∇^2 is the Laplacian of 3-dimensional *flat* geometry with coordinates (ρ, θ, ϕ) . This is just Laplace's equation. The solution with the correct asymptotic behavior is

$$h(\mathbf{x}) = 1 + \sum_{i=1} \frac{M_i}{|\mathbf{x} - \mathbf{x}_i|}.$$

Another interesting property of the extreme Reissner-Nordström solution is that the any proper distance from a point outside the horizon to the horizon is infinite. Let's look at radial trajectories. In a constant time slice ($dt = 0$), the proper distance from $r = r_1 > M$ to the horizon $r = M$ is given by

$$s = \int \sqrt{g_{rr}} dr = \int_M^{r_1} \frac{r dr}{r - M} = \infty.$$

This does not mean that it takes infinite proper time to reach the horizon because the proper distance is a spacelike quantity. Any timelike or null geodesics can reach the horizon easily.

8.4 Rotating (Kerr) Black Holes

8.4.1 The Kerr Metric

Unlike charged black holes, rotating black holes are expected to be much more common in the universe. However, the solution to rotating black holes is more difficult to find. It was almost after 50 years of the invention of general relativity that this exact solution was found. The solution is given by

$$ds^2 = -\left(1 - \frac{2Mr}{\rho^2}\right) dt^2 - \frac{4Mar \sin^2 \theta}{\rho^2} dt d\phi + \frac{\rho^2}{\Delta} dr^2 + \rho^2 d\theta^2 + \frac{(r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta}{\rho^2} \sin^2 \theta d\phi^2, \quad (8.10)$$

where

$$\rho^2 \equiv r^2 + a^2 \cos^2 \theta \quad \text{and} \quad \Delta \equiv r^2 - 2Mr + a^2.$$

This complicated metric is known as the **Kerr metric**, discovered by Roy Kerr in 1963. (And of course, we will not derive it in this notes.) The coordinates (t, r, θ, ϕ) in which the Kerr metric is written is called **Boyer-Lindquist coordinates**. The Kerr metric keeps the vacuum and asymptotically flat assumption, but drops the static and spherical symmetry. The metric is stationary but not static because of the cross term $dt d\phi$, the so time-reversal symmetry is gone; the metric is axially symmetric, a much weaker condition than spherical symmetry. As a result, there is no analogous Birkhoff's theorem for the Kerr geometry. A stationary, axially symmetric, asymptotically flat, vacuum solution is not uniquely given by the Kerr metric. (Recall that the black hole uniqueness theorem 8.4 requires a horizon. This is the key for the Kerr metric to be unique.) Therefore, we cannot just assume that the Kerr metric models the spacetime outside a rotating star.

The angular momentum J of the black hole is defined to be

$$a = \frac{J}{M}.$$

If the black hole also possesses electric charge, the Kerr metric is generalized to the **Kerr-Newman metric** by replacing $2Mr$ with $2Mr - Q^2$. The associated 4-potential has nonzero components

$$A_t = \frac{Qr}{\rho^2}, \quad A_\phi = -\frac{Qar \sin^2 \theta}{\rho^2}.$$

We will make analysis of the Kerr black hole assuming that $Q = 0$. Let's now take some limit of the Kerr metric:

- As $a \rightarrow 0$ (no angular momentum), it is easy to check that the Kerr metric reduces to the Schwarzschild metric.
- As $M \rightarrow 0$ while keeping a fixed (no mass but nonzero angular momentum), the metric becomes

$$ds^2 = -dt^2 + \frac{r^2 + a^2 \cos^2 \theta}{r^2 + a^2} dr^2 + (r^2 + a^2 \cos^2 \theta)^2 d\theta^2 + (r^2 + a^2) \sin^2 \theta d\phi^2. \quad (8.11)$$

This is the Minkowski metric in **ellipsoidal coordinates**, related to Cartesian coordinates by

$$x = (r^2 + a^2)^{1/2} \sin \theta \cos \phi, \quad y = (r^2 + a^2)^{1/2} \sin \theta \sin \phi, \quad z = r \cos \theta.$$

In fact, the Kerr metric is already written in ellipsoidal coordinates. The coordinate radius $r = 0$ is not a point, but a disk with radius a .

- As $r \rightarrow \infty$ (asymptotic region), the spacetime is Minkowski. Something interesting happens if we keep the leading order terms in $1/r$,

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 - \frac{2Ma \sin^2 \theta}{r} dt d\phi + \left(1 + \frac{2M}{r}\right) dr^2 + r^2 d\Omega^2.$$

Everything *looks like* Schwarzschild to leading order except the cross term. Indeed, if we wish to study dynamics far away from the black hole, it is common to set up an orthonormal inertial frame. The basis 1-forms in the orthonormal frame are a mix of dt , dr , $r d\theta$, and $r \sin \theta d\phi$. Hence the metric to leading order should be more appropriately written as

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 - \frac{2Ma \sin \theta}{r^2} dt(r \sin \theta d\phi) + \left(1 + \frac{2M}{r}\right) dr^2 + r^2 d\Omega^2.$$

The dynamical effect created by $g_{t\phi}$ is to (at least) $1/r^2$. In conclusion, the Kerr metric reduces to Schwarzschild metric to leading order of $1/r$, and to Minkowski metric asymptotically.

There are two Killing vectors in the Kerr geometry, $\xi = \partial_t$ and $\eta = \partial_\phi$, because the metric components are independent of t and ϕ . They correspond to conserved energy E and conserved angular momentum L_z for a particle moving in this spacetime. Specifically, for a particle with 4-momentum p^μ ,

$$E = -g_{\mu\nu} \delta^\mu_t p^\nu = -g_{tt} p^t - g_{t\phi} p^\phi \quad \text{and} \quad L_z = g_{\mu\nu} \delta^\mu_\phi p^\nu = g_{\phi t} p^t + g_{\phi\phi} p^\phi.$$

There is a hidden conserved quantity known as the **Carter constant** Q , given by

$$Q = p_\theta^2 + \left[a^2 (m^2 - E^2) + \left(\frac{L_z}{\sin \theta} \right)^2 \right] \cos^2 \theta, \quad (8.12)$$

where m is the mass of the particle and $p_\theta = g_{\theta\theta}p^\theta$. This conserved quantity results from a $(0, 2)$ Killing tensor,

$$\sigma_{\mu\nu} = 2\rho^2 l_{(\mu} n_{\nu)} + r^2 g_{\mu\nu}.$$

where l^μ and n^μ are null vectors defined as

$$l^\mu = \frac{1}{\Delta}(r^2 + a^2, \Delta, 0, a) \quad n^\mu = \frac{1}{2\rho^2}(r^2 + a^2, -\Delta, 0, a), \quad \text{with} \quad l_\mu l^\mu = n_\mu n^\mu = 0, \quad n_\mu l^\mu = -1.$$

One can check that $\nabla_{(\alpha}\sigma_{\mu\nu)} = 0$ and $\sigma_{\mu\nu}u^\mu u^\nu = Q$, the Carter constant.

8.4.2 The Event Horizon and Singularity

The Kretschmann scalar for the Kerr metric is

$$R_{\alpha\beta\mu\nu}R^{\alpha\beta\mu\nu} = \frac{48M^2(r^6 - 15r^4a^2\cos^2\theta + 15r^2a^4\cos^4\theta - a^6\cos^6\theta)}{\rho^{12}},$$

suggesting a curvature singularity at $\rho^2 = 0$. The event horizon of a Kerr black hole occurs at values of r at which $g_{rr} \rightarrow \infty$, or

$$\Delta = r^2 - 2Mr + a^2 = 0.$$

This is very similar to the Reissner-Nordström solution: the three possibilities are $M > a$, $M = a$, and $M < a$. The $M = a$ solution is an extremal Kerr black hole, while the $M < a$ solution has a naked singularity. They are considered nonphysical, so we shall focus on the $M > a$ solution. In this case, Δ has two real roots, corresponding to two event horizons

$$r_\pm = M \pm \sqrt{M^2 - a^2}.$$

The curvature singularity occurs at

$$\rho^2 = r^2 + a^2 \cos^2\theta = 0.$$

For any Kerr black hole ($a \neq 0$), this equation has a particular solution of r and θ ,

$$r = 0, \quad \theta = \frac{\pi}{2}.$$

Recall that in ellipsoidal coordinate

$$x = (r^2 + a^2)^{1/2} \sin\theta \cos\phi, \quad y = (r^2 + a^2)^{1/2} \sin\theta \sin\phi, \quad z = r \cos\theta,$$

the set of points with $r = 0$ is a disk of radius a . The set of points with $r = 0$ and $\theta = \pi/2$ is a *ring*. This is also saying that we can go below $r = 0$ into a spacetime described by the Kerr metric with $r < 0$. In this spacetime, Δ is always nonzero, giving no horizons.

8.4.3 Ergosphere

The physics inside a Kerr black hole is not very interesting compared to the region outside it. Outside the black hole, there exists an effect often called the **dragging of inertial frames**. Consider a particle with zero angular momentum

$$L_z = p_\phi = g_{\phi t}p^t + g_{\phi\phi}p^\phi = 0.$$

The particle's trajectory has an angular velocity

$$\omega \equiv \frac{d\phi}{dt} = \frac{p^\phi}{p^t} = -\frac{g_{\phi t}}{g_{\phi\phi}} = \frac{2Mar}{(r^2 + a^2)^2 - a^2\Delta \sin^2\theta}. \quad (8.13)$$

The angular velocity of a particle with zero angular momentum is “dragged” under the influence of gravity by the Kerr black hole. Moreover, the denominator is always a nonzero number. Outside the event horizon, $\Delta > 0$, we have

$$(r^2 + a^2)^2 - a^2\Delta \sin^2\theta \geq (r^2 + a^2)^2 - a^2\Delta = r^4 + a^2r^2 + 2Ma^2r > 0.$$

This means that ω has the same sign as a ; the particle acquires an angular velocity in the same direction as the angular momentum of the black hole. From (8.13), the effect of this dragging is proportional to $1/r^3$ at large r .

Next, let's analyze the photon trajectory in the plane of fixed θ outside a Kerr black hole. Photons travel along null geodesics,

$$g_{tt}dt^2 + 2g_{t\phi}dt d\phi + g_{\phi\phi}d\phi^2 = 0 \implies \frac{d\phi}{dt} = -\frac{g_{t\phi}}{g_{\phi\phi}} \pm \left[\left(\frac{g_{t\phi}}{g_{\phi\phi}} \right)^2 - \frac{g_{tt}}{g_{\phi\phi}} \right]^{1/2}.$$

If $g_{tt} = 0$, the two solutions are

$$\frac{d\phi}{dt} = 0 \quad \text{and} \quad \frac{d\phi}{dt} = -\frac{2g_{t\phi}}{g_{\phi\phi}}.$$

The second solution is similar to (8.13); it represents a photon traveling in the direction of the black hole's rotation. The first solution is a photon traveling in the direction opposite to the black hole's rotation, but it appears to be not moving. The surface at which $g_{tt} = 0$ is certainly a special surface, known as the **stationary limit surface**. The region outside the event horizon but inside the stationary limit surface is called the **ergosphere**. The coordinate radius of the stationary limit surface is given by

$$r_{\text{erg}}(\theta) = M + \sqrt{M^2 - a^2 \cos^2 \theta}.$$

In the ergosphere, $g_{tt} > 0$, there is no way for a photon to move oppositely to the black hole's rotation. Material particles cannot travel faster than light, so they are inevitably dragged by the black hole no matter how much angular momentum they have. The ergosphere, however, is not the event horizon (except that it coincides with the event horizon at the poles). Particles can still move along the radial direction and escape. The overall structure of a Kerr black hole is shown in Figure 8.7.

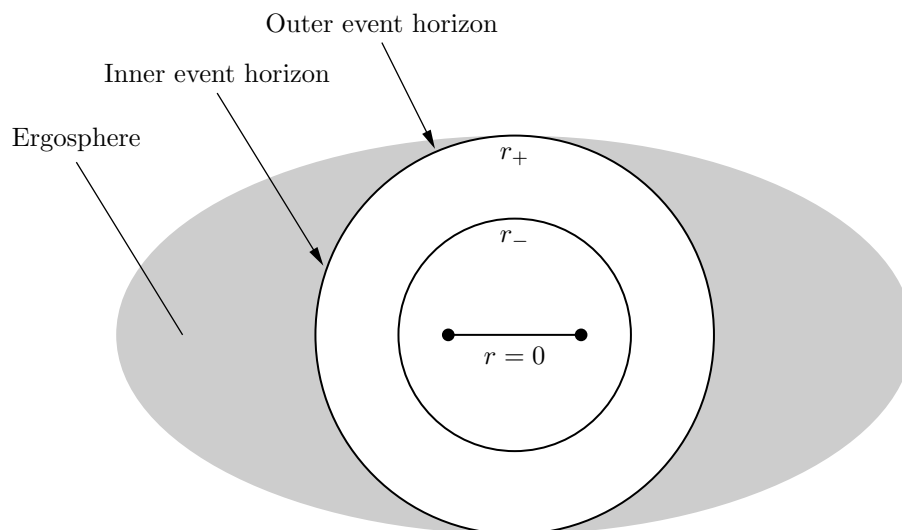


Figure 8.7: The structure of a Kerr black hole (side view) with $M > a$. There are two event horizons located at $r = r_{\pm}$. The curvature singularity $r = 0$ and $\theta = \pi/2$ is a ring. Outside the outer horizon is the ergosphere where no observer can remain stationary.

8.4.4 Equatorial Photon Trajectories

Now we want to investigate the radial motion of photon in the equatorial plane, $\theta = \pi/2$. The Kerr metric has the form

$$ds^2 = g_{tt}dt^2 + 2g_{t\phi}dt d\phi + g_{\phi\phi}d\phi^2 + g_{rr}dr^2.$$

For a photon with 4-momentum p^μ traveling along its geodesic, the conserved energy and angular momentum are

$$E = -g_{tt}p^t - g_{t\phi}p^\phi \quad \text{and} \quad L_z = g_{\phi t}p^t + g_{\phi\phi}p^\phi.$$

Solving for p^t and p^ϕ in terms of E and L_z ,

$$p^t = \frac{g_{\phi\phi}E + g_{\phi t}L_z}{g_{\phi t}^2 - g_{tt}g_{\phi\phi}} \quad \text{and} \quad p^\phi = -\frac{g_{\phi t}E + g_{tt}L_z}{g_{\phi t}^2 - g_{tt}g_{\phi\phi}}.$$

There is a useful identity for the Kerr metric:

$$g_{\phi t}^2 - g_{tt}g_{\phi\phi} = \Delta \sin^2 \theta, \quad \Delta = r^2 - 2Mr + a^2.$$

In the equatorial plane $\theta = \pi/2$, we just have $g_{\phi t}^2 - g_{tt}g_{\phi\phi} = \Delta$. Thus, p^t and p^ϕ can be written as

$$p^t = \frac{g_{\phi\phi}E + g_{\phi t}L_z}{\Delta} \quad \text{and} \quad p^\phi = -\frac{g_{\phi t}E + g_{tt}L_z}{\Delta}.$$

The normalization condition $g_{\mu\nu}p^\mu p^\nu = 0$ for a massless particle gives

$$0 = g_{tt} \left(\frac{g_{\phi\phi}E + g_{\phi t}L_z}{\Delta} \right)^2 - 2g_{\phi t} \frac{g_{\phi\phi}E + g_{\phi t}L_z}{\Delta} \frac{g_{\phi t}E + g_{tt}L_z}{\Delta} + g_{rr}(p^r)^2 + g_{\phi\phi} \left(\frac{g_{\phi t}E + g_{tt}L_z}{\Delta} \right)^2.$$

Isolating $g_{rr}(p^r)^2$ to one side, and simplifying the other side,

$$\begin{aligned} g_{rr}(p^r)^2 &= -g_{tt} \left(\frac{g_{\phi\phi}E + g_{\phi t}L_z}{\Delta} \right)^2 + 2g_{\phi t} \frac{g_{\phi\phi}E + g_{\phi t}L_z}{\Delta} \frac{g_{\phi t}E + g_{tt}L_z}{\Delta} - g_{\phi\phi} \left(\frac{g_{\phi t}E + g_{tt}L_z}{\Delta} \right)^2 \\ &= \frac{1}{\Delta^2} [-g_{tt}(g_{\phi\phi}^2 E^2 + 2g_{\phi\phi}g_{\phi t}EL_z + g_{\phi t}^2 L_z^2) + 2g_{\phi t}(g_{\phi\phi}g_{\phi t}E^2 + g_{\phi\phi}g_{tt}EL_z + g_{\phi t}^2 EL_z + g_{\phi t}g_{tt}L_z^2) - g_{\phi\phi}(g_{\phi t}^2 E^2 + 2g_{\phi t}g_{tt}EL_z + g_{tt}^2 L_z^2)] \\ &= \frac{1}{\Delta^2} [(-g_{tt}g_{\phi\phi} + g_{\phi t}^2)g_{\phi\phi}E^2 + 2(g_{\phi t}^2 - g_{tt}g_{\phi\phi})g_{\phi t}EL_z + (g_{\phi t}^2 - g_{tt}g_{\phi\phi})g_{tt}L_z^2] \\ &= \frac{1}{\Delta} (g_{\phi\phi}E^2 + 2g_{\phi t}EL_z + g_{tt}L_z^2). \end{aligned}$$

The radial equation of motion is

$$\left(\frac{dr}{d\lambda} \right)^2 = \frac{1}{g_{rr}\Delta} (g_{\phi\phi}E^2 + 2g_{\phi t}EL_z + g_{tt}L_z^2) = \frac{1}{r^2} (g_{\phi\phi}E^2 + 2g_{\phi t}EL_z + g_{tt}L_z^2).$$

According to this equation of motion, under what condition can the photon have a radial turning point along its trajectory? In order to get a radial turning point, the photon must have $dr/d\lambda = 0$ somewhere. In other words, the quadratic $g_{\phi\phi}E^2 + 2g_{\phi t}EL_z + g_{tt}L_z^2$ needs to have real roots. The discriminant of the quadratic is $4g_{\phi t}^2 - 4g_{tt}g_{\phi\phi} = 4\Delta$. (This is probably why Δ is denoted as “ Δ ”, the same letter for discriminant, in the Kerr metric.) We know that $\Delta = r^2 - 2Mr + a^2 = (r - r_+)(r - r_-)$ is negative when $r_- < r < r_+$, and as a result $g_{\phi\phi}E^2 + 2g_{\phi t}EL_z + g_{tt}L_z^2$ has no real roots. This is the region where the photon cannot turn back, proving that $r_\pm$ are event horizons.

The above analysis assumes that $\theta = \pi/2$. It is easy to generalize to any fixed polar angle θ ; just replace $\Delta \rightarrow \Delta \sin^2 \theta$ and $r^2 \rightarrow \rho^2$, and the conclusion will be the same. The analysis for arbitrary geodesics with $d\theta \neq 0$ are more complicated, but the fact that $r_\pm$ are event horizons is still true.

8.4.5 The Penrose Process

Consider a particle with 4-momentum p_0^μ moving along a geodesic. It is able to decay into two particles somehow with 4-momentum p_1^μ and p_2^μ , satisfying 4-momentum conservation,

$$p_0^\mu = p_1^\mu + p_2^\mu.$$

In the ergosphere, the Killing vector $\xi = \partial_t$ becomes spacelike because $g_{tt} > 0$. Consequently, it turns out that it is possible for a particle to have $E = -\xi_\mu p^\mu < 0$ in this region. Hence the following process can happen: a particle with positive energy $E_0 = -\xi_\mu p_0^\mu > 0$ enters the ergosphere through its geodesic. Then it decays into particle 1 and 2, with particle 2 traveling along a specific geodesic such that $E_2 = -\xi_\mu p_2^\mu < 0$. Due to 4-momentum conservation, particle one will carry an energy $E_1 = E_0 - E_2 > E_0$, more than the energy of the initial particle, and leave the ergosphere. That is, we can extract energy, if done properly, from the ergosphere. (*Ergon* means *work* in Greek, the origin of the word *ergosphere*.) This process is known as the **Penrose process**.

The ergosphere is a vacuum region, and obviously we cannot gain energy out of nothing. We are extracting the energy from the black hole, and the result is a decrease in its energy and angular momentum. Here is one way to view this. Define a new Killing vector, a linear combination of $\xi = \partial_t$ and $\eta = \partial_\phi$,

$$\chi^\mu = \xi^\mu + \Omega_H \eta^\mu,$$

where Ω_H is called the **angular velocity of the horizon**, defined as the angular velocity of photons at $r = r_+$.

$$\Omega_H = \left. \frac{d\phi}{dt} \right|_{r_+} = \left\{ -\frac{g_{t\phi}}{g_{\phi\phi}} \pm \left[\left(\frac{g_{t\phi}}{g_{\phi\phi}} \right)^2 - \frac{g_{tt}}{g_{\phi\phi}} \right]^{1/2} \right\} \Big|_{r_+}.$$

The square root contains $g_{t\phi}^2 - g_{tt}g_{\phi\phi} \sim \Delta$. At $r = r_+$, $\Delta = 0$ and $r_+^2 + a^2 = 2Mr_+$, so

$$\Omega_H = -\frac{g_{t\phi}}{g_{\phi\phi}} \Big|_{r_+} = \frac{2Mar_+}{(r_+^2 + a^2)^2 - a^2 \Delta \sin^2 \theta} = \frac{a}{2Mr_+}. \quad (8.14)$$

It is straightforward to check that the Killing vector χ is null on the horizon, timelike outside, and spacelike inside. Using the fact that the inner product of two future-directed timelike vectors is negative, we have

$$\chi_\mu p^\mu < 0$$

for any particle outside the horizon where χ^μ is timelike. But χ_μ is a Killing vector, so $\chi_\mu p^\mu$ is conserved and $\chi_\mu p^\mu < 0$ holds everywhere along the particle's geodesic. Plugging in $\chi^\mu = \xi^\mu + \Omega_H \eta^\mu$ and the conserved energy and angular momentum, we find that

$$L_z < \frac{E}{\Omega_H}.$$

In particular, $L_{z2} < E_2/\Omega_H$. If $E_2 < 0$ and $\Omega_H > 0$, the particle must have a negative angular momentum; it is moving against the black hole's rotation. After particle 2 falls into the black hole, the black hole's mass and angular momentum will decrease by $\delta M = E_2$ and $\delta J = L_2$. Then the inequality $L_{z2} < E_2/\Omega_H$ becomes an inequality for the black hole's mass and angular momentum,

$$\delta J < \frac{\delta M}{\Omega_H}. \quad (8.15)$$

It turns out that not all of the energy of a Kerr black hole can be extracted due to this inequality. There is an **irreducible mass** of a black hole, defined by the part of the black hole's mass that cannot be reduced by any *classical* process. It is related to **Hawking's area theorem**: the area of the event horizon is non-decreasing. (To be proved later.) The area of the event horizon is given by the integral over (θ, ϕ) at $r = r_+$ in a constant time slice. The induced metric on the horizon is

$$\gamma_{ij} dx^i dx^j = (r_+^2 + a^2 \cos^2 \theta) d\theta^2 + \frac{(r_+^2 + a^2)^2 \sin^2 \theta}{r_+^2 + a^2 \cos^2 \theta} d\phi^2.$$

The horizon area is then

$$A = \int d\theta d\phi \sqrt{|\gamma|} = \int_0^\pi d\theta \int_0^{2\pi} d\phi (r_+^2 + a^2) \sin \theta = 4\pi(r_+^2 + a^2) = 8\pi Mr_+.$$

Since the horizon area cannot decrease, it is believed that there should be an irreducible mass M_{irr} related to A . By dimensional analysis, the area is length squared, while the mass has the same dimension as length, so we expect

$$M_{\text{irr}}^2 = kA = 8\pi kMr_+ = 8\pi k(M^2 + \sqrt{M^4 - J^2}).$$

for some constant k . Now vary the irreducible mass squared M_{irr}^2 ,

$$\begin{aligned} 2M_{\text{irr}} \delta M_{\text{irr}} &= 8\pi k \left(2M \delta M + \frac{2M^3 \delta M - J \delta J}{\sqrt{M^4 - J^2}} \right) \\ &= \frac{8\pi k}{\sqrt{M^4 - J^2}} \left(2M \sqrt{M^4 - J^2} \delta M + 2M^3 \delta M - J \delta J \right) \\ &= \frac{8\pi k}{\sqrt{M^2 - a^2}} \left[2M \left(\sqrt{M^2 - a^2} + 2M \right) \delta M - a \delta J \right] \\ &= \frac{8\pi ka}{\sqrt{M^2 - a^2}} \left[\frac{2M (M + \sqrt{M^2 - a^2})}{a} \delta M - \delta J \right]. \end{aligned}$$

We recognize the factor $2M (M + \sqrt{M^2 - a^2})/a$ as the angular velocity of the horizon (8.14), so

$$\delta M_{\text{irr}} = \frac{4\pi ka}{M_{\text{irr}} \sqrt{M^2 - a^2}} \left(\frac{\delta M}{\Omega_H} - \delta J \right).$$

The inequality (8.15) says $\delta M/\Omega_H > \delta J$, which is equivalent to

$$\delta M_{\text{irr}} > 0.$$

The irreducible mass, suggested by the horizon area and Hawking's area theorem, is indeed irreducible. Then how do we determine the proportionality constant k in $M_{\text{irr}}^2 = kA$? Note that a Schwarzschild black hole does not have an ergosphere because of $J = 0$. The irreducible mass of a Kerr black hole, when all the angular momentum of the hole is gone, must be equal to the mass of a Schwarzschild black hole. The horizon area of the Schwarzschild black hole is just $A = 4\pi R_S^2 = 16\pi M^2$, where $R_S = 2M$ is the Schwarzschild radius. This gives the proportionality constant

$$M_{\text{irr}}^2 = M^2 = kA = 16\pi kM^2 \implies k = \frac{1}{16\pi}.$$

In conclusion, the irreducible mass of a Kerr black hole is related to its initial mass M and angular momentum J by

$$M_{\text{irr}}^2 = \frac{A}{16\pi} = \frac{1}{2} \left(M^2 + \sqrt{M^4 - J^2} \right). \quad (8.16)$$

The maximum energy we can extract from a black hole is the total mass minus the irreducible mass,

$$M - M_{\text{irr}} = M - \frac{1}{\sqrt{2}} \left(M^2 + \sqrt{M^4 - J^2} \right)^{1/2}.$$

For an extreme Kerr black hole, $M = J$, approximately 29% of its total energy can be extracted through an ideal Penrose process.

Example 8.1. Coalescence of two Kerr black holes

Consider two widely separated Kerr black holes with parameters (M_1, J_1) and (M_2, J_2) , initially at rest with their rotation axes aligned along the direction of their separation. The black hole then fall together and coalesce into a single black hole with parameters (M, J) . The spacetime is axially symmetric, the total angular momentum of black holes is conserved in this process, $J_1 + J_2 = J$. We want to derive an upper limit for the amount of energy that can be radiated away.

We know that the area of event horizons cannot decrease,

$$A_f \geq A_1 + A_2.$$

By (8.16), the area of the two black holes are

$$A_1 = 16\pi M_{\text{irr},1}^2 = 8\pi \left(M_1^2 + \sqrt{M_1^4 - J_1^2} \right), \quad A_2 = 16\pi M_{\text{irr},2}^2 = 8\pi \left(M_2^2 + \sqrt{M_2^4 - J_2^2} \right).$$

After the merger, the area of the final black hole is

$$A_f = 16\pi M_{\text{irr},f}^2 = 8\pi \left(M^2 + \sqrt{M^4 - J^2} \right), \quad \text{where } J = J_1 + J_2.$$

Using the area theorem,

$$\frac{1}{2} \left(M^2 + \sqrt{M^4 - J^2} \right) \geq M_{\text{irr},1}^2 + M_{\text{irr},2}^2.$$

Solving for M , we find that

$$M \geq \left[\frac{J^2 + 4(M_{\text{irr},1}^2 + M_{\text{irr},2}^2)^2}{4(M_{\text{irr},1}^2 + M_{\text{irr},2}^2)} \right]^{1/2}.$$

Since the black hole is initially at rest and they are far away, the initial energy is the total rest mass. The energy radiated is then given by

$$E_{\text{rad}} = M_1 + M_2 - M \leq M_1 + M_2 - \left[\frac{(J_1 + J_2)^2 + 4(M_{\text{irr},1}^2 + M_{\text{irr},2}^2)^2}{4(M_{\text{irr},1}^2 + M_{\text{irr},2}^2)} \right]^{1/2}.$$

Notice that when J_1 and J_2 are antiparallel, the energy radiated is the largest.

8.5 Black Hole Thermodynamics

8.5.1 Killing Horizons

A **null surface** Σ is a hypersurface where the normal vector at every point is a null vector. If a Killing vector χ is null along a null surface Σ , then Σ is called a **Killing horizon** of χ . The concept of a Killing horizon is independent of an event horizon, but they are closely related when the spacetime is static or stationary. It can be shown that Killing horizons have the following properties:

- Every event horizon in a stationary, asymptotically flat spacetime is a Killing horizon for some Killing vector χ .
- For static black holes, the event horizon is a killing horizon for the Killing vector $\xi = \partial_t$.
- For axisymmetric, stationary black holes, if there is a rotational Killing vector $\eta = \partial_\phi$, the Killing vector associated with the Killing horizon will be $\chi = \xi + \Omega_H \eta$ for some constant Ω_H .

Example 8.2. Rindler Spacetime

Consider the 2-dimensional Rindler spacetime

$$ds^2 = -X^2 dT^2 + dX^2$$

Apparently, $X = 0$ is a Killing horizon for the Killing vector ∂_T . Mathematically, Rindler spacetime is just a coordinate transformed version of Minkowski spacetime $ds^2 = -dt^2 + dx^2$ via

$$x = X \sinh T, \quad t = X \cosh T.$$

Example 8.3. Kerr spacetime

Kerr spacetime is an axisymmetric, stationary black holes with a timelike Killing vector ∂_t and a rotational Killing vector ∂_ϕ . In the previous section we found that neither ∂_t or ∂_ϕ is null on the event horizon r_+ . Instead, their linear combination $\chi = \xi + \Omega_H \eta$ is null on the event horizon, where Ω_H is the angular velocity of the horizon given by (8.14),

$$\Omega_H = \frac{a}{2Mr_+}.$$

8.5.2 Surface Gravity

Consider a Killing horizon Σ for a Killing vector χ . The object $\Phi = \chi_\mu \chi^\mu$ is a scalar function of spacetime. Along the constant surfaces of Φ , the gradient $\nabla_\mu \Phi$ is normal to these surfaces. This is because for any tangent vector T^μ , we have $T^\mu \nabla_\mu \Phi = 0$ along the constant surfaces because $T^\mu \nabla_\mu$ is the total derivative along T^μ . Thus, $\nabla_\mu \Phi$ is normal to any tangent vector, showing that it is normal to the constant surfaces.

Theorem 8.5.

Two null vectors ξ and η are orthogonal, $\xi_\mu \eta^\mu = 0$, iff they are proportional to each other, $\xi^\mu = c\eta^\mu$ for some constant c .

Proof. Orthogonality of two vectors is a local property. At each point in spacetime, we can construct a local inertial frame that is Minkowski with 6 additional degrees of freedom for Lorentz boosts or rotations. Choose a particular Lorentz boost and rotation such that $\xi^\mu = (\xi^0, \xi^0, 0, 0)$, where $\xi^0 \neq 0$. Taking the inner product of ξ^μ with $\eta^\mu = (\eta^0, \eta^1, \eta^2, \eta^3)$ and using orthogonality,

$$0 = \xi_\mu \eta^\mu = -\xi^0 \eta^0 + \xi^0 \eta^1 \implies \eta^0 = \eta^1.$$

Because η is also null,

$$0 = \eta_\mu \eta^\mu = -(\eta^0)^2 + (\eta^1)^2 + (\eta^2)^2 + (\eta^3)^2 = (\eta^2)^2 + (\eta^3)^2,$$

this means $\eta^2 = \eta^3 = 0$, leaving $\eta^\mu = (\eta^0, \eta^0, 0, 0)$, proportional to ξ^μ . The converse is straightforward to prove. $\square$

By definition, the Killing horizon Σ is a null surface with also $\Phi = 0$. A null surface has every normal vector being null, so $\nabla_\mu \Phi$ is a null vector at the Killing horizon. Meanwhile, $\Phi = \chi_\mu \chi^\mu = 0$ says that χ^μ is also a null vector at the Killing horizon. Now, $\nabla^\mu \Phi$ and χ^μ are orthogonal because

$$\chi^\mu \nabla_\mu (\chi_\nu \chi^\nu) = 2\chi^\mu \chi^\nu \nabla_\mu \chi_\nu = 2\chi^\mu \chi^\nu \nabla_{(\mu} \chi_{\nu)} = 0.$$

In the last equality, we used the fact that χ^μ is a Killing vector so that it satisfies Killing's equation $\nabla_{(\mu} \chi_{\nu)} = 0$. This means χ^μ , orthogonal to the normal vector $\nabla^\mu \Phi$, is tangent to the Killing horizon. Moreover, because χ^μ and $\nabla^\mu \Phi$ are both null vectors and they are orthogonal to each other, χ^μ must be proportional to $\nabla^\mu \Phi$. In other words, the Killing vector χ^μ is *both normal and tangent* to the Killing horizon.

Now we argue that χ^μ obeys the non-affinely parametrized geodesic equation along the Killing horizon. Evaluating $\nabla_\mu \Phi$,

$$\nabla_\mu \Phi = \nabla_\mu (\chi_\nu \chi^\nu) = 2\chi^\nu \nabla_\mu \chi_\nu = -2\chi^\nu \nabla_\nu \chi_\mu,$$

where again we used Killing's equation in the last equality. Since χ^μ is proportional to $\nabla^\mu \Phi$, we have the non-affinely parametrized geodesic equation

$$\chi^\nu \nabla_\nu \chi^\mu = -\kappa \chi^\mu, \quad (8.17)$$

where the constant of proportionality κ is called the **surface gravity** on the Killing horizon.

Theorem 8.6. Frobenius's theorem

A vector field ξ^μ is normal to a family of hypersurfaces if and only if on these hypersurfaces,

$$\xi_{[\alpha} \nabla_\mu \xi_{\nu]} = 0. \quad (8.18)$$

Proof. To be written. $\square$

Using Frobenius's theorem $\chi_{[\alpha} \nabla_\mu \chi_{\nu]}$ and Killing's equation $\nabla_{(\mu} \chi_{\nu)} = 0$, we can derive a formula for the surface gravity. Starting from Frobenius's theorem,

$$0 = \chi_{[\alpha} \nabla_\mu \chi_{\nu]} = \frac{1}{3}(\chi_\alpha \nabla_\mu \chi_\nu + \chi_\mu \nabla_\nu \chi_\alpha + \chi_\nu \nabla_\alpha \chi_\mu).$$

Moving $\chi_\alpha \nabla_\mu \chi_\nu$ to one side such that

$$\chi_\alpha \nabla_\mu \chi_\nu = -\chi_\mu \nabla_\nu \chi_\alpha - \chi_\nu \nabla_\alpha \chi_\mu.$$

Then contract both sides with $\nabla^\mu \chi^\nu$ and use (8.17),

$$\begin{aligned} \chi_\alpha (\nabla^\mu \chi^\nu) (\nabla_\mu \chi_\nu) &= -\chi_\mu (\nabla^\mu \chi^\nu) (\nabla_\nu \chi_\alpha) - \chi_\nu (\nabla^\mu \chi^\nu) (\nabla_\alpha \chi_\mu) \\ &= -(\chi_\mu \nabla^\mu \chi^\nu) \nabla_\nu \chi_\alpha - (\chi_\nu \nabla^\nu \chi^\mu) (\nabla_\mu \chi_\alpha) \\ &= \kappa \chi^\nu \nabla_\nu \chi_\alpha + \kappa \chi^\mu \nabla_\mu \chi_\alpha \\ &= -2\kappa^2 \chi_\alpha. \end{aligned}$$

This gives a formula for the surface gravity:

$$\kappa^2 = -\frac{1}{2}(\nabla_\mu \chi_\nu) (\nabla^\mu \chi^\nu). \quad (8.19)$$

The RHS should be evaluated at the Killing horizon for everything to be valid. We know that a Killing vector scaled by a constant is still a Killing vector. From (8.19), we see that κ scales linearly with χ^μ . To fix the surface gravity, we normalize the Killing vector χ^μ such that the time-translation Killing vector $\xi = \partial_t$ satisfies

$$\xi_\mu \xi^\mu(r \rightarrow \infty) = -1.$$

Note that for stationary spacetime with $\chi = \partial_t + \Omega_H \partial_\phi$, normalizing ∂_t also fixes this linear combination and hence the surface gravity.

The quantity κ is called the “surface gravity” because of the following reason: in a static, asymptotically flat spacetime, the surface gravity is the acceleration of a static observer near the horizon, as measured by a static observer at infinity. Consider a static observer with 4-velocity $u^\mu = (1/V, 0, 0, 0) = \xi^\mu/V$. Due to the normalization $u_\mu u^\mu = -1$, the function V is

$$V = \sqrt{-\xi_\mu \xi^\mu}.$$

For static spacetime, ξ^μ is the Killing vector associated with the Killing horizon. The quantity V is often called the **redshift factor**. This is because the energy of a photon is $E = -p_\mu \xi^\mu$; the energy measured by the static observer is $\omega = -p_\mu u^\mu = -p_\mu V \xi^\mu = E/V$. If a photon is emitted at x_1 with energy ω_1 with redshift factor $V(x_1)$, and the photon energy ω_∞ is measured by a static observer at infinity with redshift factor 1, the energy appears to be lowered by the redshift factor

$$\frac{\omega_\infty}{\omega_1} = \frac{E}{E/V(x_1)} = V(x_1) < 1.$$

The static observer near the black hole is obviously not following a geodesic. The 4-acceleration on the observer to remain static is

$$a^\mu = u^\nu \nabla_\nu u^\mu = \frac{\xi^\nu}{V} \nabla_\nu \left(\frac{\xi^\mu}{V} \right) = \frac{\xi^\nu}{V^2} \nabla_\nu \xi^\mu - \frac{\xi^\mu \xi^\nu}{V} \nabla_\nu \left(\frac{1}{V} \right).$$

The second term is zero because $\xi^\nu \nabla_\nu = \partial_t$ for a scalar function $1/V$, but $V = \sqrt{-\xi_\mu \xi^\mu} = \sqrt{-g_{tt}}$ is time independent, or otherwise $\xi = \partial_t$ would not be a Killing vector. Thus,

$$a^\mu = \frac{\xi^\nu}{V^2} \nabla_\nu \xi^\mu = -\frac{\xi_\nu}{V^2} \nabla^\mu \xi^\nu = -\frac{1}{2V^2} \nabla^\mu (\xi_\nu \xi^\nu) = \frac{1}{2V^2} \nabla^\mu V^2 = \nabla^\mu \ln V.$$

The magnitude of the acceleration is then

$$a = \sqrt{a_\mu a^\mu} = V^{-1} \sqrt{\nabla_\mu V \nabla^\mu V}.$$

As $V \rightarrow 0$ at the Killing horizon, the acceleration needed for an observer to remain static is infinite. Now, how can we interpret this acceleration for an observer at infinity? Consider a static observer at infinity holding an infinitely long (and stiff) rope connected to a static particle of mass m near the horizon. Let the 4-momentum of the mass be $p^\mu = m u^\mu = m \xi^\mu/V$. For an observer at infinity with 4-velocity $u_\infty^\mu = \xi^\mu$, it observes the energy of the mass to be

$$E_\infty = -p_\mu u_\infty^\mu = m \frac{\xi_\mu}{V} \xi^\mu = mV.$$

Define the outward radial unit vector n^μ such that $n_\mu n^\mu = 1$ and $n_\mu u^\mu = 0$. Suppose the observer move the rope outward by a proper distance $d\ell$. The work done on the particle in this process as measured at infinity is $dW = dE_\infty$, so the force measured at infinity is

$$F_\infty = \frac{dE_\infty}{d\ell} = m \frac{dV}{d\ell} = m n^\mu \nabla_\mu V = m a_\infty.$$

This gives the acceleration on the particle measured at infinity,

$$a_\infty = n^\mu \nabla_\mu V = V n^\mu \nabla_\mu \ln V = V n^\mu a_\mu.$$

Since a_μ of the particle is also radial, $n^\mu a_\mu$ is just the magnitude of the 4-acceleration a near the horizon. That is,

$$a_\infty = Va.$$

The surface gravity is the acceleration measured at infinity as the particle approaches the horizon while keeping it static:

$$\kappa = \lim_{r \rightarrow r_+} Va = \lim_{r \rightarrow r_+} \sqrt{\nabla_\mu V \nabla^\mu V}. \quad (8.20)$$

This physical interpretation holds only when the spacetime is static. As we have seen in Kerr spacetime (which is not static), a particle within the ergosphere cannot remain static but have to rotate. Nevertheless, the name “surface gravity” is still adopted.

Note that (8.20) is consistent with the definition of the surface gravity (8.17) if we replace $\xi^\mu \rightarrow \chi^\mu$. This can be proved by a direct

calculation of $\nabla_\mu V \nabla^\mu V$ at the horizon,

$$\nabla_\mu V \nabla^\mu V = \nabla_\mu (-\chi_\alpha \chi^\alpha)^{1/2} \nabla^\mu (-\chi_\nu \chi^\nu)^{1/2} = \left[-\frac{\chi^\alpha \nabla_\mu \chi_\alpha}{(-\chi_\beta \chi^\beta)^{1/2}} \right] \left[-\frac{\chi_\nu \nabla^\mu \chi^\nu}{(-\chi_\beta \chi^\beta)^{1/2}} \right] = \frac{\chi^\alpha \nabla_\alpha \chi_\mu}{V} \frac{\chi_\nu \nabla^\mu \chi^\nu}{V}.$$

Using the definition of the surface gravity (8.17), $\chi^\alpha \nabla_\alpha \chi_\mu = -\kappa \chi_\mu$,

$$\nabla_\mu V \nabla^\mu V = \frac{\kappa \chi_\mu}{V} \frac{\kappa \chi^\mu}{V} = \frac{\kappa^2 \chi_\mu \chi^\mu}{V^2} = \kappa^2.$$

In general, we would use (8.20) because the covariant derivative of a scalar function V is just its partial derivative. We don't have to worry about the Christoffel symbols in (8.19).

Example 8.4. Surface gravity of a Schwarzschild black hole

To be written.

Example 8.5. Surface gravity of a Kerr black hole

To be written.

8.5.3 Laws of Black Hole Mechanics

Consider a black hole with mass M , angular momentum J .

Zeroth law of black hole thermodynamics

The surface gravity κ is a constant on the event horizon for any stationary black holes.

Proof. To be written. □

First law of black hole thermodynamics

The change in the mass of a black hole is given by

$$dM = \frac{\kappa}{8\pi} dA + \Omega_H dJ,$$

where Ω_H is the angular velocity of the event horizon.

Proof. To be written. □

Second law of black hole thermodynamics: Hawking's area theorem

The area of a black hole always increase,

$$\delta A \geq 0.$$

Third law of black hole thermodynamics

It is impossible to achieve $\kappa = 0$ by a physical process.

Proof. To be written.

□

9 HOMOGENEOUS COSMOLOGY

9.1 Homogeneous, Isotropic Universe

9.1.1 Scales, Units, and the Cosmological Principle

We use special units on the extreme scale of cosmology

- **Length.** The astronomical unit (AU) is the mean distance between the Earth and Sun: $1 \text{ AU} = 1.50 \times 10^{11} \text{ m}$. This is only for our solar system. A larger unit that is useful for interstellar distances is the parsec (pc), the distance at which 1 AU subtends an angle of 1 arcsecond, $1 \text{ pc} = 3.09 \times 10^{16} \text{ m}$. The distance from the Sun to the nearest star Proxima Centauri is about 1.30 pc. To measure intergalactic distances, we often use the megaparsec (Mpc), $1 \text{ Mpc} = 10^6 \text{ pc} = 3.09 \times 10^{22} \text{ m}$. The distance between the Milky Way galaxy and the nearest galaxy (M31, or the Andromeda galaxy) is about 0.76 Mpc.
- **Mass and Luminosity.** The standard unit of mass and luminosity in astrophysics and cosmology are the mass and luminosity of the sun: $1 M_{\odot} = 1.998 \times 10^{30} \text{ kg}$ and $1 L_{\odot} = 3.85 \times 10^{24} \text{ W}$. The mass of the Milky Way galaxy is of the order $M_{\text{gal}} \sim 10^{12} M_{\odot}$, and its luminosity is about $L_{\text{gal}} \approx 3 \times 10^{10} L_{\odot}$.
- **Time.** For evolution of the universe, 1 year is too short. (This is not always the case. If we are studying the early universe, 1 second is a very long time.) We often use megayears (Myr), $1 \text{ Myr} = 10^6 \text{ yr} = 3.16 \times 10^{13} \text{ s}$, or even gigayears (Gyr), $1 \text{ Gyr} = 3.16 \times 10^{16} \text{ s}$. The age of the Earth is about 4.57 Gyr.

The universe is observed to be isotropic and homogeneous *on a large scale* ($\gtrsim 100 \text{ Mpc}$). This is known as the **cosmological principle**, and is the fundamental principle we will assume in this chapter. The universe is of course not truly isotropic or homogeneous, but observations indicate that the cosmological principle works as an extremely good approximation on a large scale.

9.1.2 Mathematical Definition of Homogeneity and Isotropy

A spacetime is said to be spatially **homogeneous** if there exists a one-parameter family of spacelike hypersurfaces Σ_t foliating the spacetime such that for each t the geometry of spacetime is the same. A spacetime is said to be spatially **isotropic** if there exists a family of observers with a tangent vector u^{μ} to the worldline and rotational symmetry about each observer, where $u_{\mu}u^{\mu} = -1$. In other words, there is no preferred tangent vector orthogonal to u^{μ} . Loosely speaking, at a given instant of time, homogeneity every point in space looks the same, while isotropy means every direction in space looks the same.

Let the induced metric on the hypersurfaces Σ_t be $h_{\mu\nu}$. There are several consequences of homogeneity and isotropy:

- Spatial isotropy then forces these hypersurfaces to be orthogonal to u^{μ} . If not, $h^{\mu\nu}u_{\mu} = v^{\nu}$ defines a preferred vector in space that violates isotropy. The induced metric orthogonal to u^{μ} can be defined as

$$h_{\mu\nu} = g_{\mu\nu} - u_{\mu}u_{\nu}. \quad (9.1)$$

It is easy to check that $h_{\mu\nu}u^{\mu} = 0$.

- Isotropic spacetime have vanishing Weyl tensor: $C_{\mu\nu\alpha\beta} = 0$. To show this, define $E_{\mu\nu} \equiv C_{\mu\alpha\nu\beta}u^{\alpha}u^{\beta}$ to be the **electric part** of the Weyl tensor. By the symmetry of the Weyl tensor, $E_{\mu\nu}$ is symmetric, orthogonal to u^{μ} , and traceless; in 4 dimensions, there are 5 independent components in $E_{\mu\nu}$ because of these constraints. Because there is no preferred direction in space, all vectors must be eigenvectors of E^{μ}_{ν} . This means E^{μ}_{ν} is proportional to the identity. But $E_{\mu\nu}$ is traceless, so all its components vanish. Similarly, define $B_{\mu\nu} = *C_{\mu\alpha\nu\beta}u^{\alpha}u^{\beta}$ to be the **magnetic part** of the Weyl tensor. $B_{\mu\nu}$ has the same property as $E_{\mu\nu}$, so $B_{\mu\nu} = 0$ by the same reasoning. In 4 dimensions, the Weyl tensor has 10 independent components. It can be reconstructed entirely using its electric and magnetic parts. Hence the Weyl tensor is zero in isotropic spacetime.

- The fact that the Weyl tensor is zero means that the spacetime is conformally flat. Vacuum means $R_{\mu\nu} = 0$. With the Weyl tensor also zero, the whole curvature is zero. This means the only isotropic vacuum solution is Minkowski. In other words, we have nothing to study in cosmology if we don't add matter to the universe.
- In isotropic universe, the only symmetric rank 2 tensors that can be constructed has the general form

$$T_{\mu\nu} = Au_\mu u_\nu + Bg_{\mu\nu},$$

so that every vector s^μ orthogonal to u^μ is an eigenvector of $T^\mu{}_\nu$,

$$T^\mu{}_\nu s^\nu = B\delta^\mu{}_\nu s^\nu.$$

Let e^μ be any spacelike unit vector orthogonal to u^μ . For an observer with 4-velocity u^μ , the energy density and pressure observed are $\rho = T_{\mu\nu}u^\mu u^\nu$ and $P = T_{\mu\nu}e^\mu e^\nu$, respectively. It is straightforward to solve for A and B . We can find that the only stress-energy tensor compatible with homogeneous, isotropic universe is a perfect fluid,

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + Pg_{\mu\nu}.$$

9.2 The Robertson-Walker Metric

9.2.1 Spaces of Constant Curvature

The homogeneity requirement implies that the curvature is the same throughout the universe. In particular, the Ricci scalar ${}^{(3)}R$ on the homogeneous hypersurface must be constant everywhere. This gives only three choices of the possible 3-dimensional space we can live in: constant positive curvature ($\kappa = +1$), zero curvature ($\kappa = 0$), or negative curvature ($\kappa = -1$). The space with $\kappa = +1$ is a 3-sphere. When embedded in a 4-dimensional flat Euclidean space with Cartesian coordinates (w, x, y, z) , a 3-sphere is defined by

$$w^2 + x^2 + y^2 + z^2 = R^2,$$

where R is the radius of curvature. In spherical coordinates of a unit 3-sphere,

$$x = \sin \chi \sin \theta \cos \phi, \quad y = \sin \chi \sin \theta \sin \phi, \quad z = \sin \chi \cos \theta, \quad w = \cos \chi.$$

The metric of a unit 3-sphere and its Ricci scalar is

$$ds^2 = d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2), \quad {}^3R = 6.$$

The space with $\kappa = 0$ is the ordinary Euclidean flat space. In Cartesian coordinates,

$$ds^2 = dx^2 + dy^2 + dz^2, \quad {}^3R = 0.$$

The space with $\kappa = -1$ is a 3-dimensional hyperboloid, defined by

$$w^2 - x^2 - y^2 - z^2 = R^2.$$

In hyperbolic coordinates of a unit hyperboloid,

$$x = \sinh \chi \sin \theta \cos \phi, \quad y = \sinh \chi \sin \theta \sin \phi, \quad z = \sinh \chi \cos \theta, \quad w = \cosh \chi,$$

and the metric and its Ricci scalar is

$$ds^2 = d\chi^2 + \sinh^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2), \quad {}^3R = -6.$$

9.2.2 The Robertson-Walker Metric

We have obtained the metric for the homogeneous hypersurfaces. The job now is to connect them in a 4-dimensional spacetime. Recall that each hypersurface Σ_t is labeled by the parameter t . We define t to be the **cosmic time**, the time on the clock carried by any isotropic observers. By homogeneity, all isotropic observers agree on the time difference between two hypersurfaces. The general form of the metric is given by

$$ds^2 = -dt^2 + R^2(t)[d\chi^2 + \sigma_\kappa^2(\chi)d\Omega^2], \quad \text{where} \quad \sigma_\kappa(\chi) = \begin{cases} \sin \chi, & \kappa = +1, \\ \chi, & \kappa = 0, \\ \sinh \chi, & \kappa = -1. \end{cases} \quad (9.2)$$

It is clear that the metric is spatially homogeneous and isotropic, as the spatial part is proportional to the spaces of constant curvature. Besides this, we have the freedom to multiply the spatial metric by a function $R^2(t)$ that does not affect spatial symmetry. The function $R(t)$ has dimensions of length. Define the coordinate $r \equiv R_0\chi$, where R_0 is the radius of curvature now. This gives the metric

$$ds^2 = -dt^2 + a^2(t)[dr^2 + S_\kappa^2(r)d\Omega^2], \quad \text{where} \quad S_\kappa(r) = \begin{cases} R_0 \sin(r/R_0), & \kappa = +1, \\ r, & \kappa = 0, \\ R_0 \sinh(r/R_0), & \kappa = -1. \end{cases} \quad (9.3)$$

The dimensionless variable $a(t) \equiv R(t)/R_0$ is called the **scale factor**. The evolution of the scale factor is what we study in the dynamics of the universe. The metric (9.3) is known as the **Robertson-Walker (RW) metric**, or the Friedmann-Robertson-Walker (FRW) metric, or the Friedmann-Lemaître-Robertson-Walker (FLRW) metric. . . . We will call it the FRW metric throughout this notes.

There is an equivalent way to write the FRW metric:

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - \kappa r^2/R_0^2} + r^2 d\Omega^2 \right].$$

This is sometimes useful because of the angular part $r^2 d\Omega^2$ in the usual spherical coordinates.

9.3 The Friedmann Equation

9.3.1 The Curvature Tensors

With the metric given in (9.3), it is straightforward to compute Christoffel symbols and components of the curvature tensors. For convenience, we set the radius of curvature $R_0 = 1$. It is easy to recover it based on dimensionality.

- Christoffel symbols:

$$\begin{array}{lll} \Gamma^t_{rr} = a\dot{a} & \Gamma^t_{\theta\theta} = a\dot{a}S_\kappa^2 & \Gamma^t_{\phi\phi} = a\dot{a}S_\kappa^2 \sin^2 \theta \\ \Gamma^r_{tr} = \frac{\dot{a}}{a} & \Gamma^r_{\theta\theta} = -S_\kappa \partial_r S_\kappa & \Gamma^r_{\phi\phi} = -S_\kappa \partial_r S_\kappa \sin^2 \theta \\ \Gamma^\theta_{t\theta} = \frac{\dot{a}}{a} & \Gamma^\theta_{t\theta} = \frac{\partial_r S_\kappa}{S_\kappa} & \Gamma^\theta_{\phi\phi} = -\cos \theta \sin \theta \\ \Gamma^\phi_{t\phi} = \frac{\dot{a}}{a} & \Gamma^\phi_{r\phi} = \frac{\partial_r S_\kappa}{S_\kappa} & \Gamma^\phi_{\theta\phi} = \cot \theta \end{array}$$

- Riemann curvature tensor:

$$\begin{aligned}
R^t_{trt} &= a\ddot{a} & R^t_{\theta t\theta} &= a\ddot{a}S_\kappa^2 & R^t_{\phi t\phi} &= a\ddot{a}S_\kappa^2 \sin^2 \theta \\
R^r_{trt} &= -\frac{\ddot{a}}{a} & R^r_{\theta r\theta} &= \dot{a}^2 S_\kappa^2 - S_\kappa \partial_r^2 S_\kappa & R^r_{\phi r\phi} &= (\dot{a}^2 S_\kappa^2 - S_\kappa \partial_r^2 S_\kappa) \sin^2 \theta \\
R^\theta_{t\theta t} &= -\frac{\ddot{a}}{a} & R^\theta_{r\theta r} &= \dot{a}^2 - \frac{\partial_r^2 S_\kappa}{S_\kappa} & R^\theta_{\phi\theta\phi} &= [\dot{a}^2 - (\partial_r S_\kappa)^2 + 1] \sin^2 \theta \\
R^\phi_{t\phi t} &= -\frac{\ddot{a}}{a} & R^\phi_{r\phi r} &= \dot{a}^2 - \frac{\partial_r^2 S_\kappa}{S_\kappa} & R^\phi_{\theta\phi\theta} &= \dot{a}^2 S_\kappa^2 - (\partial_r S_\kappa)^2 + 1
\end{aligned}$$

- Ricci tensor:

$$R_{tt} = -3\frac{\ddot{a}}{a}, \quad R_{rr} = a\ddot{a} + 2\dot{a}^2 + 2\kappa, \quad R_{\theta\theta} = R_{rr}S_\kappa^2, \quad R_{\phi\phi} = R_{\theta\theta} \sin^2 \theta.$$

- Ricci scalar:

$$R = 6 \left[\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a} \right)^2 + \frac{\kappa}{a^2} \right].$$

9.3.2 Evolution of Energy Density

Before we dive into Einstein's equation for homogeneous, isotropic cosmology, a **fluid equation** can be obtained from the conservation laws. We showed that the stress-energy tensor of a homogeneous, isotropic universe must be a perfect fluid, $T_{\mu\nu} = (\rho + P)u_\mu u_\nu + P g_{\mu\nu}$. Using the t -component of the conservation of stress-energy tensor,

$$0 = \nabla_\mu T^\mu_t = \partial_\mu T^\mu_t + \Gamma^\mu_{\mu\nu} T^\nu_t - \Gamma^\nu_{\mu t} T^\mu_\nu,$$

we can get a fluid equation:

$$\dot{\rho} = -3\frac{\dot{a}}{a}(\rho + P). \quad (9.4)$$

In cosmology, the relevant perfect fluids are assumed to obey the **equation of state**

$$P = w\rho, \quad (9.5)$$

where w is a dimensionless constant. Substituting the equation of state (9.5) into the fluid equation (9.4) gives

$$\frac{\dot{\rho}}{\rho} = -3(1+w)\frac{\dot{a}}{a}.$$

This equation is easy to integrate. The result is that the energy density takes a power law of the scale factor:

$$\rho = \rho_0 a^{-3(1+w)}. \quad (9.6)$$

The energy density of the universe essentially comes from three sources: matter, radiation, and dark energy. We shall analyze their energy density evolution related to the scale factor here.

- **Matter.** The **matter** component is composed of low-density, non-relativistic particles. This is probably the most familiar component to us. The equation of state of matter is the famous ideal gas law,

$$P = \frac{\rho}{\bar{m}} kT,$$

where $\bar{m}$ is the average rest mass of the particles, k is the Boltzmann constant, and T is the temperature. It is well-known from the equipartition theorem of thermodynamics that the average thermal (or kinetic) energy of particles in a gas is $\sim kT$. For example, a monatomic gas has average thermal energy of particle $\langle E_k \rangle = \frac{3}{2}kT$. Since we define matter as non-relativistic, the kinetic energy should be much less than the rest mass energy of the particles, $E_k \ll \bar{m}$. Therefore, matter obeys the equation of state with $w = 0$, i.e. matter is pressureless. The matter energy density then scales like

$$\rho_m = \rho_{m,0} a^{-3}.$$

This makes sense physically. As the universe expands, the volume scales like a^3 , so matter energy density is diluted as a^{-3} .

- **Radiation.** The **radiation** component is composed of relativistic particles. One example is the photon gas. It can be derived from statistical mechanics that a photon gas has energy density and pressure

$$\rho = a_{\text{rad}} T^4 \quad \text{and} \quad P = \frac{1}{3} a_{\text{rad}} T^4,$$

where in ordinary units $a_{\text{rad}} = \pi^2 k^4 / 60 \hbar^3 c^2$ is the radiation constant. Clearly, radiation obeys the equation of state with $w = 1/3$. The factor $w = 1/3$ can also be derived from the stress-energy tensor of the Maxwell field. See Section 2.6.2. (For a fermionic relativistic gas, such as a neutrino gas, its energy density and pressure may be different from those of a bosonic gas by a factor of 7/8. This is a factor from quantum statistics: Fermi-Dirac vs. Bose-Einstein statistics of particles. Nevertheless, the ratio between the pressure and energy density is still $w = 1/3$.) The radiation energy density scales like

$$\rho_r = \rho_{r,0} a^{-4}.$$

It drops one factor of a faster than matter energy density, which comes from the cosmological redshift. We can interpret this as the following: the photon number density scales like a^{-3} , just like matter. Additionally, the wavelength of individual photons is stretched due to expansion of the universe, $\lambda \propto a$, and the photon energy is proportional to λ^{-1} , or a^{-1} . Together, they give an $\rho_r \propto a^{-4}$ power law. Because radiation energy density has an extra a^{-1} factor compared to matter energy density, we expect the early universe (when a is small) to be dominated by radiation. Also, because the photon gas has energy density proportional to T^4 , there is a simple relation between the radiation temperature and the expansion of the universe, $T \propto a^{-1}$. When we talk about the temperature of the universe, we are referring to the temperature of the **cosmic microwave background (CMB)**, the grandest photon gas that fills up everywhere in the universe. It is also the most perfect blackbody spectrum that we found. The temperature fluctuation in the CMB is of the order $\sim 10^{-5}$, a good indication of homogeneity and isotropy of our universe.

- **Dark energy.** It is well-known that the major contribution of energy density in our universe now is dark energy. Any source with $w < -1/3$ is generally called dark energy. The dark energy in our universe is a result from putting a cosmological constant Λ in Einstein's equation. The equation of state of a cosmological constant corresponds to $w = -1$, a constant energy density. We will see this energy density contribution directly from Einstein's equation in the next section.

9.3.3 The Friedmann Equation

Einstein's equation says

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}.$$

If we plug in the Ricci tensor and Ricci scalar calculated above, Einstein's equation gives two independent equations that govern the dynamics of the universe: the tt -component is

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3} \rho - \frac{\kappa}{a^2 R_0^2} + \frac{\Lambda}{3}, \quad (9.7)$$

and the $\chi\chi$ -, $\theta\theta$ -, and $\phi\phi$ -components all give

$$\frac{\ddot{a}}{a} = -\frac{4\pi}{3} (\rho + 3P) + \frac{\Lambda}{3}. \quad (9.8)$$

The first equation is usually referred to just as **the Friedmann equation**. It is the governing equation of cosmological dynamics. The second equation is sometimes called the **acceleration equation** or the **second Friedmann equation**. A universe with the scale factor obeying the Friedmann equations is called an **FRW universe**. In the Friedmann equation (9.7), the cosmological constant can be seen as an energy density term. It corresponds to a time-independent energy density of

$$\rho_\Lambda = \frac{\Lambda}{8\pi},$$

Because we can view the cosmological constant as an energy density, it is sometimes absorbed in the total energy density ρ and does not appear in the Friedmann equation explicitly. (In fact, we can even view the curvature term as a fictitious energy density with $w = -1/3$.) Notice that the fluid equation (9.4) can be derived from the two Friedmann equations, since the conservation of stress-energy tensor is already inside Einstein's equation. This means we have $2 + 1 = 3$ independent equations (2 of the Friedmann equation, acceleration equation, or fluid equation, plus 1 from the equation of state) for 3 variables: $a(t)$, $\rho(t)$, and $P(t)$. One of the goals of cosmology is to solve for the evolution of the scale factor using the Friedmann equation.

9.3.4 Cosmological Parameters

Here we define some terminology frequently used in cosmology. First, the cosmic time at *present* is denoted as t_0 , and by convention we set $a(t_0) = 1$. Every quantity measured at t_0 will have a subscript 0. The spatial coordinates (r, θ, ϕ) are called **comoving coordinates**. The comoving coordinates of any point is constant if the expansion of the universe is perfectly homogeneous and isotropic. The only way to change the comoving coordinates of an object is to account for the object's own motion in the universe. For example, a galaxy may be attracted by another galaxy. Their relative comoving distance is changing. On the other hand, the **proper distance** is the physical distance between two points at the same cosmic time. Consider two galaxies with the same coordinates (θ, ϕ) , the proper distance between them is

$$d_p(t) = a(t) \int_0^r dr' = ar.$$

Suppose we want to determine the proper distance between us and a galaxy at time t . If we know the time of emission t_e of the light from the galaxy, the proper distance at time t can be calculated by

$$d_p(t) = a(t) \int_0^r dr' = a(t) \int_{t_e}^t \frac{dt'}{a(t')}. \quad (9.9)$$

This comes from the fact that light travels along null geodesics, $dt^2 = a^2(t)dr^2$. The **horizon distance** is defined as the farthest object an observer can observe in its light cone. All we need to do is to extend $t_e \rightarrow 0$,

$$d_{\text{hor}}(t) = a(t) \int_0^t \frac{dt'}{a(t')}. \quad (9.10)$$

A reasonable question to ask is what is the rate of change of the proper distance between two objects? Taking the time-derivative of the proper distance,

$$\dot{d}_p = \dot{a}r = \frac{\dot{a}}{a}d_p.$$

This means in a homogeneous, isotropic universe, an observer will see the speed of an object to be proportional to their relative proper distance at a fixed cosmic time. This is the famous **Hubble's law**. We define the **Hubble parameter** to be a measure of the rate of expansion of the universe.

$$H(t) \equiv \frac{\dot{a}}{a}. \quad (9.11)$$

Due to historical reasons, the units of the Hubble parameter is usually given in “speed per distance”. The Hubble parameter now $H_0 = H(t_0)$ is the **Hubble constant**. There is still some debate in the value of the Hubble constant, known as the *Hubble tension*. The approximate value is

$$H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1} \approx 2.3 \times 10^{-18} \text{ s}^{-1}.$$

Typical cosmological scales are set by the **Hubble length** and the **Hubble time** (which are the same when $c = 1$), defined as

$$d_H = \frac{c}{H_0} = 2.1 \times 10^3 \text{ Mpc} \quad \text{and} \quad t_H = H_0^{-1} = 14 \text{ Gyr}.$$

In terms of the Hubble parameter, the Friedmann equation can be written as

$$H^2 = \frac{8\pi}{3}\rho - \frac{\kappa}{a^2 R_0^2} \quad (9.12)$$

with the cosmological constant absorbed in ρ . We now define the **critical density** ρ_c to be the density at which the universe becomes flat ($\kappa = 0$):

$$\rho_c \equiv \frac{3H^2}{8\pi}. \quad (9.13)$$

Another useful quantity to define is the **density parameter**, Ω , which is the ratio of the energy density to the critical density,

$$\Omega \equiv \frac{8\pi\rho}{3H^2} = \frac{\rho}{\rho_c}. \quad (9.14)$$

The Friedmann equation in terms of the density parameter is then

$$\Omega - 1 = \frac{\kappa}{H^2 a^2 R_0^2}. \quad (9.15)$$

The critical density does evolve with time. However, notice that the sign of the RHS of (9.15) is fixed once κ is set. This means whether Ω is greater than, equal to, or less than one cannot be changed as the universe evolves. This is why ρ_c is called the *critical* density. Observation nowadays indicates that $0.995 < \Omega < 1.005$; the curvature is still unknown, but our universe is very close to flat!

Finally, we define the **cosmological redshift** z to be the redshift to compare the wavelength of light observed today and that emitted by some source. Let the 4-velocity of a comoving observer to be $u^\mu = (1, 0, 0, 0)$. Then it is straightforward to check that

$$K_{\mu\nu} = a^2 h_{\mu\nu} = a^2 (g_{\mu\nu} + u_\mu u_\nu)$$

is a Killing tensor, i.e. $\nabla_{(\alpha} K_{\mu\nu)} = 0$. Then there is a conserved quantity $K^2 = K_{\mu\nu} p^\mu p^\nu$ for a particle with 4-momentum p^μ along its geodesic. For null geodesics, $p_\mu p^\mu = 0$, we should find that

$$K = a u_\mu p^\mu = -aE,$$

where E is the energy of the photon. In other words, if $E(t_0)$ is the observed photon energy today, and $E(t_e)$ is the photon energy emitted by some galaxy,

$$\frac{E(t_0)}{E(t_e)} = \frac{\lambda(t_e)}{\lambda(t_0)} = \frac{a(t_e)}{a(t_0)} = a(t_e).$$

The redshift is defined as

$$z = \frac{\lambda(t_0) - \lambda(t_e)}{\lambda(t_e)}.$$

This implies a formula relating the scale factor at the time of emission to the redshift observed:

$$\boxed{1 + z = \frac{1}{a(t_e)}}. \quad (9.16)$$

The cosmological redshift is different from Doppler effect. It occurs because the photon traveling through space loses energy due to the expanding universe. In cosmology, the redshift is a common indicator the time of an event. For example, we say that the first atom forms at a redshift $z = 1380$. This is an equivalent way to say that the scale factor at the time of the first atom formation is 1381 times smaller than it is nowadays.

9.4 Model Universes

In this section, we explore various models of the universe and solve the Friedmann equation

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3}\rho - \frac{\kappa}{a^2 R_0^2}.$$

We discuss the empty universe, single-component universes, multiple-component universes, and finally the standard model of cosmology we use today. The 4 possible components are matter, radiation, cosmological constant, and *curvature*. For the most part, we only state the result and skip the derivations. For a more detailed discussion and plots of the scale factor, please consult Chapter 7 of **ASTROPHYSICS** notes or other textbooks.

9.4.1 Empty (Milne) Universe

An empty universe is a universe with $\rho = 0$ (including the cosmological constant). The Friedmann equation for an empty universe is

$$\dot{a}^2 = -\frac{\kappa}{R_0^2}.$$

An immediate but uninteresting solution is Minkowski spacetime with constant scale factor. Notice that the LHS is non-negative, so an empty universe cannot have positive curvature. A negatively curved universe has solution (assuming expanding universe)

$$a(t) = \frac{t}{t_0}, \quad \text{where } t_0 = R_0.$$

The normalization factor $1/t_0$ is to ensure that $a(t_0) = 1$. This universe is known as the **Milne universe**. The redshift of a light source is given by

$$1 + z = \frac{1}{a(t_e)} = \frac{t_0}{t_e}$$

and the current proper distance (9.9) from us to a galaxy is

$$d_p(t_0) = t_0 \int_{t_e}^{t_0} \frac{dt}{t} = t_0 \ln \left(\frac{t_0}{t_e} \right) = \frac{1}{H_0} \ln(1 + z).$$

The horizon distance as $t_e \rightarrow 0$ is infinite, which means we can see objects currently at an arbitrarily large distance in the Milne universe. If we are interested in the proper distance between us to a galaxy at the time of emission, all we need to do is to divide $d_p(t_0)$ by $1 + z$:

$$d_p(t_e) = \frac{a(t_e)}{a(t_0)} d_p(t_0) = a(t_e) d_p(t_0) = \frac{1}{H_0} \frac{\ln(1 + z)}{1 + z}.$$

The maximum $d_p(t_e)$ occurs at redshift $z = e - 1$. Interestingly, objects with higher redshift has smaller proper distance at the time of emission.

9.4.2 Single-Component Universes

A single-component universe is spatially flat ($\kappa = 0$), with only one component with a certain value of $w \neq -1$, i.e. not cosmological constant. The Friedmann equation reduces to

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi\rho_0}{3} a^{-(1+3w)}.$$

The solution is

$$a(t) = \left(\frac{t}{t_0} \right)^{2/(3+3w)}, \quad \text{where} \quad t_0 = \frac{1}{1+w} \frac{1}{\sqrt{6\pi\rho_0}}.$$

From the solution, it is reasonable to see that why we need $w \neq -1$. The Hubble constant of a single-component universe is

$$H_0 = \left(\frac{\dot{a}}{a} \right) \Big|_{t=t_0} = \frac{2}{3t_0(1+w)}.$$

We see that if $w > -1/3$, the universe is younger than the Hubble time, and otherwise if $w < -1/3$. This makes sense because any component with $w < -1/3$ is considered dark energy that accelerates the expansion of the universe. Given the scale factor, we can find the redshift and its relation to t_e ,

$$1 + z = \frac{1}{a(t_e)} = \left(\frac{t_0}{t_e} \right)^{2/(3+3w)} \implies t_e = \frac{t_0}{(1+z)^{3(1+w)/2}} = \frac{2(1+z)^{-3(1+w)/2}}{3(1+w)H_0}.$$

The current proper distance from an object and the horizon distance are

$$d_p(t_0) = \frac{2c}{H_0(1+3w)} \left[1 - \frac{1}{(1+z)^{(1+3w)/2}} \right] \quad \text{and} \quad d_{\text{hor}}(t_0) = \frac{2c}{H_0(1+3w)}.$$

The horizon distance is finite if $w > -1/3$, and is infinite for $w \leq -1/3$.

Example 9.1. Matter-only universe

The matter-only universe with $w = 0$ has cosmological parameters

$$t_0 = \frac{2}{3H_0}, \quad d_{\text{hor}}(t_0) = \frac{2}{H_0}, \quad a(t) = \left(\frac{t}{t_0} \right)^{2/3}.$$

A galaxy with redshift z is at proper distance at the time of observation/emission

$$d_p(t_0) = \frac{2}{H_0} \left[1 - \frac{1}{\sqrt{1+z}} \right], \quad d_p(t_e) = \frac{2}{H_0(1+z)} \left[1 - \frac{1}{\sqrt{1+z}} \right].$$

$d_p(t_e)$ is at maximum value $8/27H_0$ when $z = 5/4$. The matter-only universe is a good approximation to our universe in the matter-dominated era.

Example 9.2. Radiation-only universe

The radiation-only universe with $w = 1/3$ has cosmological parameters

$$t_0 = \frac{1}{2H_0}, \quad d_{\text{hor}}(t_0) = \frac{1}{H_0}, \quad a(t) = \left(\frac{t}{t_0}\right)^{1/2}.$$

Note that the horizon distance of a radiation-only universe is exactly equal to the Hubble distance. The proper distance at the time of observation/emission are

$$d_p(t_0) = \frac{1}{H_0} \frac{z}{1+z}, \quad d_p(t_e) = \frac{1}{H_0} \frac{z}{(1+z)^2}.$$

$d_p(t_e)$ is at maximum value $1/4H_0$ at $z = 1$. The radiation-only universe is useful model our universe in the radiation-dominated era—the early universe.

Example 9.3. Λ -only universe

The Λ -only universe has $w = -1$, and we cannot use the formula derived above. Fortunately, the Friedmann equation is even easier to solve in this case:

$$\dot{a}^2 = \frac{8\pi\rho_\Lambda}{3}a^2,$$

where ρ_Λ is constant with time. Also, the cosmological constant needs to be positive for the Friedmann equation to work. (The scale factor needs to be real.) For an expanding universe, the equation can be written as

$$\frac{\dot{a}}{a} = H_0, \quad \text{where} \quad H_0 = \left(\frac{8\pi\rho_\Lambda}{3}\right)^{1/2}.$$

The solution is an exponential growth,

$$a(t) = e^{H_0(t-t_0)}.$$

The other cosmological parameters are

$$z = e^{H_0(t_0-t_e)} - 1, \quad d_p(t_0) = \frac{1}{H_0} \left[e^{H_0(t_0-t_e)-1} \right] = \frac{z}{H_0}, \quad d_p(t_e) = \frac{1}{H_0} \frac{z}{1+z}.$$

The Λ -only universe with a positive cosmological constant is called the **de Sitter universe**. It does not have a well-defined age of the universe because there is no time at which $a(t) = 0$. In other words, there is no Big Bang in this universe. Our universe in the far future can be modeled by a Λ -only universe.

9.4.3 Multiple-Component Universes

For multiple-component universes, we start with the Friedmann equation in the form (9.15):

$$1 - \Omega(t) = -\frac{\kappa}{H^2 a^2 R_0^2}.$$

Let Ω_0 be the density parameter today. We can express the curvature as

$$\frac{\kappa}{R_0^2} = H_0^2(\Omega_0 - 1).$$

Then use the Friedmann equation in the form (9.12):

$$H^2 = \frac{8\pi}{3}\rho - \frac{H_0^2}{a^2}(\Omega_0 - 1) \implies \frac{H^2}{H_0^2} = \frac{\rho}{\rho_{c,0}} + \frac{1 - \Omega_0}{a^2},$$

where $\rho_{c,0} \equiv 3H_0^2/8\pi$ is the current critical density. The total energy density is the sum of the energy density between individual components, $\rho = \rho_m + \rho_r + \rho_\Lambda$. Using $\rho_m = \rho_{m,0}a^{-3}$, $\rho_r = \rho_{r,0}a^{-4}$, and $\rho_\Lambda = \rho_{\Lambda,0}$, the Friedmann equation for a general multiple-component universe is

$$\frac{H^2}{H_0^2} = \frac{\Omega_{r,0}}{a^4} + \frac{\Omega_{m,0}}{a^3} + \Omega_{\Lambda,0} + \frac{1 - \Omega_0}{a^2}. \quad (9.17)$$

The sum of all density parameters is fixed by $\Omega_{r,0} + \Omega_{m,0} + \Omega_{\Lambda,0} = \Omega_0$. Below we show some examples of analytical solutions, but remember that equation (9.17) usually requires numerical integration if all components are present.

Example 9.4. Radiation + Matter

The spatially flat, radiation + matter universe is an appropriate model for our universe when radiation and matter contributes to the energy density roughly equally. The Friedmann equation is

$$\frac{H^2}{H_0^2} = \frac{\Omega_{r,0}}{a^4} + \frac{\Omega_{m,0}}{a^3}.$$

The exact solution is

$$H_0 t = \frac{4a_{rm}^2}{3\sqrt{\Omega_{r,0}}} \left[1 - \left(1 - \frac{a}{2a_{rm}} \right) \left(1 + \frac{a}{a_{rm}} \right)^{1/2} \right], \quad \text{where} \quad a_{rm} \equiv \frac{\Omega_{r,0}}{\Omega_{m,0}}.$$

There is not much interesting consequence in radiation + matter universe.

Example 9.5. Matter + Curvature

The Friedmann equation for a universe with matter and curvature is

$$\frac{H^2}{H_0^2} = \frac{\Omega_0}{a^3} + \frac{1 - \Omega_0}{a^2},$$

where $\Omega_{m,0} = \Omega_0$ since there is only one component except the curvature. The exact solution is often written in a parametric form. For $\Omega_0 > 1$ (positive curvature),

$$a(\eta) = \frac{\Omega_0}{2(\Omega_0 - 1)}(1 - \cos \eta), \quad t(\eta) = \frac{\Omega_0}{2H_0(\Omega_0 - 1)^{3/2}}(\eta - \sin \eta),$$

where $\eta \in [0, 2\pi]$. Interestingly, there are two points where the scale factor goes to zero: $\eta = 0$ defines a Big Bang of the universe, and $\eta = 2\pi$ defines a **Big Crunch**. The cosmic time of the Big Crunch is

$$t_{\text{crunch}} = \frac{\pi\Omega_0}{H_0(\Omega_0 - 1)^{3/2}}.$$

For $\Omega_0 < 1$ (negative curvature),

$$a(\eta) = \frac{\Omega_0}{2(1 - \Omega_0)}(\cosh \eta - 1), \quad t(\eta) = \frac{\Omega_0}{2H_0(1 - \Omega_0)^{3/2}}(\sinh \eta - \eta),$$

where $\eta \in [0, \infty)$. The negatively curved universe does not have a Big Crunch. (As an aside, the variable η is called the **conformal time**, defined as $d\eta = a dt$. It will be used more frequently in advanced cosmology.)

Example 9.6. Radiation + Curvature

The radiation + curvature universe is very similar to the matter + curvature universe. The Friedmann equation for a universe with radiation and curvature is

$$\frac{H^2}{H_0^2} = \frac{\Omega_0}{a^4} + \frac{1 - \Omega_0}{a^2},$$

where $\Omega_{r,0} = \Omega_0$. For $\Omega_0 > 1$ (positive curvature),

$$a(\eta) = \left(\frac{\Omega_0}{\Omega_0 - 1} \right)^{1/2} \sin \eta, \quad t(\eta) = \frac{1}{H_0} \left(\frac{\Omega_0}{\Omega_0 - 1} \right)^{1/2} (1 - \cos \eta),$$

where $\eta \in [0, \pi]$. The Big Crunch occurs at

$$t_{\text{crunch}} = \frac{2}{H_0} \left(\frac{\Omega_0}{\Omega_0 - 1} \right)^{1/2}.$$

For $\Omega_0 < 1$ (negative curvature),

$$a(\eta) = \left(\frac{\Omega_0}{1 - \Omega_0} \right)^{1/2} \sinh \eta, \quad t(\eta) = \frac{1}{H_0} \left(\frac{\Omega_0}{1 - \Omega_0} \right)^{1/2} (\cosh \eta - 1),$$

where $\eta \in [0, \infty)$.

Example 9.7. Matter + Λ

The spatially flat, matter + Λ universe models our current universe well. In a flat universe, $\Omega_{\Lambda,0} = 1 - \Omega_{m,0}$. The Friedmann equation (9.17) reads

$$\frac{H^2}{H_0^2} = \frac{\Omega_{m,0}}{a^3} + (1 - \Omega_{m,0}).$$

The solution depends on whether the cosmological constant is positive or negative. If the cosmological constant is positive, then $\Omega_{m,0} < 1$. The RHS of the Friedmann equation is always positive. The universe will expand forever if it is expanding at $t = t_0$, which is basically the fate of our universe. The exact solution is

$$a(t) = \left(\frac{\Omega_{m,0}}{1 - \Omega_{m,0}} \right)^{1/3} \sinh^{2/3} \left[\frac{3}{2} H_0 \sqrt{1 - \Omega_{m,0}} t \right].$$

If the cosmological constant is negative, then $\Omega_{m,0} > 1$. At some time the RHS of the Friedmann equation will be zero and then becomes negative. The maximum scale factor occurs at $H = 0$,

$$a_{\text{max}} = \left(\frac{\Omega_{m,0}}{\Omega_{m,0} - 1} \right)^{1/3}.$$

The exact solution is

$$a(t) = \left(\frac{\Omega_{m,0}}{\Omega_{m,0} - 1} \right)^{1/3} \sin^{2/3} \left[\frac{3}{2} H_0 \sqrt{\Omega_{m,0} - 1} t \right].$$

This solution also admits a big crunch, which occurs at

$$t_{\text{crunch}} = \frac{2\pi}{3H_0\sqrt{\Omega_{m,0} - 1}}.$$

9.4.4 The Λ CDM Model

The **Lambda Cold Dark Matter (Λ CDM)** is the benchmark model we use today. It is assumed to be flat, and composed of primarily dark energy (Λ), cold (meaning non-relativistic) dark matter, ordinary matter, and very small amount of radiation. It is not guaranteed to be correct, but it matches the observation such as the cosmic microwave background, the accelerating universe, the large-scale structures, and the abundances of baryonic matter. The Λ CDM model is considered the “standard model of cosmology”. However, it is obviously not contained in the Standard Model of elementary particles. The Standard Model contains neither Λ nor CDM. Until now, we don’t have a clear understanding of dark matter or dark energy.

Table 9.1 summarizes the density parameters of components in the Λ CDM model.

Components	Density parameters
Total radiation	$\Omega_{r,0} = 9.0 \times 10^{-5}$
Photons	$\Omega_{\gamma,0} = 5.35 \times 10^{-5}$
Neutrinos	$\Omega_{\nu,0} = 3.65 \times 10^{-5}$
Total matter	$\Omega_{m,0} = 0.31$
Baryonic matter	$\Omega_{\text{bary},0} = 0.048$
Nonbaryonic dark matter	$\Omega_{\text{dm},0} = 0.262$
Cosmological Constant	$\Omega_{\Lambda,0} = 0.69$

Table 9.1: Properties of the Λ CDM model.

The scale factor $a(t)$ can be computed numerically via the Friedmann equation (9.17). Figure 9.1 shows the evolution of the scale factor.

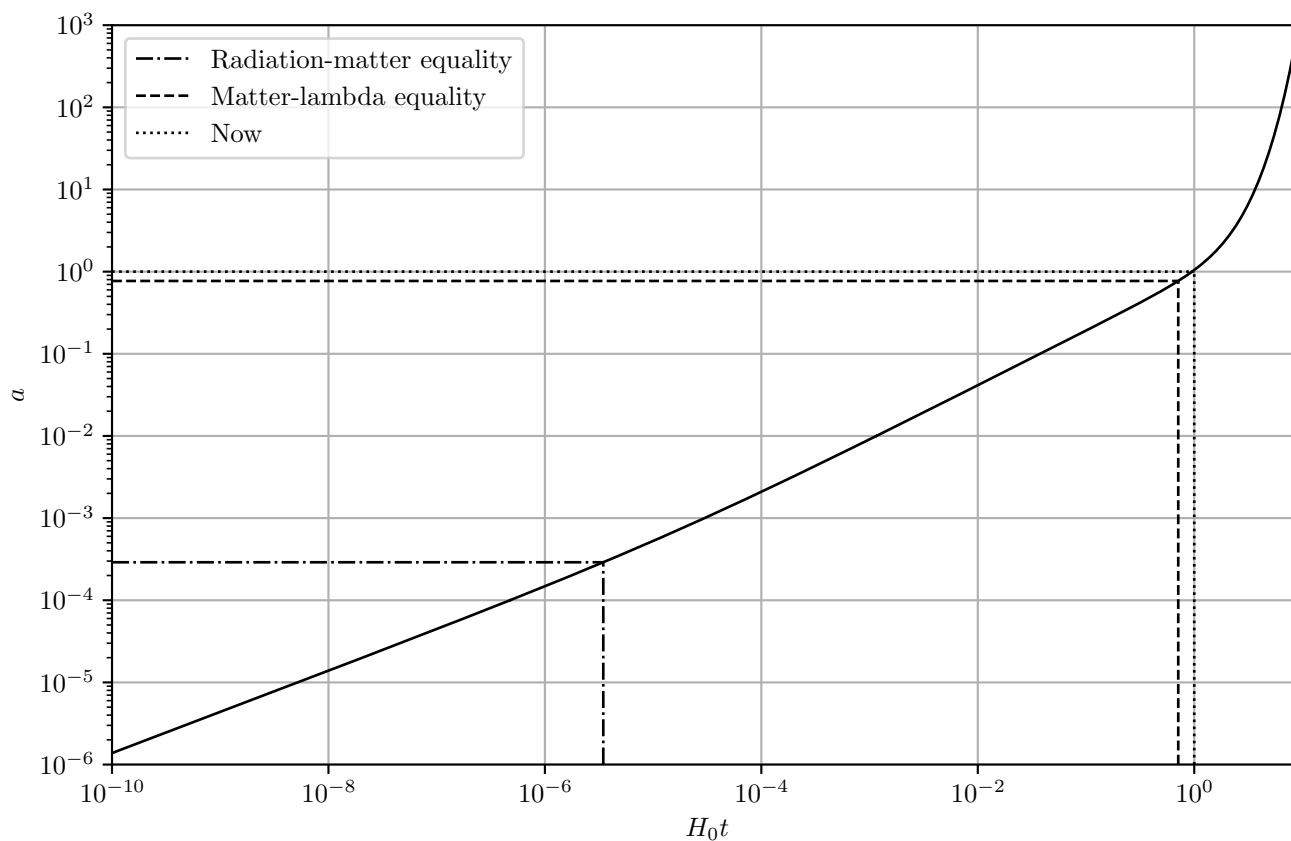


Figure 9.1: The scale factor as a function of t in units of the Hubble time.

The current horizon distance of the Λ CDM is

$$d_{\text{hor}}(t_0) = 3.20/H_0 = 14000 \text{ Mpc}.$$

9.5 The History of the Universe

We conclude this chapter by going through the history of our universe backward by discussing important epochs and events happened in the past. The Λ CDM model is adopted.

- **Present.** The universe now is $t_0 = 13.8$ Gyr old, with a (CMB) temperature of 2.725 K. The scale factor is set to $a(t_0) = 1$. The density parameter now is dominated by the cosmological constant, accelerating the expansion of the universe.
- **Matter- Λ equality.** It is known that $\Omega_m = \Omega_{m,0}a^{-3}$ and $\Omega_\Lambda = \Omega_{\Lambda,0}$. We expect that at some time in the past, the energy density of matter (including dark matter) was equal to that of the cosmological constant. This moment is called the **matter- Λ equality**. It is not very far away in the past (in cosmological sense). The matter- Λ equality occurred at

$$a_{m\Lambda} = \left(\frac{\Omega_{m,0}}{\Omega_{\Lambda,0}} \right)^{1/3} = \left(\frac{0.31}{0.69} \right)^{1/3} = 0.77,$$

which corresponds to the age of the universe $t_{m\Lambda} = 10.2$ Gyr. Near this event, the matter + Λ universe model is a good approximation.

- **Matter era.** In the matter-dominated era, stars (including the Sun), galaxies, and large-scale structures formed. The scale factor grew like $a \sim t^{2/3}$.
- **Recombination.** The epoch of **recombination** is the time at which the baryonic component of the universe transition from an ionized plasma to a gas of neutral elements. It is defined to the instant when the number density of ions is equal to that of neutral atoms. Before the epoch of recombination, the baryonic matter and radiation were in thermal equilibrium. Thus, when a proton captured an electron and tried to form an atom, energetic photons could frequently ionize the electron, preventing atom formation. To determine the time of recombination requires an analysis to the reaction $\text{H} + \gamma \leftrightarrow p + e^-$ using the **Saha equation**:

$$\frac{n_p n_e}{n_H} = \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-\chi/kT},$$

where n_p , n_e , n_H are the number density of proton, electron, and hydrogen, respectively, $h = 2\pi\hbar$ is the Planck constant, and $\chi = 13.6$ eV is the binding energy of hydrogen atom. After some careful analysis, we can find the temperature at recombination was $T_{\text{rec}} = 3760$ K. Using the fact that $T \propto a^{-1}$, the scale factor at recombination was $a_{\text{rec}} = 7.2 \times 10^{-4}$, and the universe was 250000 years old. When the age of the universe was $t_{\text{ls}} = 380000$ yr, a typical photon decoupled from electrons after undergoing its last scattering (mainly Thomson scattering). It was the time when the rate at which photons scatter from electrons becomes smaller than the Hubble parameter. Note that many textbooks may define “recombination” to be this epoch instead of 250000 years after the Big Bang.

- **Radiation-matter equality.** The radiation and matter energy densities are related to the scale factor by $\Omega_r = \Omega_{r,0}a^{-4}$ and $\Omega_m = \Omega_{m,0}a^{-3}$, respectively. The moment at which the energy density of radiation is equal to that of matter is called the **radiation-matter equality**. It occurred at

$$a_{rm} = \frac{\Omega_{r,0}}{\Omega_{m,0}} = \frac{9.0 \times 10^{-5}}{0.31} = 2.9 \times 10^{-4},$$

which corresponds to the age of the universe $t_{rm} = 47000$ yr. The radiation + matter universe model is a good approximation here.

- **Radiation era.** Before the radiation-matter equality, the universe was in a radiation-dominated era. The scale factor grew like $a \sim t^{1/2}$.
- **Big Bang Nucleosynthesis.** The epoch of **Big Bang nucleosynthesis (BBN)** is the nuclear counterpart of recombination. It is the time at which protons and neutrons combine to form nuclei like deuteron (deuterium nucleus) and helium nucleus. In a similar analysis to the reaction $\text{D} + \gamma \leftrightarrow p + n$ and we need Saha-like equations. The temperature at BBN can be calculated as $T_{\text{BBN}} \approx 7.6 \times 10^8$ K. The scale factor was $a_{\text{BBN}} \approx 3.5 \times 10^{-9}$ and the age of the universe was 200 s.
- **Neutrino decoupling/freeze-out.** At even higher temperature, neutrinos were in equilibrium with other matter even though they interact only weakly. The **neutrino freeze-out** time is when the rate of the reaction $n + \nu_e \rightarrow p + e^-$ becomes smaller than the Hubble parameter. The freeze-out temperature was about $T_{\text{freeze}} \sim 9 \times 10^9$ K, corresponding to the scale factor $a_{\text{freeze}} \sim 3 \times 10^{-10}$ and the age of the universe $t_{\text{freeze}} \sim 1$ s. We expect that after neutrinos were decoupled from other matter, they formed a **cosmic neutrino background (C ν B)** like the CMB. However, they cannot be detected by any detection method now because of their low energy. These neutrinos might even be non-relativistic by now because neutrinos do have masses, unlike photons. In the future, if we could detect C ν B, we are then able to infer about the universe just 1 s after the Big Bang!

- **Inflation.** The standard Hot Big Bang model is consistent with many observations, but it is not perfect. There are several “problems” that are logically and mathematically valid, but are usually viewed as “unacceptable coincidence”. The two famous ones are the horizon problem and the flatness problem.

The **horizon problem** says that patches on the surface of last scattering observed from CMB are not causally related ever. For example, consider a patch on the surface of last scattering. Its current proper distance to us is

$$d_p(t_0) = \int_{t_{\text{ls}}}^{t_0} \frac{dt}{a(t)} = 0.98d_{\text{hor}}.$$

This means the patch at $(\theta, \phi) = (\pi/2, 0)$ and the patch at $(\theta, \phi) = (\pi/2, \pi)$ are separated by $1.96d_{\text{hor}} > d_{\text{hor}}$. They are not causally related throughout the history of the universe, yet they have the same temperature, suggesting that they were in thermal equilibrium. It is hard to explain it by a coincidence.

The **flatness problem** says that if the universe is very flat today, $|1 - \Omega_0| \leq 0.005$, it should be flatter in the past. The Friedmann equation is

$$1 - \Omega = -\frac{\kappa}{H^2 a^2 R_0^2}, \quad \text{with} \quad 1 - \Omega_0 = -\frac{\kappa}{H_0^2 R_0^2}.$$

Combining them gives

$$1 - \Omega = \frac{H_0^2(1 - \Omega_0)}{H^2 a^2}.$$

Since we are studying the early universe, consider the Friedmann equation of a radiation + matter universe,

$$\frac{H^2}{H_0^2} = \frac{\Omega_{r,0}}{a^4} + \frac{\Omega_{m,0}}{a^3} \quad \implies \quad 1 - \Omega = \frac{(1 - \Omega_0)a^2}{\Omega_{r,0} + a\Omega_{m,0}}.$$

Clearly if we go back in time when the scale factor gets smaller, $1 - \Omega$ also gets smaller—the universe was flatter in the past. At Planck time $t_P \approx 5 \times 10^{-44}$ s just after the birth of the universe, it would be unbelievably flat, $1 - \Omega_P \leq 2 \times 10^{-62}$.

The theory of **inflation** in modern cosmology solves these problems. The epoch of inflation is a period in the very early universe $t_i \sim 10^{-36}$ s driven by a large cosmological constant, leading to an exponential scale factor $a \propto e^{H_i t}$. How this cosmological constant emerged is yet unknown. After the inflation, this cosmological constant transformed into matter and radiation through a process called **reheating**, with a mechanism also unknown. The precise formalism of inflation is beyond the scope of this notes. The theory is based on general relativity with a scalar field called the **inflaton**. Understanding the universe before inflation is believed to require the theory of quantum gravity.

10 GLOBAL STRUCTURES AND SINGULARITIES

So far we have discussed many important solutions to Einstein's equation: weak field solution, gravitational waves, black holes, and cosmology. In the strong field limit (black holes and cosmology), the exact solutions involve lots of symmetries. In this chapter, we will discuss features of general spacetimes without imposing any symmetry.

10.1 Futures and Pasts

Consider a manifold M on which a metric $g_{\mu\nu}$ is defined, and let $(M, g_{\mu\nu})$ be a spacetime. At each event $p \in M$, we can define light cones of p . One of the light cones is assigned as the “future”, and the other is assigned as the “past”. A spacetime $(M, g_{\mu\nu})$ is said to be **time-orientable** if a continuous choice of one light cone as the future light cone can be made everywhere on M . Let p be a fixed point in a spacetime $(M, g_{\mu\nu})$. Consider any closed curve γ in M that begins and ends at p . Fix a time-orientation at p and carry it continuously along γ until back to p . The final time-orientation is either the same or the opposite as the initial one. If same, we say that γ is **time-preserving**, and if not, then γ is **time-reversing**.

Theorem 10.1.

A spacetime is time-orientable if and only if every closed curve through p is time-preserving.

A differentiable curve $\gamma(t)$ is said to be a **future-directed timelike curve** if at each point $p \in \gamma$ the tangent vector ξ^μ is a future-directed timelike vector. If ξ^μ is null at some point along γ , then γ is said to be a **future-directed causal curve**.

10.1.1 Causality

If a spacetime is **causally violated**, it has solutions with closed timelike curves.

A spacetime is **stably causal** if there exists a timelike vector field ξ^μ such that $\tilde{g}_{\mu\nu} \equiv g_{\mu\nu} - \xi_\mu \xi_\nu$ has no closed timelike curves. If $g_{\mu\nu} \ell^\mu \ell^\nu = 0$, then $\tilde{g}_{\mu\nu} \ell^\mu \ell^\nu = -(\ell \cdot \xi)^2 < 0$.

Example 10.1. Stably causal spacetimes

Schwarzschild, Reissner-Nordström, FRW solutions.

A **time-function** on a spacetime is a scalar field t such that $\nabla_\mu t$ is everywhere timelike. This implies no closed timelike curves since t increases along every future-directed timelike curve.

Theorem 10.2.

A spacetime admits a time-function iff it is stably causal.

Proof. If there exists a time function t , then the new metric $g_{\mu\nu} - \nabla_\mu t \nabla_\nu t$ has no closed timelike curves since t is still a time-function for the new metric. The other way: see Wald. $\square$

10.1.2 Domain of Influence

Let M be a stably causal spacetime and let a point $p \in M$. Define the **chronological future**, denoted $I^+(p)$, to be

$$I^+(p) \equiv \{ \text{all } q \in M \mid \text{There exist a future directed timelike curve from } p \text{ to } q \}.$$

Similar definitions can be made for the **chronological past** of p , denoted $I^-(p)$.

Some properties of $I^\pm(p)$:

- $q \in I^-(p)$ iff $p \in I^+(q)$.
- $q \in I^-(p)$ leads to $I^-(q) \subset I^-(p)$. The proof is to combine timelike curves. The converse is false. Counterexample: q is on the light cone of p .
- $I^\pm(p)$ are open.
- $q \notin I^-(p)$ iff $I^+(q) \cap I^-(p) = \emptyset$

Theorem 10.3.

Through every point on $\partial I^-(p)$, there exists a future-directed null geodesics which stays on boundary and either reaches p or is inextendible.

Proof. To be written. □

Define $I^-[U] = \bigcup_{p \in U} I^-(p)$. An asymptotically flat spacetime contains a **black hole** if $I^-(\mathcal{I}^+) \neq M$. The boundary $\partial I^-(\mathcal{I}^+)$ is an **event horizon**.

10.1.3 Domain of Dependence

Can initial data on Σ determine what happens at $p \in M$?

A spacelike surface Σ is said to be **achronal** if no two points are timelike related. That is, $I^+[\Sigma] \cap \Sigma = \emptyset$. Let Σ be a closed 3-dimensional achronal surface. Define **domain of dependence** $D^+[\Sigma]$ to be

$$D^+[\Sigma] \equiv \{ p \in M \mid \text{every past directed timelike curve that reaches } \Sigma \}.$$

$D^-[\Sigma]$ is defined in a similar way.

Example 10.2.

To be written.

Properties of the domain of dependence:

1. $D^\pm[\Sigma]$ are closed. Proof: let $p \in \partial D^+[\Sigma] - \Sigma$. Suppose $p \notin D^+[\Sigma]$. Then there exists γ that does not reach Σ . Let $r \in \gamma$, and clearly $r \notin D^+[\Sigma]$. Then any point in $I^+(r)$ is not in $D^+[\Sigma]$. But $I^+(r)$ is an open set and contains p , so p is not on the boundary.
2. If $p \in D^+[\Sigma]$ and $q \in I^-(p)$ and $q \in I^+[\Sigma]$, then $q \in D^+[\Sigma]$. Proof: if not, combine curves from q to p and get contradiction.
3. If $p \in \partial D^+[\Sigma] - \Sigma$, then $I^+(p) \cap D^+[\Sigma] = \emptyset$. Proof: if $q \in I^+(p) \cap D^+[\Sigma]$ then $p \in I^-(q)$ and $p \in I^+[\Sigma]$. Then p is in an open set in $D^+[\Sigma]$. This contradicts $p \in \partial D^+[\Sigma] - \Sigma$.

4. If $p \in \partial D^+[\Sigma] - \Sigma$, then there exists a past directed null geodesic from p which stays on the boundary. Proof: let $p_i \notin D^+[\Sigma]$ that converge to p . Let γ_i be timelike curves from p_i that do not reach Σ . γ cannot enter $I^-(p)$. Otherwise some of γ_i can enter $I^-(p)$, and this will contradict property 2. Then γ must be a null geodesic, the same argument as for $I^\pm(p)$. γ stays on $\partial I^-(p) \subset D^+[\Sigma]$, so γ is on $\partial D^+[\Sigma]$.

10.1.4 Cauchy Surface

A **Cauchy surface** is an achronal surface Σ such that $M = D^+[\Sigma] \cup D^-[\Sigma]$.

Theorem 10.4.

Every spacetime with a Cauchy surface is stably causal. Also, there exists a global time function t such that every $t = \text{const.}$ surface is a Cauchy surface, and $M = \Sigma \times \mathbb{R}$.

10.2 Singularity Theorems

10.2.1 Local Argument

$$\theta = \nabla_\alpha t^\alpha.$$

$$\dot{\theta} = t^\nu \nabla_\nu \theta = t^\nu \nabla_\nu (\nabla_\mu t^\mu) = t^\nu (\nabla_\mu \nabla_\nu t^\mu - R_{\mu\nu} t^\mu) = \nabla_\mu (t^\nu \nabla_\nu t^\mu) - \nabla_\mu t^\nu \nabla_\nu t^\mu - R_{\mu\nu} t^\mu t^\nu = -\nabla_\mu t^\nu \nabla_\nu t^\mu - R_{\mu\nu} t^\mu t^\nu.$$

$\nabla_\mu t_\nu$ is symmetric.

Define metric on $t = \text{const.}$ surface, $h_{\mu\nu} = g_{\mu\nu} + t_\mu t_\nu$. Define

$$\sigma_{\mu\nu} = \nabla_\mu t_\nu - \frac{1}{3}\theta h_{\mu\nu}, \quad \sigma^\mu{}_\mu = 0.$$

$$\nabla_\mu t_\nu = \sigma_{\mu\nu} + \frac{1}{3}\theta h_{\mu\nu}$$

$$\nabla_\mu t_\nu \nabla_\mu t^\nu = \sigma_{\mu\nu} \sigma^{\mu\nu} + \frac{1}{3}\theta^2.$$

$$\dot{\theta} = -\frac{1}{3}\theta^2 - \sigma_{\mu\nu} \sigma^{\mu\nu} - R_{\mu\nu} t^\mu t^\nu \tag{10.1}$$

$$R_{\mu\nu} t^\mu t^\nu = 8\pi \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right) t^\mu t^\nu.$$

If the RHS is non-negative for all timelike t^μ , we say that the matter satisfies the **strong energy condition**.

The vacuum and Maxwell field satisfy the strong energy condition. For perfect fluid, the strong energy condition is violated if $\rho + P \geq 0$ and $\rho + 3P \geq 0$. The strong energy condition is violated by a positive cosmological constant, $\Lambda > 0$.

When the strong energy condition is satisfied, it implies that

$$\dot{\theta} + \frac{1}{3}\theta^2 \leq 0 \implies \frac{d}{dt}(\theta^{-1}) \geq \frac{1}{3}.$$

$$\frac{1}{\theta} \geq \frac{1}{\theta_0} + \frac{t}{3}.$$

If $\theta_0 < 0$, then $\theta \rightarrow -\infty$ in finite time. This means the geodesics cross.

10.2.2 Global Argument

Theorem 10.5.

Let Σ be an achronal surface. Through each point p in the interior of $D^+[\Sigma]$, there exists a timelike curve of maximal proper time from p to Σ .

The curve of maximum length is a geodesic, and must be orthogonal to Σ .

10.2.3 Singularity Theorems

Theorem 10.6.

Let Σ be a Cauchy surface in a spacetime satisfying the strong energy condition. Assume that $\theta \leq \theta_0 < 0$ everywhere. Then there exists no future directed timelike curve from Σ with length greater than $3/|\theta_0|$.

Proof. Let $\tilde{\gamma}$ be a timelike curve with length greater than $3/|\theta_0|$. Let γ_1 be a timelike curve of maximal length. From local argument, there is a timelike geodesic γ_2 that cross γ_1 at point r . γ_2 has the same length as γ_1 to second order from Σ to r . $\square$

Theorem 10.7.

Let Σ be a compact achronal surface in a stably causal spacetime satisfying strong energy condition. Assume that $\theta \leq \theta_0 < 0$ everywhere. Then there exists a future directed timelike geodesic with length less than $3/|\theta_0|$.

Proof. Let $H = \partial D^+[\Sigma] - \Sigma$ (the Cauchy horizon). H can't be compact, otherwise there exist past directed null geodesics on H which wrap around and become arbitrarily close to themselves. This violates stable causality.

Let $p_i \in H$ without an accumulation point. Move p_i into the interior of $D^+[\Sigma]$. Let γ_i be curves of maximum length to Σ hit Σ at q_i . Since Σ is compact, there is an accumulation point $q = \lim q_i$. Let γ be a timelike geodesic through q that is orthogonal to Σ . Then γ must have length less than $3/|\theta_0|$ or else it would provide limit point for p_i . $\square$

Consider a family of null geodesics with tangent vector k^μ . If $\xi^\mu k_\mu = 0$, then $\xi^\mu + ck^\mu$ is also orthogonal to k^μ . Choose ℓ^μ such that $\ell^\mu k_\mu = -1$ and $k^\mu \nabla_\mu \ell^\nu = 0$ and require that $\xi^\mu \ell_\mu = 0$ also. Then the metric on the 2-dimensional subspace orthogonal to k^μ and ℓ^μ has the form

$$q_{\mu\nu} = g_{\mu\nu} + \ell_\mu k_\nu + k_\mu \ell_\nu.$$

$$k^\mu \nabla_\mu g_{\alpha\beta} = 0.$$

Let $B_{\mu\nu} = q^\alpha{}_\mu q^\beta{}_\nu \nabla_\alpha k_\beta$. Its trace is

$$q^{\mu\nu} B_{\mu\nu} = \nabla_\mu k^\mu \equiv \theta.$$

$$k^\gamma \nabla_\gamma B_{\alpha\beta} = -B_{\alpha\gamma} B^\gamma{}_\beta + q^\alpha{}_\mu q^\nu{}_\beta k^\gamma R_{\gamma\mu\nu\delta} k^\delta.$$

$$\sigma_{\mu\nu} = B_{\mu\nu} - \frac{1}{2}\theta q_{\mu\nu}.$$

Raychaudhuri equation for null geodesics

$$\dot{\theta} = k^\mu \nabla_\mu \theta = - \left(\frac{1}{2}\theta^2 + \sigma_{\mu\nu}\sigma^{\mu\nu} + R_{\mu\nu}k^\mu k^\nu \right). \quad (10.2)$$

$$R_{\mu\nu}k^\mu k^\nu = 8\pi T_{\mu\nu}k^\mu k^\nu.$$

null energy condition: $T_{\mu\nu}k^\mu k^\nu \geq 0$. It is satisfied by all known forms of classical matter. If the null energy condition is satisfied, then

$$\begin{aligned} \dot{\theta} + \frac{1}{2}\theta^2 \leq 0 &\implies \frac{d}{d\lambda}(\theta^{-1}) \geq \frac{1}{2}. \\ \frac{1}{\theta} &\geq \frac{1}{\theta_0} + \frac{\lambda}{2}. \end{aligned}$$

If $\theta_0 < 0$, then $\theta \rightarrow -\infty$ in finite λ . Null geodesics cross. Broken null curves can be lengthened if you connect the end points of two crossing null geodesics and rounding the corner at the crossing.

A **trapped surface** is a compact spacelike 2-surface such that both outgoing and ingoing orthogonal null geodesics are converging.

If there exists a continuous one-to-one map with continuous inverse $M \rightarrow N$, then M and N are homeomorphic.

Theorem 10.8. Singularity Theorem (Penrose, 1965)

If a spacetime has a non-compact Cauchy surface Σ , a trapped surface T , and satisfies null energy condition, then it must be null geodesic incomplete.

Proof. To be written. □

A singularity in $I^-[\mathscr{J}^+]$ is called **naked**. The conjecture that naked singularities don't exist is called **cosmic censorship**.

If $I^-[\mathscr{J}^+]$ has a Cauchy surface, there exists no naked singularity.

Theorem 10.9.

Given a spacetime such that $I^-[\mathscr{J}^+]$ has a Cauchy surface and satisfies the null energy condition. Then every trapped surface is outside $I^-[\mathscr{J}^+]$.

Theorem 10.10. Hawking's Area Theorem

Given a spacetime that satisfies the null energy condition and either 1. $I^-[\mathscr{J}^+]$ has a Cauchy surface, or 2. null generators of $\partial I^-[\mathscr{J}^+]$ are complete. Then the area of any cross section of event horizon cannot decrease.

11 QUANTUM EFFECTS IN CURVED SPACETIME

To be written.

Acknowledgements

I thank my Physics 131 instructor, professor Omer Blaes, and my Physics 231A and 231B instructor, professor Gary Horowitz.

Reference

A First Course in General Relativity. 2nd ed. (Schutz).

A First Course in General Relativity. 3rd ed. (Schutz).

Gravity: An Introduction to Einstein's General Relativity. (Hartle).

Spacetime and Geometry: An Introduction to General Relativity. (Carroll).

General Relativity. (Wald).

A PROOFS AND DERIVATIONS

A.1 Special Relativity

A.1.1 Lorentz Transformation

To derive the Lorentz transformation, we need to use both postulates of special relativity:

1. The laws of physics take the same form in all inertial reference frames. All inertial frames are fundamentally indistinguishable based on experiments in that frame only.
2. The speed of light (and other massless particles) c is the maximum speed in the universe and has the same value in all inertial reference frames.

Consider two frames S and S' , moving at a velocity v relative to S , with parallel coordinate axes. The x -axis and x' -axis are aligned. When origins of S and S' coincide (call them $\mathcal{O}$ and $\mathcal{O}'$ respectively), the time in both frames is set to zero, $t = t' = 0$. Since the x -axes are aligned, we will ignore y - and z -components here because they are not interesting. Suppose the Lorentz transformation from inertial frame S to another inertial frame S' takes the form

$$t' = f(x, t), \quad x' = g(x, t),$$

and f and g are linearly independent. Consider a free particle moving with constant velocity $v_x = dx/dt$ in S frame. By postulate 1, this particle observed in S' must also move at a constant velocity $v'_x = dx'/dt'$, which means

$$v'_x = \frac{dx'}{dt'} = \frac{dx'/dt}{dt'/dt} = \frac{\frac{\partial g}{\partial x} \frac{dx}{dt} + \frac{\partial g}{\partial t}}{\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial t}} = \frac{\frac{\partial g}{\partial x} v_x + \frac{\partial g}{\partial t}}{\frac{\partial f}{\partial x} v_x + \frac{\partial f}{\partial t}}.$$

Since v_x and v'_x can be arbitrary constants, all partial derivatives must also be constants. This means the Lorentz transformation must be linear:

$$t' = a_s + a_0 t + a_1 x,$$

$$x' = b_s + b_0 t + b_1 x.$$

The goal is to find the constants a_s , a_0 , a_1 , b_s , b_0 , and b_1 . When $\mathcal{O}$ and $\mathcal{O}'$ coincide, we have $x = x' = 0$ at $t' = t = 0$. This synchronization event sets $a_s = b_s = 0$. Now, the frame S' is moving at a velocity v relative to S , then the position of $\mathcal{O}'$ in S is $x = vt$. In S' , its origin $\mathcal{O}'$ is always at $x' = 0$, so

$$0 = x' = b_0 t + b_1(vt) \implies b_0 = -b_1 v \implies x' = b_1(x - vt).$$

Moreover, by symmetry, S is moving at a velocity of $-v$ relative to S' . If the transformation works for all frames (by postulate 1), then we should obtain

$$x = b_1(x' + vt').$$

The transformation equations now look like

$$t' = a_0 t + a_1 x,$$

$$x' = b_1(x - vt),$$

$$x = b_1(x' + vt').$$

It's time to use postulate 2. Suppose a light pulse is emitted at $(t, x) = (0, 0)$ and $(t', x') = (0, 0)$, i.e., at synchronization. By the constancy of the speed of light, the event $(t, x) = (t, ct)$ in S should have coordinates $(t', x') = (t', ct')$ in S' . Thus,

$$x' = ct' = b_1(x - vt) = b_1(ct - vt) = b_1(1 - \beta)ct,$$

$$x = ct = b_1(x' + vt') = b_1(ct' + vt') = b_1(1 + \beta)ct',$$

where $\beta \equiv v/c$. Substituting ct into ct' , we have

$$ct' = b_1^2(1 - \beta^2)ct \implies b_1 = \pm \frac{1}{\sqrt{1 - \beta^2}}.$$

We will keep the positive solution as the negative solution does not satisfy $x = x'$ and $t = t'$ at $v = 0$. This b_1 is the Lorentz factor,

$$\gamma \equiv \frac{1}{\sqrt{1 - \beta^2}}.$$

Using $x' = \gamma(x - vt)$ and $x = \gamma(x' + vt')$ to solve for t' and we will get

$$t' = \gamma t - \gamma \beta (x/c).$$

The Lorentz transformation is

$$\begin{aligned} ct' &= \gamma(ct - \beta x), \\ x' &= \gamma(x - \beta ct), \\ y' &= y, \\ z' &= z. \end{aligned}$$

The line element $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ is invariant under Lorentz transformation:

$$\begin{aligned} -c^2 dt'^2 + dx'^2 + dy'^2 + dz'^2 &= -\gamma^2(c dt - \beta dx)^2 + \gamma^2(dx - \beta c dt)^2 + dy^2 + dz^2 \\ &= \gamma^2(-c^2 dt^2 + 2\beta c dt dx - \beta^2 dx^2 + dx^2 - 2\beta c dt dx + \beta^2 c^2 dt^2) + dy^2 + dz^2 \\ &= \gamma^2 c^2 (\beta^2 - 1) dt^2 + \gamma^2 (1 - \beta^2) dx^2 + dy^2 + dz^2 \\ &= -dt^2 + dx^2 + dy^2 + dz^2. \end{aligned}$$

A.2 Differential Geometry and Curvature

A.2.1 Determinant

To be written.

A.3 The Geodesic Equation

The Lagrangian and Euler-Lagrange equation are given by

$$L = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} = \frac{d\tau}{d\lambda} \quad \text{and} \quad \frac{\partial L}{\partial x^\alpha} - \frac{d}{d\lambda} \frac{\partial L}{\partial(dx^\alpha/d\lambda)} = 0.$$

We want the derivatives,

$$\begin{aligned} \frac{\partial L}{\partial x^\alpha} &= \frac{1}{2L} \left(-\frac{\partial g_{\mu\nu}}{\partial x^\alpha} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right) = -\frac{1}{2} \frac{d\lambda}{d\tau} \left(\frac{\partial g_{\mu\nu}}{\partial x^\alpha} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right) = -\frac{1}{2} \partial_\alpha g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}. \\ \frac{\partial L}{\partial(dx^\alpha/d\lambda)} &= \frac{1}{2L} \left(-2g_{\alpha\nu} \frac{dx^\nu}{d\lambda} \right) = -\frac{d\lambda}{d\tau} g_{\alpha\nu} \frac{dx^\nu}{d\lambda} = -g_{\alpha\nu} \frac{dx^\nu}{d\tau}. \end{aligned}$$

Note the factor of 2 in the first equality comes from

$$\left[-g_{\mu\nu} \frac{\partial}{\partial(dx^\alpha/d\lambda)} \left(\frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right) \right] = -g_{\mu\nu} \delta^\mu_\alpha \frac{dx^\nu}{d\lambda} - g_{\mu\nu} \delta^\nu_\alpha \frac{dx^\mu}{d\lambda} = -g_{\nu\alpha} \frac{dx^\nu}{d\lambda} - g_{\mu\alpha} \frac{dx^\mu}{d\lambda} = -2g_{\nu\alpha} \frac{dx^\nu}{d\lambda}.$$

Multiply $d\lambda/d\tau$ on both sides of the Euler-Lagrange equation

$$\frac{d\lambda}{d\tau} \frac{\partial L}{\partial x^\alpha} - \frac{d}{d\tau} \frac{\partial L}{\partial(dx^\alpha/d\lambda)} = 0.$$

Substituting the derivatives,

$$-\frac{1}{2} \partial_\alpha g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} - \frac{d}{d\tau} \left(-g_{\alpha\nu} \frac{dx^\nu}{d\tau} \right) = 0.$$

Expanding the τ -derivative using the product rule and chain rule,

$$-\frac{1}{2} \partial_\alpha g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} + \partial_\mu g_{\alpha\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + g_{\alpha\nu} \frac{d^2 x^\nu}{d\tau^2} = 0.$$

Multiplying both sides by $g^{\alpha\beta}$, the inverse of $g_{\alpha\beta}$,

$$-\frac{1}{2} g^{\alpha\beta} \partial_\alpha g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + g^{\alpha\beta} \partial_\mu g_{\alpha\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \delta^\beta_\nu \frac{d^2 x^\nu}{d\tau^2} = 0.$$

The second term can be split into two parts,

$$\partial_\mu g_{\alpha\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = \frac{1}{2} \partial_\mu g_{\alpha\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \frac{1}{2} \partial_\nu g_{\alpha\mu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = \frac{1}{2} (\partial_\mu g_{\alpha\nu} + \partial_\nu g_{\alpha\mu}) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau},$$

where we interchanged the dummy indices μ and ν . Hence

$$-\frac{1}{2} g^{\alpha\beta} \partial_\alpha g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \frac{1}{2} (\partial_\mu g_{\alpha\nu} + \partial_\nu g_{\alpha\mu}) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \delta^\beta_\nu \frac{d^2 x^\nu}{d\tau^2} = 0.$$

Rearranging some terms, we get the geodesic equation,

$$\frac{d^2 x^\beta}{d\tau^2} + \frac{1}{2} g^{\alpha\beta} (\partial_\mu g_{\alpha\nu} + \partial_\nu g_{\alpha\mu} - \partial_\alpha g_{\mu\nu}) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0,$$

or equivalently,

$$\boxed{\frac{d^2 x^\alpha}{d\tau^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0, \quad \text{where} \quad \Gamma^\alpha_{\mu\nu} = \frac{1}{2} g^{\alpha\beta} \left(\frac{\partial g_{\beta\nu}}{\partial x^\mu} + \frac{\partial g_{\beta\mu}}{\partial x^\nu} - \frac{\partial g_{\mu\nu}}{\partial x^\beta} \right).}$$

A.3.1 Affinely Parametrized Geodesic Equation

There is a subtlety in the previous derivation, where we let $L = d\tau/d\lambda$. Consider another Lagrangian, which is the minus square of the previous one:

$$L = g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}.$$

This time we do not even mention the proper time $d\tau$, but use the Euler-Lagrange equation to derive the geodesic equation.

The two Lagrangians,

$$L = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} \quad \text{and} \quad L' = \frac{1}{2} g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda},$$

yield two different geodesic equations. Varying L gives the general geodesic equation,

$$\frac{d^2 x^\alpha}{d\lambda^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \propto \frac{dx^\alpha}{d\lambda}.$$

Varying L' gives the affinely parametrized geodesic equation,

$$\frac{d^2 x^\alpha}{d\lambda^2} + \Gamma^\alpha_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0,$$

even though the Lagrangian L seems to be the more physical one.

To be written.

A.4 Gravitational Waves

A.4.1 Circularly Polarized Gravitational Waves

To be written.

A.5 The Schwarzschild Solution

A.5.1 The Periapsis Precession

Direct Integration The periapsis precession integral is

$$\Delta\phi = 2L \int_{r_1}^{r_2} \frac{dr}{r^2} \left(c^2 e^2 - c^2 + \frac{2GM}{r} - \frac{L^2}{r^2} + \frac{2GML^2}{r^3 c^2} \right)^{-1/2}.$$

We restore the G 's and c 's because then we can track the order of the effects. We want the lowest order ($1/c^2$) effects from the integral. Rearrange the integral,

$$\Delta\phi = 2L \int_{r_1}^{r_2} \frac{dr}{r^2} \left[c^2 e^2 - \left(1 - \frac{2GM}{rc^2} \right) \left(c^2 + \frac{L^2}{r^2} \right) \right]^{-1/2} = 2L \int_{r_1}^{r_2} \frac{dr}{r^2} \left(1 - \frac{2GM}{rc^2} \right)^{-1/2} \left[c^2 e^2 \left(1 - \frac{2GM}{rc^2} \right)^{-1} - c^2 - \frac{L^2}{r^2} \right]^{-1/2}.$$

Now expand to leading order of c^2 ,

$$\left(1 - \frac{2GM}{rc^2} \right)^{-1/2} \simeq 1 + \frac{GM}{rc^2}, \quad c^2 e^2 \left(1 - \frac{2GM}{rc^2} \right)^{-1} \simeq c^2 e^2 \left(1 + \frac{2GM}{rc^2} + \frac{4G^2 M^2}{r^2 c^4} \right).$$

Substituting them into $\Delta\phi$,

$$\begin{aligned} \Delta\phi &\simeq 2L \int_{r_1}^{r_2} \frac{dr}{r^2} \left(1 + \frac{GM}{rc^2} \right) \left[c^2 (e^2 - 1) + \frac{2GMe^2}{r} + \left(\frac{4G^2 M^2 e^2}{c^2 L^2} - 1 \right) \frac{L^2}{r^2} \right]^{-1/2} \\ &= 2L \int_{r_1}^{r_2} \frac{dr}{r^2} \left(1 + \frac{GM}{rc^2} \right) \left[2E_{\text{Newt}} + \frac{2GM}{r} \left(1 + \frac{2E_{\text{Newt}}}{c^2} \right) - \frac{L^2}{r^2} \left(1 - \frac{4G^2 M^2 e^2}{c^2 L^2} \right) \right]^{-1/2}, \end{aligned}$$

where $E_{\text{Newt}} = c^2(e^2 - 1)/2$. In fact, if $e^2 = 1 + 2E_{\text{Newt}}/c^2$, the last term in the square bracket will be

$$-\frac{L^2}{r^2} \left(1 - \frac{4G^2 M^2 e^2}{c^2 L^2} \right) = -\frac{L^2}{r^2} \left(1 - \frac{4G^2 M^2}{c^2 L^2} - \frac{8G^2 M^2 E_{\text{Newt}}}{c^4 L^2} \right) \simeq -\frac{L^2}{r^2} \left(1 - \frac{4G^2 M^2}{c^2 L^2} \right),$$

so we have no e 's in all expressions. Making a change of variables $u = 1/r$, $du = -dr/r^2$,

$$\begin{aligned} \Delta\phi &\simeq 2L \int_{u_2}^{u_1} du \left(1 + \frac{GMu}{c^2} \right) \left[2E_{\text{Newt}} + 2GMu \left(1 + \frac{2E_{\text{Newt}}}{c^2} \right) - L^2 u^2 \left(1 - \frac{4G^2 M^2}{c^2 L^2} \right) \right]^{-1/2} \\ &= \frac{2L}{L(1 - 4G^2 M^2/c^2 L^2)^{1/2}} \int_{u_2}^{u_1} du \left(1 + \frac{GMu}{c^2} \right) \left[\frac{2E_{\text{Newt}}}{L^2(1 - 4G^2 M^2/c^2 L^2)} + \frac{2GMu(1 + 2E_{\text{Newt}}/c^2)}{L^2(1 - 4G^2 M^2/c^2 L^2)} - u^2 \right]^{-1/2}. \end{aligned}$$

The terms in the square bracket is a mess, but recall that r_1 and r_2 are periapsis and apoapsis where $dr/d\tau = 0$, while $dr/d\tau$ is just the square bracket (to the power of $-1/2$). Thus, u_1 and u_2 are the roots of this quadratic expression, so it can be written as $(u_1 - u)(u - u_2)$. The integral looks a lot nicer now,

$$\Delta\phi \simeq 2 \left(1 + \frac{2G^2 M^2}{c^2 L^2} \right) \int_{u_2}^{u_1} du \frac{1 + GMu/c^2}{\sqrt{(u_1 - u)(u - u_2)}} = 2 \left(1 + \frac{2G^2 M^2}{c^2 L^2} \right) \int_{u_2}^{u_1} \frac{du}{\sqrt{(u_1 - u)(u - u_2)}} + \frac{2GM}{c^2} \int_{u_2}^{u_1} \frac{u du}{\sqrt{(u_1 - u)(u - u_2)}}.$$

Completing the square,

$$(u_1 - u)(u - u_2) = -u_1 u_2 + (u_1 + u_2)u - u^2 = -u_1 u_2 - \left(u - \frac{u_1 + u_2}{2} \right)^2 + \frac{(u_1 + u_2)^2}{4} = \frac{(u_1 - u_2)^2}{4} - \left(u - \frac{u_1 + u_2}{2} \right)^2.$$

Let

$$u - \frac{u_1 + u_2}{2} = \frac{u_1 - u_2}{2} \sin \theta, \quad du = \frac{u_1 - u_2}{2} \cos \theta d\theta \quad \Rightarrow \quad \left(u - \frac{u_1 + u_2}{2}\right)^2 = \frac{(u_1 - u_2)^2}{4} \sin^2 \theta,$$

and

$$(u_1 - u)(u - u_2) = \frac{(u_1 - u_2)^2}{4} - \frac{(u_1 - u_2)^2}{4} \sin^2 \theta = \left(\frac{u_1 - u_2}{2} \cos \theta\right)^2.$$

As a check if u covers $[u_2, u_1]$, we can find that $u = u_1$ when $\theta = \pi/2$ and $u = u_2$ when $\theta = -\pi/2$. The integrals are now almost trivial because

$$\frac{du}{\sqrt{(u_1 - u)(u - u_2)}} = \frac{[(u_1 - u_2)/2] \cos \theta}{[(u_1 - u_2)/2] \cos \theta} d\theta = d\theta.$$

The first integral becomes

$$\int_{u_2}^{u_1} \frac{du}{\sqrt{(u_1 - u)(u - u_2)}} = \int_{-\pi/2}^{\pi/2} d\theta = \pi.$$

The second integral is

$$\int_{u_2}^{u_1} \frac{u du}{\sqrt{(u_1 - u)(u - u_2)}} = \int_{u_2}^{u_1} u d\theta = \int_{-\pi/2}^{\pi/2} \left(\frac{u_1 + u_2}{2} + \frac{u_1 - u_2}{2} \sin \theta\right) d\theta = \frac{1}{2}(u_1 + u_2)\pi.$$

Substituting these integral into $\Delta\phi$,

$$\Delta\phi \simeq 2\pi \left(1 + \frac{2G^2 M^2}{c^2 L^2}\right) + \frac{GM}{c^2}(u_1 + u_2)\pi.$$

The second term is already in the order $1/c^2$, so we need the sum of the roots in the Newtonian limit of the quadratic equation

$$\frac{2E_{\text{Newt}}}{L^2} + \frac{2GMu}{L^2} - u^2 = 0.$$

The sum of roots of a quadratic in the form $au^2 + bu + c$ is just $-b/a$. In our case, $u_1 + u_2 = 2GM/L^2$, so

$$\Delta\phi \simeq 2\pi \left(1 + \frac{2G^2 M^2}{c^2 L^2}\right) + \frac{2G^2 M^2}{L^2 c^2} \pi = 2\pi + \frac{6G^2 M^2}{c^2 L^2} \pi.$$

In conclusion, the apsidal angle $\delta\phi = \Delta\phi - 2\pi$ is

$$\delta\phi = \frac{6\pi G^2 M^2}{c^2 L^2}.$$

Perturbation in Circular Orbit To be written.

A.5.2 The Shapiro Delay Integral

The goal is to solve the integral

$$t(r, r_1) = \int_{r_1}^r dr' \frac{1}{b} \left(1 - \frac{2M}{r'}\right)^{-1} \left[\frac{1}{b^2} - Q_{\text{eff}}(r')\right]^{-1/2}, \quad \text{where} \quad Q_{\text{eff}}(r) = \frac{1}{r^2} \left(1 - \frac{2M}{r}\right).$$

We want to approximate each factor to first order in M/r or M/r_1 , provided that $r_1 \gg 2M$. The first factor: impact parameter b is related to the radius of closest approach r_1 by

$$\frac{1}{b^2} = \frac{1}{r_1^2} \left(1 - \frac{2M}{r_1}\right) \quad \Rightarrow \quad \frac{1}{b} \simeq \frac{1}{r_1} \left(1 - \frac{M}{r_1}\right).$$

The second factor:

$$\left(1 - \frac{2M}{r}\right)^{-1} \simeq 1 + \frac{2M}{r}.$$

The third factor:

$$\frac{1}{b^2} - Q_{\text{eff}}(r) \simeq \frac{1}{r_{\text{min}}^2} \left(1 - \frac{2M}{r_1}\right) - \frac{1}{r^2} \left(1 - \frac{2M}{r}\right) = \frac{r^2 - r_1^2}{r^2 r_1^2} - \frac{2M(r^3 - r_1^3)}{r^3 r_1^3}.$$

Then

$$\begin{aligned}
\left[\frac{1}{b^2} - Q_{\text{eff}}(r) \right]^{-1/2} &\simeq \frac{rr_1}{\sqrt{r^2 - r_1^2}} \left[1 - \frac{2M(r^3 - r_1^3)}{rr_1(r^2 - r_1^2)} \right]^{-1/2} \\
&\simeq \frac{rr_1}{\sqrt{r^2 - r_1^2}} \left[1 + \frac{M(r^3 - r_1^3)}{rr_1(r^2 - r_1^2)} \right] \\
&= \frac{r'r_1}{\sqrt{r'^2 - r_1^2}} + \frac{M(r^3 - r_1^3)}{(r^2 - r_1^2)^{3/2}}.
\end{aligned}$$

Putting these terms into the integral, and keep terms to the first order in M/r or M/r_1 ,

$$\begin{aligned}
t(r, r_1) &\simeq \int_{r_1}^r \frac{1}{r_1} \left(1 - \frac{M}{r_1} \right) \left(1 + \frac{2M}{r'} \right) \left[\frac{r'r_1}{\sqrt{r'^2 - r_1^2}} + \frac{M(r'^3 - r_1^3)}{(r'^2 - r_1^2)^{3/2}} \right] dr' \\
&= \int_{r_1}^r dr' \left[\frac{r'}{\sqrt{r'^2 - r_1^2}} \left(1 - \frac{M}{r_1} \right) + \frac{2M}{\sqrt{r'^2 - r_1^2}} + \frac{M(r'^3 - r_1^3)}{r_1(r'^2 - r_1^2)^{3/2}} \right].
\end{aligned}$$

The last term can be simplified as

$$\begin{aligned}
\frac{M(r'^3 - r_1^3)}{r_1(r'^2 - r_1^2)^{3/2}} &= \frac{M}{r_1 \sqrt{r'^2 - r_1^2}} \left(\frac{r'^3 - r_1^3}{r'^2 - r_1^2} \right) \\
&= \frac{M}{r_1 \sqrt{r'^2 - r_1^2}} \left[\frac{r'(r'^2 - r_1^2) + r_1^2(r' - r_1)}{r'^2 - r_1^2} \right] \\
&= \frac{Mr'}{r_1 \sqrt{r'^2 - r_1^2}} + \frac{Mr_1}{(r' + r_1) \sqrt{r'^2 - r_1^2}}.
\end{aligned}$$

Substituting it back into the integral cancels one term:

$$t(r, r_1) \simeq \int_{r_1}^r dr' \left[\frac{r'}{\sqrt{r'^2 - r_1^2}} \left(1 - \cancel{\frac{M}{r_1}} \right) + \frac{2M}{\sqrt{r'^2 - r_1^2}} + \cancel{\frac{Mr'}{r_1 \sqrt{r'^2 - r_1^2}}} + \frac{Mr_1}{(r' + r_1) \sqrt{r'^2 - r_1^2}} \right].$$

Now we can evaluate each term one by one: the first integral is

$$\int_{r_1}^r \frac{r' dr'}{\sqrt{r'^2 - r_1^2}} = \frac{1}{2} \int_{r_1}^r \frac{d(r'^2)}{\sqrt{r'^2 - r_1^2}} = \sqrt{r'^2 - r_1^2} \Big|_{r_1}^r = \sqrt{r^2 - r_1^2}.$$

The second integral is

$$\int_{r_1}^r dr' \frac{2M}{\sqrt{r'^2 - r_1^2}}.$$

Let $r' = r_1 \cosh \theta$, $dr' = r_1 \sinh \theta d\theta$. Using the identity $\cosh^2 \theta - 1 = \sinh^2 \theta$,

$$\int_{r_1}^r dr' \frac{2M}{\sqrt{r'^2 - r_1^2}} = 2M \int_0^{\cosh^{-1}(r/r_1)} \frac{r_1 \sinh \theta}{r_1 \sinh \theta} d\theta = 2M \cosh^{-1} \left(\frac{r}{r_1} \right).$$

The $\cosh^{-1}$ can be written in terms of a natural log,

$$\int_{r_1}^r dr' \frac{2M}{\sqrt{r'^2 - r_1^2}} = 2M \ln \left(\frac{r}{r_1} + \sqrt{\frac{r^2}{r_1^2} - 1} \right) = 2M \ln \left(\frac{r + \sqrt{r^2 - r_1^2}}{r_1} \right).$$

The third integral is

$$\int_{r_1}^r dr' \frac{Mr_1}{(r' + r_1) \sqrt{r'^2 - r_1^2}}.$$

It turns out that the integrand can be written as a total derivative,

$$\frac{r_1}{\sqrt{r^2 - r_1^2}(r + r_1)} = \frac{r_1}{\sqrt{r - r_1}(r + r_1)^{3/2}} = \frac{1}{2} \left(\frac{r + r_1}{r - r_1} \right)^{1/2} \left[\frac{r + r_1 - (r - r_1)}{(r + r_1)^2} \right] = \frac{d}{dr} \left(\frac{r - r_1}{r + r_1} \right)^{1/2}.$$

Hence

$$\int_{r_1}^r dr' \frac{Mr_1}{(r' + r_1) \sqrt{r'^2 - r_1^2}} = M \left(\frac{r - r_1}{r + r_1} \right)^{1/2}.$$

The Shapiro delay is

$$t(r, r_1) \simeq \sqrt{r^2 - r_1^2} + 2M \ln \left(\frac{r + \sqrt{r^2 - r_1^2}}{r_1} \right) + M \left(\frac{r - r_1}{r + r_1} \right)^{1/2}.$$

A.5.3 Static, Spherically Symmetric Perfect Fluid; Conservation of Stress-Energy Tensor

The stress-energy tensor for a static, spherically symmetric perfect fluid is (7.18)

$$T_{\mu\nu} = \begin{bmatrix} \rho e^{2\alpha} & & & \\ & P e^{2\beta} & & \\ & & P r^2 & \\ & & & P r^2 \sin^2 \theta \end{bmatrix}, \quad T^{\mu\nu} = \begin{bmatrix} \rho e^{-2\alpha} & & & \\ & P e^{-2\beta} & & \\ & & P r^{-2} & \\ & & & P r^{-2} \sin^{-2} \theta \end{bmatrix}.$$

Everything in the stress-energy tensor depends only on r . For reference, the nonzero Christoffel symbols are

$$\begin{aligned} \Gamma_{tr}^t &= \partial_r \alpha \\ \Gamma_{tt}^r &= e^{2(\alpha-\beta)} \partial_r \alpha & \Gamma_{rr}^r &= \partial_r \beta & \Gamma_{\theta\theta}^r &= -r e^{-2\beta} & \Gamma_{\phi\phi}^r &= -r e^{-2\beta} \sin^2 \theta \\ \Gamma_{r\theta}^\theta &= \frac{1}{r} & \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta \\ \Gamma_{r\phi}^\phi &= \frac{1}{r} & \Gamma_{\theta\phi}^\phi &= \cot \theta. \end{aligned}$$

The conservation of stress-energy tensor $\nabla_\mu T^{\mu\nu} = 0$ can be written in terms of the Christoffel symbols,

$$\partial_\mu T^{\mu\nu} + \Gamma_{\mu\alpha}^\mu T^{\alpha\nu} + \Gamma_{\mu\alpha}^\nu T^{\mu\alpha} = 0.$$

For $\nu = t$, the equation is trivial:

$$\partial_t T^{tt} + \Gamma_{\mu t}^\mu T^{tt} + \Gamma_{\mu\alpha}^t T^{\mu\alpha} = 0$$

because each term is zero. For $\nu = \theta$, the equation is

$$\partial_\theta T^{\theta\theta} + \Gamma_{\mu\theta}^\mu T^{\theta\theta} + \Gamma_{\mu\alpha}^\theta T^{\mu\alpha} = 0.$$

The first term is zero, while the second and third terms cancel each other,

$$\Gamma_{\phi\theta}^\phi T^{\theta\theta} + \Gamma_{\phi\phi}^\theta T^{\phi\phi} = \cot \theta (P r^{-2}) + \sin \theta \cos \theta (P r^{-2} \sin^{-2} \theta) = 0,$$

so the $\nu = \theta$ equation is also trivial. For $\nu = \phi$,

$$\partial_\phi T^{\phi\phi} + \Gamma_{\mu\phi}^\mu T^{\phi\phi} + \Gamma_{\mu\alpha}^\phi T^{\mu\alpha} = 0$$

is trivial because again, each term is zero. Finally, the nontrivial equation is the $\nu = r$ equation. We calculate $\nabla_r T^{rr}$,

$$\begin{aligned} \nabla_r T^{rr} &= \partial_r T^{rr} + \Gamma_{\mu r}^\mu T^{rr} + \Gamma_{\mu\alpha}^r T^{\mu\alpha} \\ &= \partial_r T^{rr} + (\Gamma_{tr}^t + \Gamma_{rr}^r + \Gamma_{\theta r}^\theta + \Gamma_{\phi r}^\phi) T^{rr} + \Gamma_{tt}^r T^{tt} + \Gamma_{rr}^r T^{rr} + \Gamma_{\theta\theta}^r T^{\theta\theta} + \Gamma_{\phi\phi}^r T^{\phi\phi} \\ &= \partial_r (P e^{-2\beta}) + \left(\partial_r \alpha + \partial_r \beta + \frac{1}{r} + \frac{1}{r} \right) P e^{-2\beta} + e^{2(\alpha-\beta)} \partial_r \alpha (\rho e^{-2\alpha}) + \partial_r \beta (P e^{-2\beta}) - r e^{-2\beta} (P r^{-2}) - r e^{-2\beta} \sin^2 \theta (P r^{-2} \sin^{-2} \theta) \\ &= \frac{dP}{dr} e^{-2\beta} - 2P e^{-2\beta} \partial_r \beta + \left(\partial_r \alpha + \partial_r \beta + \frac{2}{r} \right) P e^{-2\beta} + \rho e^{-2\beta} \partial_r \alpha + P e^{-2\beta} \partial_r \beta - \frac{2}{r} P e^{-2\beta} \\ &= e^{-2\beta} \left[\frac{dP}{dr} + (\rho + P) \partial_r \alpha \right]. \end{aligned}$$

Setting this equal to zero gives

$$(\rho + P) \frac{d\alpha}{dr} = -\frac{dP}{dr}.$$

A.6 Black Holes

A.6.1 Kruskal-Szekeres Metric

The coordinates are defined as

$$u = \left(\frac{r}{2M} - 1\right)^{1/2} e^{r/4M} \cosh\left(\frac{t}{4M}\right), \quad v = \left(\frac{r}{2M} - 1\right)^{1/2} e^{r/4M} \sinh\left(\frac{t}{4M}\right),$$

for $r > 2M$ and

$$u = \left(1 - \frac{r}{2M}\right)^{1/2} e^{r/4M} \sinh\left(\frac{t}{4M}\right), \quad v = \left(1 - \frac{r}{2M}\right)^{1/2} e^{r/4M} \cosh\left(\frac{t}{4M}\right),$$

for $r < 2M$. The implicit definition of r and t in terms of u and v are

$$\left(\frac{r}{2M} - 1\right) e^{r/2M} = u^2 - v^2, \quad \tanh\left(\frac{t}{4M}\right) = \begin{cases} v/u, & r > 2M, \\ u/v, & r < 2M. \end{cases}$$

Differentiating the r -equation on both sides:

$$\left[\frac{1}{2M} + \left(\frac{r}{2M} - 1\right) \frac{1}{2M}\right] e^{r/2M} dr = 2u du - 2v dv,$$

$$dr = \frac{8M^2}{r} e^{-r/2M} (u du - v dv).$$

This works for all $r \in (0, \infty)$.

The t -equation has two cases, $r > 2M$ and $r < 2M$, but we will find that they produce the same result. For $r > 2M$,

$$\frac{dt}{4M \cosh^2(t/4M)} = \frac{u dv - v du}{u^2} = \frac{u dv - v du}{(r/2M)(1 - 2M/r) e^{r/2M} \cosh^2(t/4M)},$$

$$dt = \frac{8M^2}{r} e^{-r/2M} \left(1 - \frac{2M}{r}\right)^{-1} (u dv - v du).$$

For $r < 2M$,

$$\frac{dt}{4M \cosh^2(t/4M)} = \frac{v du - u dv}{v^2} = \frac{v du - u dv}{(r/2M)(2M/r - 1) e^{r/2M} \cosh^2(t/4M)},$$

$$dt = \frac{8M^2}{r} e^{-r/2M} \left(1 - \frac{2M}{r}\right)^{-1} (u dv - v du),$$

which is the same as in $r > 2M$. We will focus on the r and t (u and v) part of the metric because the θ and ϕ part are the same as in Schwarzschild coordinates. Computing the line element,

$$\begin{aligned} -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 &= \left(1 - \frac{2M}{r}\right)^{-1} \frac{64M^4}{r^2} e^{-r/2M} (-u^2 dv^2 - v^2 du^2 + 2uv du dv + u^2 du^2 + v^2 dv^2 - 2uv du dv) \\ &= \left(1 - \frac{2M}{r}\right)^{-1} \frac{64M^4}{r^2} e^{-r/2M} [(u^2 - v^2)(-dv^2 + du^2)] \\ &= \left(1 - \frac{2M}{r}\right)^{-1} \frac{64M^4}{r^2} e^{-r/2M} \frac{r}{2M} \left(1 - \frac{2M}{r}\right) e^{r/2M} (-dv^2 + du^2) \\ &= \frac{32M^3}{r} e^{-r/2M} (-dv^2 + du^2). \end{aligned}$$

A.6.2 Interesting Things about Black Holes

Homogeneous, Isotropic Spacetime. We keep saying that the Schwarzschild coordinates are not good coordinates. In fact, they behave well except only at $r = 2M$ and $r = 0$. The Schwarzschild coordinates are good coordinates inside the black hole. Let's look at the Schwarzschild metric,

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2.$$

Inside the black hole, $0 < r < 2M$, t becomes a spacelike coordinate while r becomes a timelike one. If we see t as “space” inside the black hole, then the metric is not only spherically symmetric, but also spatially *homogeneous* in space because the metric has no t -dependence.

Can You See Your Feet? Consider the following question: you are at the edge of a black hole. Then your feet passes through the event horizon $r = 2M$ while your head is still outside. Can you see your feet at this moment?

The answer is yes. It is very clear if we draw spacetime diagram of a person in Kruskal-Szekeres coordinates. To be written.

Crushing a Rocket. To be written.

To be written more.

A.6.3 The Reissner-Nordström Metric

Here we derive the Reissner-Nordström metric with the assumption of

- The spacetime is static and spherically symmetric.
- The only nonzero components of the Maxwell field tensor are $F_{tr} = -F_{rt}$. That is, there is no magnetic field present, and the only component of the electric field is E_r , the radial component. However, we do not assume that F_{tr} is static or spherically symmetric.

The general static, spherically symmetric metric has the form

$$ds^2 = -f(r)dt^2 + h(r)dr^2 + r^2d\Omega^2.$$

The Christoffel symbols derived from this metric are

$$\begin{aligned} \Gamma^t_{tr} &= \frac{f'}{2f} \\ \Gamma^r_{tt} &= \frac{f'}{2h} & \Gamma^r_{rr} &= \frac{h'}{2h} & \Gamma^r_{\theta\theta} &= -\frac{r}{h} & \Gamma^r_{\phi\phi} &= -\frac{r \sin^2 \theta}{h} \\ \Gamma^\theta_{r\theta} &= \frac{1}{r} & \Gamma^\theta_{\phi\phi} &= -\sin \theta \cos \theta \\ \Gamma^\phi_{r\phi} &= \frac{1}{r} & \Gamma^\phi_{\theta\phi} &= \cot \theta, \end{aligned}$$

where primes denote derivatives with respect to r . Other Christoffel symbols are either zero or related by symmetry.

The first step is to solve Maxwell’s equations in this background. If there is no external charge coupling to the electromagnetic field ($J^\mu = 0$), then Maxwell’s equations in curved spacetime are

$$g^{\mu\nu} \nabla_\mu F_{\nu\alpha} = 0 \quad \text{and} \quad \nabla_{[\mu} F_{\nu\alpha]} = 0.$$

In terms of Christoffel symbols, the two equations are

$$\begin{cases} g^{\mu\nu} (\partial_\mu F_{\nu\alpha} - \Gamma^\beta_{\mu\nu} F_{\beta\alpha} - \Gamma^\beta_{\mu\alpha} F_{\nu\beta}) = 0 & \text{(I)} \\ \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} + \partial_\alpha F_{\mu\nu} = 0 & \text{(II)}. \end{cases}$$

The second equation comes from the fact that $\nabla_{[\mu} F_{\nu\alpha]} = \partial_{[\mu} F_{\nu\alpha]}$ (4.10), then we use $F_{\mu\nu} = -F_{\nu\mu}$. From equation (I), the $\alpha = t$

equation is

$$\begin{aligned}
0 &= g^{\mu\nu} \partial_\mu F_{\nu t} - g^{\mu\nu} \Gamma_{\mu\nu}^\beta F_{\beta t} - g^{\mu\nu} \Gamma_{\mu t}^\beta F_{\nu\beta} \\
&= g^{rr} \partial_r F_{rt} - g^{\mu\nu} \Gamma_{\mu\nu}^r F_{rt} - g^{\mu r} \Gamma_{\mu t}^r F_{rt} - g^{\mu t} \Gamma_{\mu t}^r F_{tr} \\
&= g^{rr} \partial_r F_{rt} - (\cancel{g^{tt} \Gamma_{tt}^r} + g^{rr} \Gamma_{rr}^r + g^{\theta\theta} \Gamma_{\theta\theta}^r + g^{\phi\phi} \Gamma_{\phi\phi}^r) F_{rt} - g^{rr} \Gamma_{rt}^t F_{rt} - \cancel{g^{tt} \Gamma_{tt}^r F_{tr}} \\
&= \frac{1}{h} \partial_r F_{rt} - \left[\frac{1}{h} \frac{h'}{2h} + \frac{1}{r^2} \left(-\frac{r}{h} \right) + \frac{1}{r^2 \sin^2 \theta} \left(-\frac{r \sin^2 \theta}{h} \right) \right] F_{rt} - \frac{1}{h} \frac{f'}{2f} F_{rt}.
\end{aligned}$$

Simplifying, we get our main equation to solve:

$$\partial_r F_{rt} = \left(\frac{f'}{2f} + \frac{h'}{2h} - \frac{2}{r} \right) F_{rt}. \quad (\text{A.1})$$

Before we solve (A.1), we need some supplemental results from other equations to turn this partial derivative into a total derivative in r . The $\alpha = r$ equation of (I) is

$$\begin{aligned}
0 &= g^{\mu\nu} \partial_\mu F_{\nu r} - g^{\mu\nu} \Gamma_{\mu\nu}^\beta F_{\beta r} - g^{\mu\nu} \Gamma_{\mu r}^\beta F_{\nu\beta} \\
&= g^{tt} \partial_t F_{tr} - g^{\mu\nu} \Gamma_{\mu\nu}^t F_{tr} - g^{\mu t} \Gamma_{\mu r}^t F_{tr} - g^{\mu r} \Gamma_{\mu r}^t F_{rt} \\
&= g^{tt} \partial_t F_{tr} - 2g^{tr} \Gamma_{tr}^t F_{tr} - g^{rt} \Gamma_{\mu r}^r F_{tr} - g^{tr} \Gamma_{tr}^t F_{rt} \\
&= \partial_t F_{tr}.
\end{aligned}$$

This shows that F_{tr} is time-independent. The $\alpha = \theta$ and $\alpha = \phi$ equations are trivial and do not provide additional information. Then we look at equation (II): $\partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} + \partial_\alpha F_{\mu\nu} = 0$. Setting $\alpha = t$ and $\nu = r$ gives

$$0 = \partial_\mu F_{rt} + \partial_r F_{t\mu} + \partial_t F_{\mu r} = 0.$$

Further setting $\mu = \theta$ yields $\partial_\theta F_{tr} = 0$, and setting $\mu = \phi$ yields $\partial_\phi F_{tr} = 0$. Thus, even though we did not assume F_{tr} to be static or spherically symmetric, it turns out that a static, spherically symmetric metric requires it to be so. Now back to (A.1), the partial derivative can be turned into a total derivative. We are left with an ordinary differential equation:

$$\frac{d}{dr} F_{rt} = \left(\frac{f'}{2f} + \frac{h'}{2h} - \frac{2}{r} \right) F_{rt}.$$

This equation is straightforward to solve. Notice that the parenthesis on the RHS can be written as a total derivative:

$$\frac{f'}{2f} + \frac{h'}{2h} - \frac{2}{r} = \frac{d}{dr} \left(\frac{1}{2} \ln f + \frac{1}{2} \ln h - 2 \ln r \right) = \frac{d}{dr} \ln \left(\frac{\sqrt{fh}}{r^2} \right).$$

Thus, we can integrate F_{rt} directly to get

$$F_{rt} = A e^{\ln(\sqrt{fh}/r^2)} = A \frac{\sqrt{fh}}{r^2}$$

for some constant A .

With the Maxwell field $F_{\mu\nu}$, the second step is to find the stress-energy tensor and solve the full Einstein's equation. The stress-energy tensor of electromagnetic field is

$$T_{\mu\nu} = \frac{1}{4\pi} \left(F_{\mu\alpha} F_{\nu}^{\alpha} - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right).$$

(There are different conventions to define the stress-energy tensor of electromagnetic field. Some drops the factor of $1/4\pi$, just like different conventions for electric field, e.g. $E = Q/4\pi r^2$ vs. $E = Q/r^2$. We adopt the convention $E = Q/r^2$, so there is a $1/4\pi$ in the stress-energy tensor.) The tensors F_{ν}^{α} and $F^{\alpha\beta}$ can be found to be

$$\begin{aligned}
F_{\nu}^{\alpha} = F_{\nu\beta} g^{\beta\alpha} &= \begin{bmatrix} 0 & F_{tr} & 0 & 0 \\ -F_{tr} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1/f & 0 & 0 & 0 \\ 0 & 1/h & 0 & 0 \\ 0 & 0 & 1/r^2 & 0 \\ 0 & 0 & 0 & 1/r^2 \sin^2 \theta \end{bmatrix} = \begin{bmatrix} 0 & F_{tr}/h & 0 & 0 \\ F_{tr}/f & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad (\text{row} = \nu, \text{ column} = \alpha). \\
F^{\alpha\beta} = g^{\alpha\mu} F_{\mu}^{\beta} &= \begin{bmatrix} -1/f & 0 & 0 & 0 \\ 0 & 1/h & 0 & 0 \\ 0 & 0 & 1/r^2 & 0 \\ 0 & 0 & 0 & 1/r^2 \sin^2 \theta \end{bmatrix} \begin{bmatrix} 0 & F_{tr}/h & 0 & 0 \\ F_{tr}/f & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \frac{1}{fh} \begin{bmatrix} 0 & -F_{tr} & 0 & 0 \\ F_{tr} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad (\text{row} = \alpha, \text{ column} = \beta).
\end{aligned}$$

Then compute $F_{\alpha\beta}F^{\alpha\beta}$, $F_{\mu\alpha}F_{\nu}^{\alpha}$, and the stress-energy tensor $T_{\mu\nu}$,

$$F_{\alpha\beta}F^{\alpha\beta} = F_{tr}F^{tr} + F_{rt}F^{rt} = -\frac{2F_{tr}^2}{fh}.$$

$$F_{\mu\alpha}F_{\nu}^{\alpha} = -F_{\nu}^{\alpha}F_{\alpha\mu} = -\begin{bmatrix} 0 & F_{tr}/h & 0 & 0 \\ F_{tr}/f & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & F_{tr} & 0 & 0 \\ -F_{tr} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \frac{F_{tr}^2}{fh} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & -h & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{\mu\nu} = \frac{F_{tr}^2}{4\pi fh} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & -h & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \frac{F_{tr}^2}{8\pi fh} \begin{bmatrix} -f & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix} = \frac{F_{tr}^2}{8\pi fh} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & -h & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix} = \frac{A^2}{8\pi r^4} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & -h & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}.$$

One can check explicitly that this stress-energy tensor is traceless, $T^{\mu}_{\mu} = 0$, as it must be for electromagnetic field. The nonzero components of the Einstein tensor are

$$G_{tt} = \frac{f}{r^2 h^2} (rh' + h^2 - h)$$

$$G_{rr} = \frac{1}{r^2 f} (rf' - fh + f)$$

$$G_{\theta\theta} = -\frac{r}{4f^2 h^2} [rf'h + 2f^2 h' + f(rf'h' - 2f'h - 2rf''h)]$$

$$G_{\phi\phi} = G_{\theta\theta} \sin^2 \theta.$$

We are ready to solve the full Einstein's equation and obtain the Reissner-Nordström metric. Setting up Einstein's equation by $G_{\mu\nu} = 8\pi T_{\mu\nu}$. The tt -, rr -, and $\theta\theta$ -equations are

$$A^2 = \frac{r^2}{h^2} (rh' + h^2 - h),$$

$$A^2 = -\frac{r^2}{fh} (rf' - fh + f),$$

$$A^2 = -\frac{r^3}{4f^2 h^2} [rf'^2 h + 2f^2 h' + f(rf'h' - 2f'h - 2rf''h)].$$

The $\phi\phi$ -equation does not provide any new information. Combining the tt -equation and rr -equation, we find that

$$\frac{1}{h} (rh' + h^2 - h) = -\frac{1}{f} (rf' - fh + f) \implies (fh)' = 0.$$

This means $fh = B$ for some constant B . Such a constant can be removed by a change of coordinate $t \rightarrow B^{1/2}t$. For convenience, we set $B = 1$, and so $h = 1/f$ and $h' = -f'/f^2$. Plugging these results into the $\theta\theta$ -equation,

$$\frac{1}{2} r^4 f'' + r^3 f' = A^2.$$

This inhomogeneous equation is easy to solve with an ansatz that $f = \sum_n a_n r^n$. The homogeneous solution f_h and the inhomogeneous (particular) solution f_p are

$$f_h = C + \frac{D}{r}, \quad f_p = \frac{A^2}{r^2}.$$

Altogether, the function f takes the form

$$f(r) = C + \frac{D}{r} + \frac{A^2}{r^2}.$$

This function should also satisfy the other two Einstein's equations. Substituting $f(r)$ and $fh = 1$ into the rr -equation gives

$$A^2 = -r^2 \left[r \left(-\frac{D}{r^2} - \frac{2A^2}{r^3} \right) - 1 + C + \frac{D}{r} + \frac{A^2}{r^2} \right] \implies C = 1.$$

The tt -equation is then automatically satisfied. Finally, we should match these constants to physical quantities by analyzing asymptotic behaviors. Asymptotically, $f(r) = 1 + D/r + \mathcal{O}(r^{-2})$. Matching it to $g_{tt} = 1 - 2\Phi$, where $\Phi = M/r$, in the Newtonian weak-field metric, we identify the parameter $D = -2M$, where M is the total mass of the black hole. On the other hand, with $fh = 1$, the

component $F_{rt} = A/r^2 = E$ is the electric field produced by the black hole. Therefore, we identify the parameter A as the total charge Q of the black hole. In conclusion,

$$f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2},$$

and the metric is the Reissner-Nordström metric for a charged black hole:

$$ds^2 = - \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right) dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right)^{-1} dr^2 + r^2 d\Omega^2.$$

B CURVATURE TENSORS

REFERENCE

Many textbooks define the curvature tensors differently. This notes follows Carroll - Spacetime and Geometry: An Introduction to General Relativity. xAct package for Mathematica references to Wald - General Relativity.

Wald - General Relativity: defined by dual vector fields ω_β ,

$$\boxed{[\nabla_\mu, \nabla_\nu]\omega_\beta = R_{\mu\nu\beta}{}^\alpha \omega_\alpha} \quad [\nabla_\mu, \nabla_\nu]v^\alpha = -R_{\mu\nu\beta}{}^\alpha v^\beta.$$

Coordinate representation:

$$R^\alpha{}_{\beta\mu\nu} = \partial_\nu \Gamma^\alpha{}_{\mu\beta} - \partial_\mu \Gamma^\alpha{}_{\nu\beta} + \Gamma^\alpha{}_{\mu\lambda} \Gamma^\lambda{}_{\nu\beta} - \Gamma^\alpha{}_{\nu\lambda} \Gamma^\lambda{}_{\mu\beta}.$$

Carroll - Spacetime and Geometry: An Introduction to General Relativity: defined by vector fields v^α ,

$$\boxed{[\nabla_\mu, \nabla_\nu]\omega_\beta = -R^\alpha{}_{\mu\nu\beta} \omega_\alpha} \quad [\nabla_\mu, \nabla_\nu]v^\alpha = R^\alpha{}_{\beta\mu\nu} v^\beta.$$

Coordinate representation:

$$R^\alpha{}_{\beta\mu\nu} = \partial_\mu \Gamma^\alpha{}_{\nu\beta} - \partial_\nu \Gamma^\alpha{}_{\mu\beta} + \Gamma^\alpha{}_{\mu\lambda} \Gamma^\lambda{}_{\nu\beta} - \Gamma^\alpha{}_{\nu\lambda} \Gamma^\lambda{}_{\mu\beta}.$$

Most textbooks have definitions the same as that in Carroll's book. The Ricci tensors are also defined differently, but the results are the same. (As they should be, because they appear in Einstein's equations.)